

8		7		6		5		4		3		2		1		CHINAFOX
BOM Variants																
BOM NUMBER		BOM NAME				BOM OPTIONS										
685-00076		COMMON PARTS,MLB-BAFFIN,X363				X363_COMMON										
985-00126		DEV,MLB-BAFFIN,X363				X363_DEVEL:PVT										
985-00232		DEV,MLB-BAFFIN,PVT,X363				X363_DEVEL:PVT										
639 BOMs have been moved to the end of the schematic																
X363 BOM Groups																
BOM GROUP		BOM OPTIONS														
X363_COMMON		ALTERNATE,COMMON,X363_COMMON1,X363_COMMON2,X363_COMMON3,X363_COMMON4,X363_PROGPARTS														
X363_COMMON1		SOC:HYNIX,SE:PROD,SKIP_5V3V3:AUDIBLE,DIPLEXER:MURATA,T208_PROG:REV5,BOARD_ID:17,VCHDA:S0														
X363_COMMON2		XDP:YES,SAMCONN,SOC_BOOT:SPI,DPMUX_XTAL:NO,GPUCLK:OSC,BAFFIN,AP_TEMP,VCCPLLLOC:S3,WIFI_SAK:NO														
X363_COMMON3		CPUTHRM:ALRT,TBTTHRM:ALRT,LOADRC:NO,OTHERRC:YES,DDRRC:YES,TBTRC:YES,TPADRC:YES,LID_FEATURE_ON														
X363_COMMON4		EDP:YES,CPUPEG:X8X4X4,TBTTHRM_SNS,GPUTHRM_SNS,S3_STATE:YES,GPU_ROM:YES,SVID_PU:CORE														
X363_PROGPARTS		BOOTROM_PROG:DVT,BT_PROG:DVT,WIFI_PROG:DVT,UPCROM_PROG:DVT,SMC_PROG:PVT,DPMUXMCU:PROG,PCC:NO														
X363_DEVEL:ENG		ALTERNATE,ENGISNS,DBGLED,XDP_CONN,USBC_DBG,DBG_BTN,DBG_FAN,DBG_XTAL,DPMUX_DEBUG,WIFI_DBG,SSD_DEBUG,GPUROM:BLANK,PCC:YES														
X363_DEVEL:DVT		ALTERNATE,ENGISNS,DBGLED,XDP_CONN,USBC_DBG,DBG_BTN,DBG_FAN,DBG_XTAL,DPMUX_DEBUG,WIFI_DBG,SSD_DEBUG														
X363_DEVEL:PVT		ALTERNATE,XDP_CONN,USBC_DBG														
ENGISNS		TBTISNS,LOADISNS,TPADISNS,DDRISNS,OTHERISNS														
Module Parts																
PART NUMBER		QTY	DESCRIPTION				REFERENCE DES		CRITICAL		BOM OPTION					
337S00227		1	CPU,SKY,SR2FT,R1,PRQ,4/2,2.9,BGA1440				U0500		CRITICAL		CPU_SKL:2.9					
337S00228		1	CPU,SKY,SR2FU,R1,PRQ,4/2,2.7,BGA1440				U0500		CRITICAL		CPU_SKL:2.7					
337S00229		1	CPU,SKY,SR2FQ,R1,PRQ,4/2,2.6,BGA1440				U0500		CRITICAL		CPU_SKL:2.6					
998-04701		1	INTERPOSER,INTEL,BGA1440,MM940989				U0500		CRITICAL		CPU_SKL:SOCKET					
337S00258		1	IC,SKL_PCH-H,SFF,SR2NH,PRQ,D1,BGA939				U1100		CRITICAL							
353S00961		4	IC,CD3215,ACE,C0,USB_PWR_SW,BLNK,BGA96				U3100,U3200,UB300,UB400		CRITICAL							
338S00254		2	IC,TBT,ALPINE_RIDGE_DP,QT55,Q5,C1,BGA337				U2800,U0000		CRITICAL							
353S01016		1	IC,ISL9239HIZ,PMU,TUBA,WCP40,2.1X3.3MM				U7000		CRITICAL							
338S00221		1	IC,PMU,P50839,7X7MM,BGA168				U7800		CRITICAL							
338S00142		1	IC,CODEC,CLIFDEN,CS42L83A,B0,MLCSP40				U6300		CRITICAL							
337S00330		1	IC,GPU,BAFFIN,ULA,A1,PS,BGA769				UA000		CRITICAL		BAFFIN_ULA					
337S00331		1	IC,GPU,BAFFIN,PROA,A1,PS,BGA769				UA000		CRITICAL		BAFFIN_PROA					
337S00332		1	IC,GPU,BAFFIN,LEA,A1,PS,BGA769				UA000		CRITICAL		BAFFIN_LEA					
998-04866		1	INTERPOSER,AMD,C989,BGA769,VDDCI/MVDD				UA000		CRITICAL		STARDUST:VDDCI_MVDD					
998-04867		1	INTERPOSER,AMD,C988,BGA769,VDDC				UA000		CRITICAL		STARDUST:VDDC					
677-04532		2	SUBASSY (TAR) PCBA, AMR, INTERPOSER, X363				J5250,J5260		CRITICAL							
Development/Base BOMs																
PART NUMBER		QTY	DESCRIPTION				REFERENCE DES		CRITICAL		BOM OPTION					
685-00076		1	COMMON PARTS,MLB-BAF,X363				BASE		CRITICAL		BASE_BOM					
985-00126		1	DEV,MLB-BAF,X363				DEVEL		CRITICAL		DEVEL_BOM					
WIFI/BT Diplexers																
PART NUMBER		QTY	DESCRIPTION				REFERENCE DES		CRITICAL		BOM OPTION					
155S0979		3	FLTR,DIPLEXER,2.45/5.54GHZ,0805				U3810,U3820,U3830		CRITICAL		DIPLEXER:MURATA					
GPU Options																
BOM GROUP		BOM OPTIONS														
2GB_MC_BAFFIN		FB_2GB_MICRON,VRAM:GRP1														
2GB_HY_BAFFIN		FB_2GB_HYNIX,VRAM:GRP1														
2GB_SM_BAFFIN		FB_2GB_SAMSUNG,VRAM:GRP2														
4GB_SM_BAFFIN		FB_4GB_SAMSUNG,VRAM:GRP1														
4GB_MC_BAFFIN		FB_4GB_MICRON,VRAM:GRP1														
FB VDRAM Parts																
PART NUMBER		QTY	DESCRIPTION				REFERENCE DES		CRITICAL		BOM OPTION					
333S00044		4	IC,GDDR5,4Gb,70bps,1.5V,25NM,A,170 BGA				UA400,UA450,UA500,UA550		CRITICAL		FB_2GB_MICRON					
333S00043		4	IC,GDDR5,4Gb,70bps,1.5V,25NM,A,170 BGA				UA400,UA450,UA500,UA550		CRITICAL		FB_2GB_HYNIX					
333S00078		4	IC,GDDR5,8Gb,70bps,1.5V,25NM,B,170 BGA				UA400,UA450,UA500,UA550		CRITICAL		FB_2GB_SAMSUNG					
333S00074		4	IC,GDDR5,8Gb,70bps,1.5V,25NM,B,170 BGA				UA400,UA450,UA500,UA550		CRITICAL		FB_4GB_SAMSUNG					
333S00075		4	IC,GDDR5,8Gb,70bps,1.5V,25NM,A,170 BGA				UA400,UA450,UA500,UA550		CRITICAL		FB_4GB_MICRON					
Sub-BOM DIPLEXER																
BOM NUMBER		BOM NAME				BOM OPTIONS										
685-00085		DIPLEXERS,MURATA,X363G				DIPLEXER:MURATA										
8		7		6		5		4		3		2		1		

8		7		6		5		4		3		2		1		
D	Programmable Parts															
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
	338S1231	1	IC, SMC12, 40MHZ/500MIPS MCU, 7X7, 1688GA	U5000	CRITICAL	SMC_PROG:BLANK	Blank									
	341S00701	1	IC, SMC-B1, EXT (V2.37F7) PVT, X363G	U5000	CRITICAL	SMC_PROG:PVT	TI									
	335S00013	1	IC, SPI SERIAL FLASH, 64M BITS, 3V, 8P, 501C, Qe=1	U6100	CRITICAL	BOOTROM_PROG:BLANK	Blank									
	341S00699	1	IC, EFI ROM (V0193), DVT, X363G	U6100	CRITICAL	BOOTROM_PROG:DVT	Macronix/Winbond									
	353S00926	2	IC, C03215, ACE, B03, BLNK, BGA96	U2890, UB090	CRITICAL	UPCROM_PROG:BLANK	Blank									
	341S00707	1	T29, AR1 (V10.5) PVT, X363G	U2890	CRITICAL	UPCROM_PROG:DVT	Winbond									
	341S00708	1	T29, AR2 (V10.5) PVT, X363G	UB090	CRITICAL	UPCROM_PROG:DVT	Winbond									
	335S00024	1	IC, SERIAL-FLASH, 2MBIT, 4V, 8-US0N, 2x3x, 6MM	U3750	CRITICAL	BT_PROG:BLANK	Blank									
	341S00695	1	IC, BT ROM (V28), DVT, X362/X363	U3750	CRITICAL	BT_PROG:DVT	Macronix/Winbond									
	341S00709	1	WIFI ROM (P107) DVT, NEW, MM1, X362/X363	U3710	CRITICAL	WIFI_PROG:DVT	Rohm/On Semi									
	341S3565	1	IC, EDP MUX-95C, (RENESAS) V3.2.0, DVB, D2	U9800	CRITICAL	DPMUXMCU:PROG										
335S0724	1	IC, 1MD1T SERIAL FLASH 2X3X0.6MM UFD0PN8 PKG	UA701	CRITICAL	GPUROM:BLANK	Blank										
C																
	B															
A																
	8		7		6		5		4		3		2		1	

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BOM Configuration

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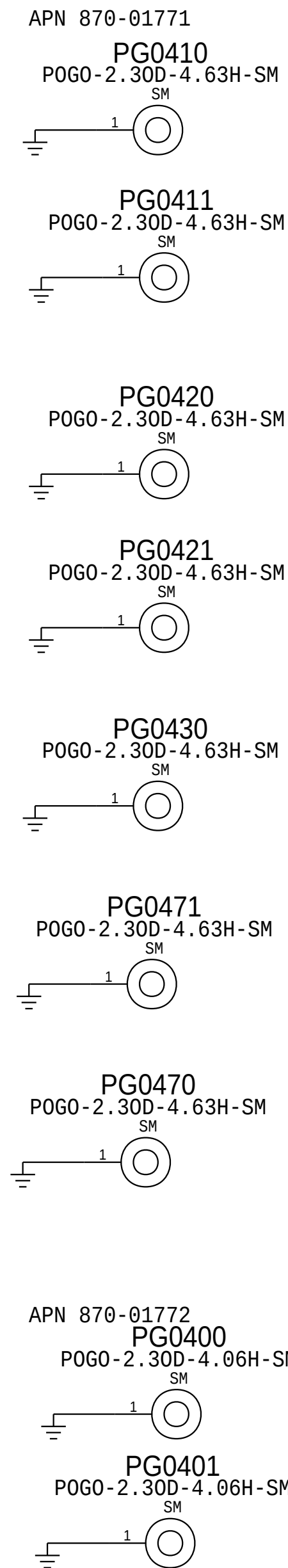
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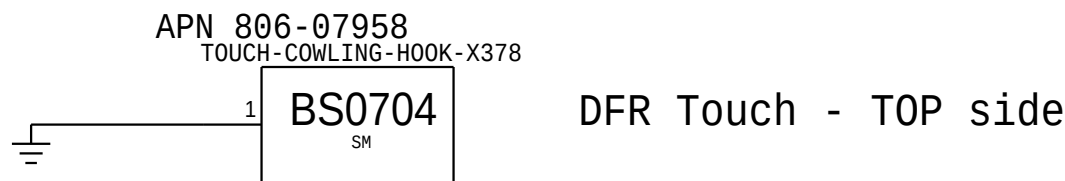
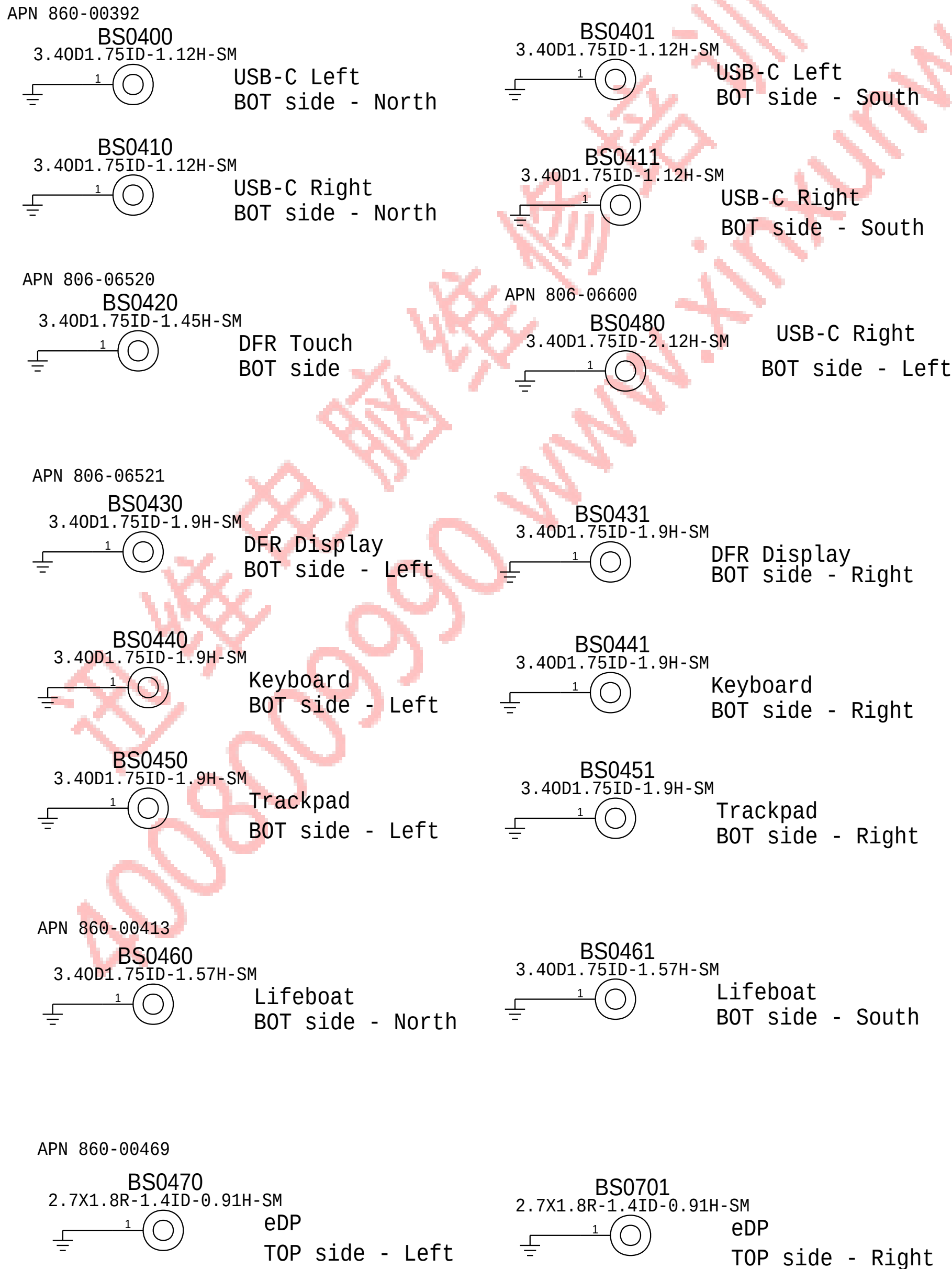
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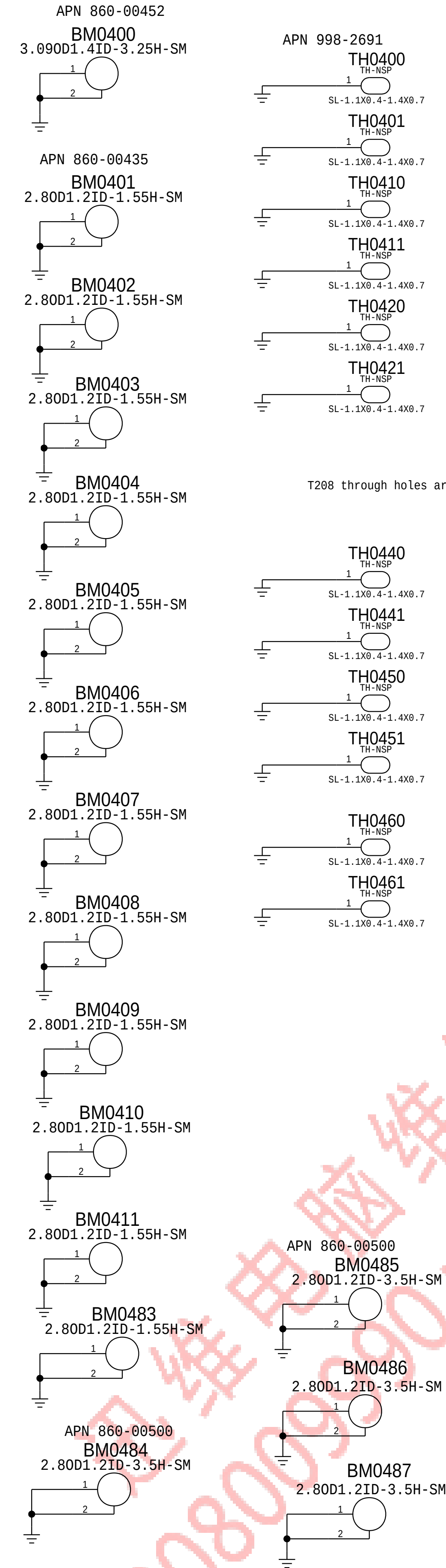
Pogo Pins



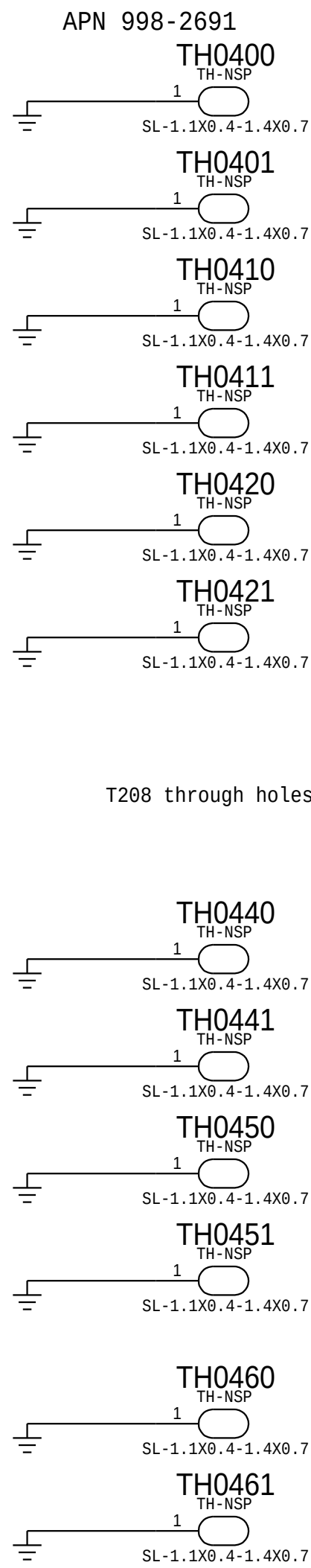
SMT Bosses



Rubber Mount Standoffs



Shield Can TH



System Memory - BOT side - Left

System Memory - BOT side - Right

TBT Left - BOT side - North

TBT Left - BOT side - South

TBT Right - BOT side - North

TBT Right - BOT side - South

T208 - TOP side - North

T208 - TOP side - South

SSD - BOT side - North

SSD - BOT side - South

SSD - TOP side - North

SSD - TOP side - South

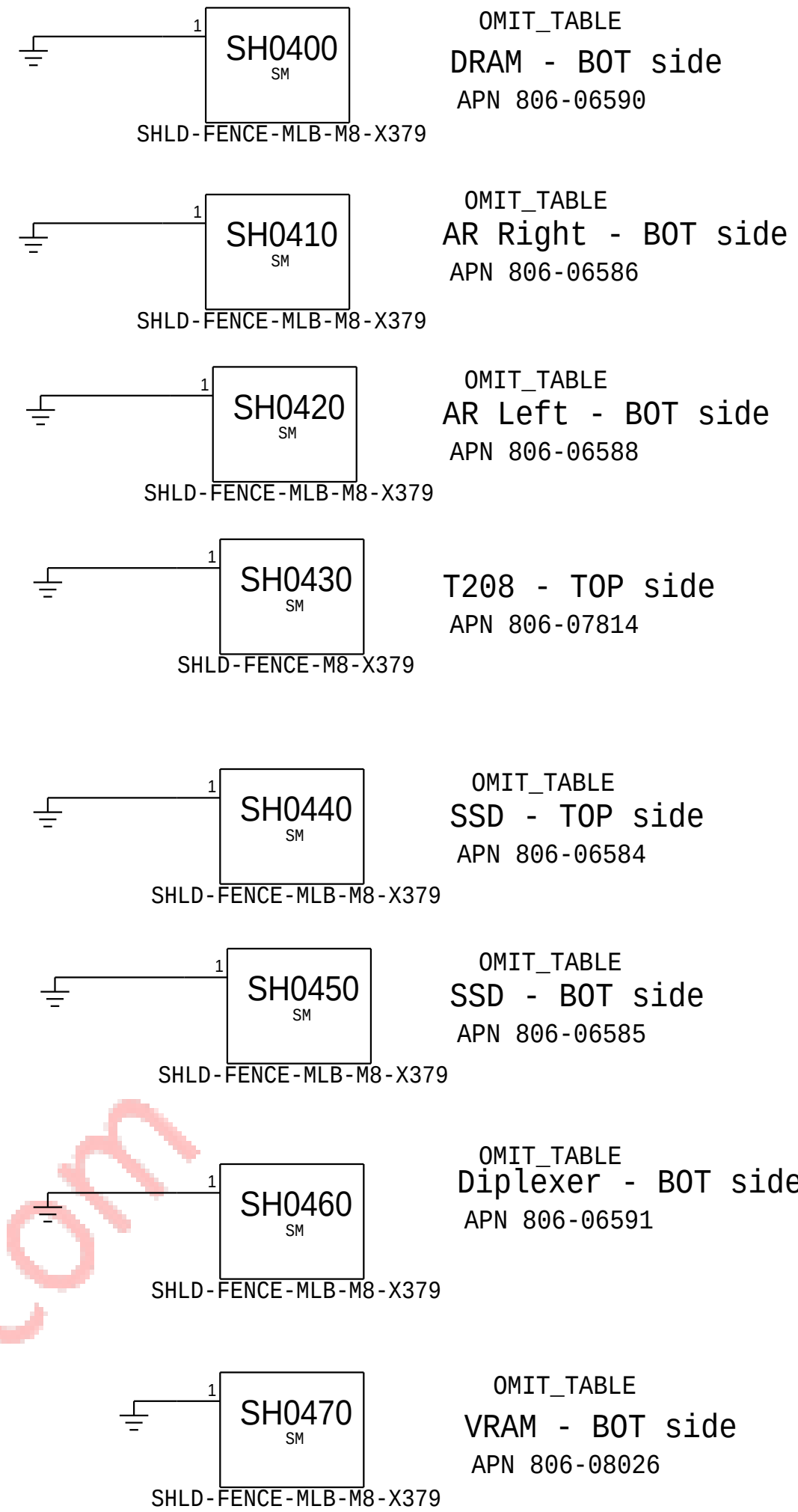
Frame Buffer Memory - BOT side - Left

Frame Buffer Memory - BOT side - Right

Shield Can Omit Table

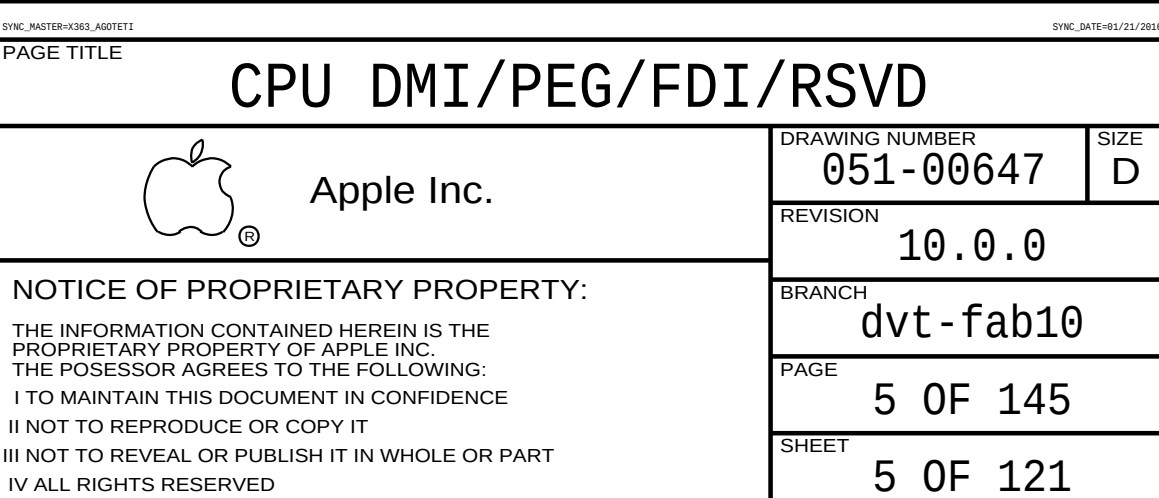
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-08023	1	SHIELD, FENCE, DRAM, X378	SH0400	CRITICAL	
806-08019	1	SHIELD, FENCE, ALPINE RIDGE, RIGHT, X378	SH0410	CRITICAL	
806-08021	1	SHIELD, FENCE, ALPINE RIDGE, LEFT, X378	SH0420	CRITICAL	
806-07918	1	SHIELD, NAND, TOP, ALT, X363	SH0440	CRITICAL	
806-07917	1	SHIELD, NAND, BOTTOM, ALT, X363	SH0450	CRITICAL	
806-08024	1	SHIELD, DIPLEX, EG, X378	SH0460	CRITICAL	
806-08026	1	FENCE, VRAM, EG, X378	SH0470	CRITICAL	

Shield Can Fence



BOM_COST_GROUP=MECHANICALS

SYNC_MASTER=J80_MLB		SYNC_DATE=11/16/2015	
PAGE TITLE			
PD Parts			
 Apple Inc.	DRAWING NUMBER	051-00647	SIZE D
	REVISION	10.0.0	
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	SHEET	4 OF 121	



D

C

B

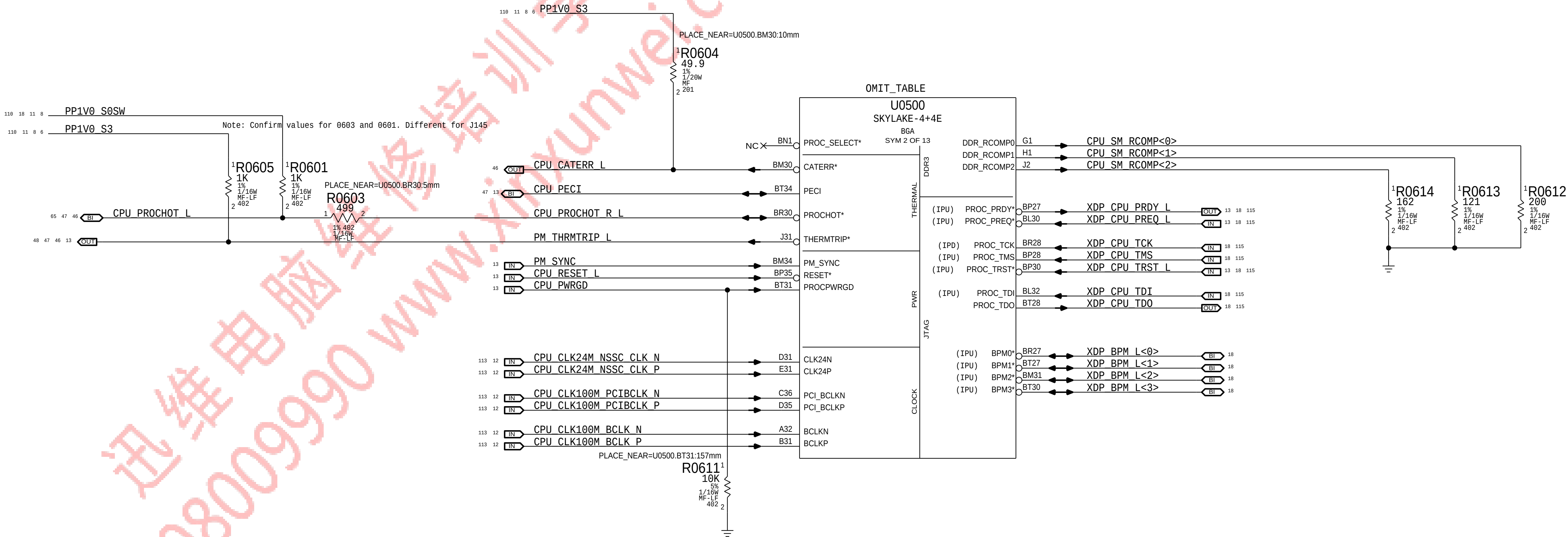
A

D

C

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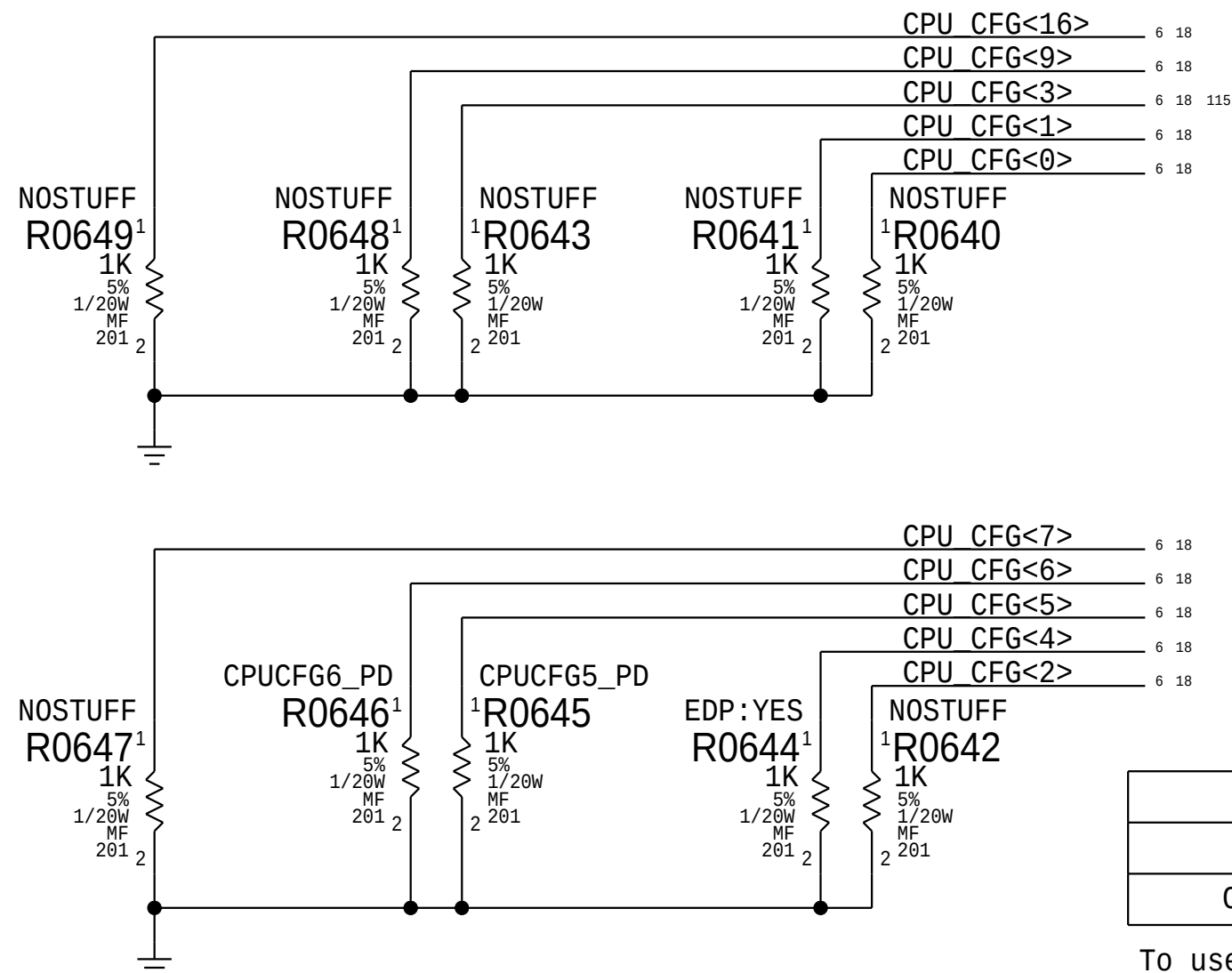
A



- PP0600 P2MM 1 TP_CPU_RSVD_R14
- PP0601 P2MM 1 TP_CPU_RSVD_N29
- PP0602 P2MM 1 TP_CPU_RSVD_AE29
- PP0603 P2MM 1 TP_CPU_RSVD_AA14

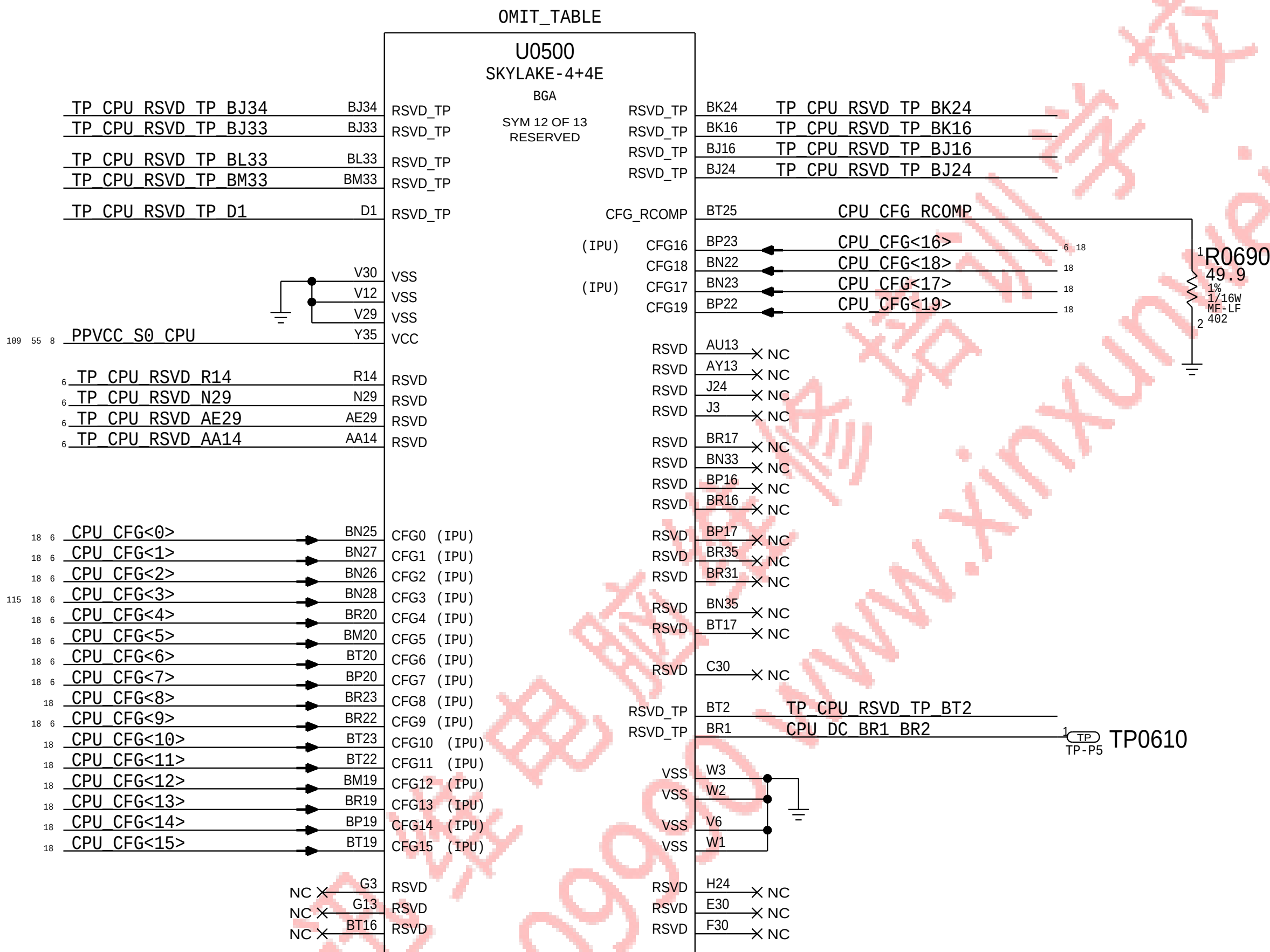
CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

These can be placed close to J1800 and only for debug access



BOM GROUP	BOM OPTIONS
CPUPEG:X8X8	CPUCFG5_PD
CPUPEG:X8X4X4	CPUCFG6_PD,CPUCFG5_PD

To use PEG X16 configuration, simply remove CPUPEG:X8X8 and CPUPEG:X8X4X4 from BOMs.



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REVISION: 10.0.0		BRANCH: dvt-fab10
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BOM_COST_GROUP=CPU & CHIPSET

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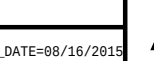
D

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8

OMIT_TABLE

U0500
SKYLAKE-4+4E
BGA
SYM 9 OF 13
GROUND

AW2	VSS	BG14
AW29	VSS	BE5
AW3	VSS	BG38
AW30	VSS	BG6
AW4	VSS	BH1
AW5	VSS	BH10
AY12	VSS	BH11
AV37	VSS	BH12
AH12	VSS	BH14
AG29	VSS	BH2
AH14	VSS	BH3
AG11	VSS	BH4
AG30	VSS	BH5
AG10	VSS	BG37
AG8	VSS	BH9
AF14	VSS	BJ12
AG7	VSS	BJ15
AG6	VSS	BJ18
AF13	VSS	BJ22
AF4	VSS	BJ25
AF3	VSS	BJ29
AF2	VSS	BH6
AF1	VSS	BJ30
AF12	VSS	BH7
AE34	VSS	BJ31
AE33	VSS	BJ32
AE6	VSS	BK13
AD30	VSS	BK14
AD11	VSS	BH8
AD29	VSS	BK29
AD10	VSS	BK6
AD12	VSS	BL13
AD8	VSS	BL29
AD7	VSS	BL35
AD9	VSS	BL38
AC6	VSS	BL6
AC5	VSS	BM11
AC4	VSS	BM12
AC3	VSS	BM13
AC2	VSS	BM18
BD10	VSS	BM2
BD5	VSS	BM21
BD37	VSS	BM23
BD38	VSS	BM25
BD6	VSS	BM26
BD7	VSS	BM27
BD8	VSS	BM28
BD9	VSS	BM29
BE1	VSS	BM3
BE2	VSS	BM5
BE29	VSS	BM6
BD11	VSS	BM7
BE3	VSS	BM8
A26	VSS	BM9
A24	VSS	BK15
A22	VSS	BK18
A20	VSS	BK22
A18	VSS	BK25
A16	VSS	U6
BF33	VSS	BN24
BF34	VSS	G8
BF6	VSS	G9
BG12	VSS	H11
BG13	VSS	H12
BN2	VSS	H18
BN19	VSS	
BN20	VSS	
BN21	VSS	

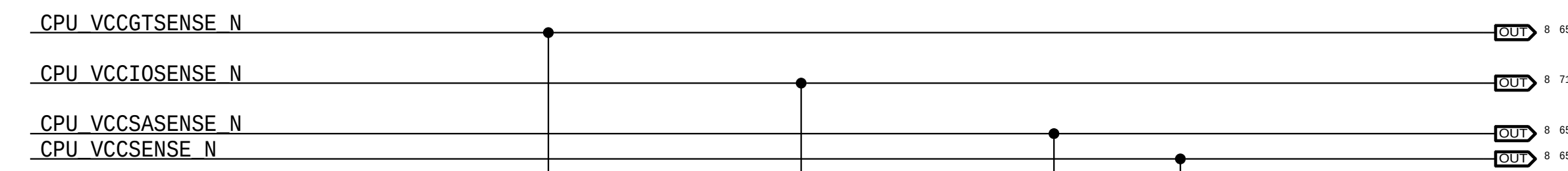
OMIT_TABLE
U0500
SKYLAKE-4+4E
BGA
SYM 8 OF 13
GROUND

AU34	VSS	AJ1
AU33	VSS	AH34
AU12	VSS	AH33
AU11	VSS	AG29
AU10	VSS	AH12
AU9	VSS	AG11
AU8	VSS	AG30
AU7	VSS	AG10
AU6	VSS	AG8
AT30	VSS	AF14
AT29	VSS	AG7
AT6	VSS	AG6
AR38	VSS	AF13
AR37	VSS	AF4
AR5	VSS	AF3
AR14	VSS	AF2
AR13	VSS	AF1
AR4	VSS	AF12
AR1	VSS	AE34
AP34	VSS	AE33
AP33	VSS	AE6
AP12	VSS	AD30
AR3	VSS	AD11
AR2	VSS	AD29
AP9	VSS	AD10
AP11	VSS	AD12
AP8	VSS	AD8
AP10	VSS	AD7
AN12	VSS	AD9
AN6	VSS	AC6
AN5	VSS	AC5
AN30	VSS	AC4
AN29	VSS	AC3
AM38	VSS	AC2
AM37	VSS	BD10
AM12	VSS	BD5
AM5	VSS	BD37
AM2	VSS	BD38
AM4	VSS	BD6
AM1	VSS	BD7
AM3	VSS	BD8
AL34	VSS	BD9
AL33	VSS	BE1
AL14	VSS	BE2
AL8	VSS	BE29
AL12	VSS	BD11
AL7	VSS	BE3
AL4	VSS	A26
AK30	VSS	A24
AK29	VSS	A22
AL10	VSS	A20
AL9	VSS	A18
AK4	VSS	A16
AJ38	VSS	A14
AJ37	VSS	A12
AJ6	VSS	A10
AJ3	VSS	A9
AJ5	VSS	A6
AJ2	VSS	AA29
AJ4	VSS	

OMIT_TABLE

U0500
SKYLAKE-4+4E
BGA
SYM 10 OF 13

BN29	VSS	H22
BN12	VSS	H25
BN30	VSS	H32
BN31	VSS	H35
BN34	VSS	H35
BN4	VSS	J10
BN7	VSS	J18
BN9	VSS	J22
BN9	VSS	G23
BP12	VSS	G24
BP14	VSS	G26
BN14	VSS	J32
BP18	VSS	J33
BP21	VSS	J35
BN18	VSS	J25
BP26	VSS	J4
BP29	VSS	J7
BP33	VSS	K1
BP34	VSS	K10
BP7	VSS	K36
BR12	VSS	K11
BR12	VSS	K2
BR14	VSS	K3
BR18	VSS	K38
BR21	VSS	K4
BR24	VSS	K5
BR24	VSS	K7
BR26	VSS	L29
BR29	VSS	L9
BR36	VSS	K9
BR7	VSS	L33
BT12	VSS	L34
BT14	VSS	M12
BT21	VSS	M13
BT24	VSS	L30
BT26	VSS	M6
BT26	VSS	M14
BT32	VSS	N1
BT5	VSS	N10
BT9	VSS	N11
C11	VSS	N12
C13	VSS	N4
C15	VSS	N4
BR34	VSS	N2
C21	VSS	N6
C23	VSS	N7
C25	VSS	N8
C27	VSS	N33
C29	VSS	N9
C31	VSS	N8
C37	VSS	P12
C5	VSS	P37
C17	VSS	P38
C8	VSS	P6
C9	VSS	R12
D10	VSS	T10
D12	VSS	R29
C19	VSS	T11
D16	VSS	T12
D18	VSS	T13
D22	VSS	R30
D24	VSS	T1
D26	VSS	T3
D28	VSS	T4
D3	VSS	T2
D30	VSS	T4
D33	VSS	T5
D6	VSS	T7
D14	VSS	T33
E34	VSS	U37
E35	VSS	T8
E38	VSS	T9
E4	VSS	A36
E9	VSS	A37
F11	VSS	BN38
F13	VSS	BN35
F15	VSS	N34
F17	VSS	G6
F19	VSS	
F2	VSS	
F21	VSS	
F23	VSS	A34
F25	VSS	A4
F27	VSS	A3
F29	VSS	B3
F3	VSS	B37
F31	VSS	BR38
F36	VSS	BT3
D9	VSS	BT35
F4	VSS	BT36
G10	VSS	BT4
F5	VSS	C2
G12	VSS	D38
G14	VSS	
G16	VSS	
F8	VSS	
G18	VSS	
G20	VSS	
G22	VSS	
F9	VSS	
G28	VSS	
G4	VSS	
G5	VSS	



BOM_COST_GROUP=CPU & CHIPSET

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SYMC_MASTER=280_PLB

SYMC_DATE=08/17/2015

PAGE TITLE

CPU Ground

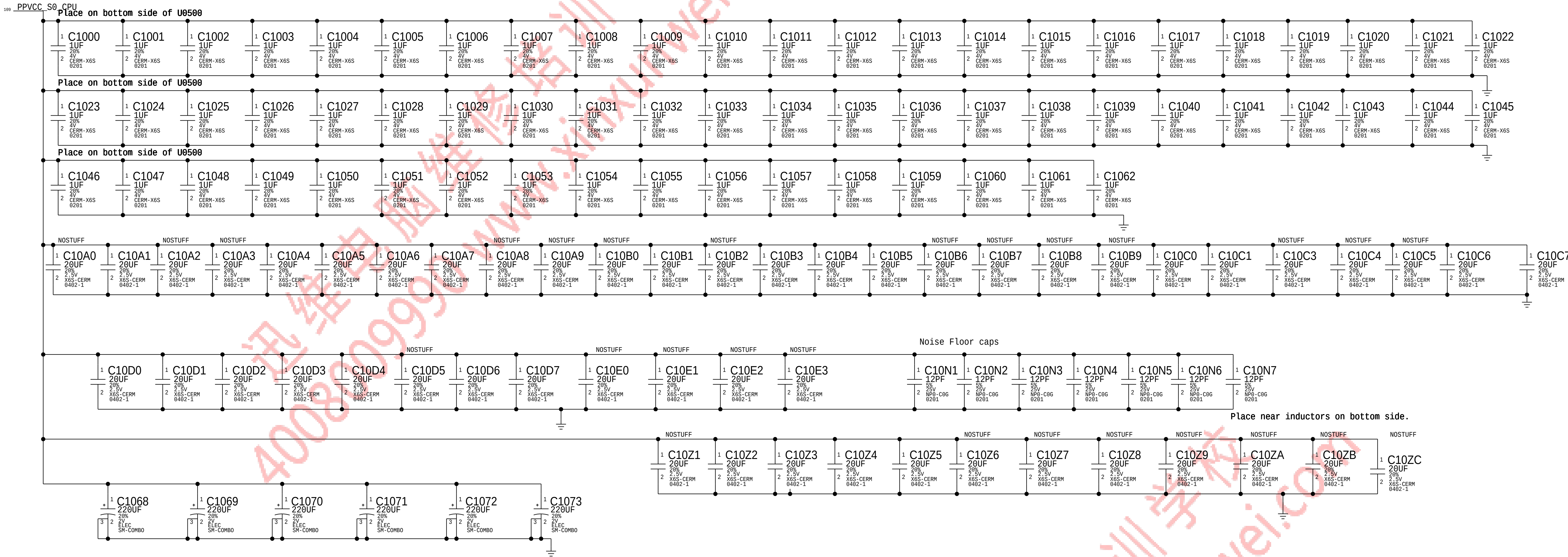


Vcc CPU Core Decoupling from 20140905 BOM

CPU VCORE Decoupling

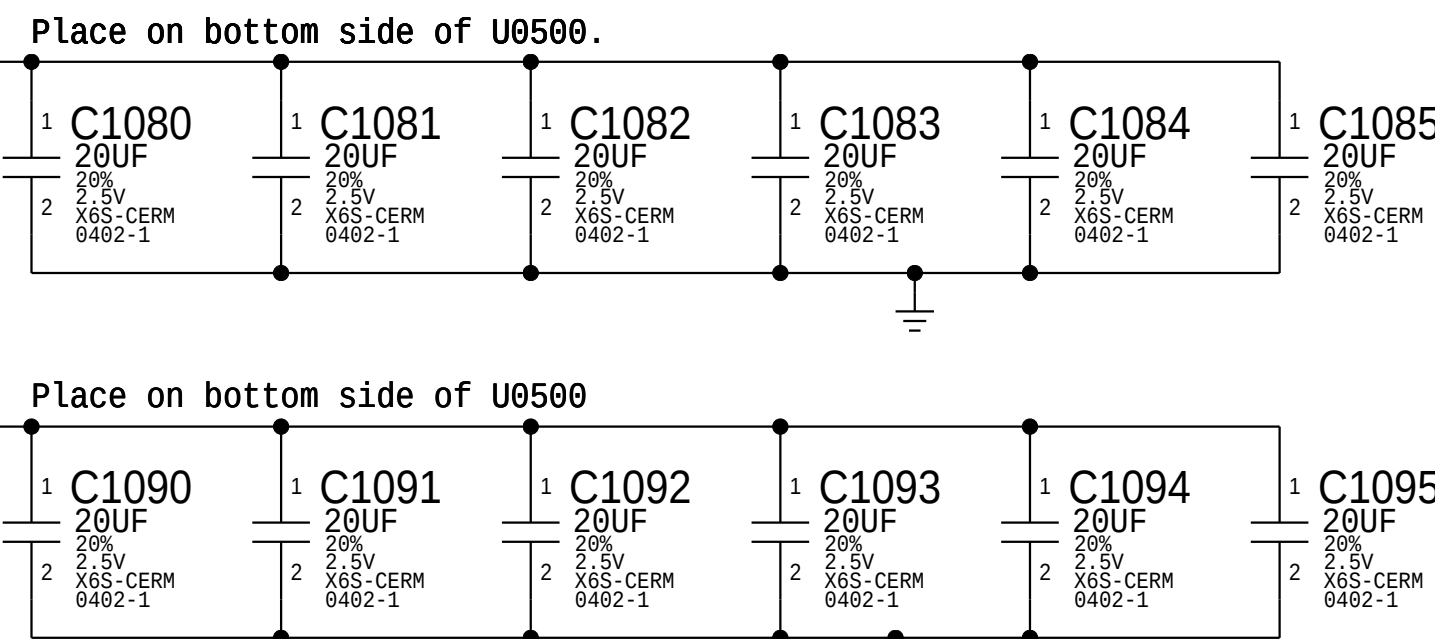
Intel recommendation: 5x 220uF ESR 5m ohms ESL 1.9nH each, 4x 47uF 0805 8x22uF 0603, 28x 10uF 0402, 3x 10uF 0402, 69x 1uF 0201
Apple Implementation:

Board Edge: 2x 220uF, 4x 47uF rest on the back side



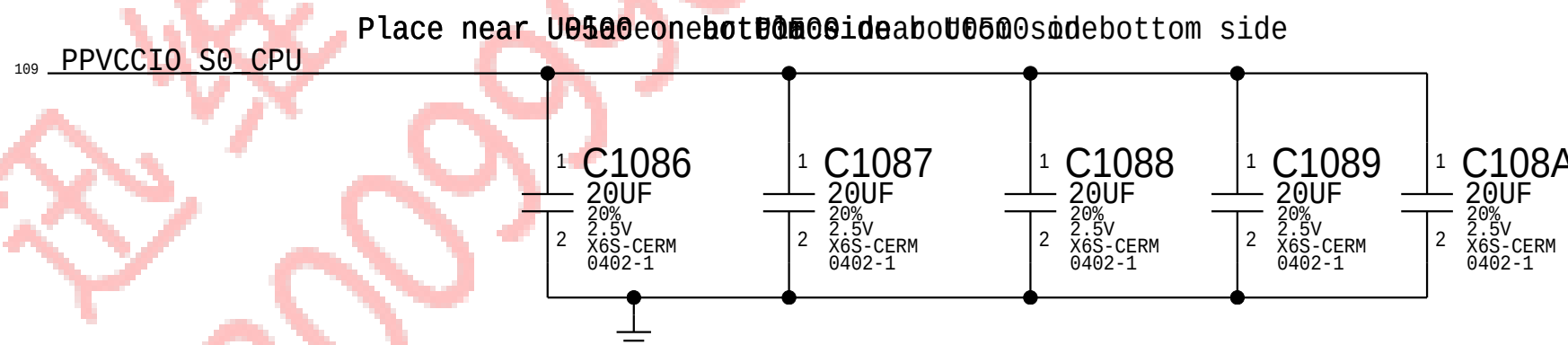
CPU VDDQ Decoupling

Intel recommendation: 10x 10uF 0402, 4x 22uF 0602
Apple Implementation:



CPU VCCIO Decoupling

Intel recommendation: 3x 10uF 0402 (opposite CPU)
Apple Implementation:



NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#55727 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

DRAWING NUMBER: 051-00647			
REVISION: 10.0.0			
BRANCH: dvt-fab10			
PAGE: 10 OF 145			
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Vcc GT Slice Core Decoupling from 20140905 BOM

CPU VGTSlice Decoupling

Intel recommendation: 7x 220uF, 6x 47uF 0805, 6x 22uF 0603, 35x 10uF 0402, 68 1uF 0201
Apple Implementation:

Board Edge: 4x220uF, 7x 47uF rest on back side

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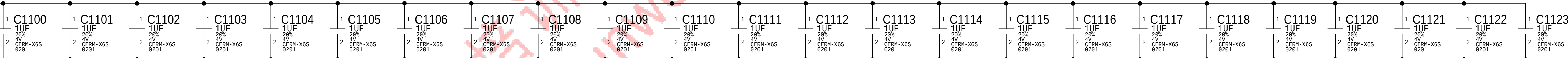
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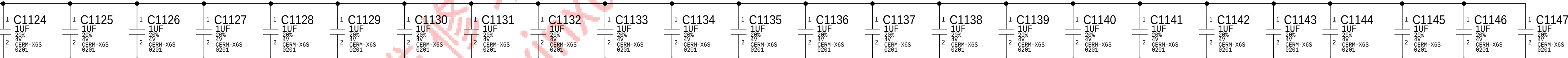
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116 109 PPVCCGT_S0 CPU

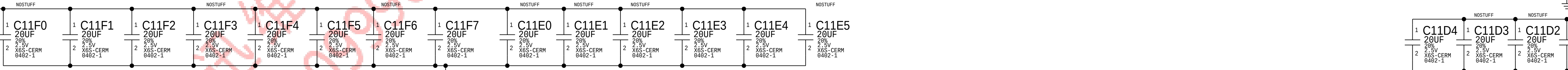
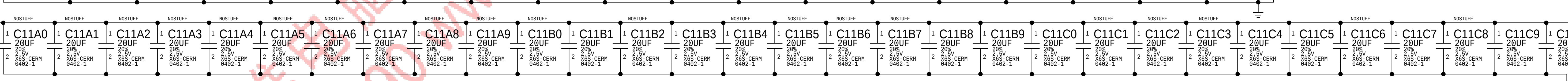
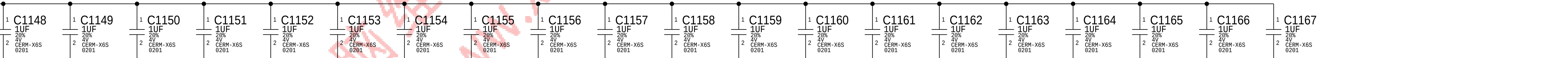
Place on bottom side of U0500



Place on bottom side of U0500

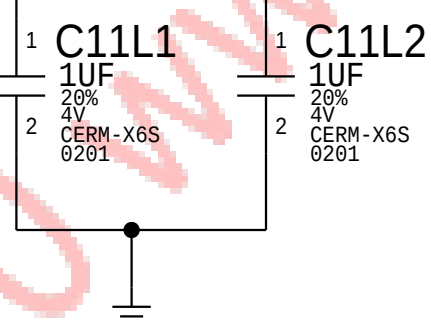


Place on bottom side of U0500

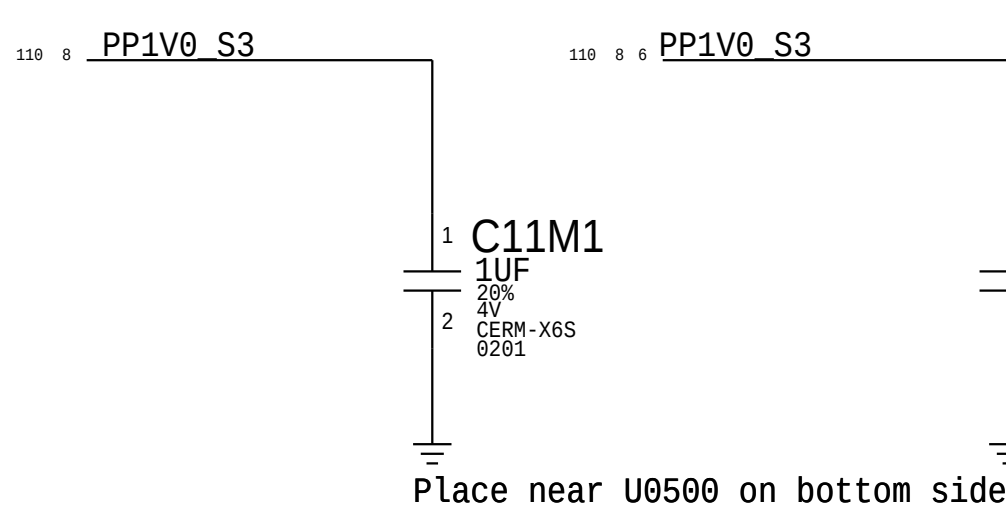


CPU VCCSTG Decoupling

119 18 8 6 PP1V0 S0SW
Place near U0500 on bottom side



CPU VCCPLL and VCCST Decoupling

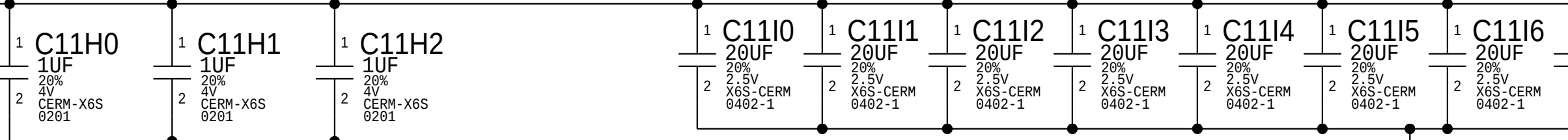


CPU VCCSA Decoupling

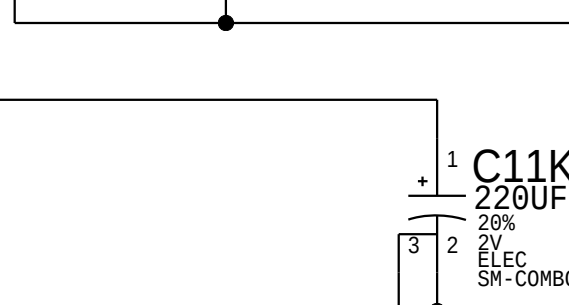
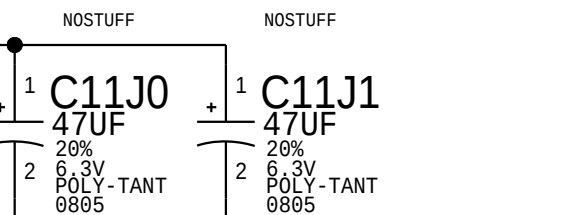
Intel recommendation: 2x 220uF, 1x 47uF 0805, 1x 22uF, 7x 10uF 0402, 3x 1uF 0201
Apple Implementation: 2x 220uF, 1x 22uF on board edge, everything else on back side

109 PPVCCSA_S0 CPU

Place on bottom side of U0000



Place on bottom side of U0000



NOTE: Intel decoupling recommendations from CBR schematics for Skylake H doc#557227 and PDG section 48.1 (document# 546884)

BOM_COST_GROUP=CPU & CHIPSET

PAGE TITLE		
CPU Decoupling 2 [11]		
	DRAWING NUMBER	051-00647
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	SHEET	11 OF 121



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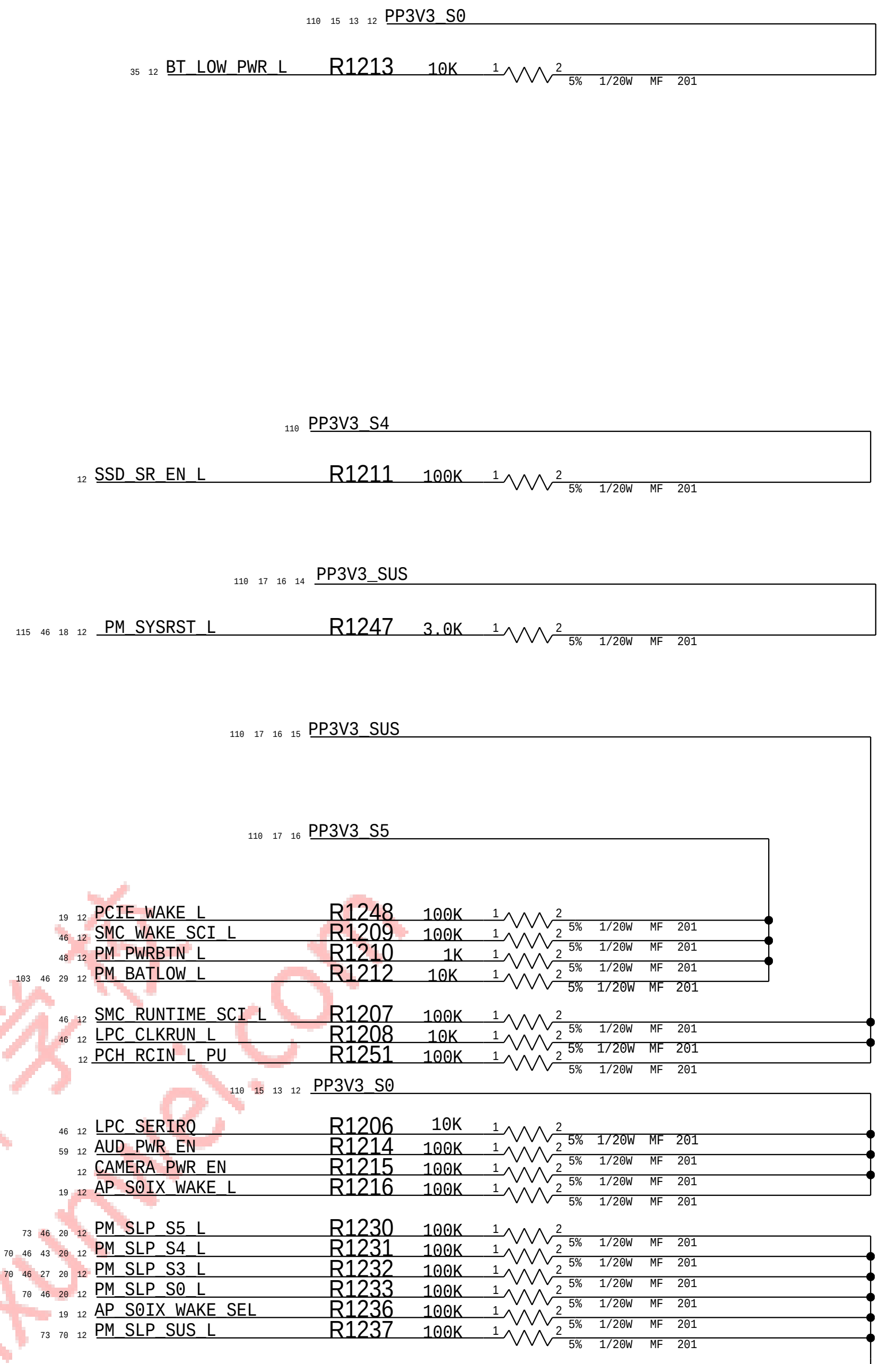
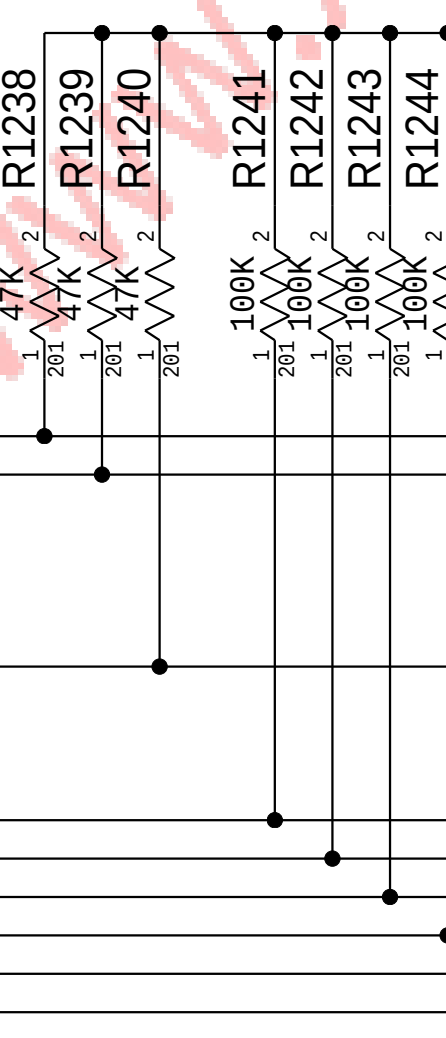
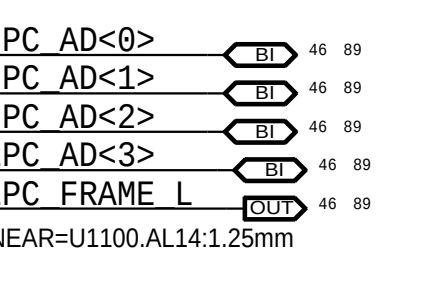
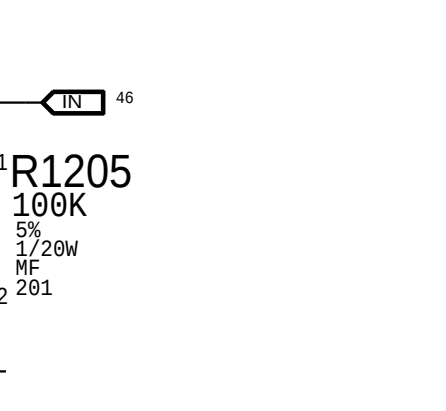
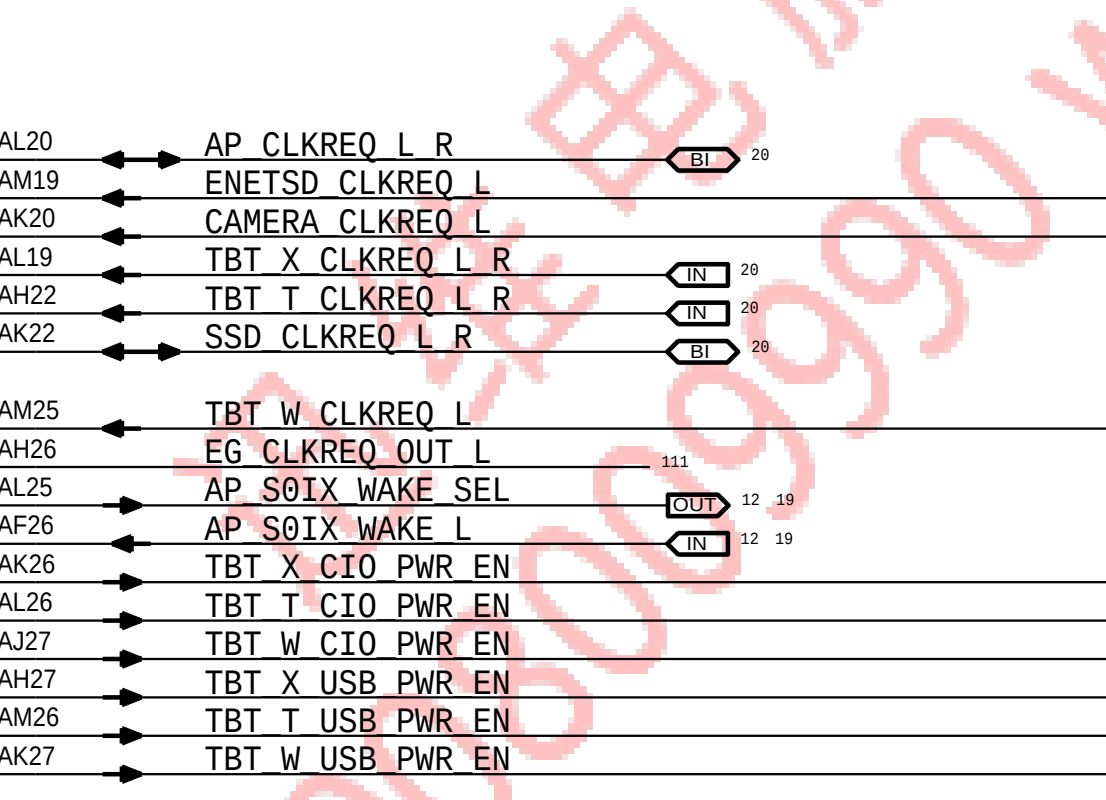
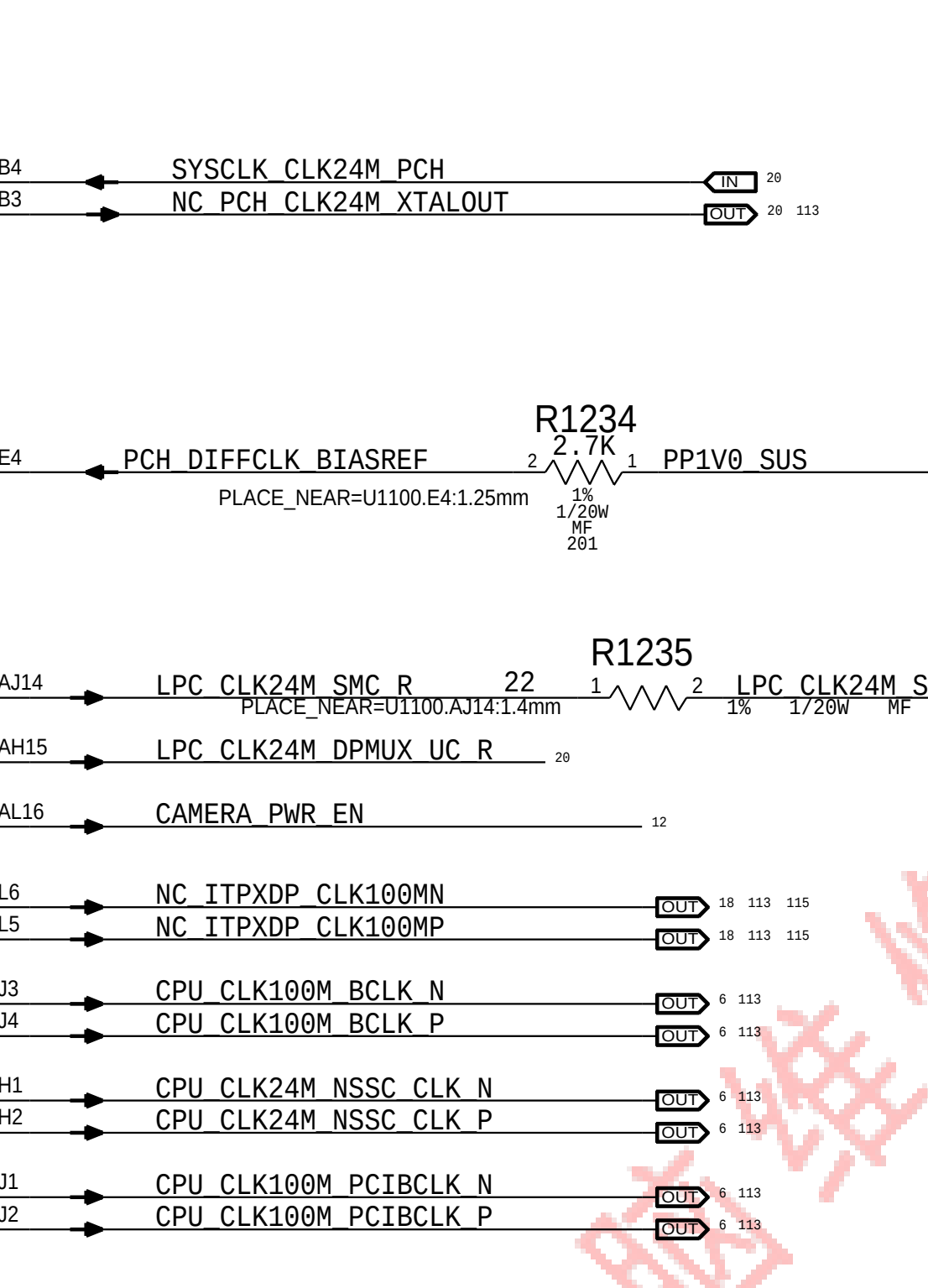
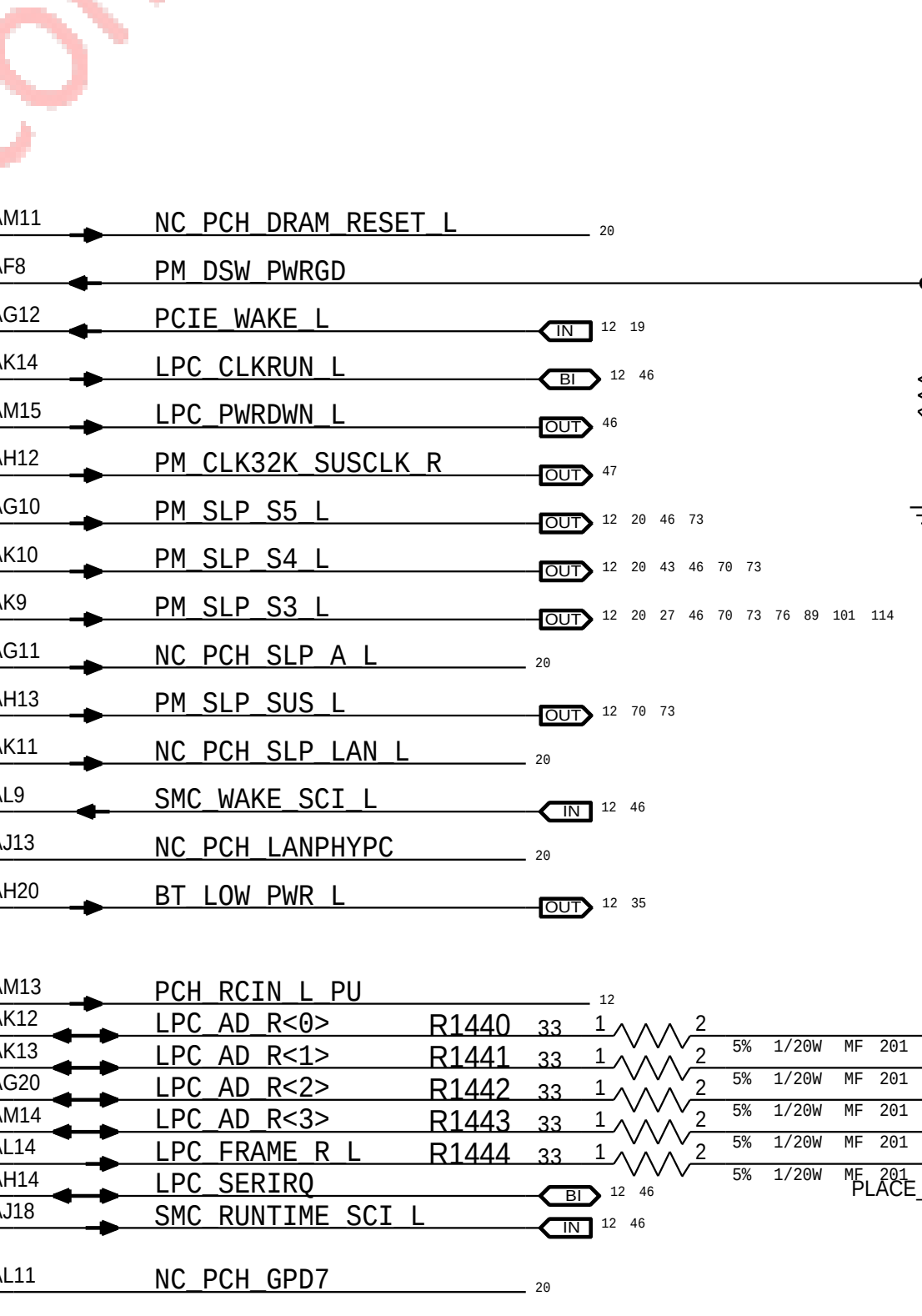
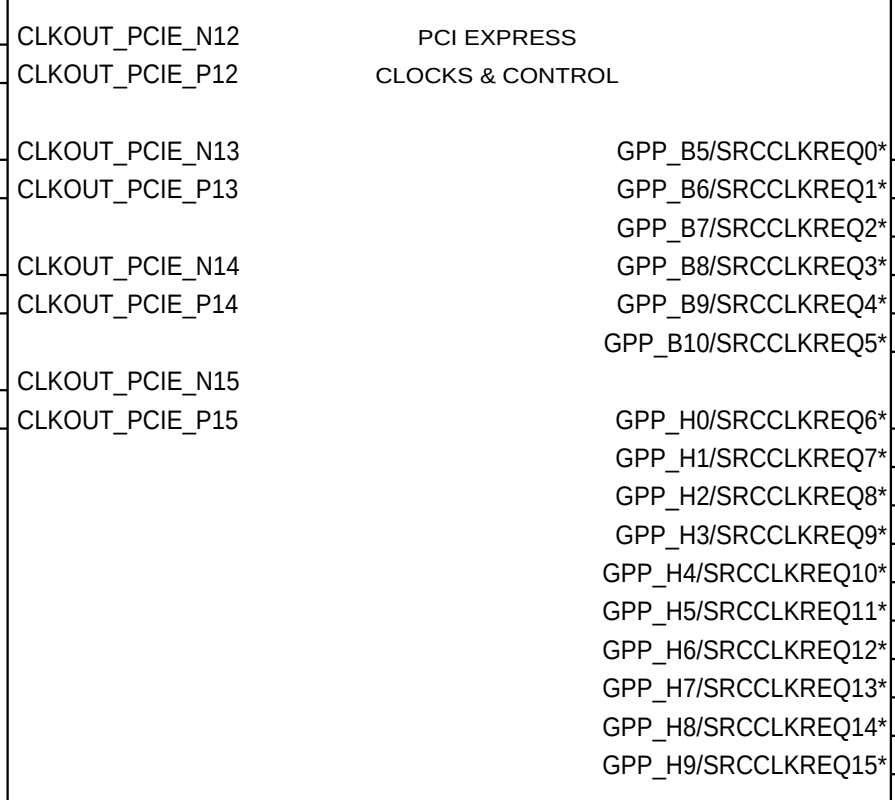
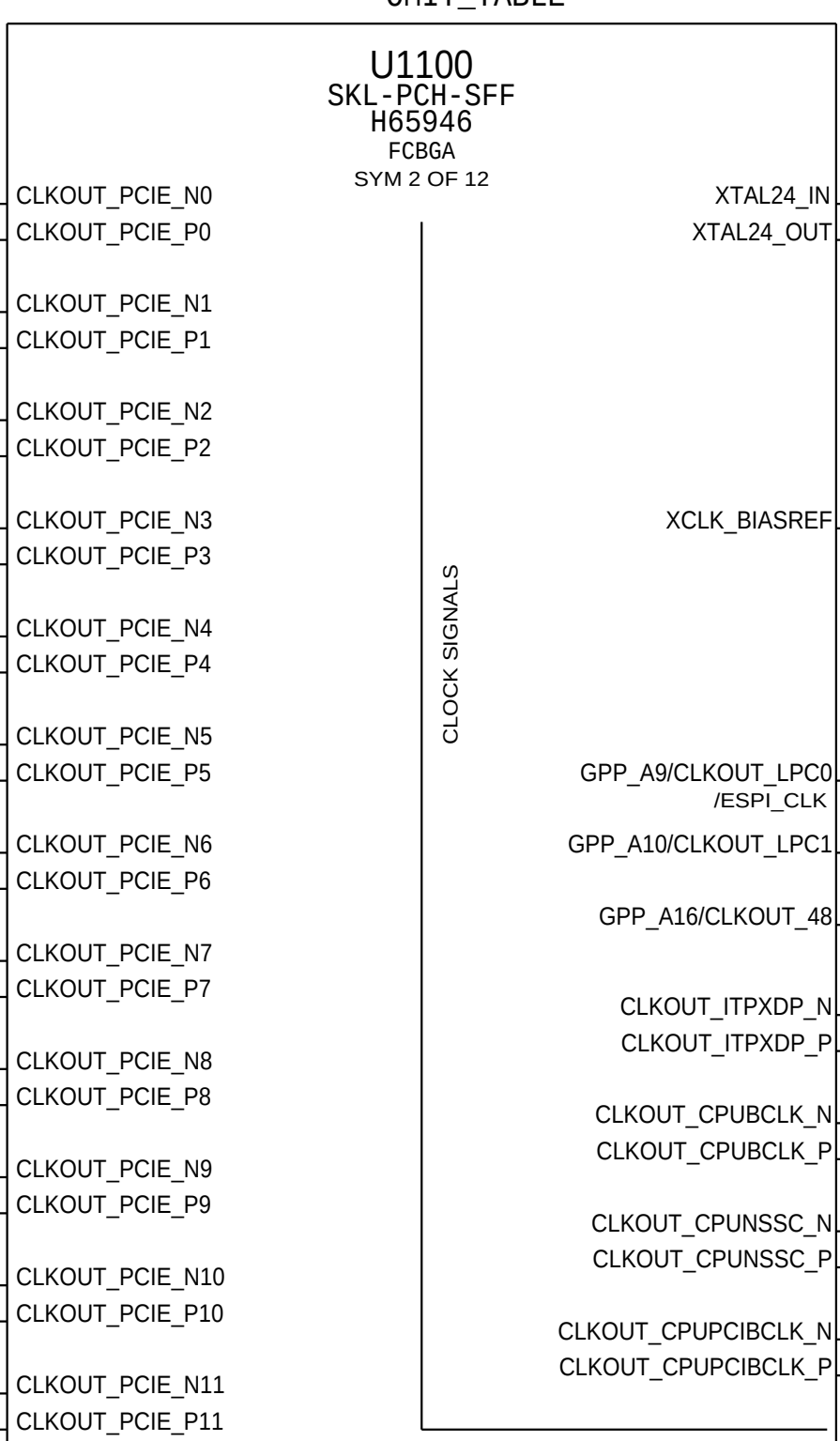
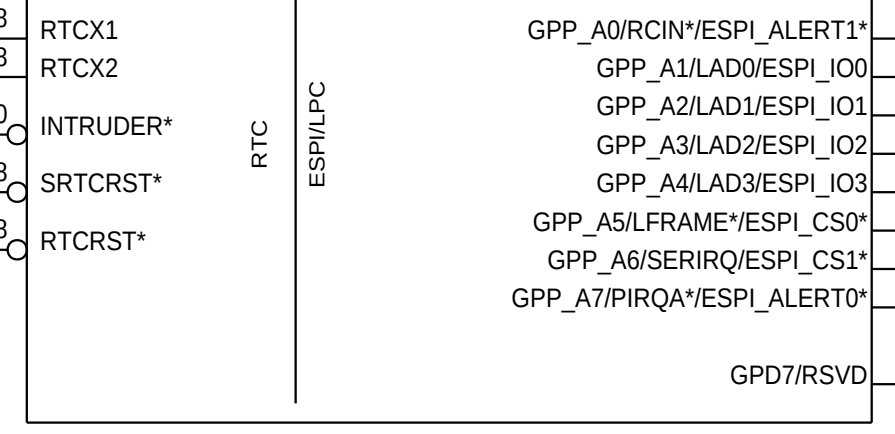
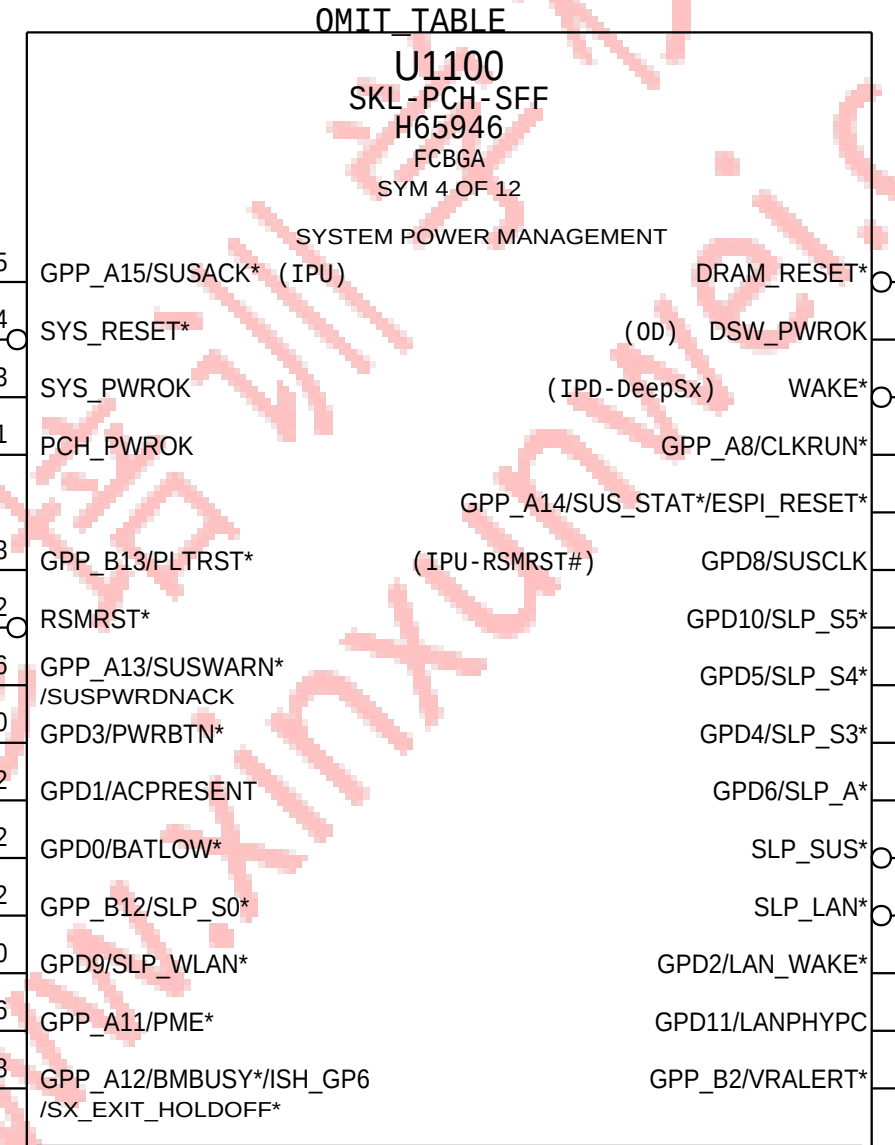
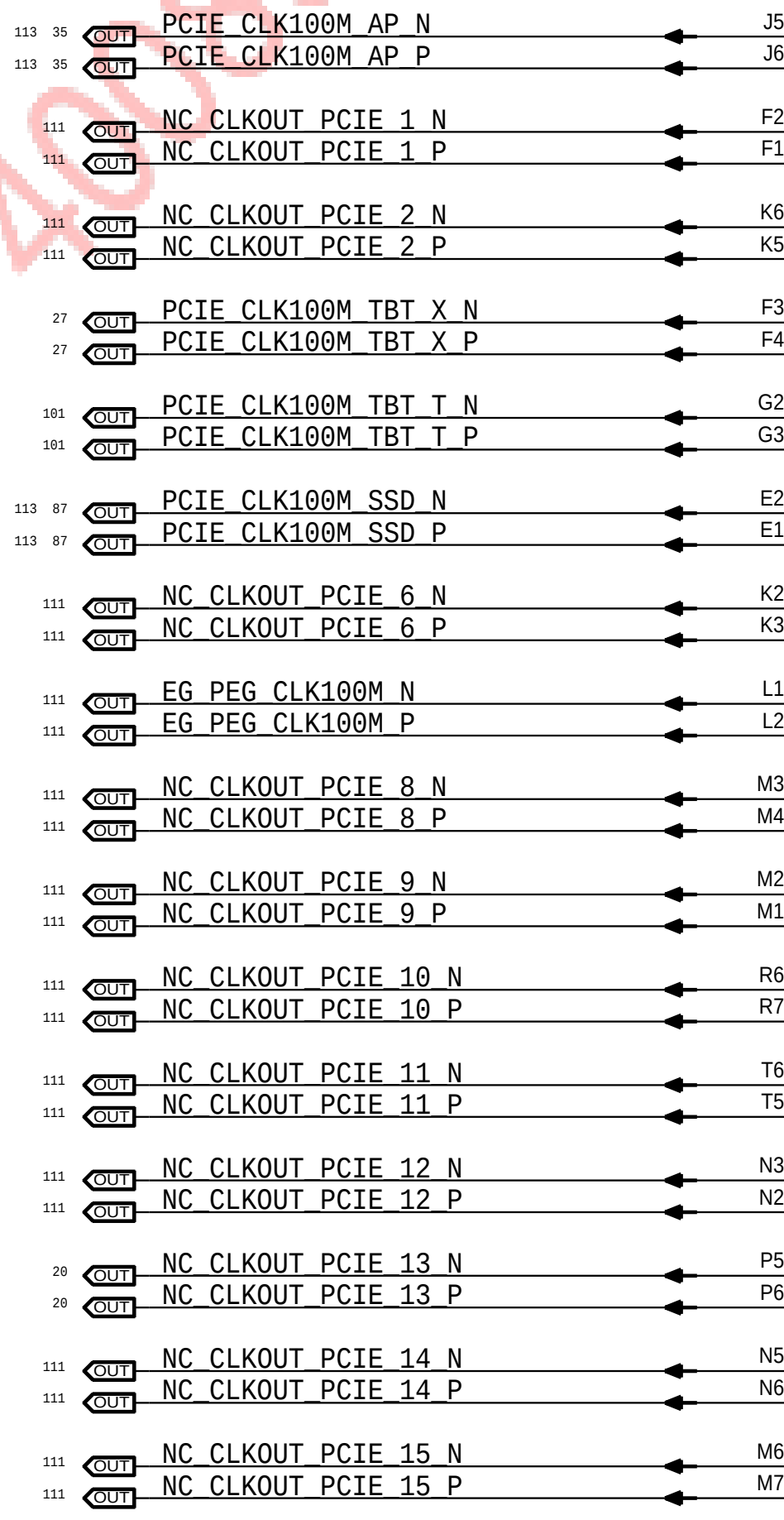
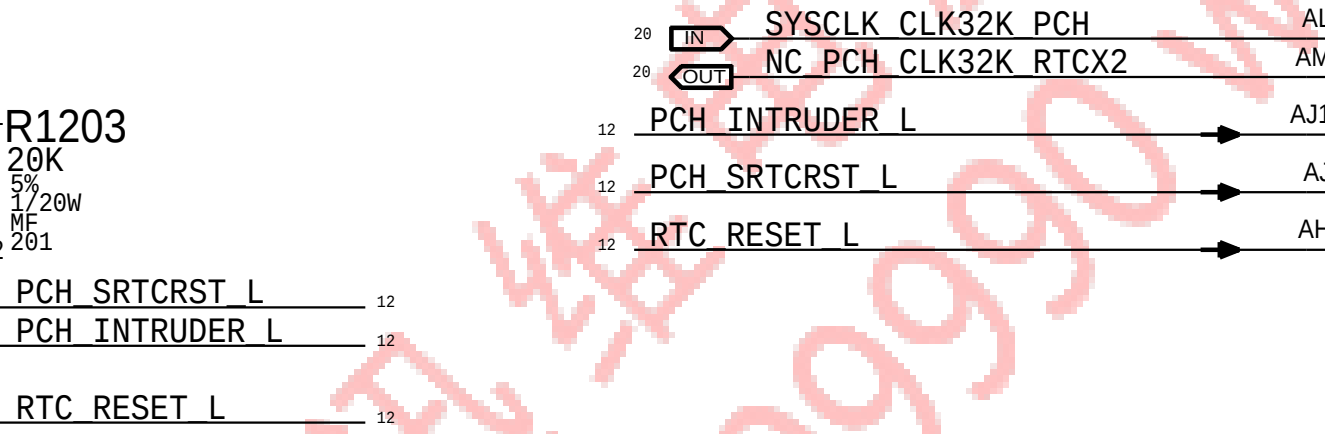
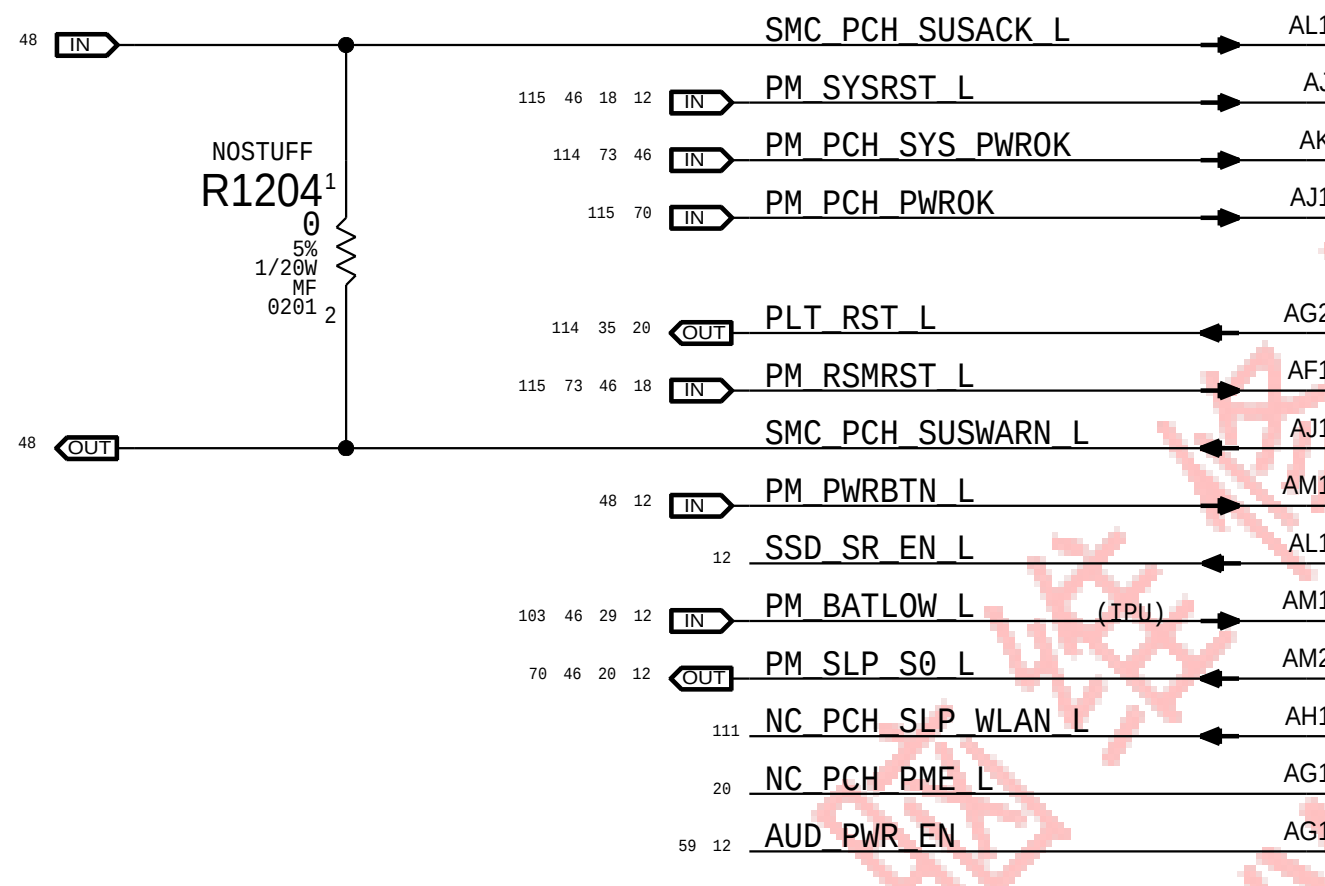
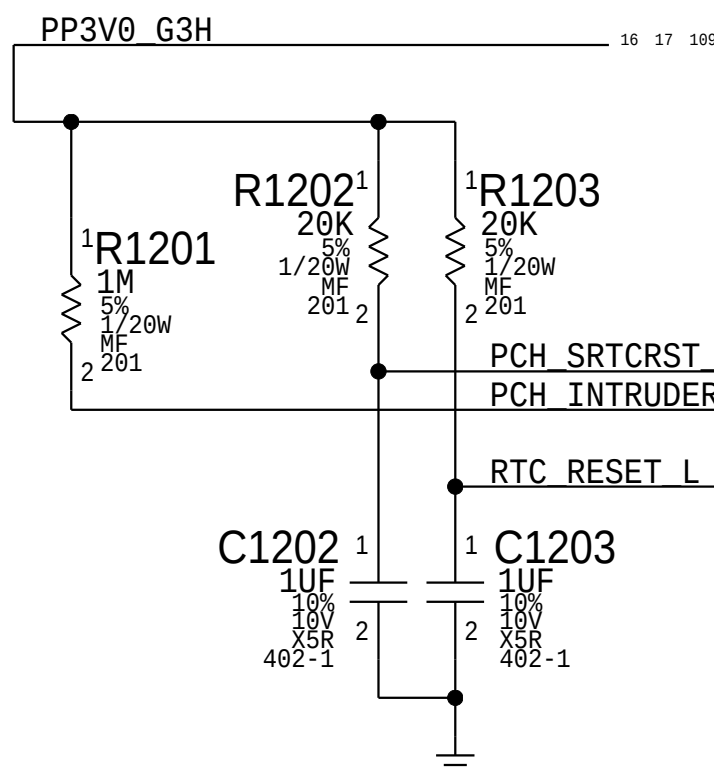
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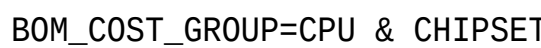
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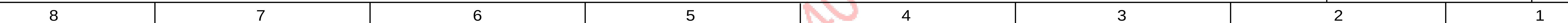
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REVISION: 10.0.0		BRANCH: dvt-fab10
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		PAGE 13 OF 145		
		SHEET 13 OF 121		





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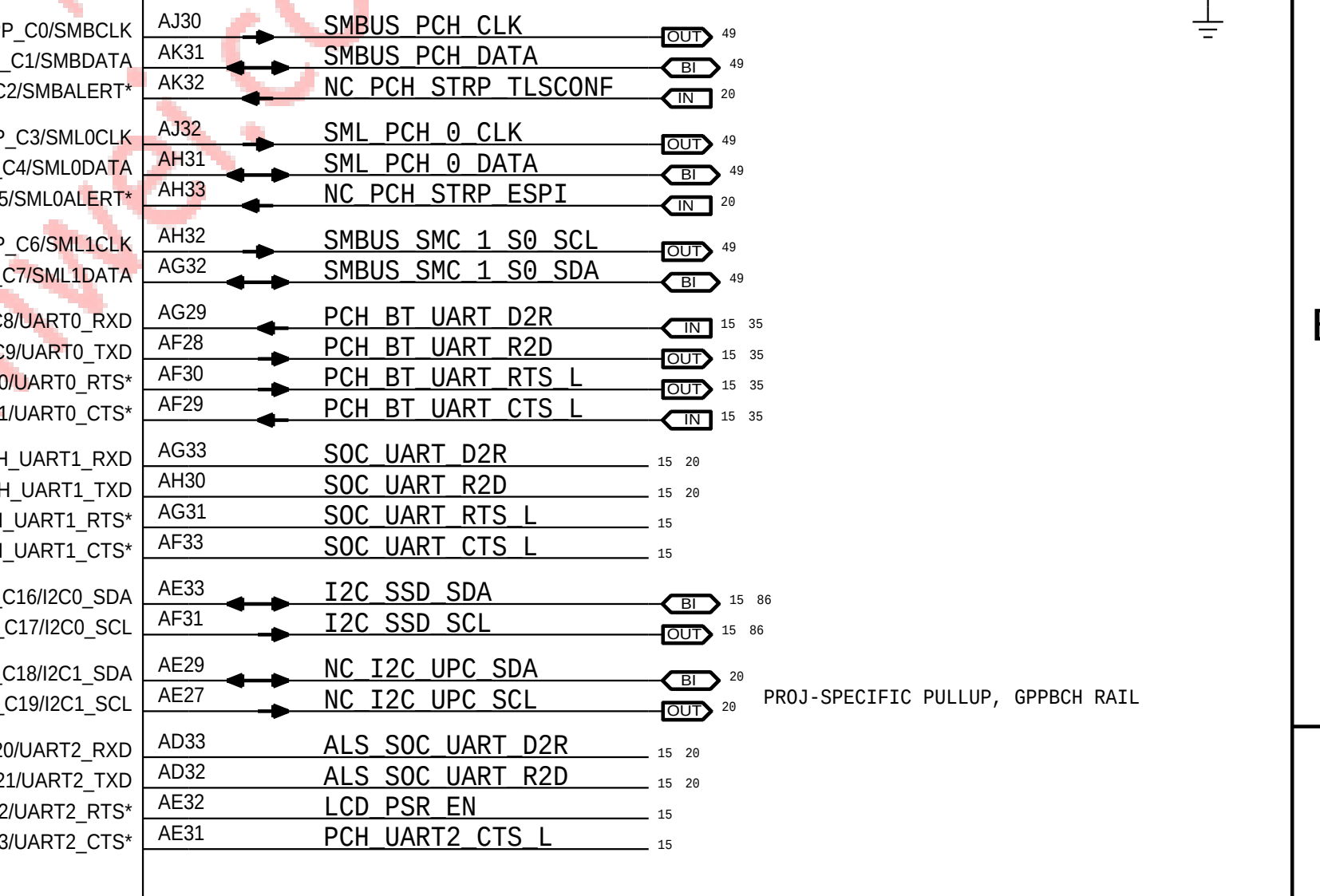
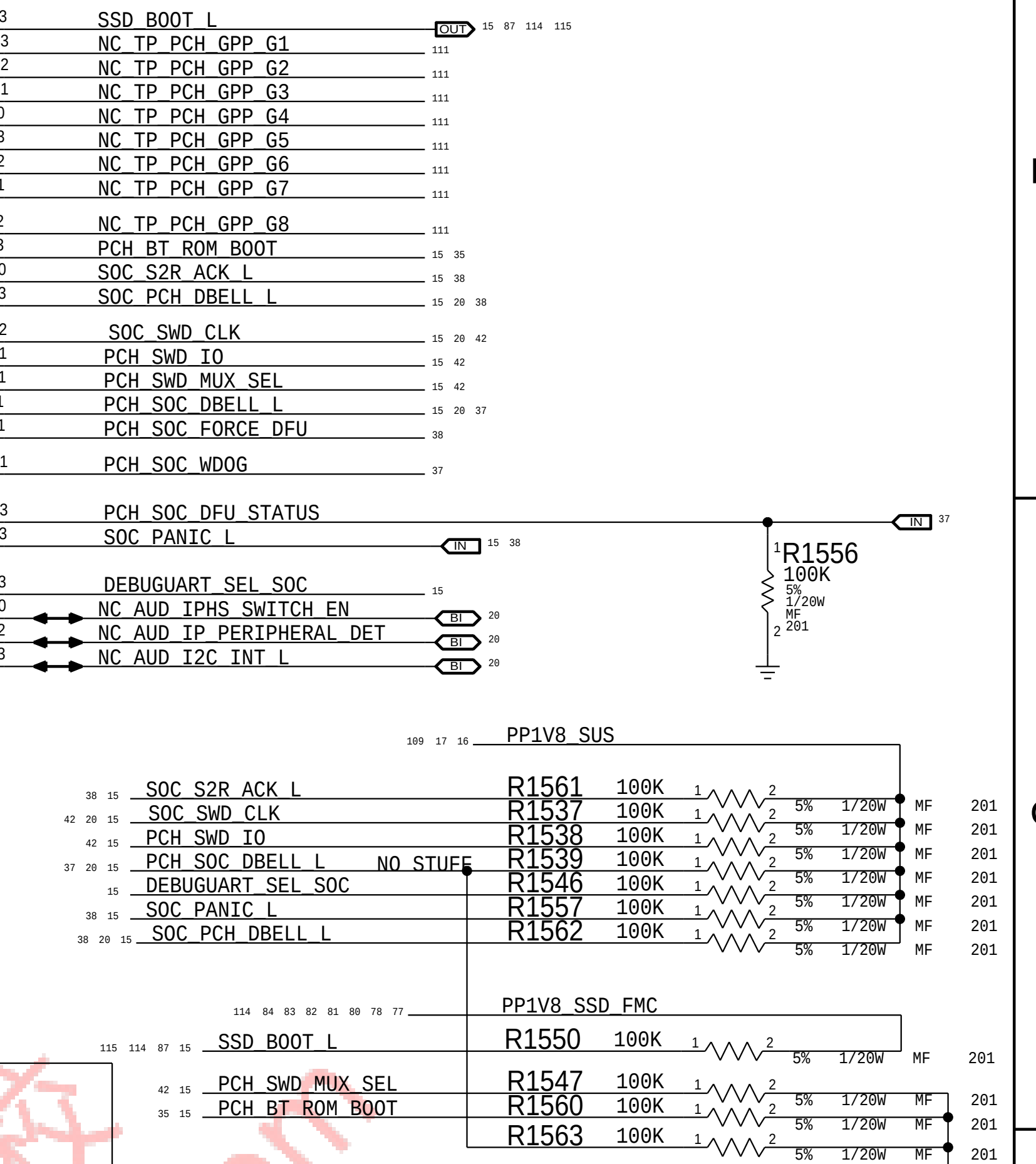
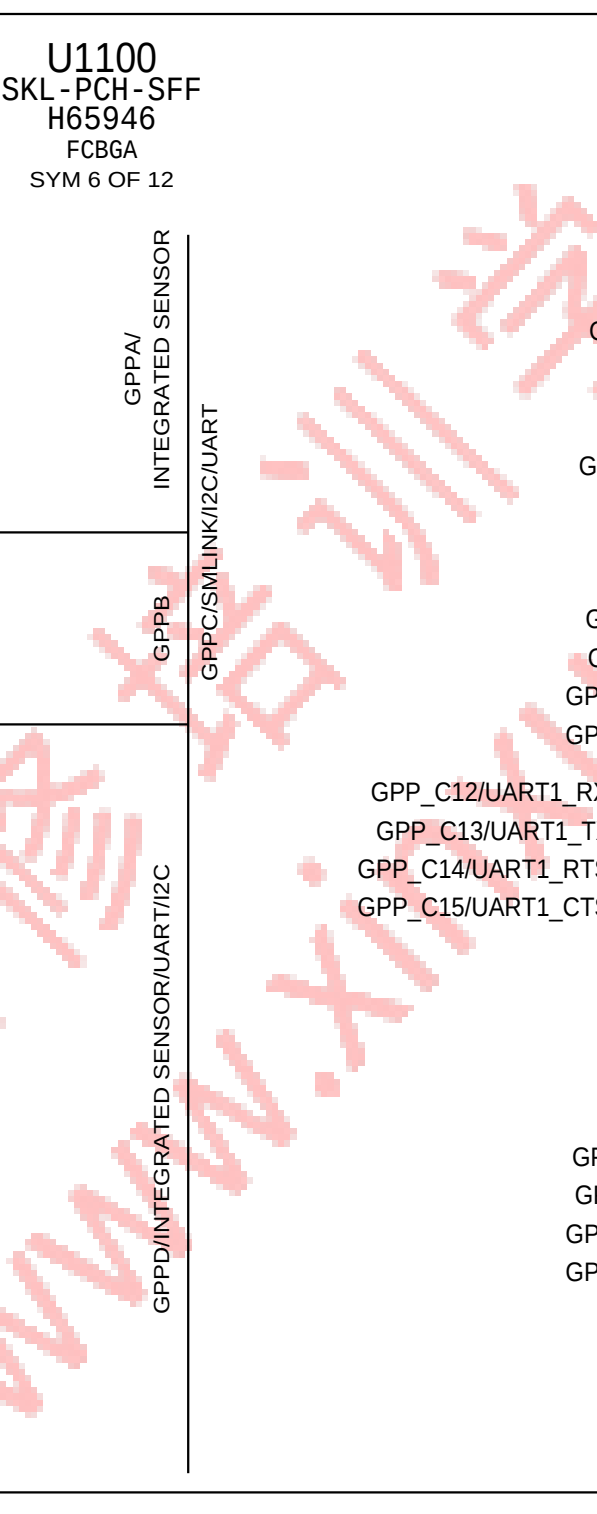
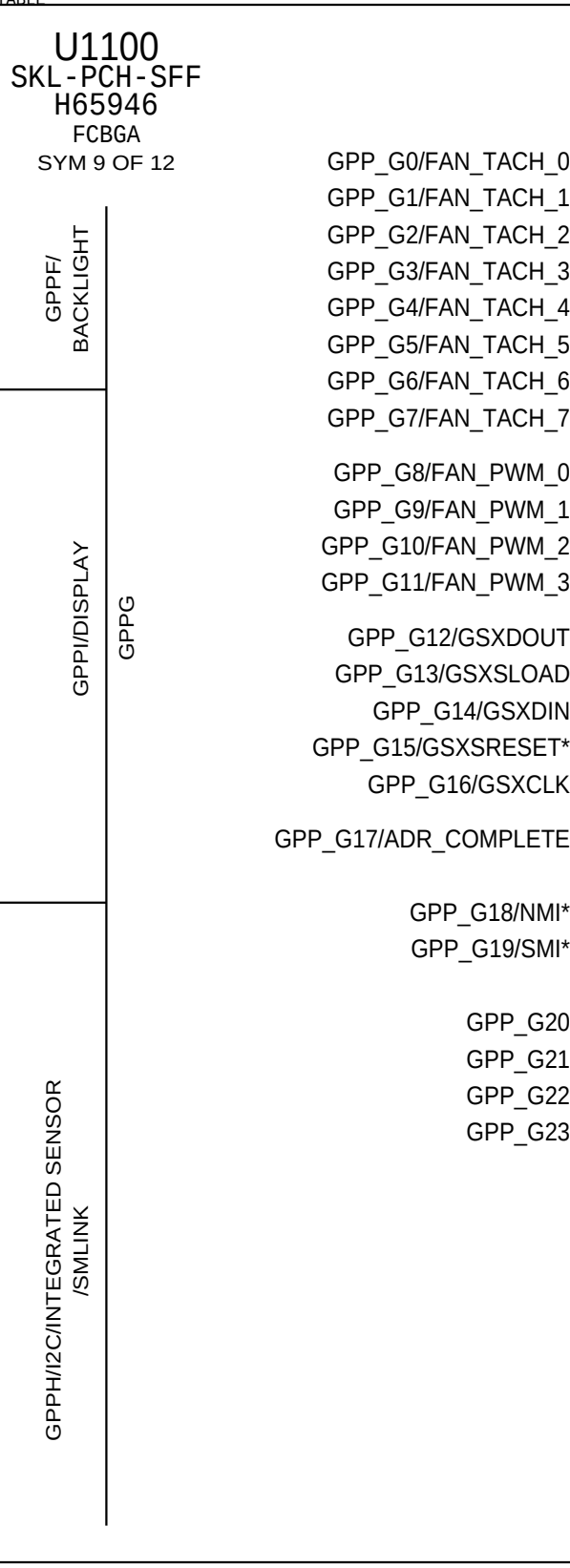
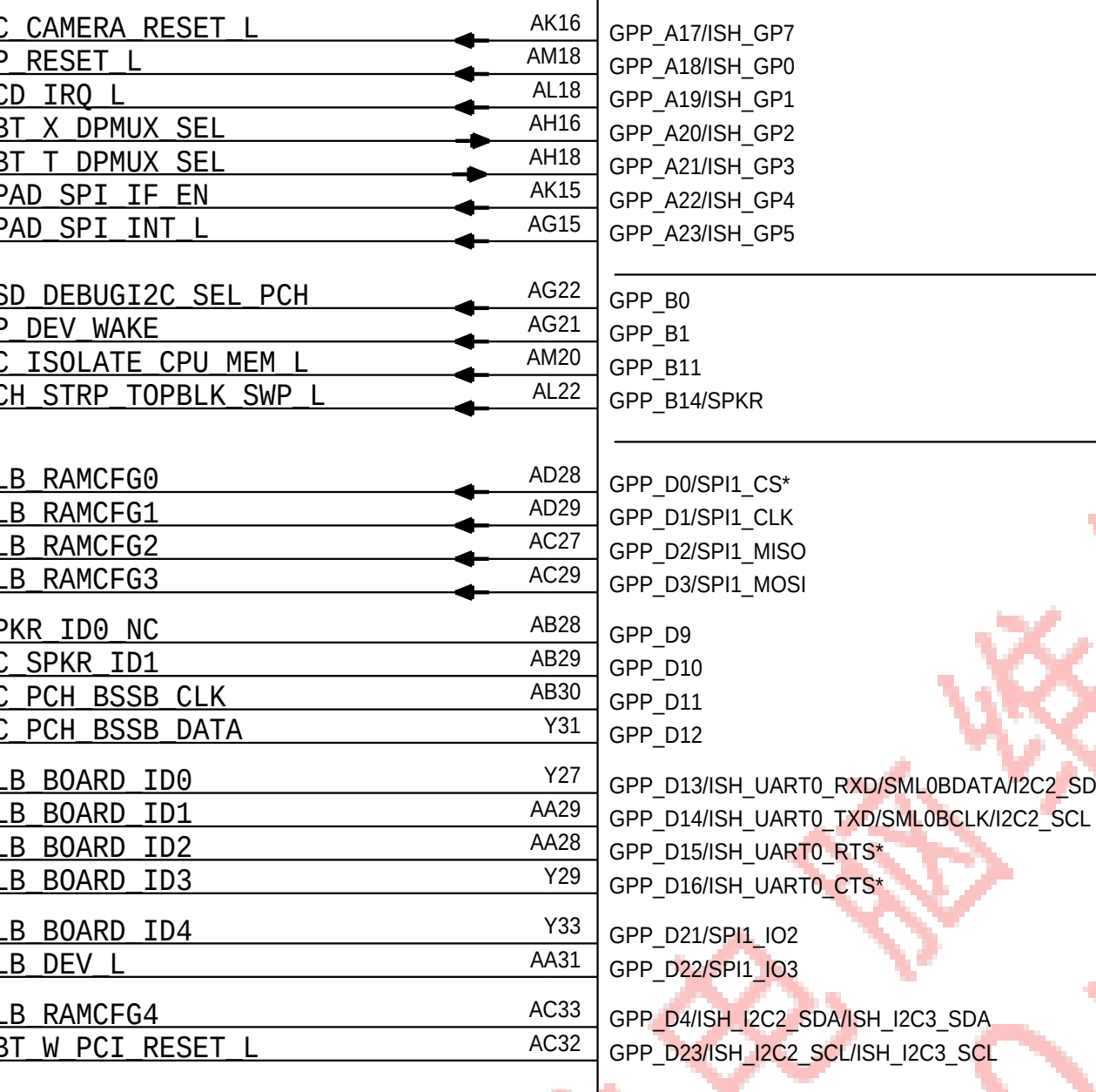
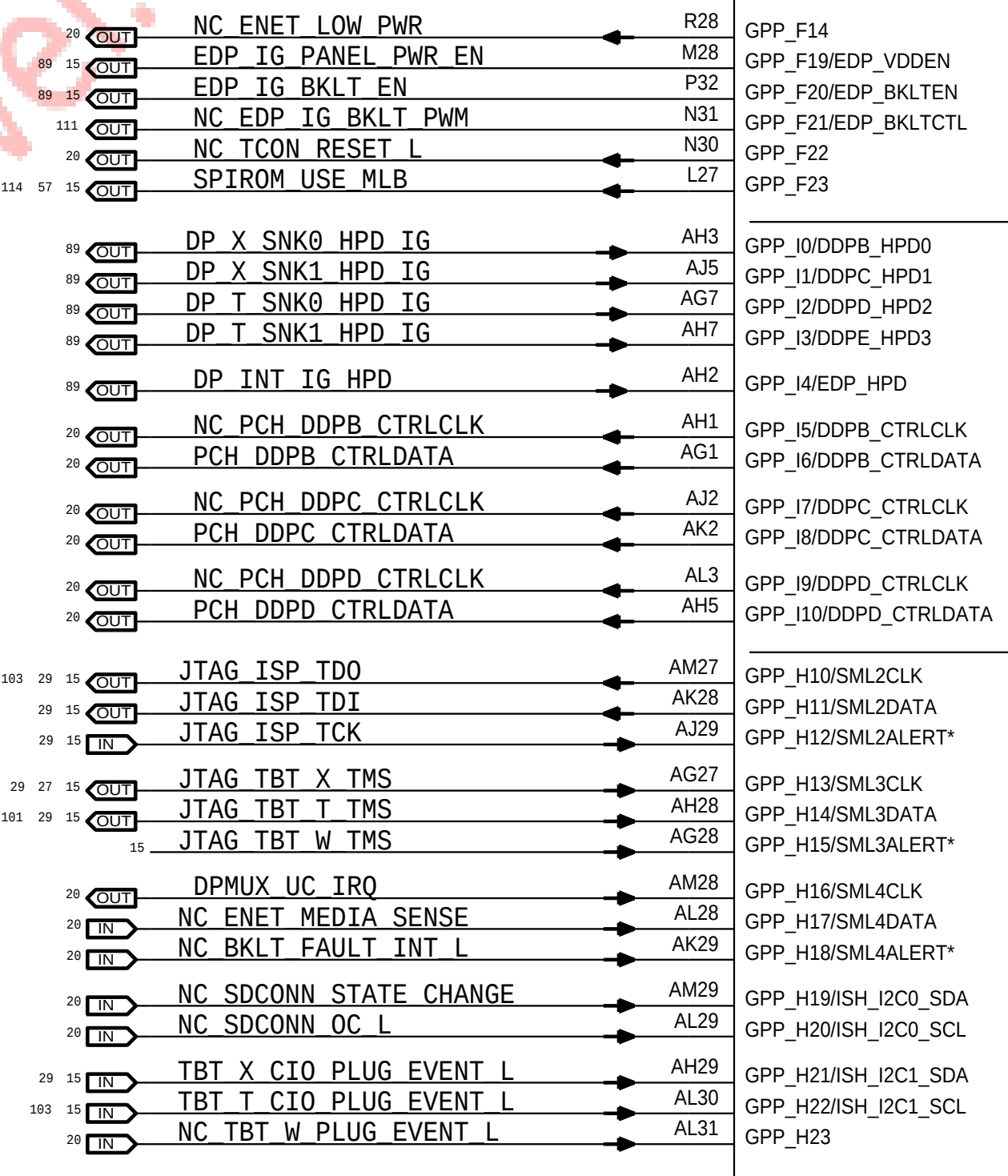
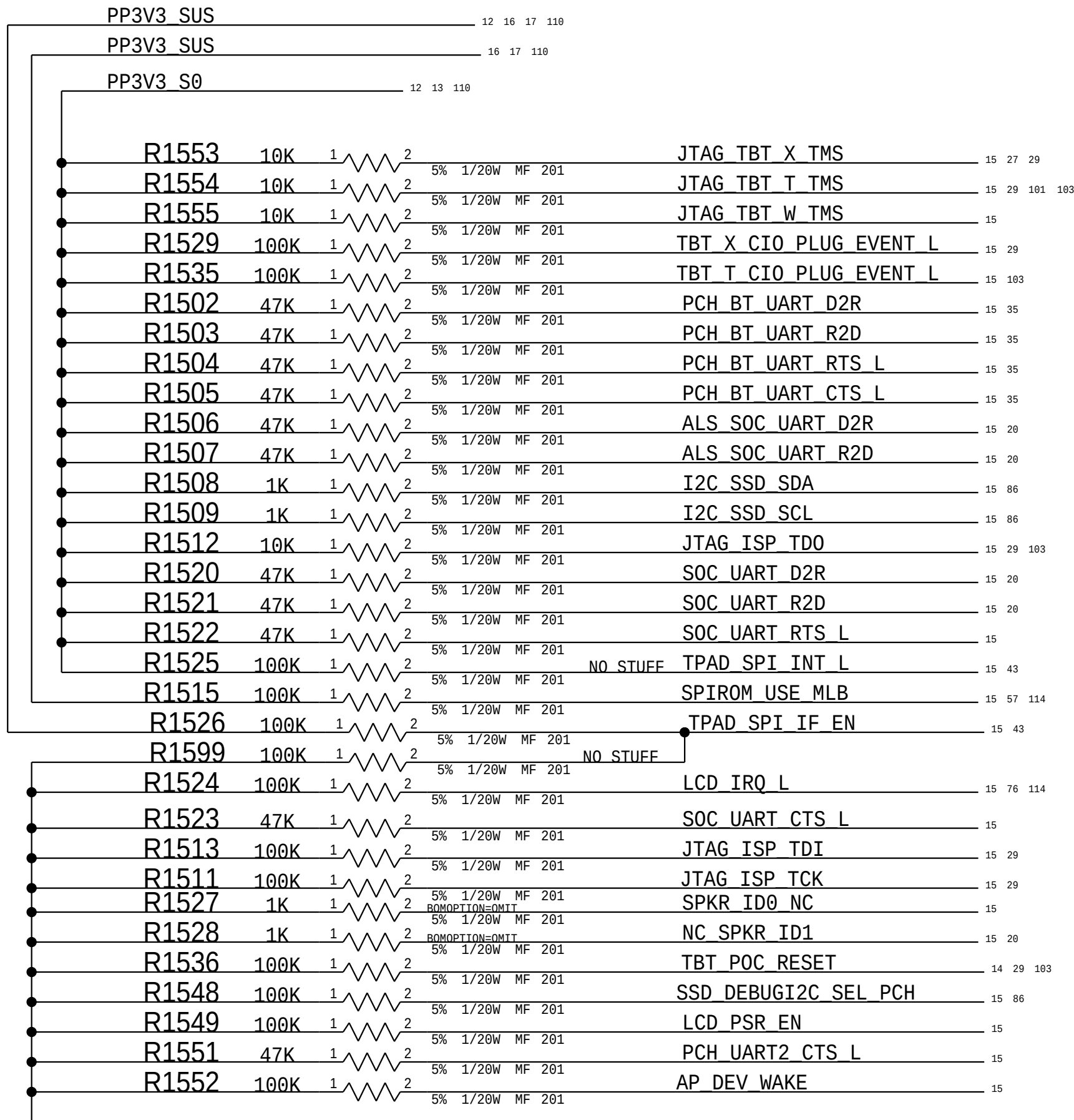
RAM Configuration Straps

BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG4:L, RAMCFG3:L, RAMCFG2:L, RAMCFG1:L, RAMCFG0:L

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0006	0	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD		BOARD_ID:1F
117S0006	1	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1540	BOARD_ID:1E
117S0006	1	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1541	BOARD_ID:1D
117S0006	2	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1541, R1540	BOARD_ID:1C
117S0006	1	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1542	BOARD_ID:1B
117S0006	2	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1542, R1540	BOARD_ID:1A
117S0006	2	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1542, R1541	BOARD_ID:19
117S0006	3	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1542, R1541, R1540	BOARD_ID:18
117S0006	1	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1543	BOARD_ID:17
117S0006	2	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1543, R1540	BOARD_ID:16
117S0006	2	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1543, R1541	BOARD_ID:15
117S0006	3	RES, MF, 1/20W, 1KOHM, 5%, 0201, SMD	R1543, R1541, R1540	BOARD_ID:14

PROTO 0 = 0x1F = 1 1 1 1 1 (-01 PCB)
PROTO 0B = 0x1E = 1 1 1 1 0 (-02 PCB)
PROTO 1 = 0x1D = 1 1 1 0 1 (-03 PCB)
PROTO 2 = 0x1C = 1 1 1 0 0 (-04 & -05 PCB)
EVT1 = 0x1B = 1 1 0 1 1 (-06 PCB)
EVT2 = 0x1A = 1 1 0 1 0 (-07 PCB)
DVT = 0x19 = 1 1 0 0 1 (-08 PCB)
DVT1-1 = 0x18 = 1 1 0 0 0 (-09 PCB)
PVT = 0x17 = 1 0 1 1 1 (-10 PCB)
0x16 = 1 0 1 1 0
0x15 = 1 0 1 0 1
0x14 = 1 0 1 0 0

MLB BOARD ID Configuration Straps



PAGE TITLE		PCH GPIO/MISC/NCTF	
Apple Inc.		DRAWING NUMBER	051-00647
		REVISION	10.0.0
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BOM_COST_GROUP=CPU & CHIPSET



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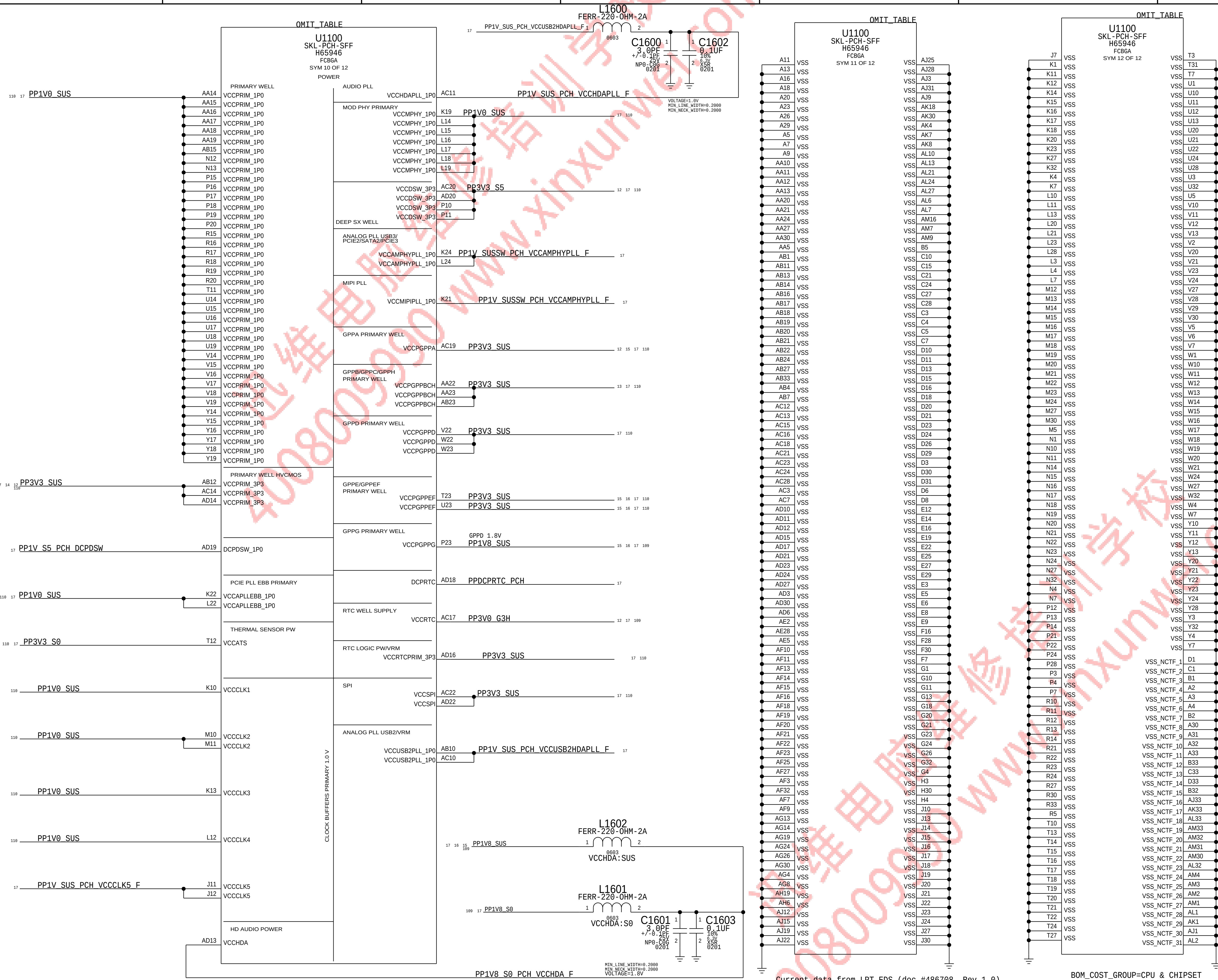
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REVISION		10.0.0		
BRANCH		dvt-fab10		
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Current data from LPT EDS (doc #486708, Rev 1.0).

BOM_COST_GROUP=CPU & CHIPSET



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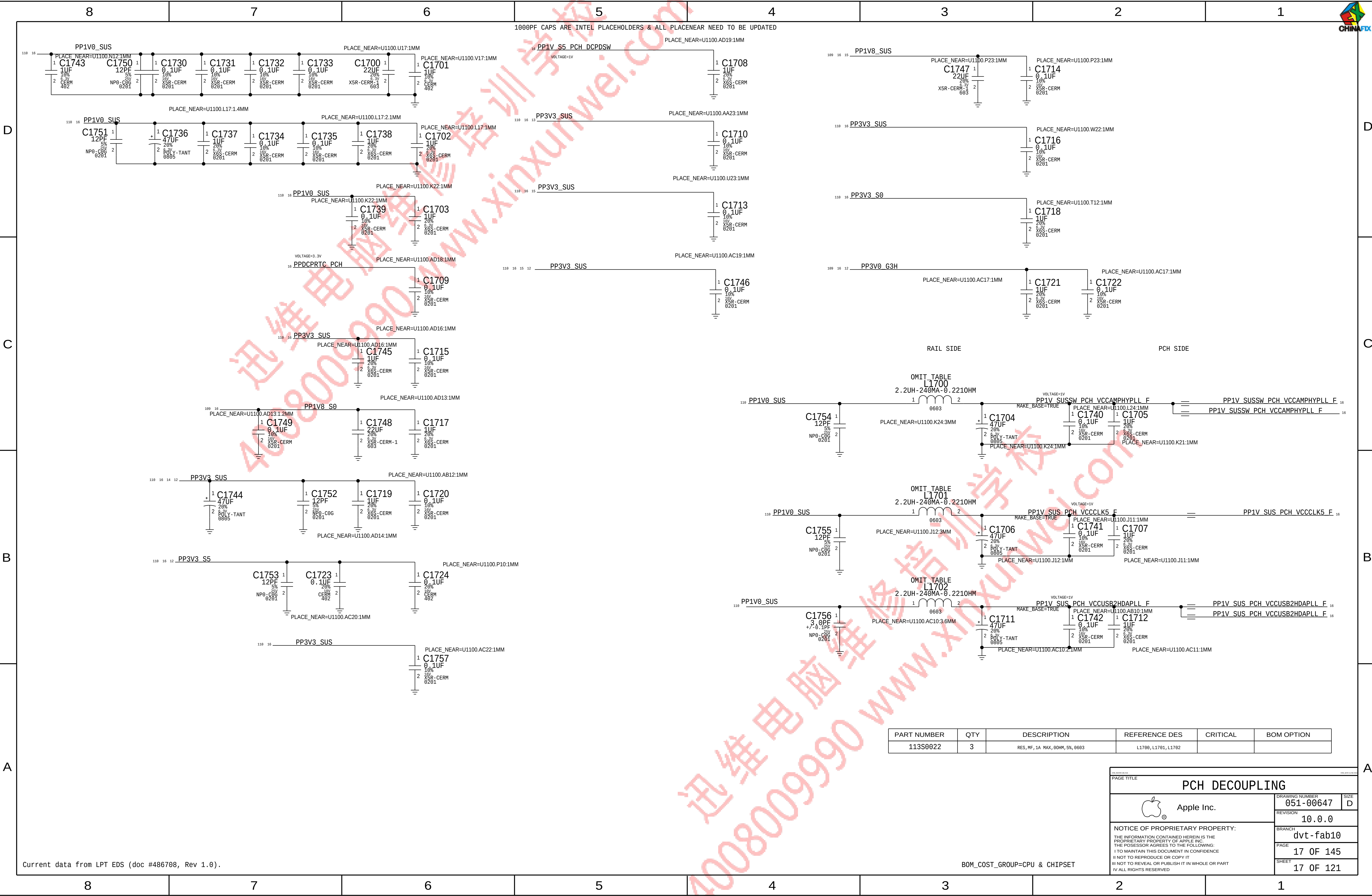
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	3	RES, MF, 1A, MAX, 00HM, 5%, 0603	L1700, L1701, L1702		

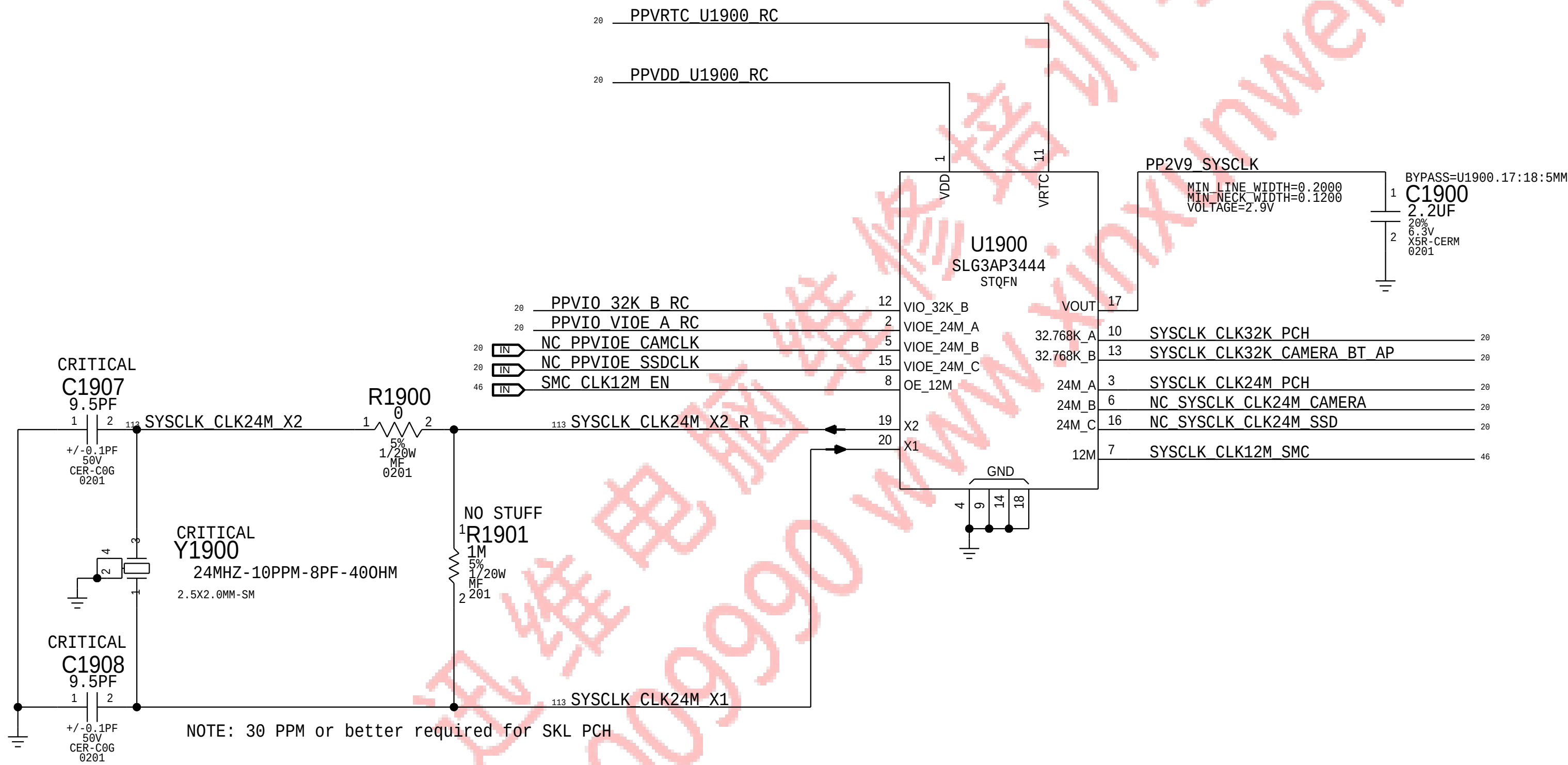
Current data from LPT EDS (doc #486708, Rev 1.0).

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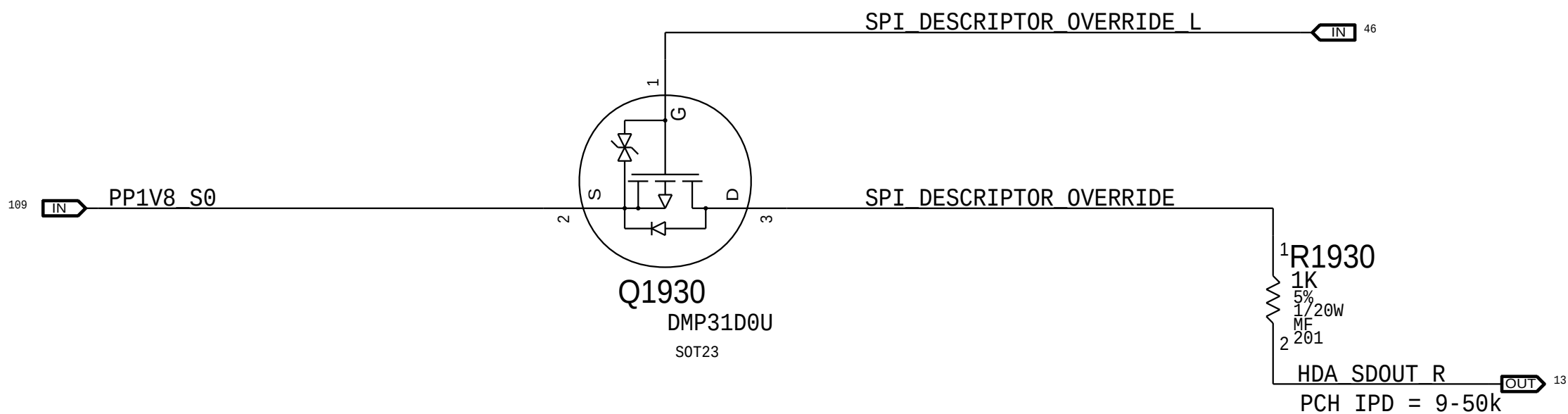
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System 32kHz / 12MHz / 24MHz Clock Generator

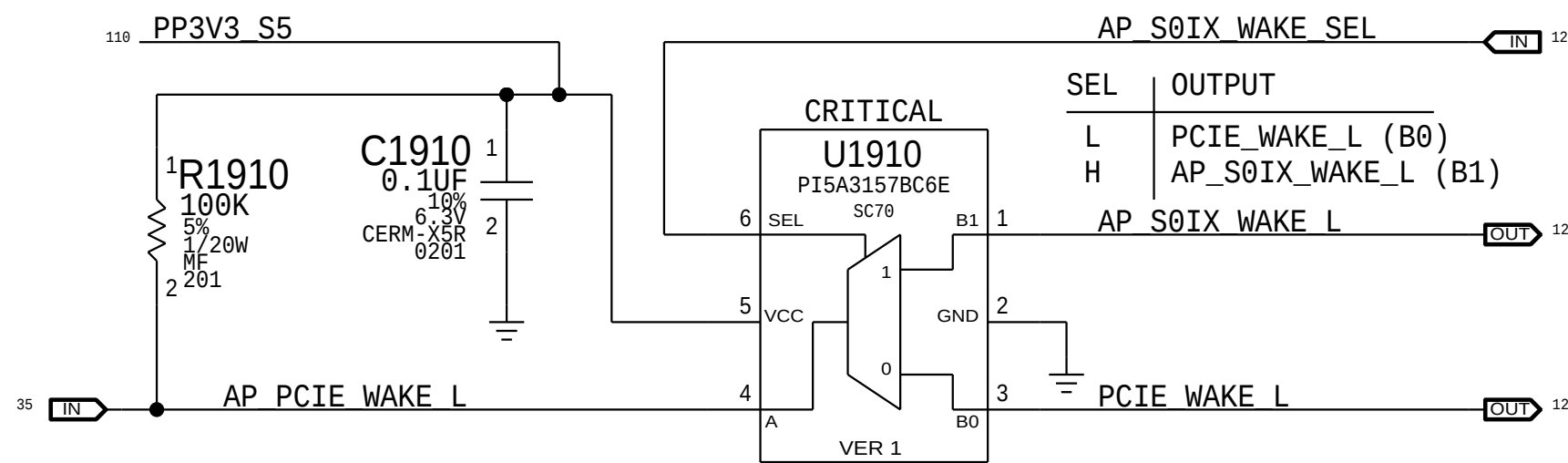


PCH ME Disable Strap



PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. ***** Circuit does not support HDA voltage >3.3V.

PCIe Wake Muxing

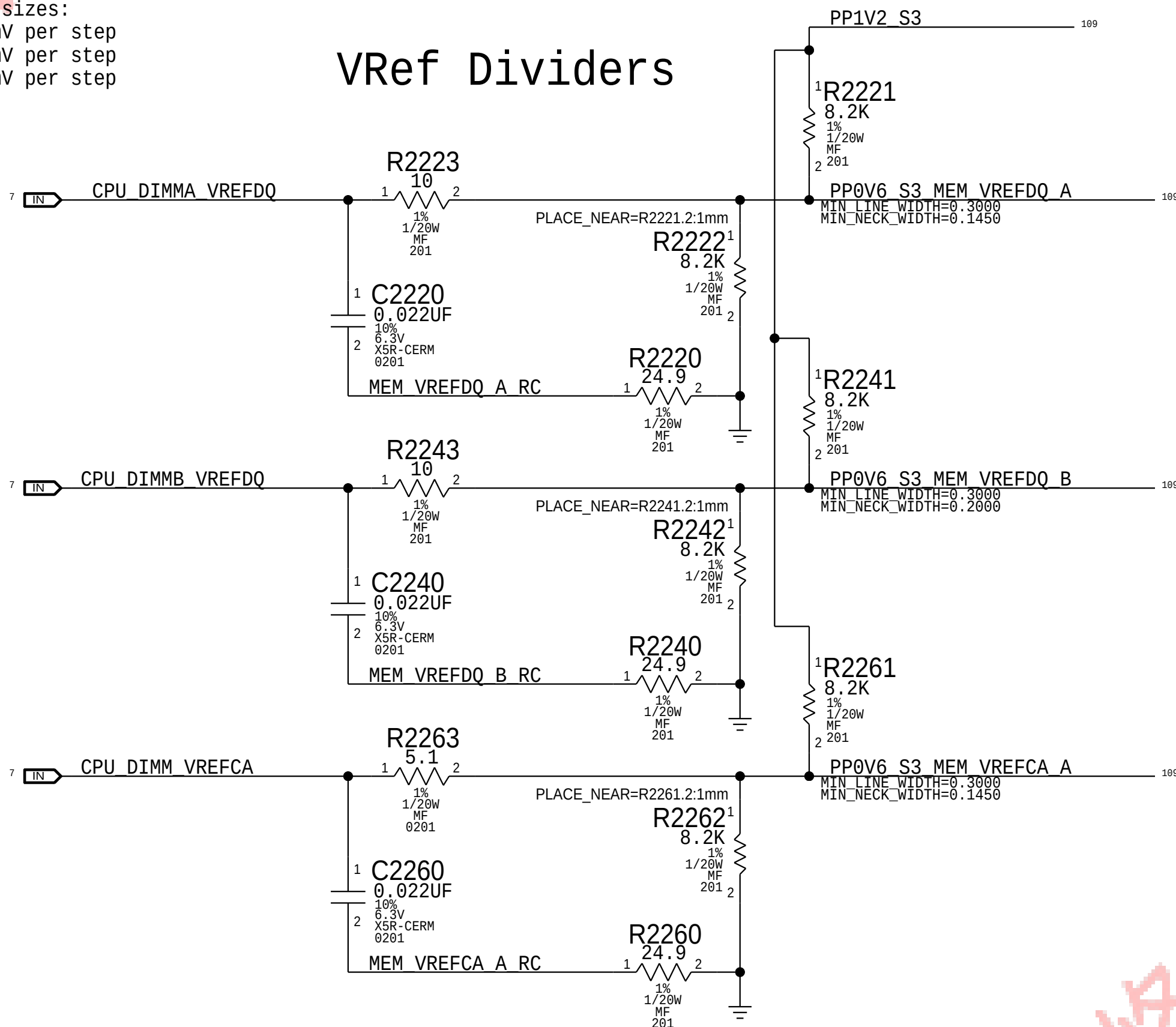


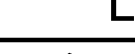
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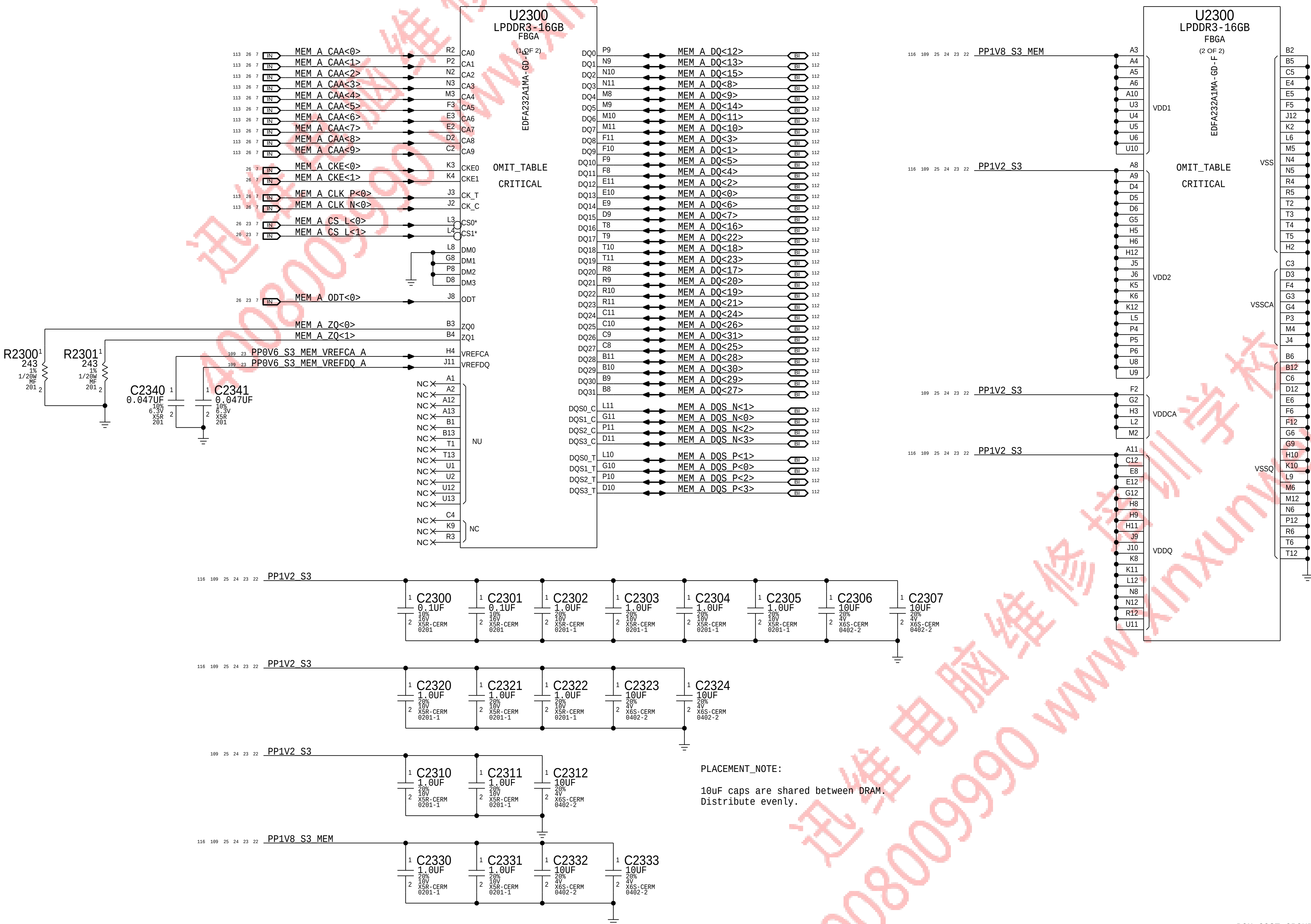
CPU-Based Margining

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step
LPDDR3 (1.2V) 7.70mV per step



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	Apple Inc.		DRAWING NUMBER	SIZE
			051-00647	D
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		PAGE		dvt-fab10
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LPDDR3 CHANNEL A (0-31)



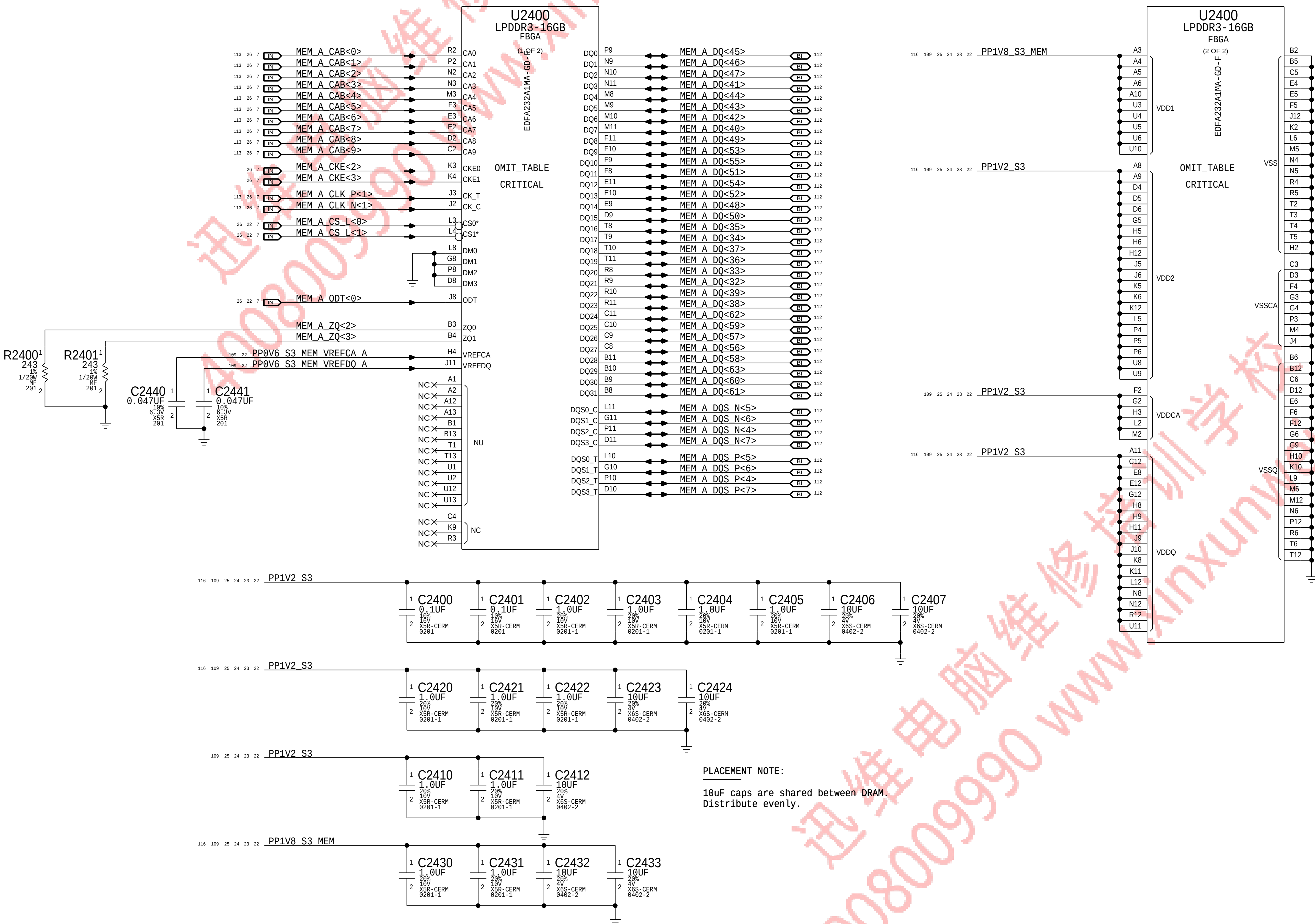
PLACEMENT_NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

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		10.0.0	
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BOM_COST_GROUP=DRAM

LPDDR3 CHANNEL A (32-63)

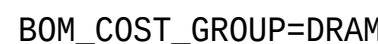


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		REVISION	10.0.0
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		PAGE	24 OF 145
		SHEET	23 OF 121

BOM_COST_GROUP=DRAM

PLACEMENT_NOTE:

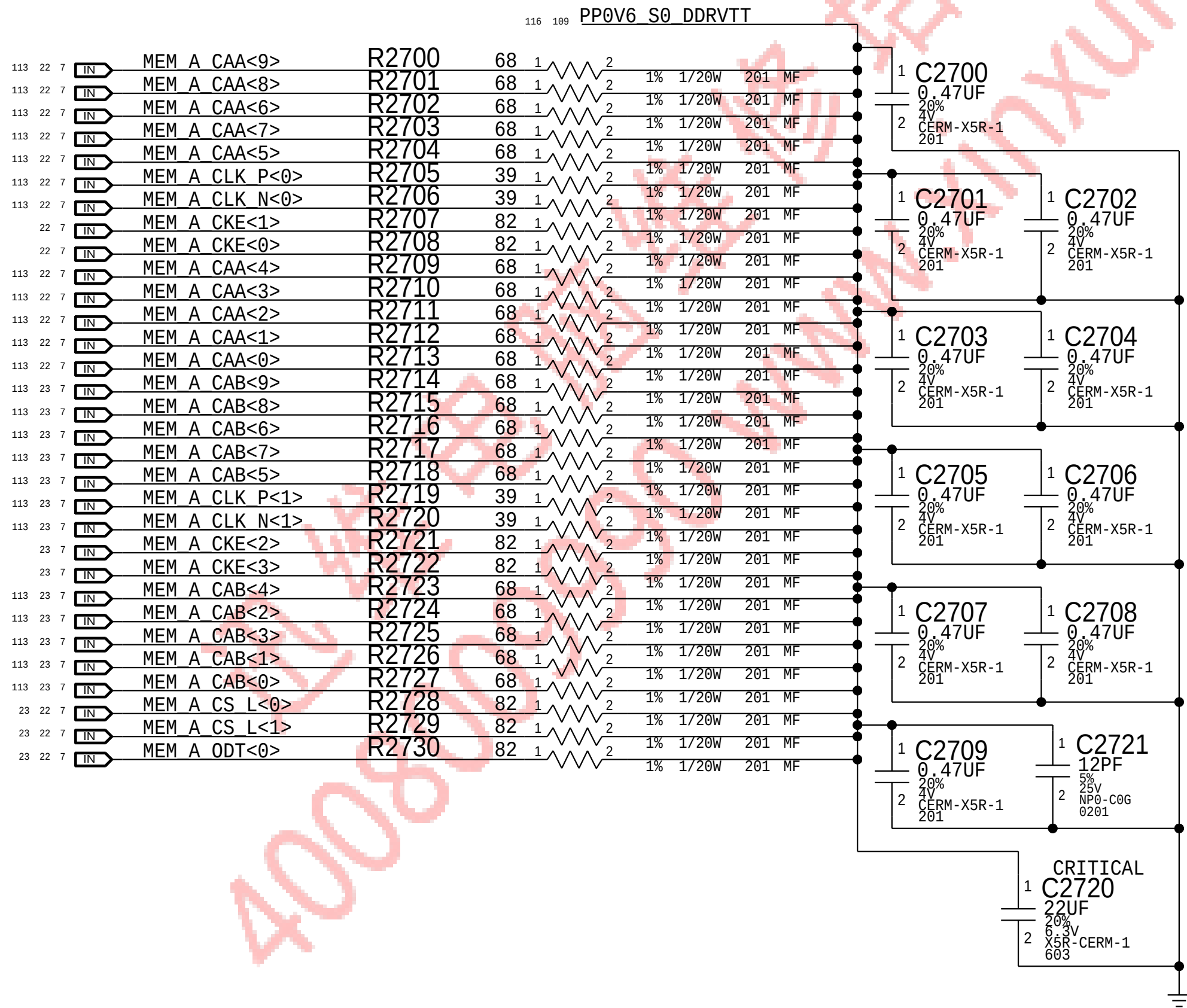
10uF caps are shared between DRAM
Distribute evenly.



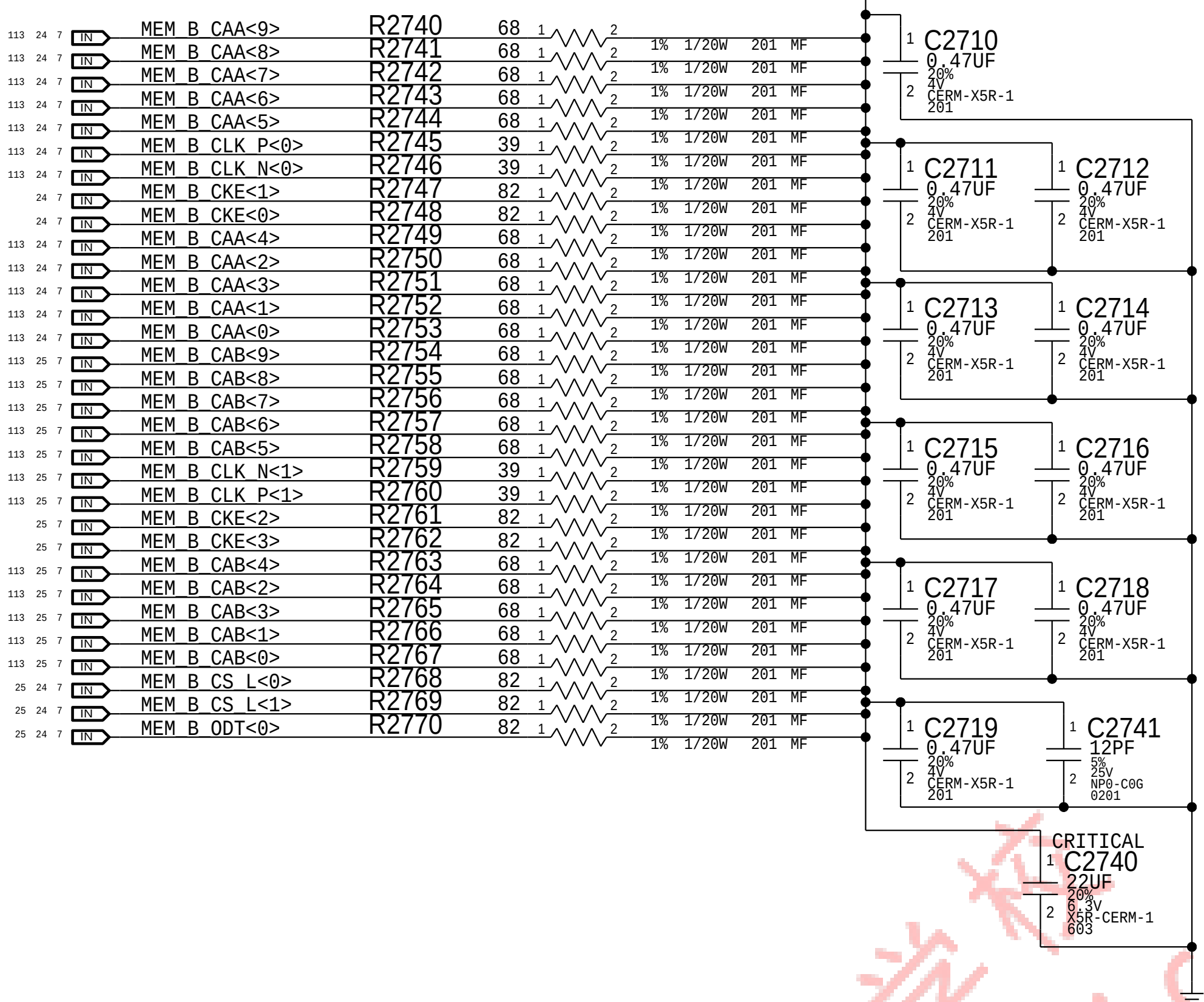
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		BRANCH	dvt-fab10
		PAGE	25 OF 145
		SHEET	24 OF 121



Intel recommends 68 Ohm for CMD/ADDR, 80 Ohm for CTRL/CKE, 38 Ohm for CLK

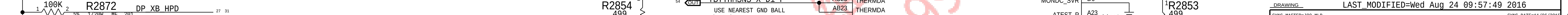


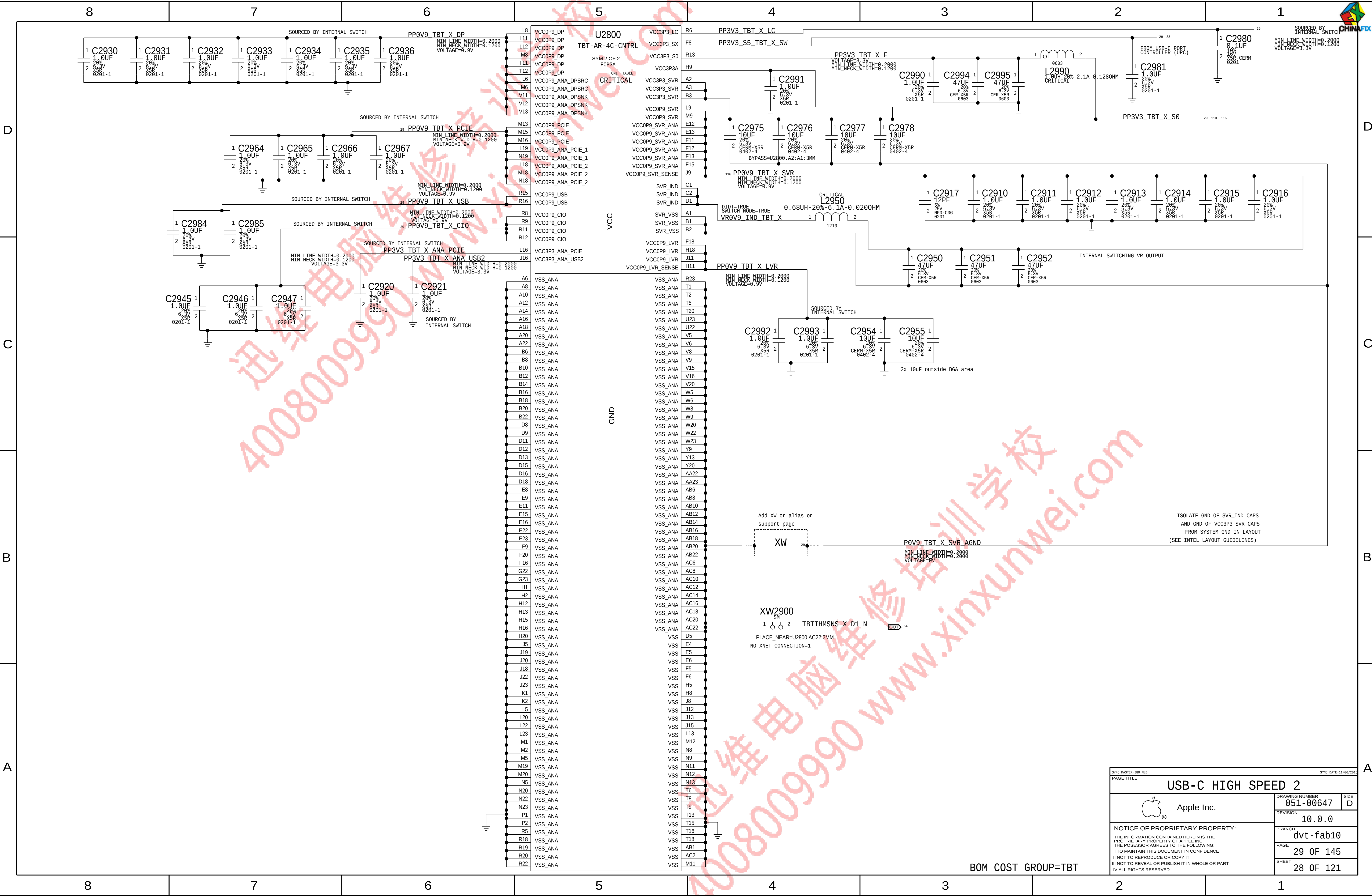
116 189 PP0V6 S0 DDRVTT



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	REVISION	10.0.0	
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BOM_COST_GROUP=DRAM





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REVISION		10.0.0		
BRANCH		dvt-fab10		
PAGE		29 OF 145		
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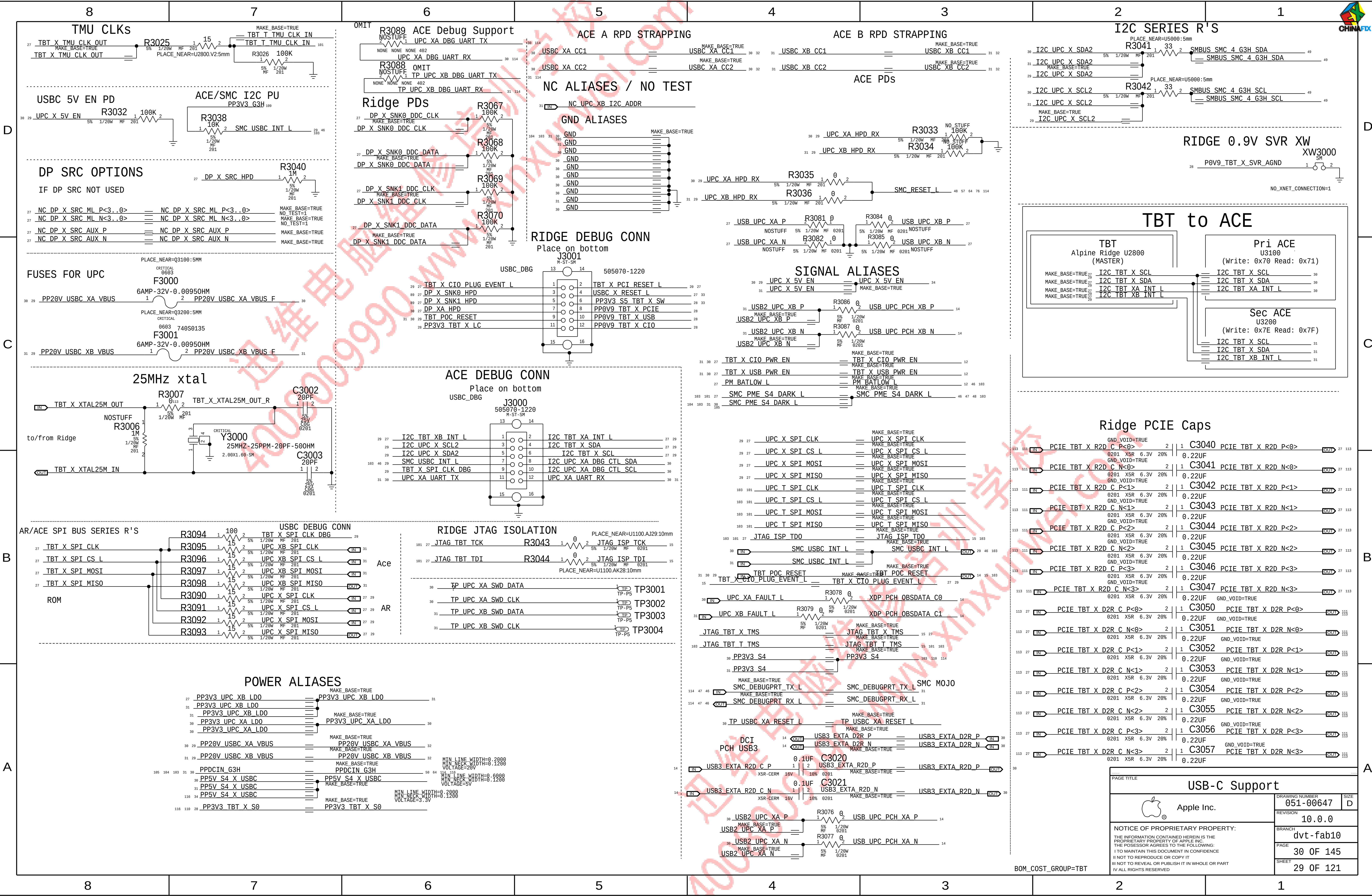
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USB-C HIGH SPEED 2



Apple Inc.

BOM_COST_GROUP=TBT



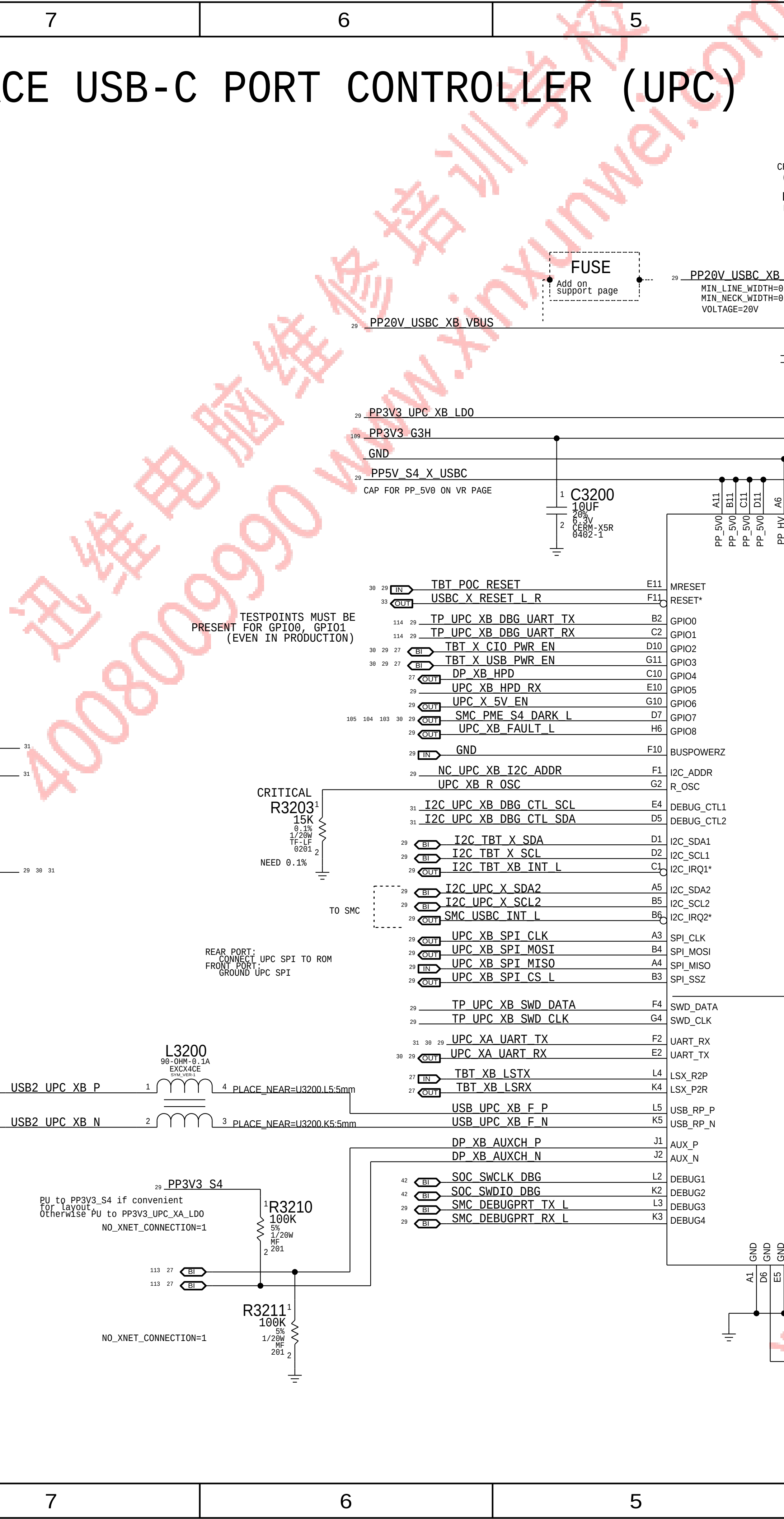
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		PAGE	30 OF 145	
		SHEET	29 OF 121	

BOM_COST_GROUP=USB-C



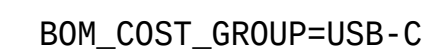
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LAST CHANGE: Wed Apr 1 22:57:37 2015			
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USB-C CONNECTOR A			
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		PAGE	33 OF 145
		SHEET	32 OF 121

BOM_COST_GROUP=USB-C



DESIGN: X502/DEV MLB U			
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BRANCH		dvt-fab10	
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SHEET		33 OF 121	



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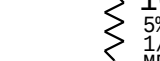
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SEP ROM

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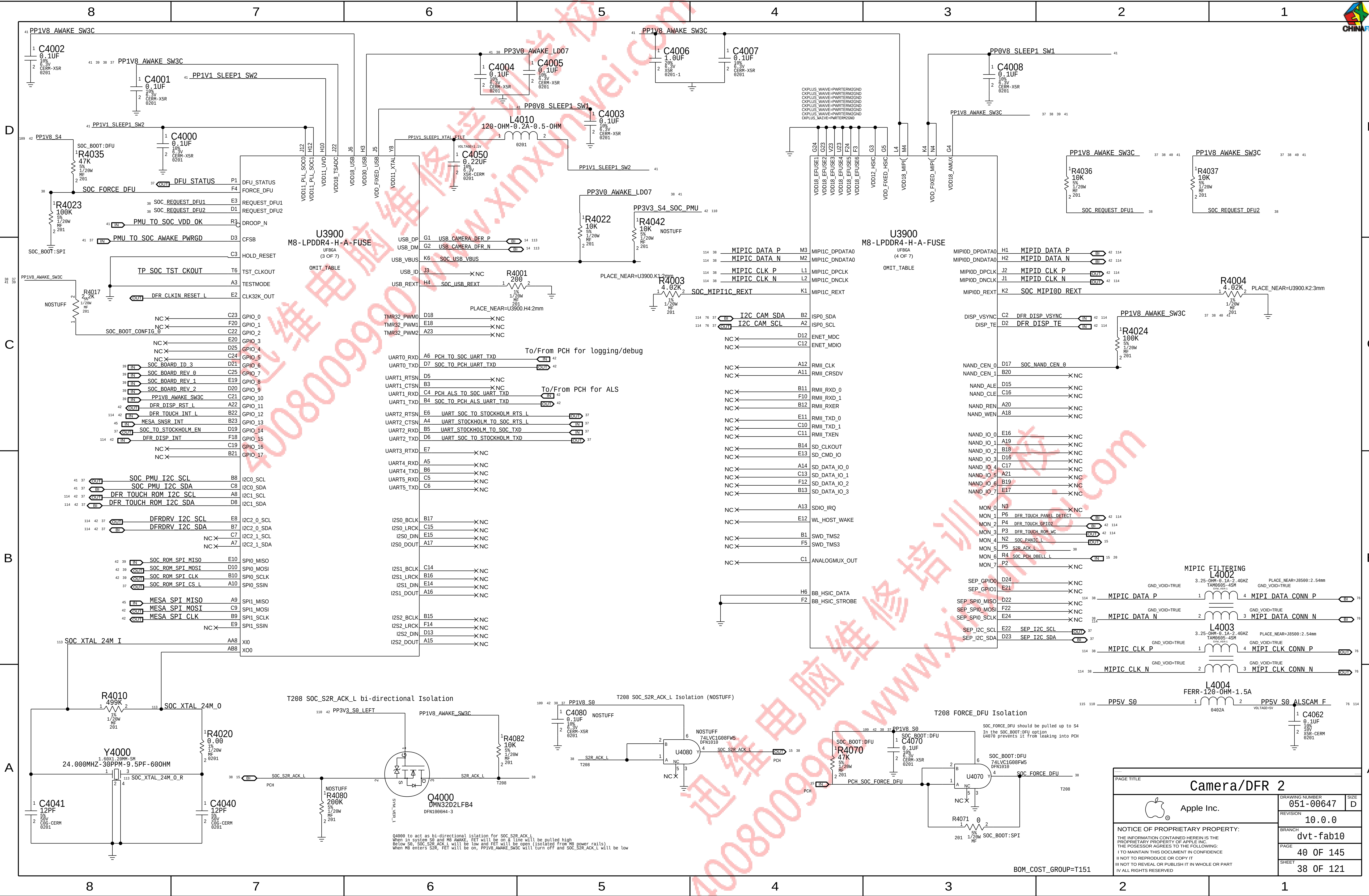
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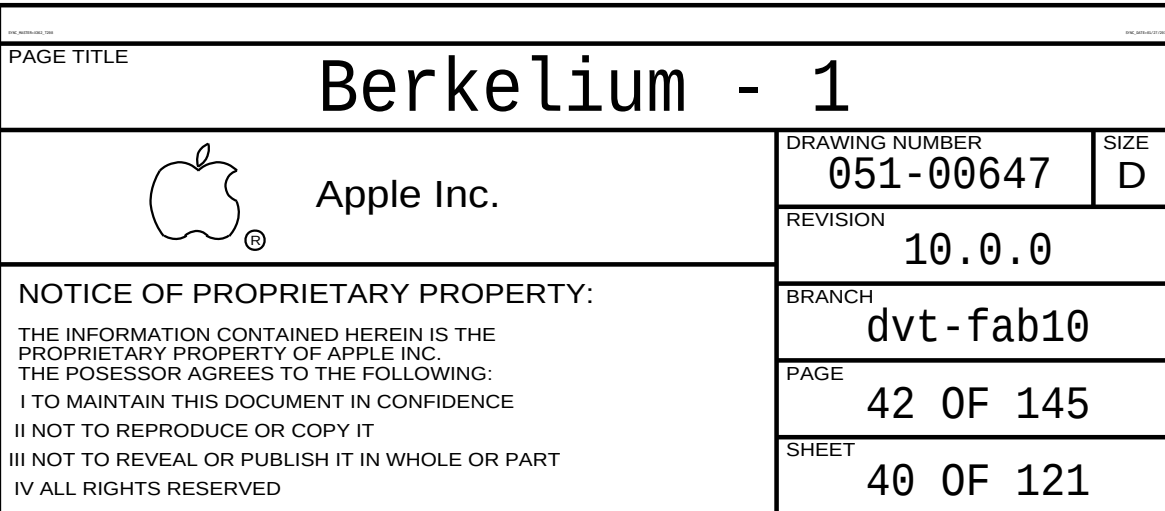
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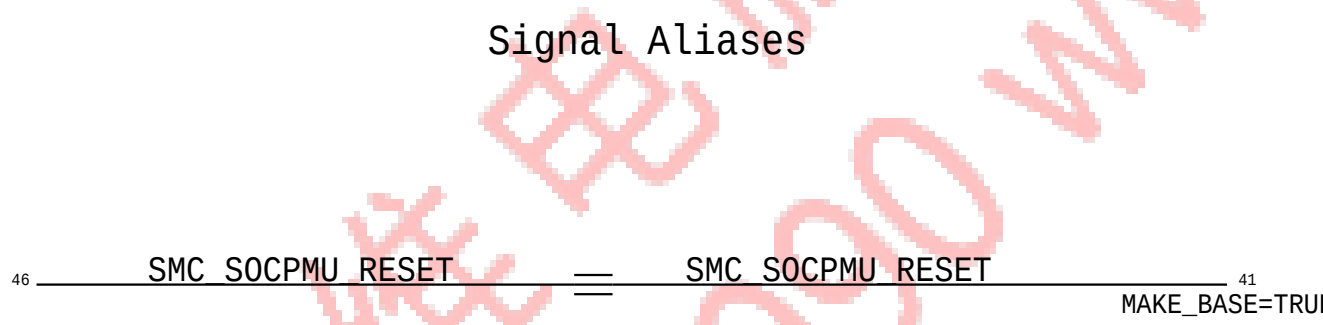
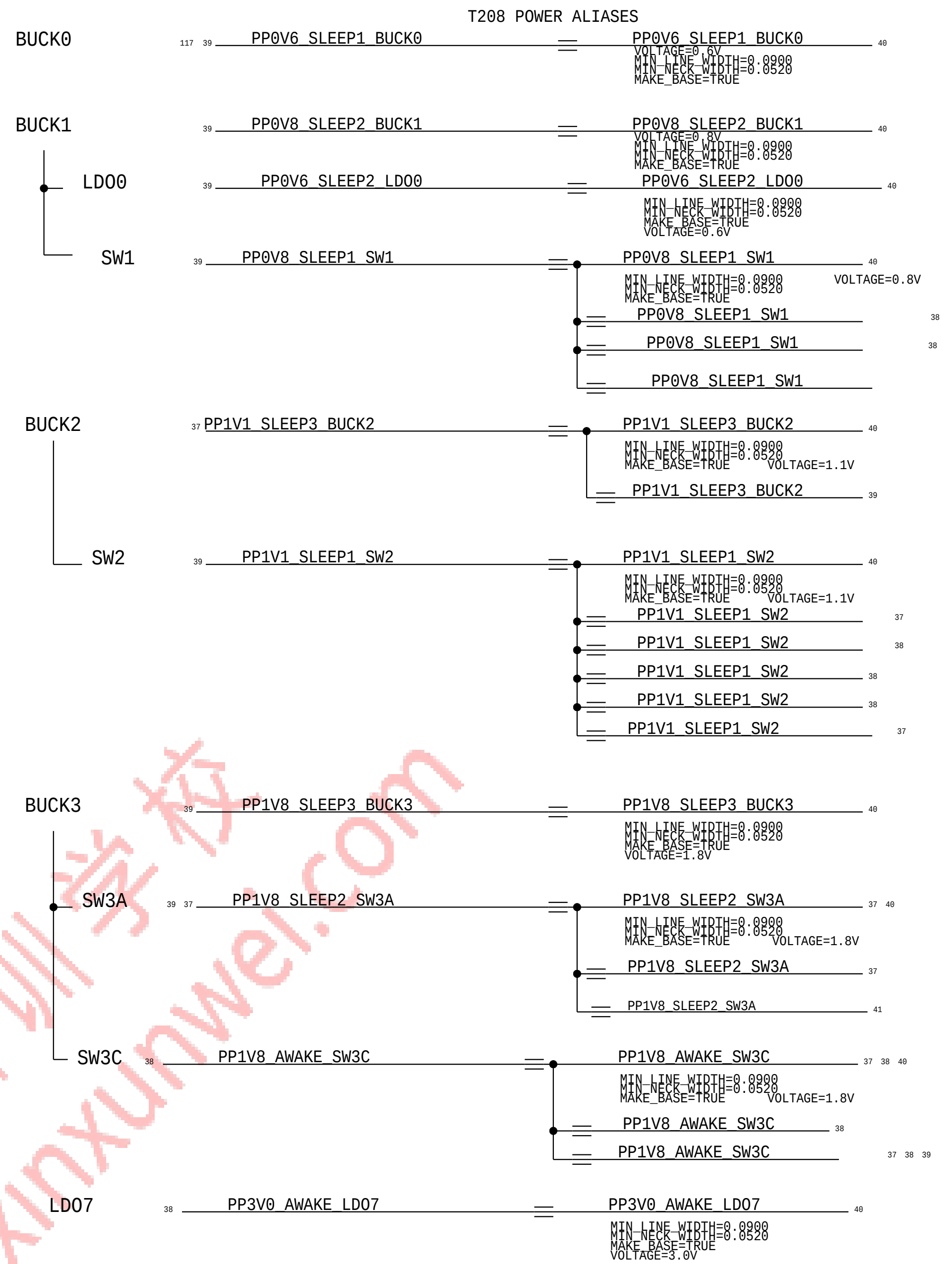
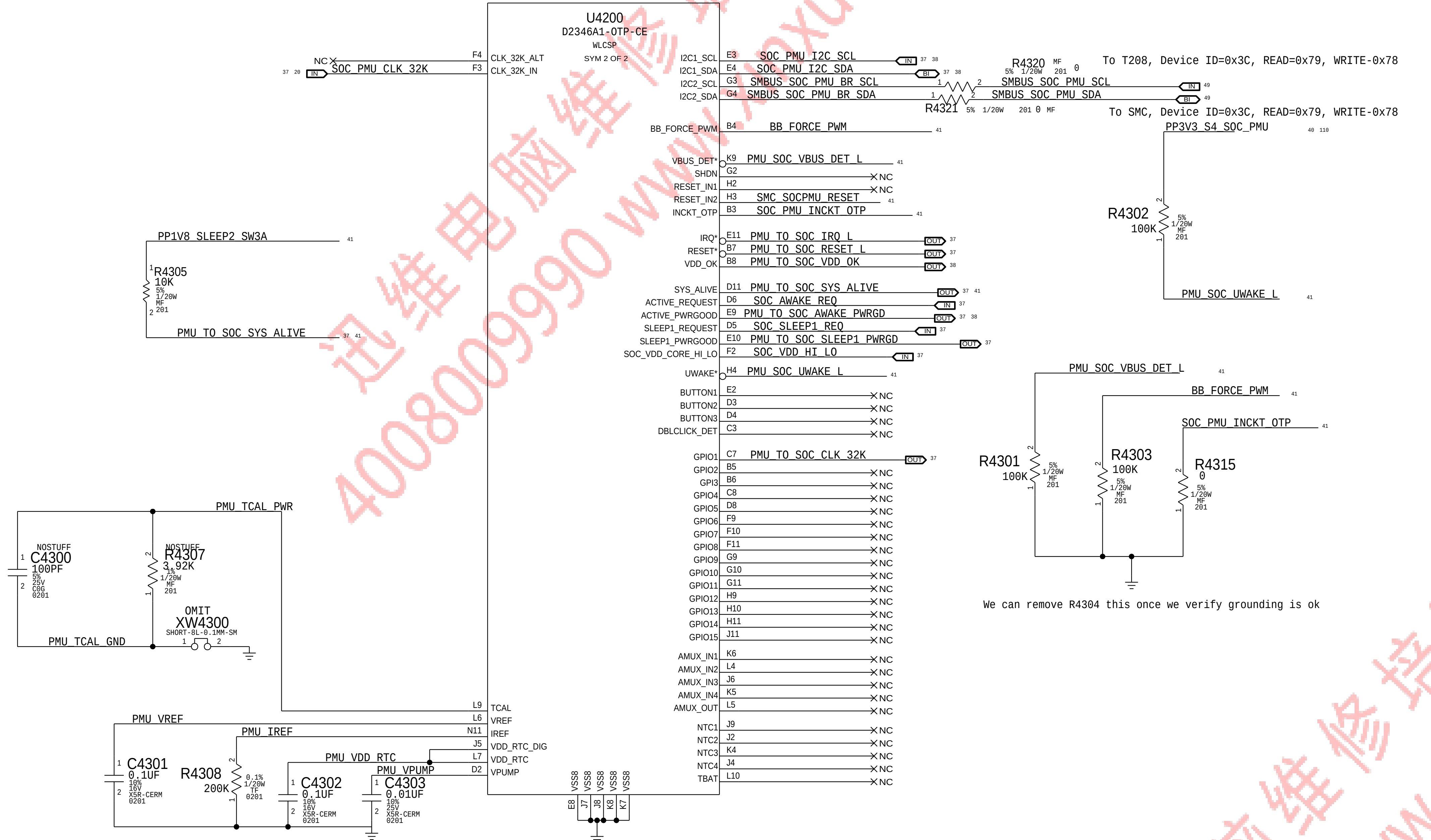




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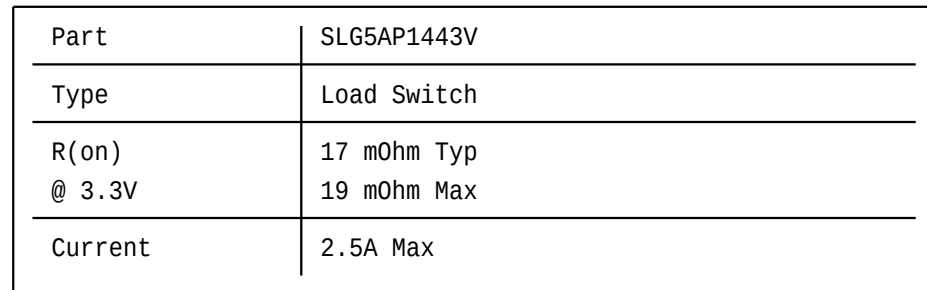


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		PAGE	43 OF 145
		SHEET	41 OF 121

T208 Support



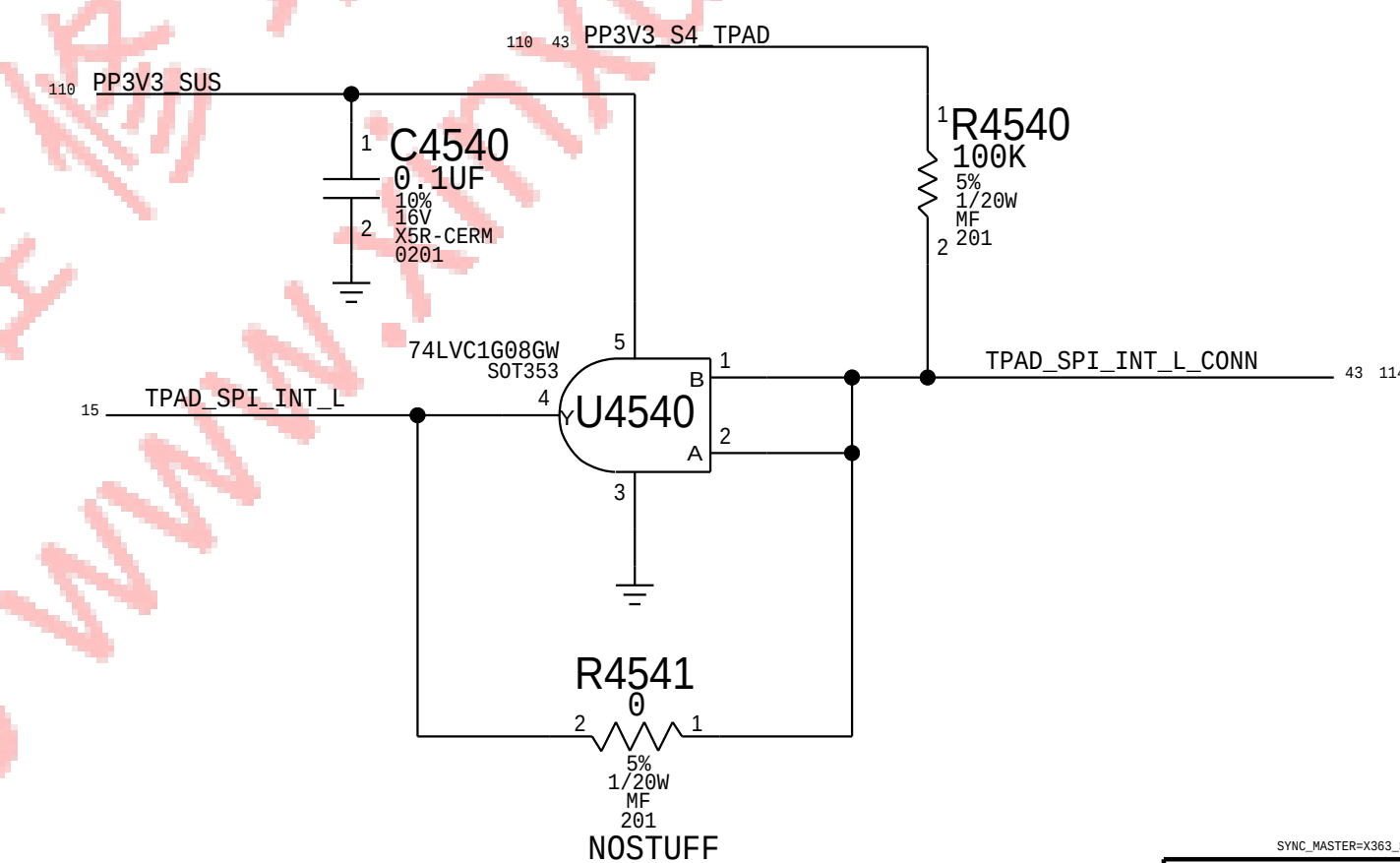
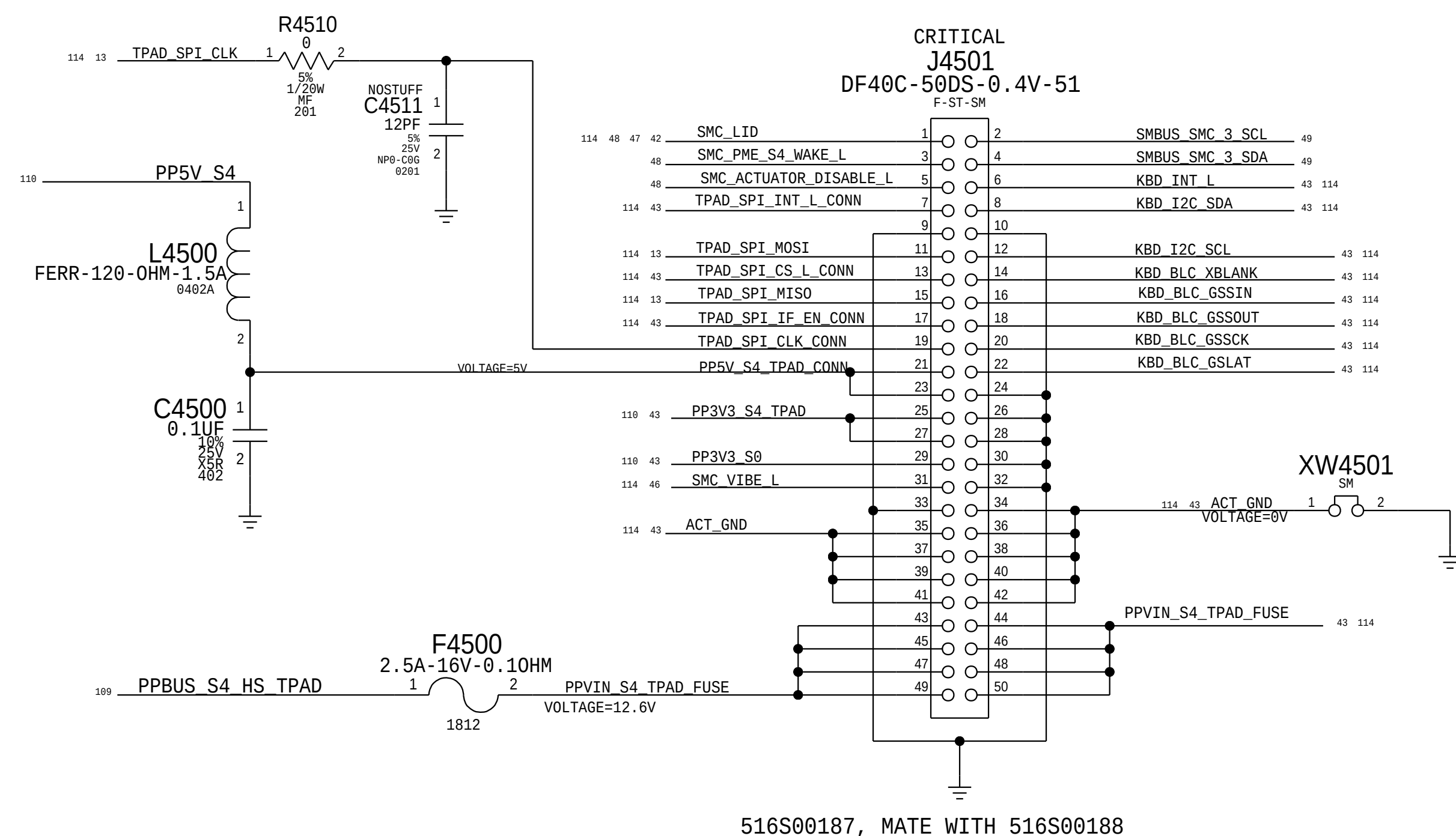
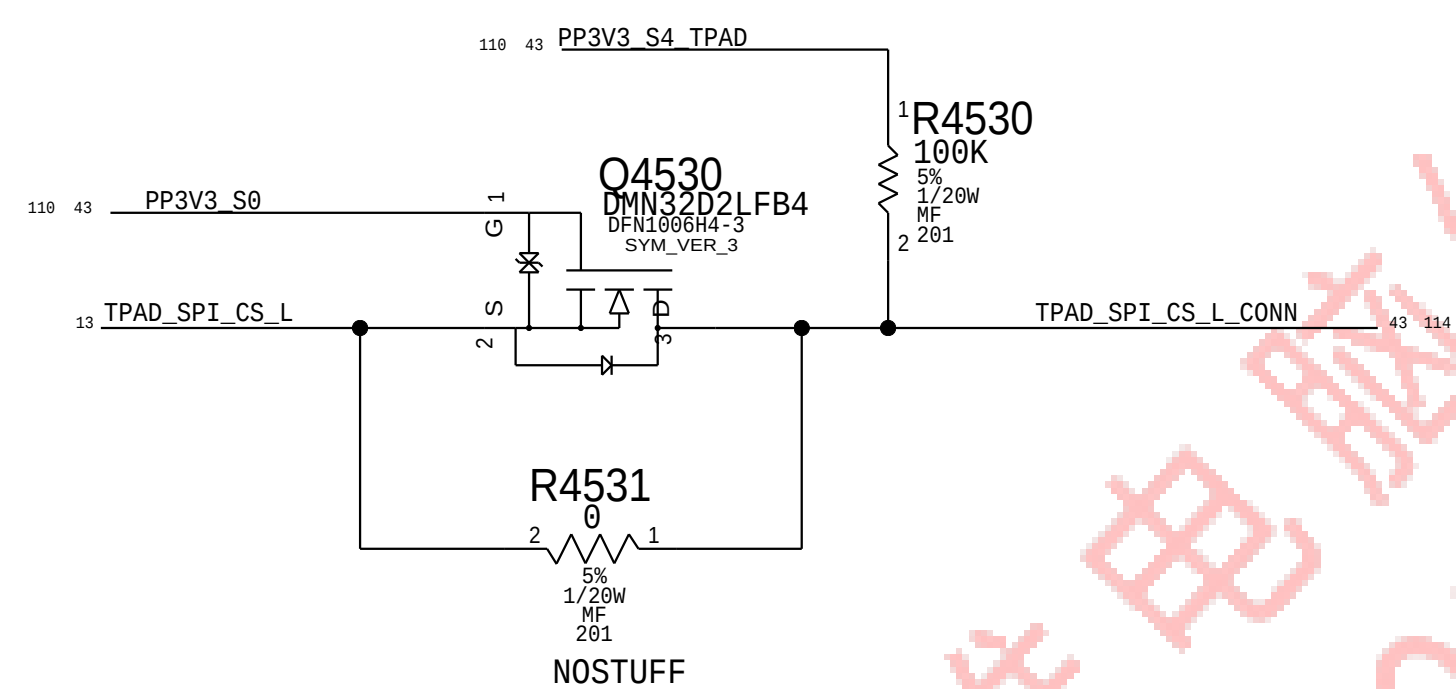
T208 I2C Mapping



T208 LEVEL SHIFTING



T208 Support



SYNCH_MSTER-X3B3_SAMANTHA		SYNCH_DATE-02/02/2006	
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Connectors&ESD			
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			43 OF 121

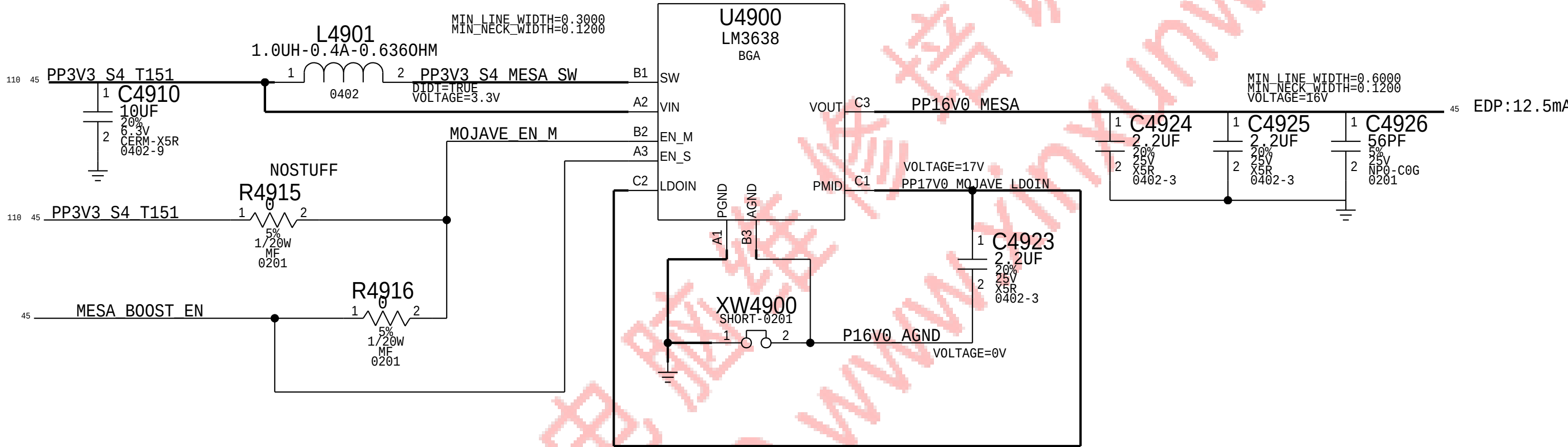
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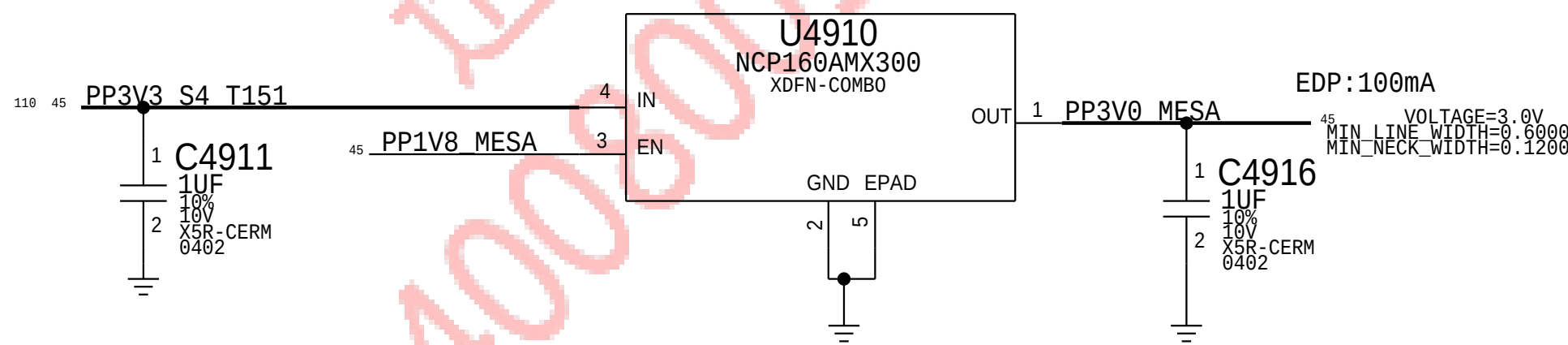
ISOLATE FROM OTHER COMPONENTS/NETS AS MUCH AS POSSIBLE

MOJAVE 16V BOOST

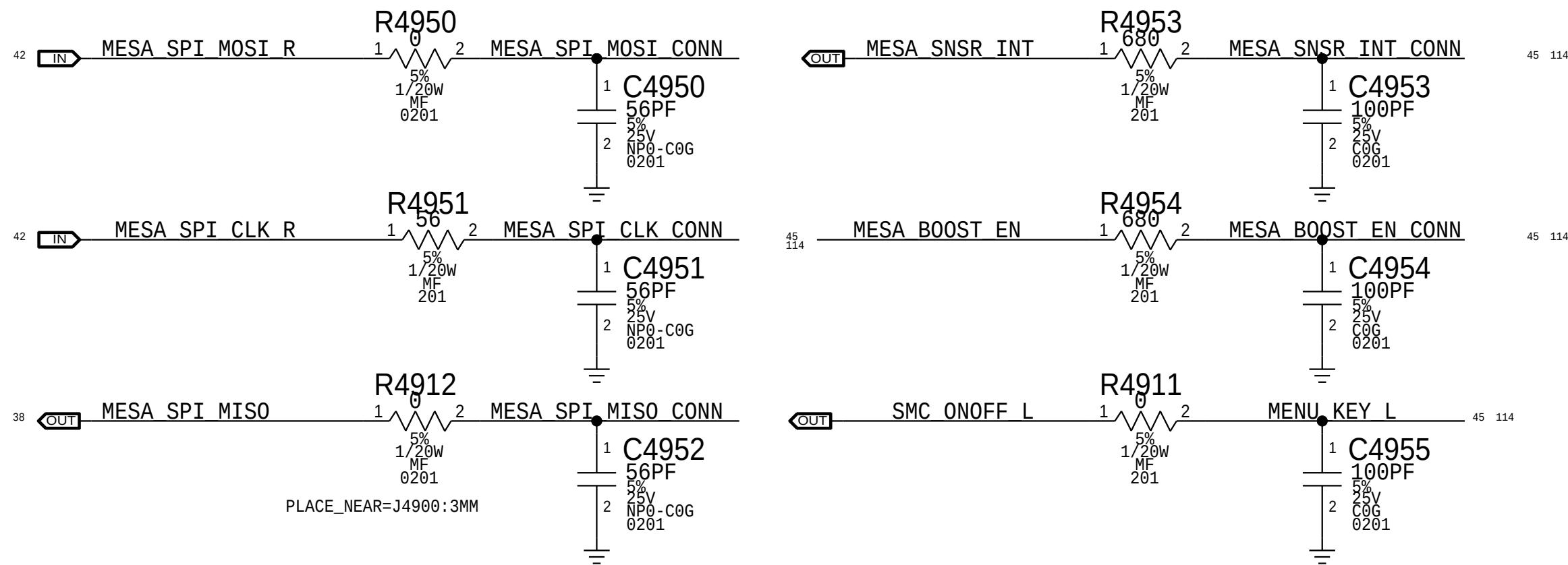
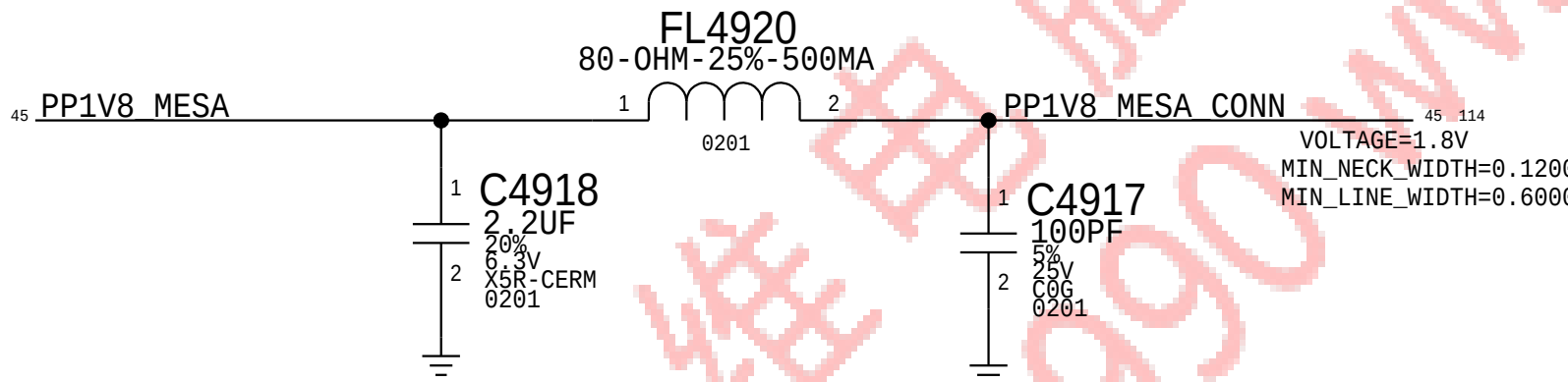
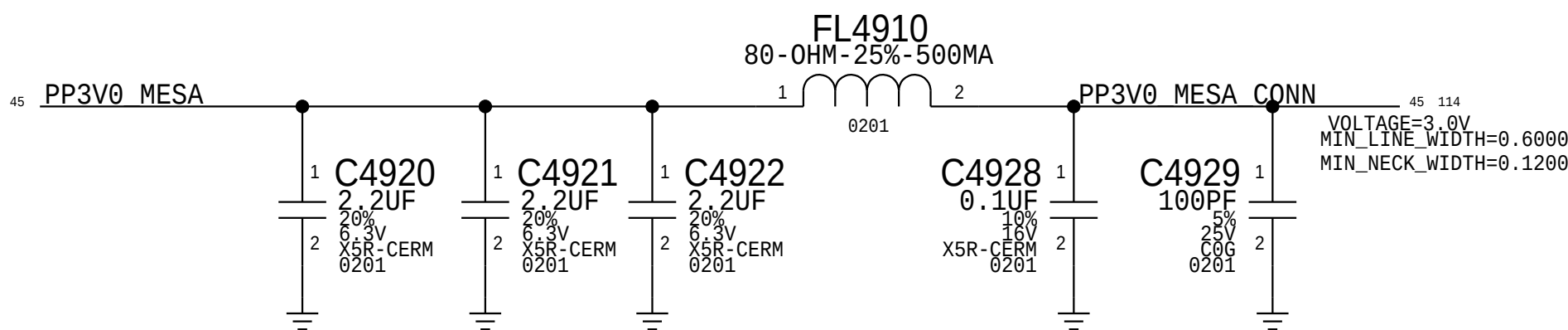
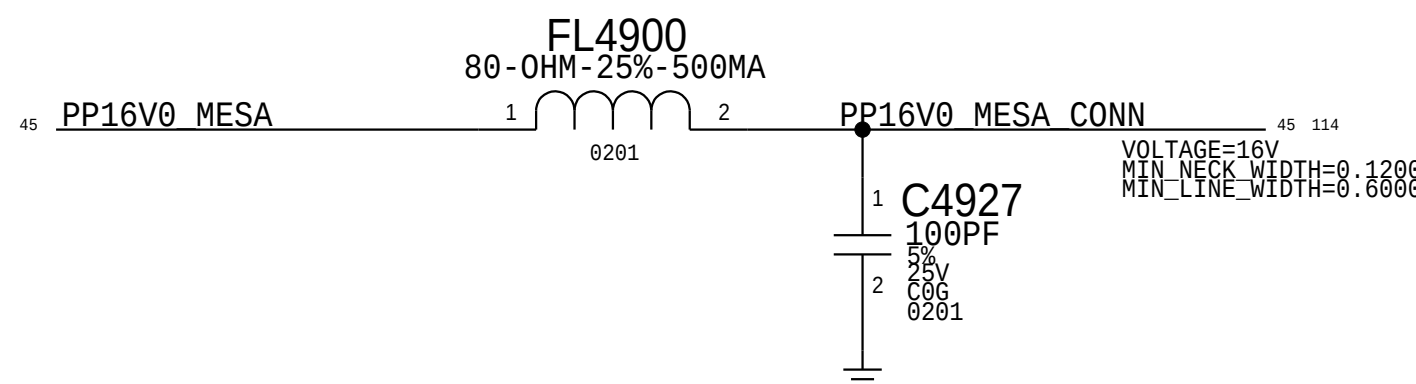
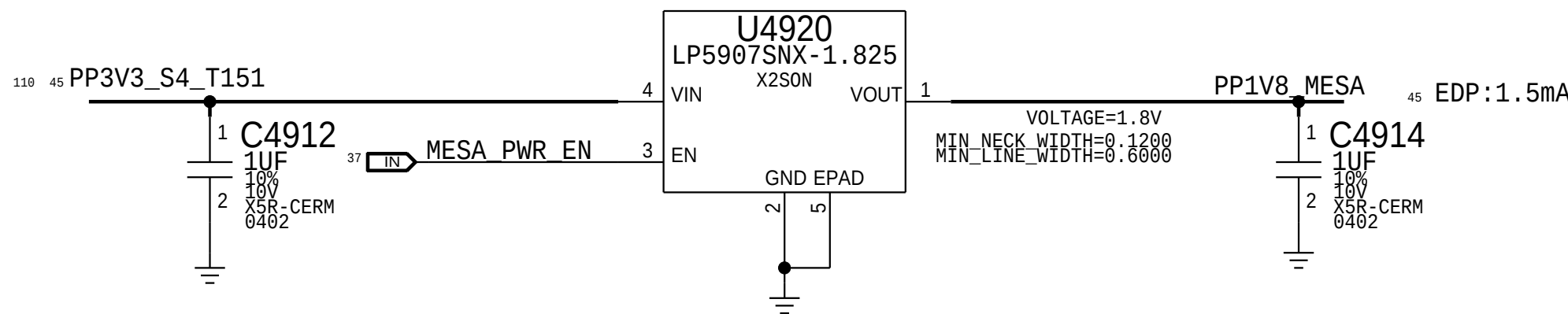


3.0V MESA

Option to feed LDO from 5V in case of dropout issue

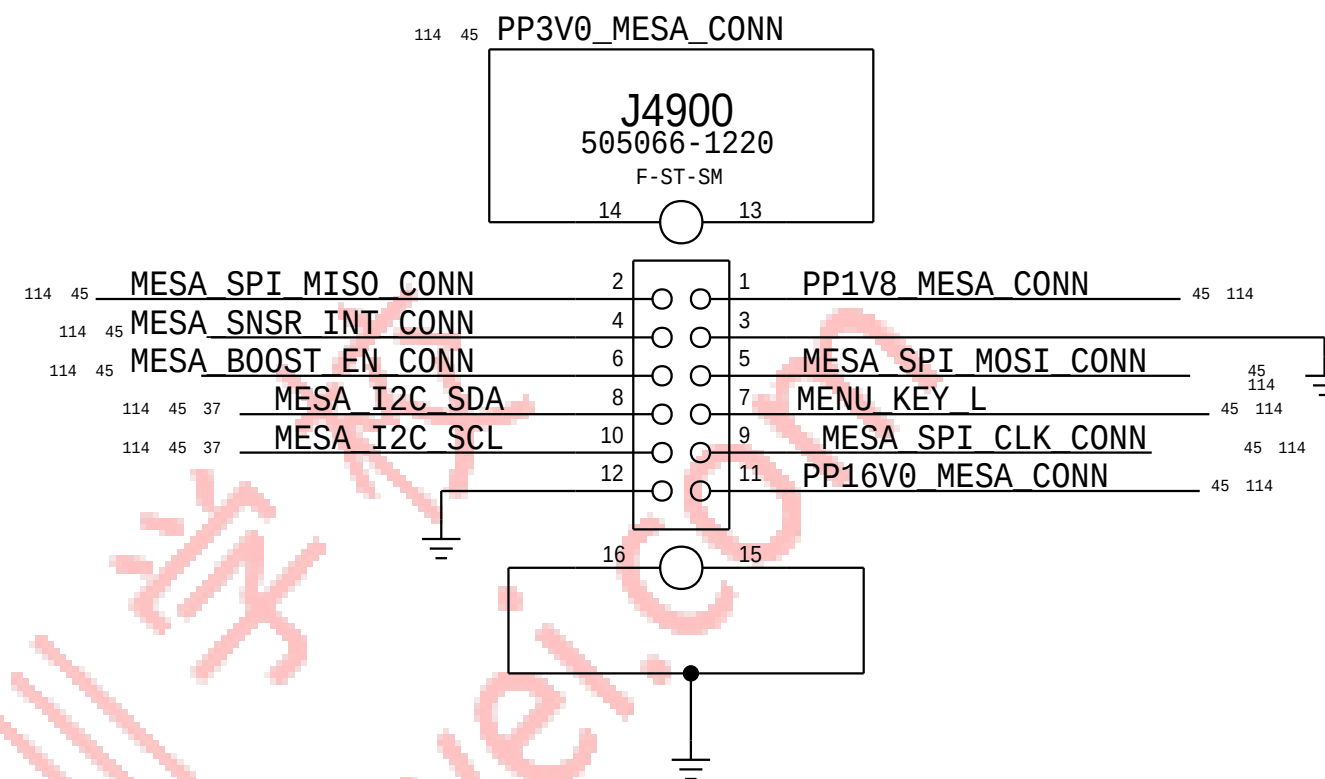


1.8V MESA



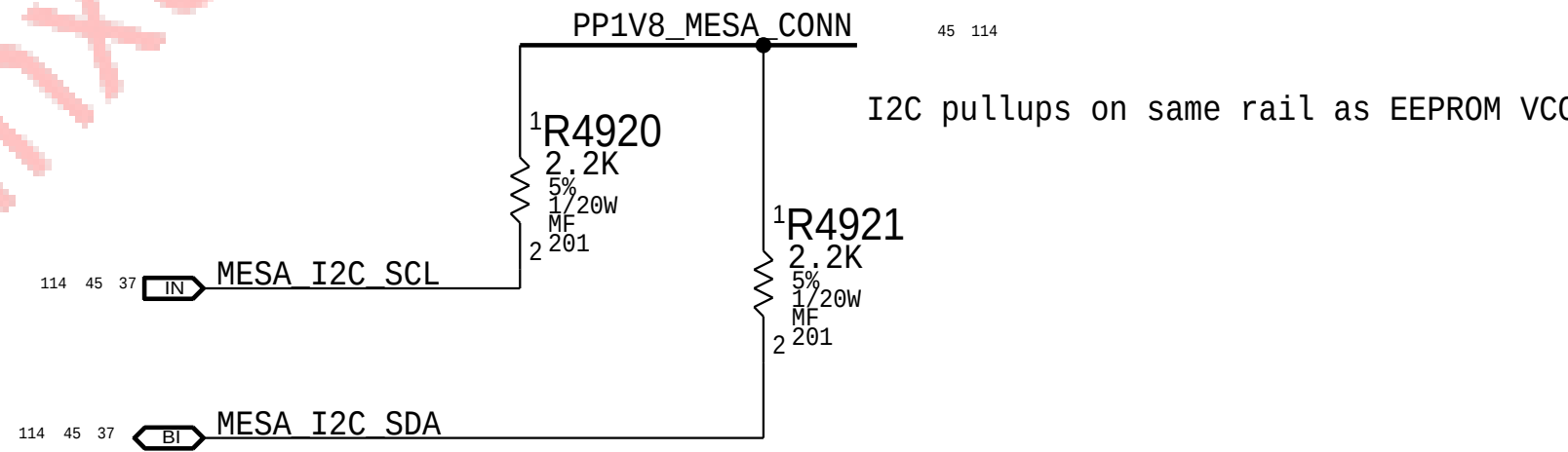
MESA FLEX CONNECTOR

Proto1 Connector for X434/X435 Support
PLUG (516S00115) - X434/ X435 Jumper
Recptacle (516S00203) - X362/X363 MLB



Mesa Power Sequencing Requirements

Power On: 1V8 -> 3V3 -> 16V0



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MESA			
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		SHEET	45 OF 121

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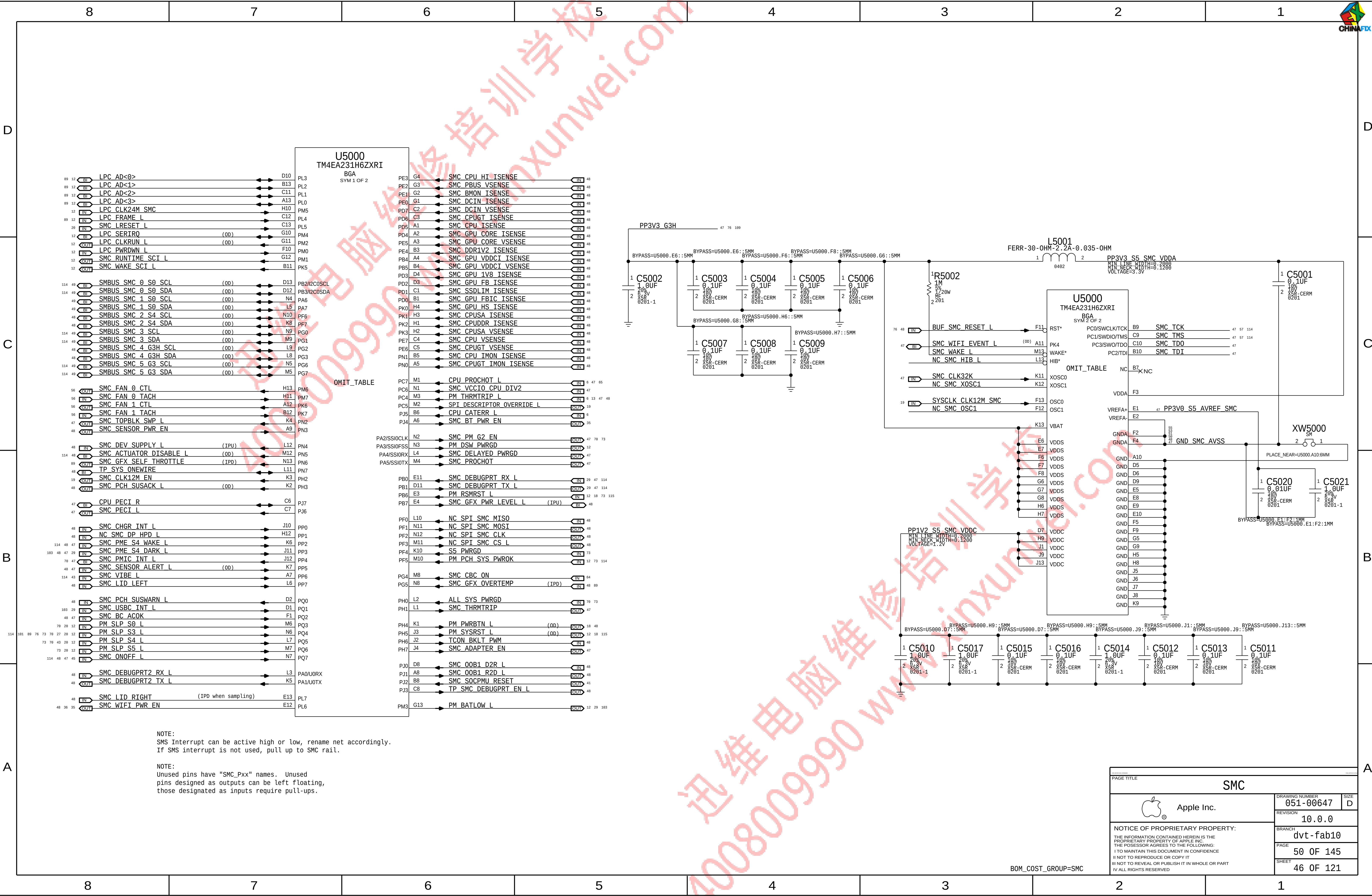
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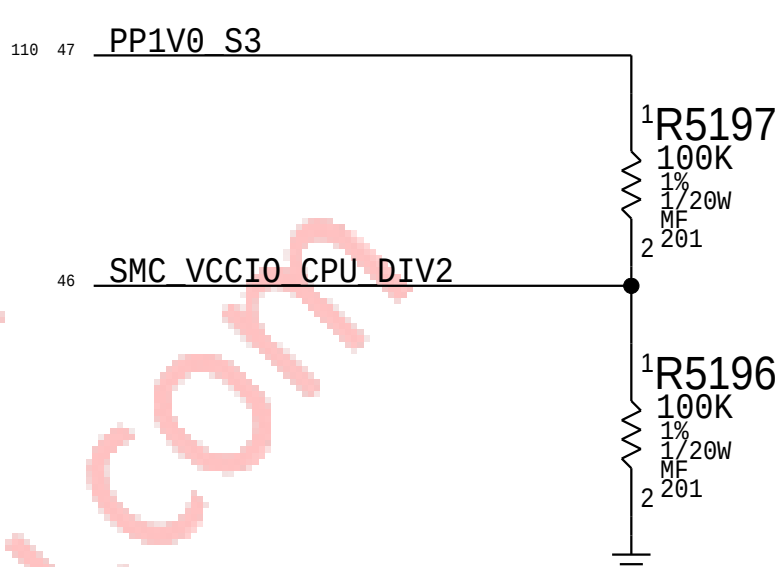
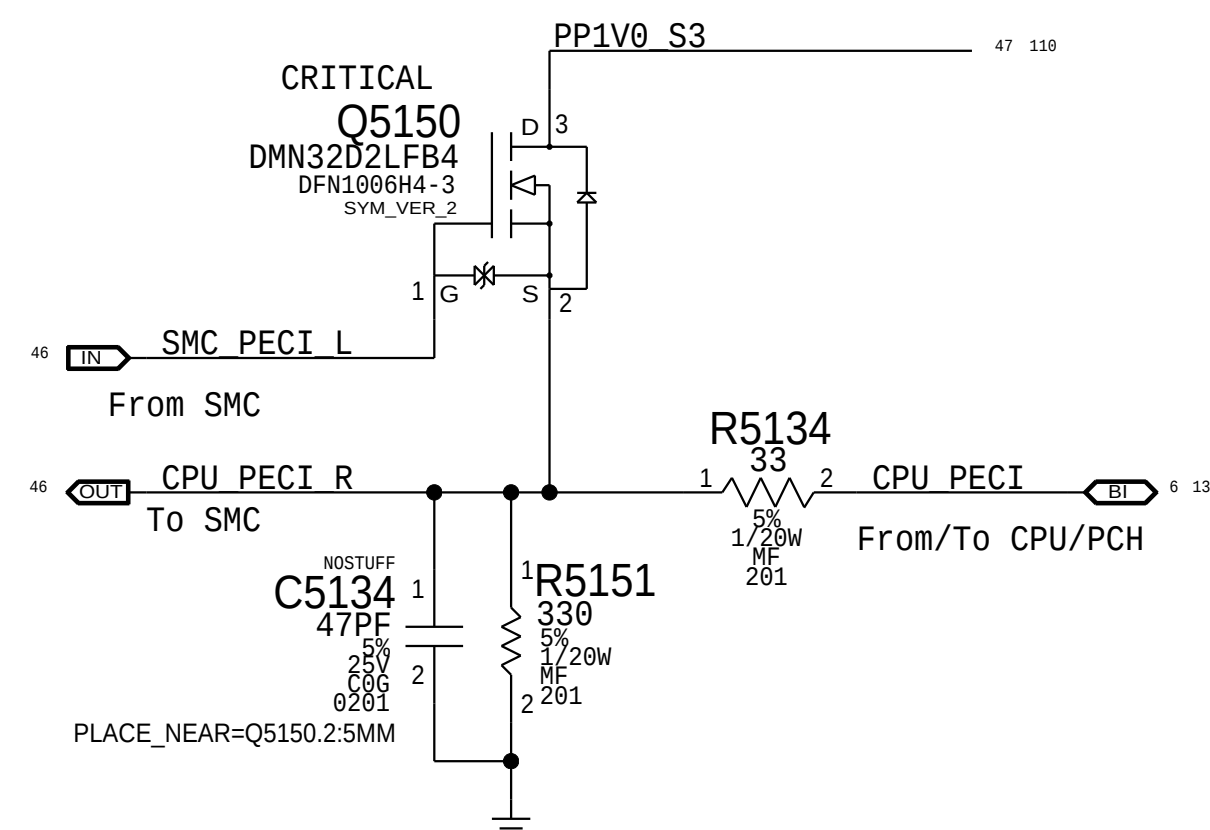
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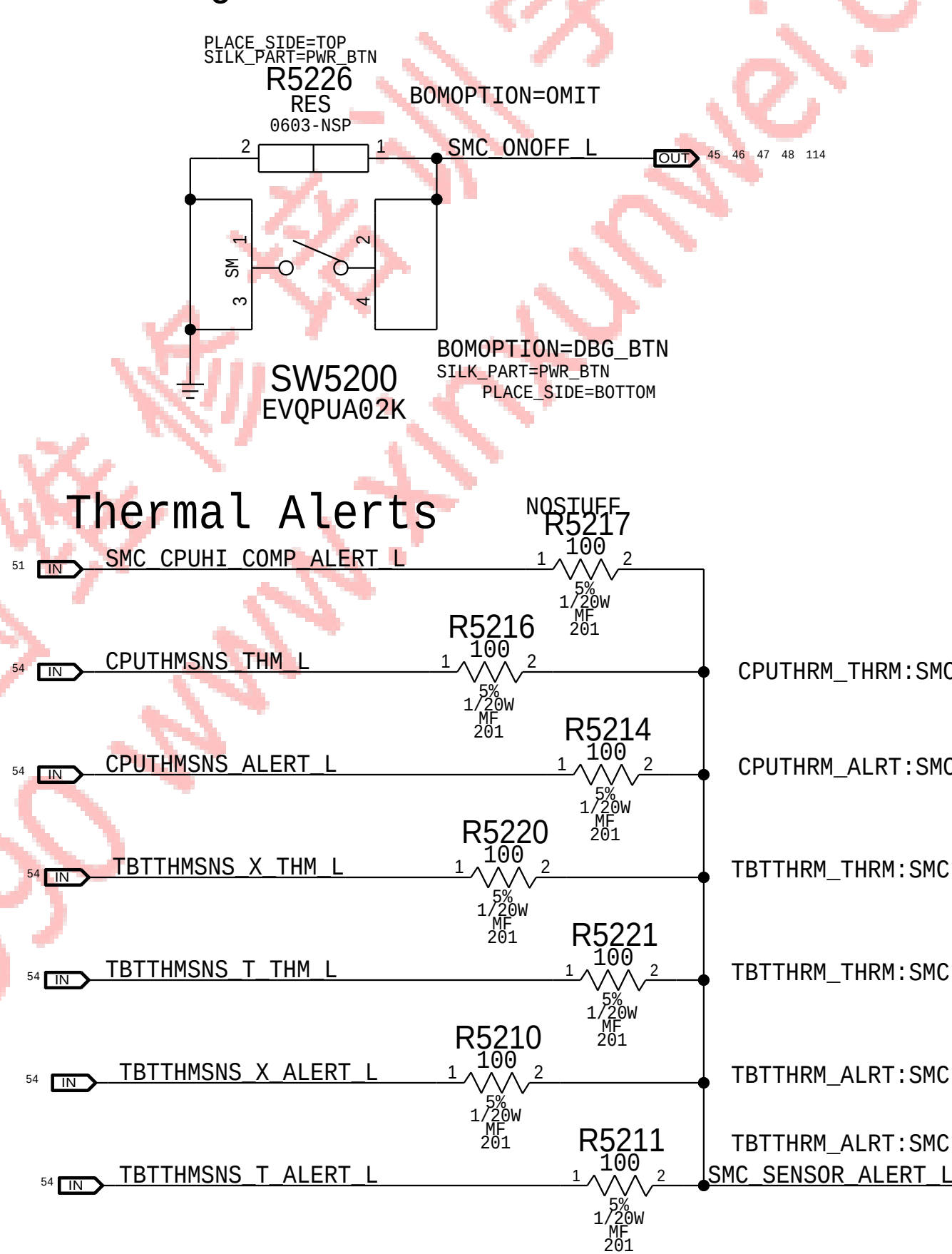


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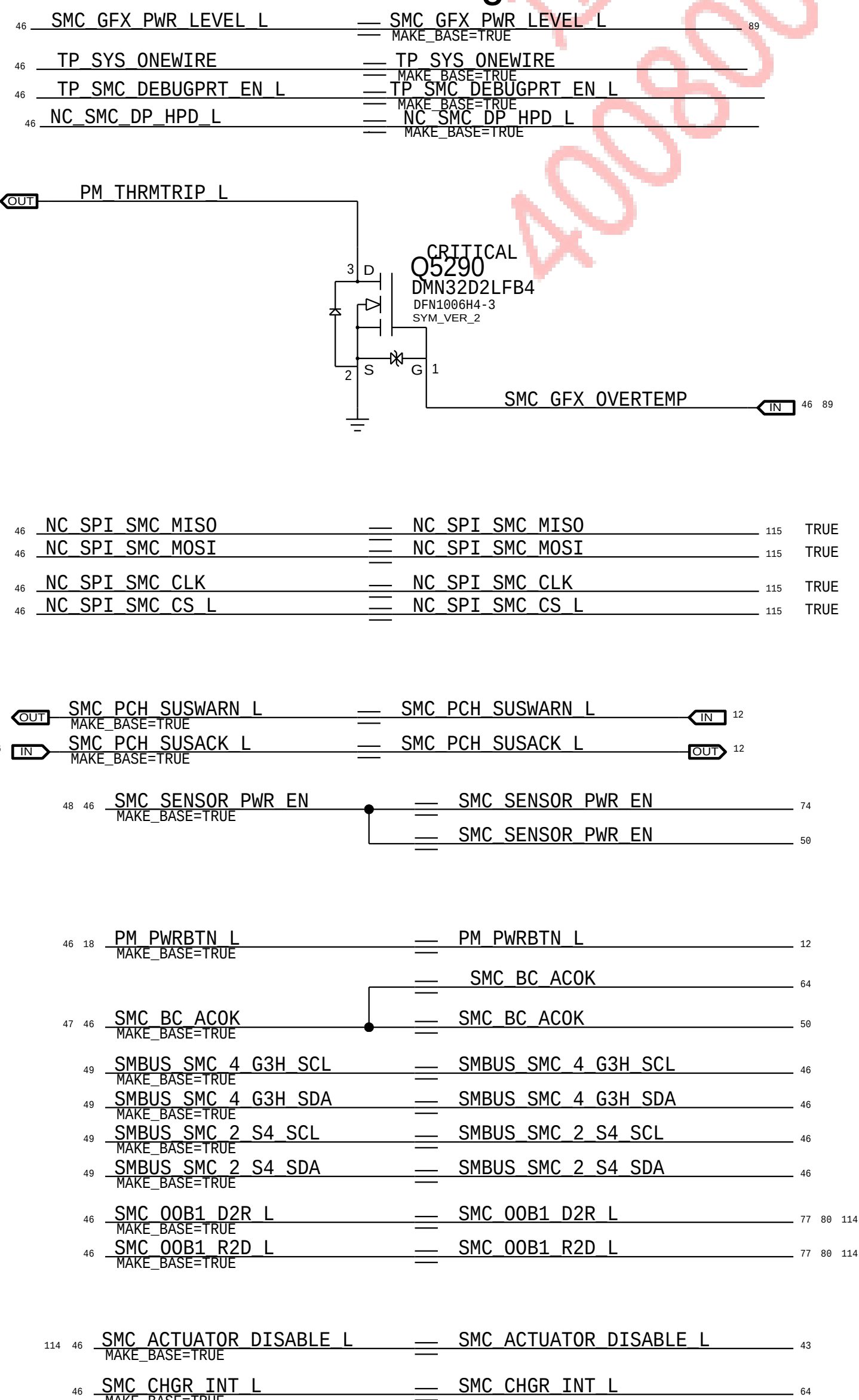
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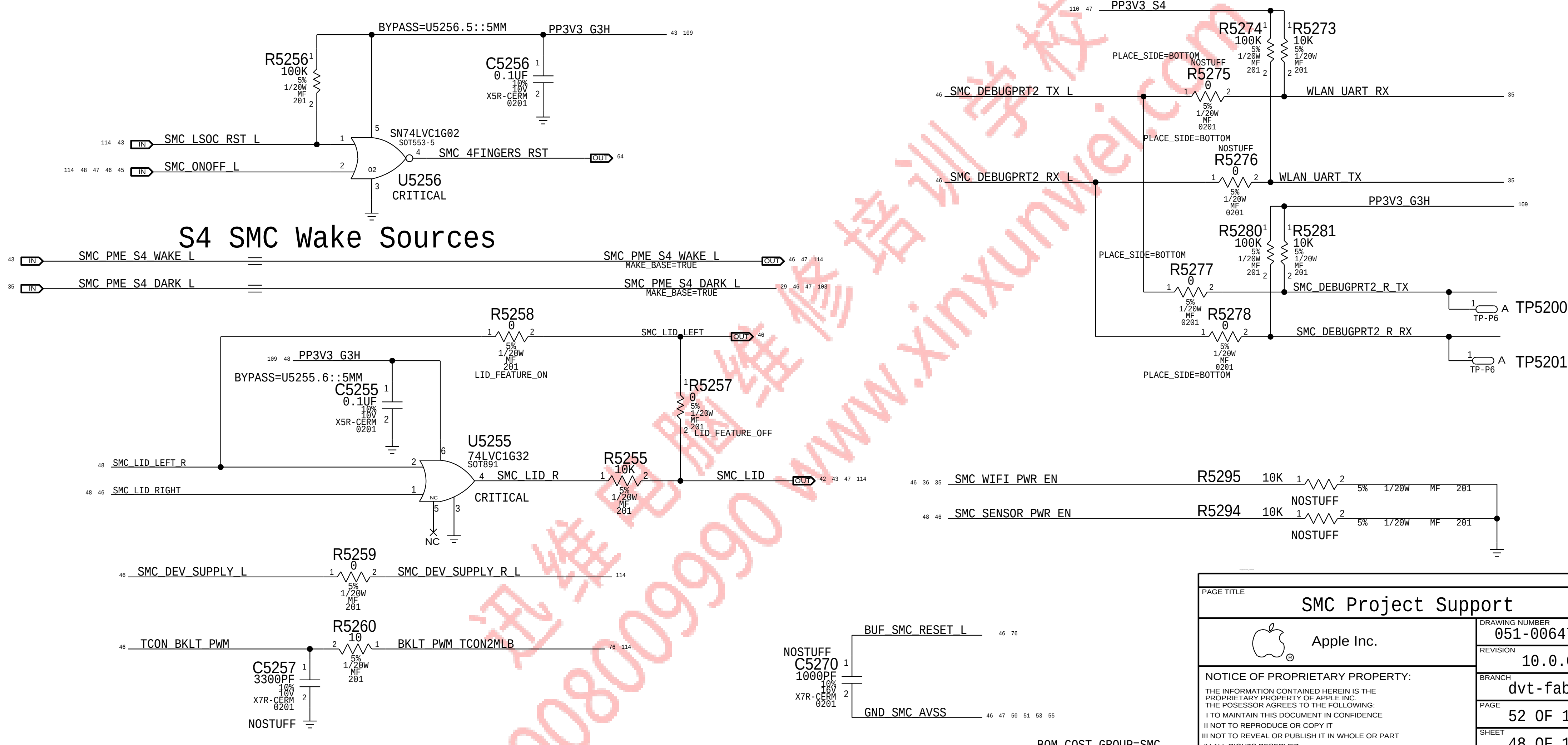
Hall Effect Pads - Left



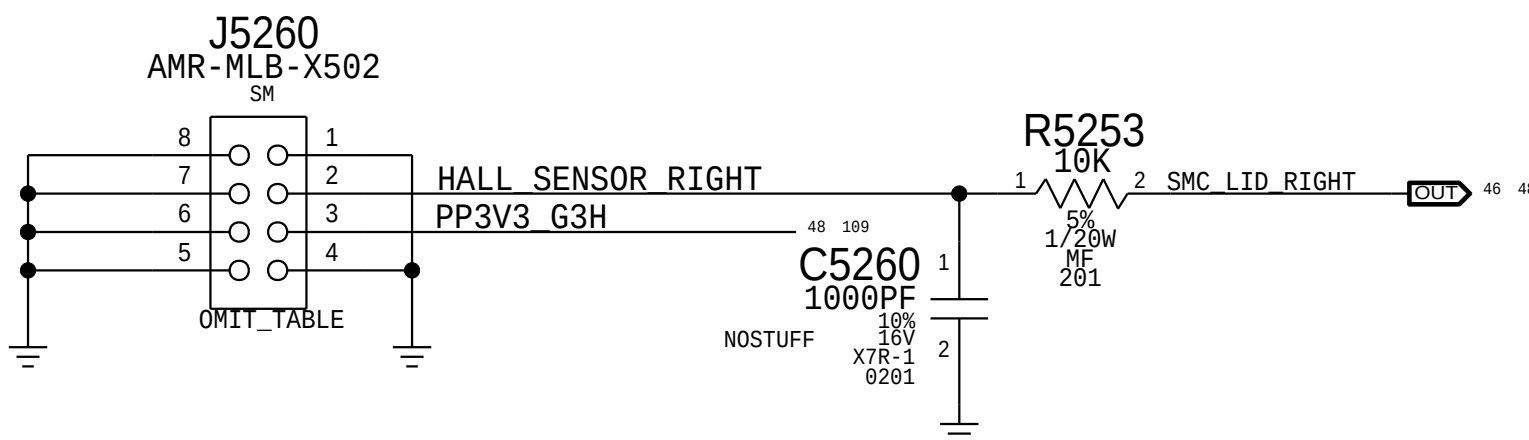
SMC12 Pin Assignments



S4 SMC Wake Sources



Hall Effect Pads - Right



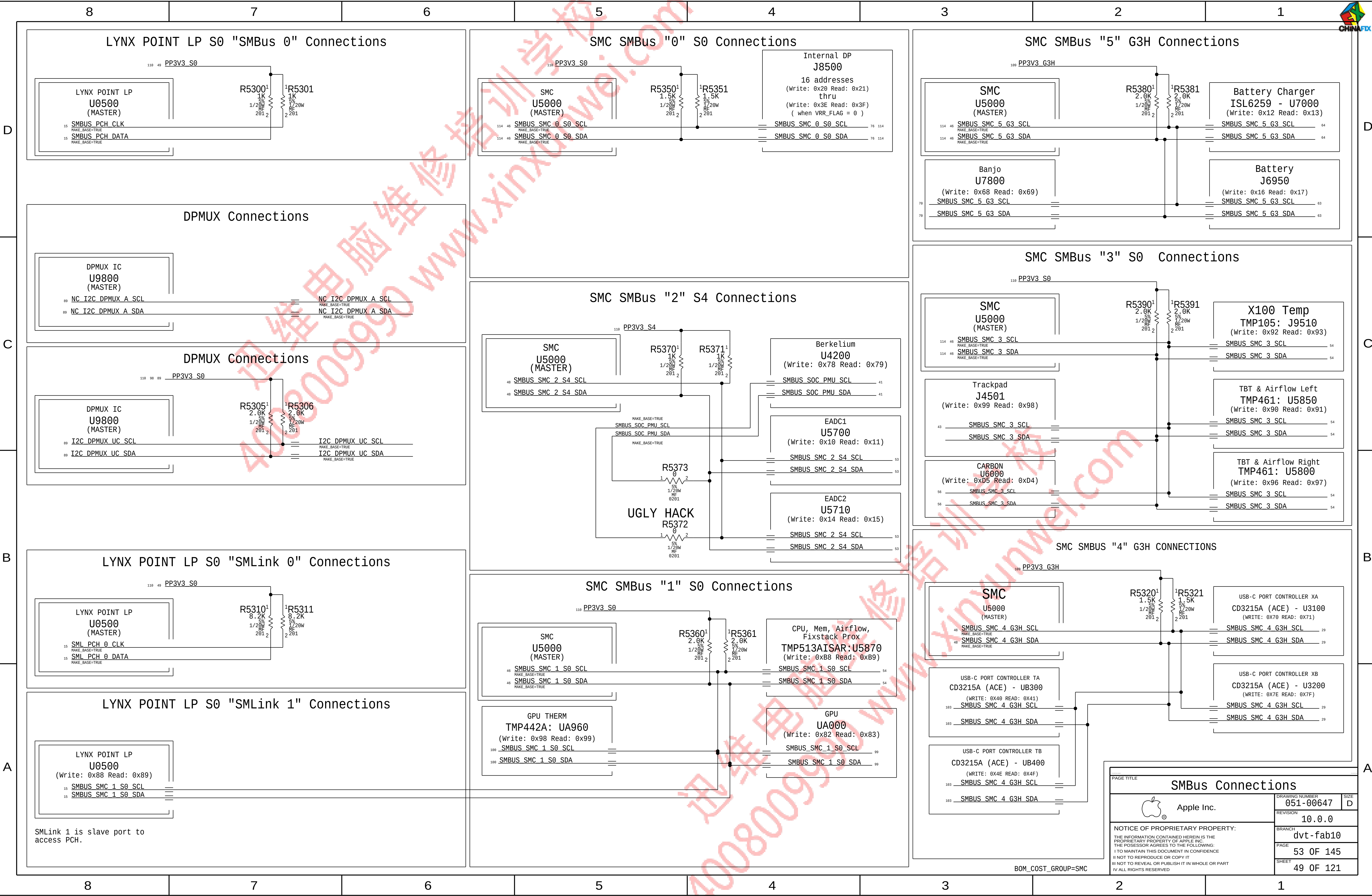
Specify one of these BOM GROUPs.

BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC
CPUTHRM:ALRT	CPUTHRM_ALRT:SMC

Specify one of these BOM GROUPs.

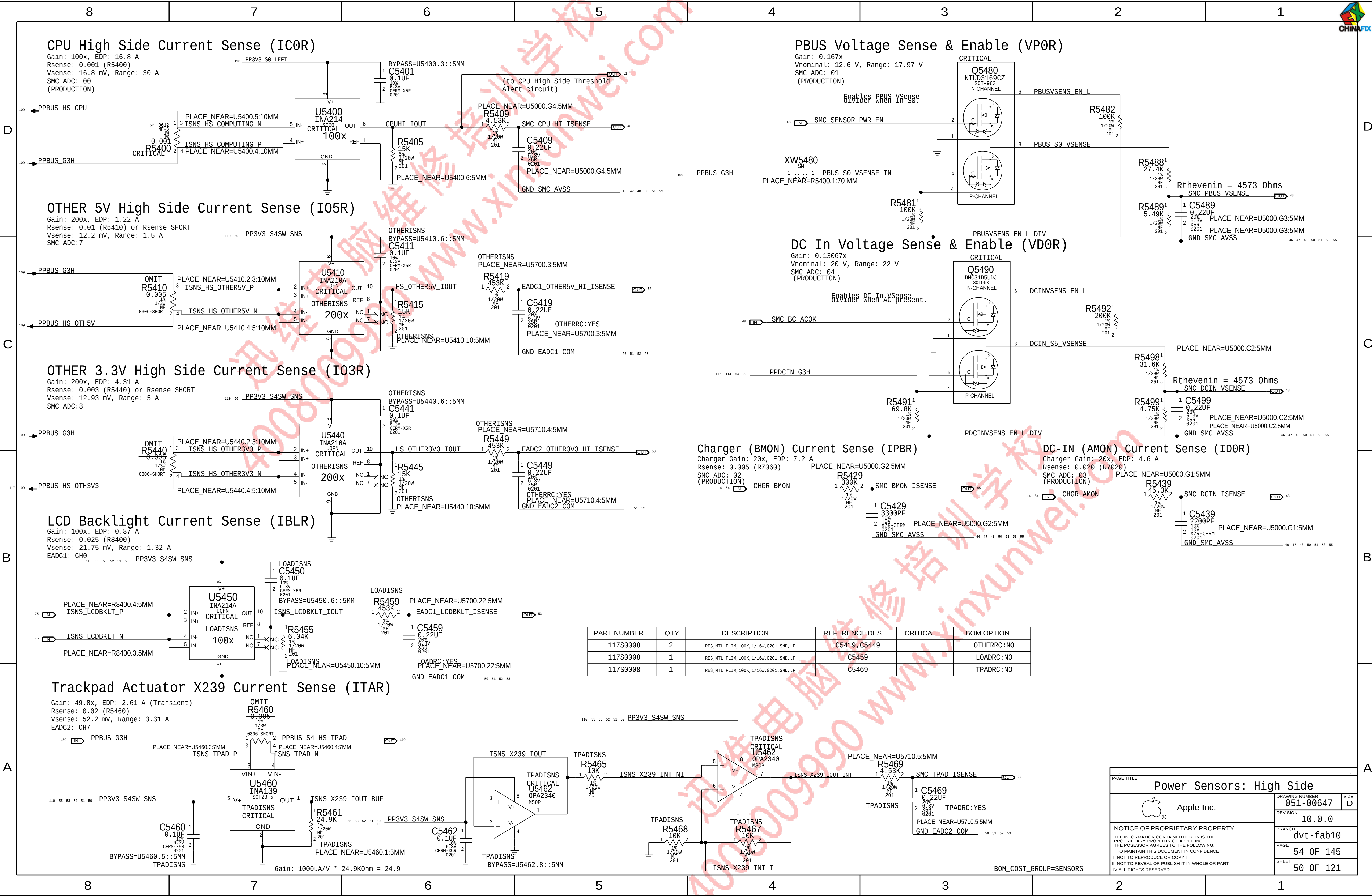
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TBTTHRM:THRM	TBTTHRM_THRM:SMC, TBTTHRM_ALRT:PU
TBTTHRM:ALRT	TBTTHRM_THRM:PU, TBTTHRM_ALRT:SMC
TBTTHRM:NONE	TBTTHRM_THRM:PU, TBTTHRM_ALRT:PU

PAGE TITLE										
SMC Project Support										
	Apple Inc.	<table><tr><td>DRAWING NUMBER</td><td>SIZE</td></tr><tr><td>051-00647</td><td>D</td></tr></table>	DRAWING NUMBER	SIZE	051-00647	D				
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PAGE	52 OF 145									
SHEET	48 OF 121									

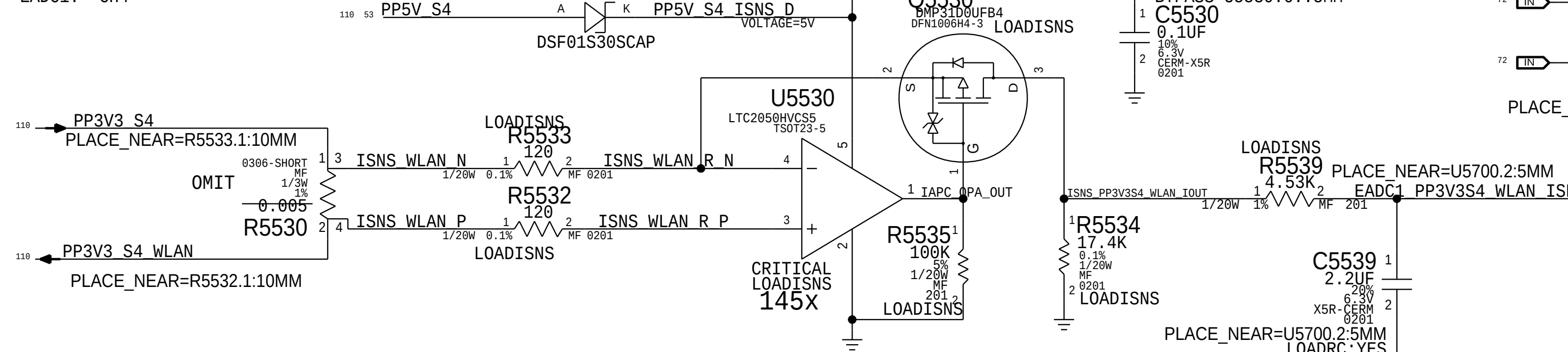


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	PAGE	53 OF 145
	SHEET	49 OF 121



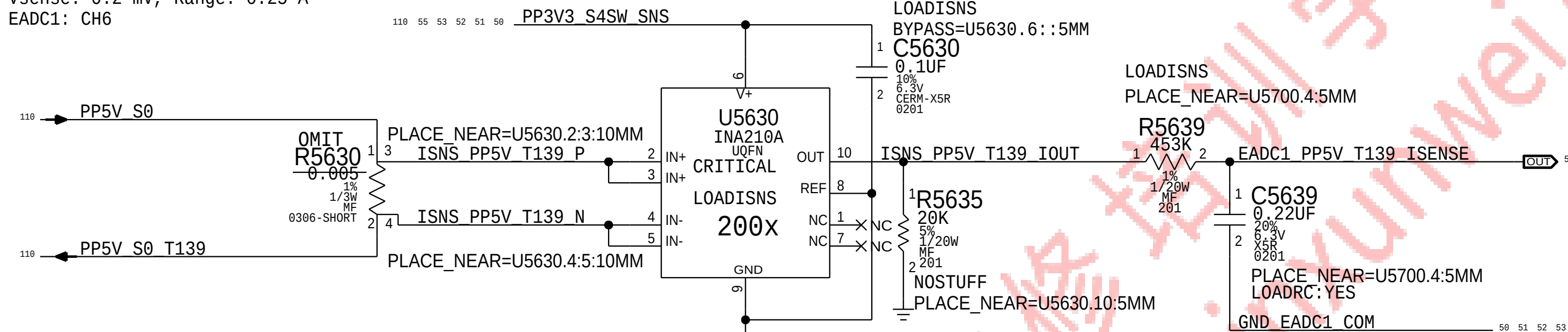
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		PAGE		
		55 OF 145		
		SHEET		
		51 OF 121		





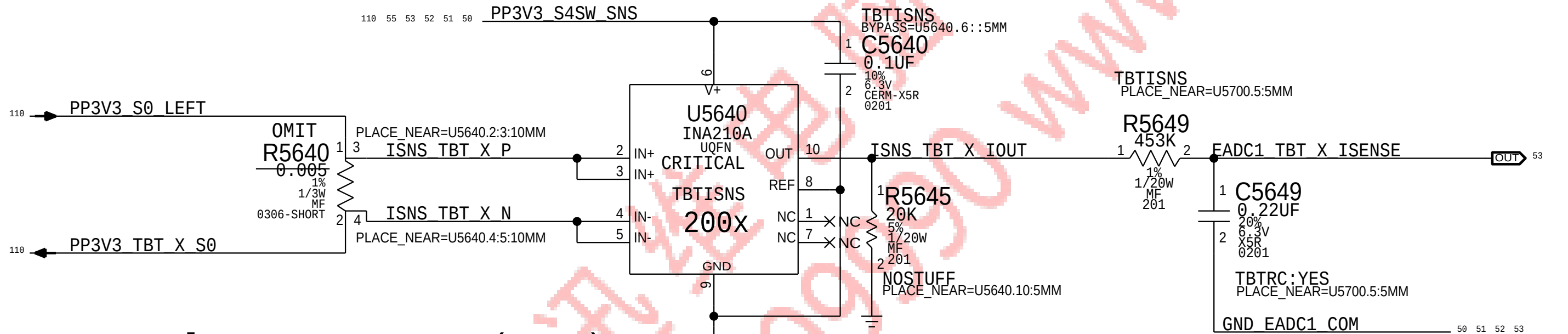
T139 5V Current Sense (IF5C)

Gain: 200x, EDP: 0.004 A
Rsense: 0.05 (R5630) or Rsense SHORT
Vsense: 0.2 mV, Range: 0.25 A
EADC1: CH6



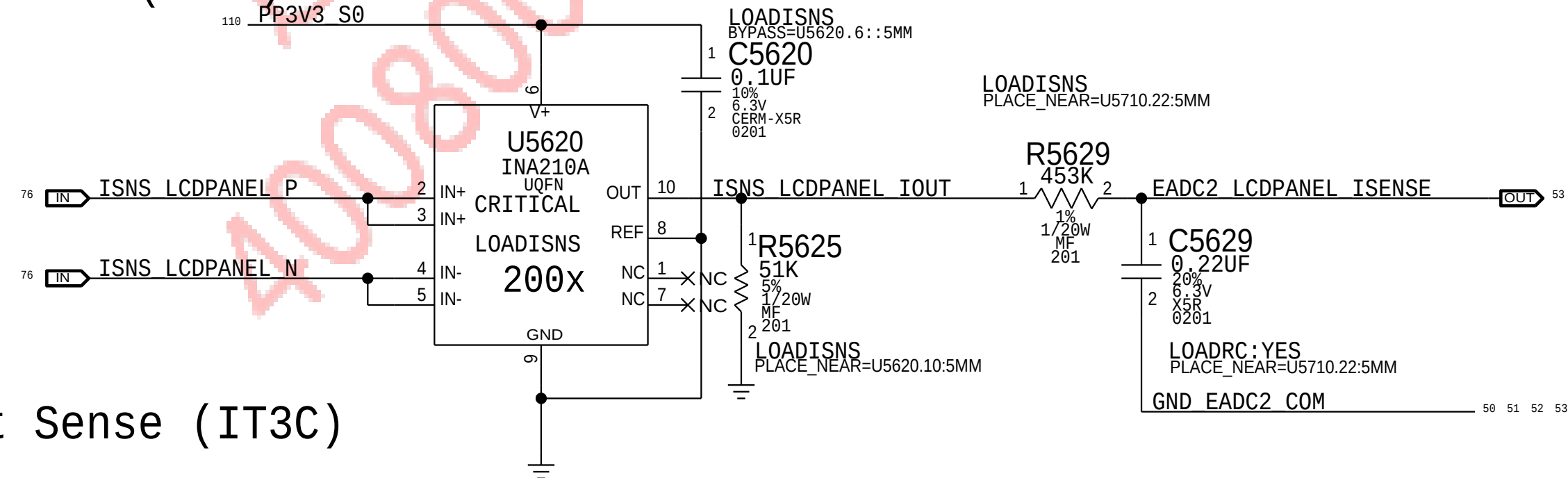
Thunderbolt TBT LEFT Current Sense (ITLC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5640) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH7



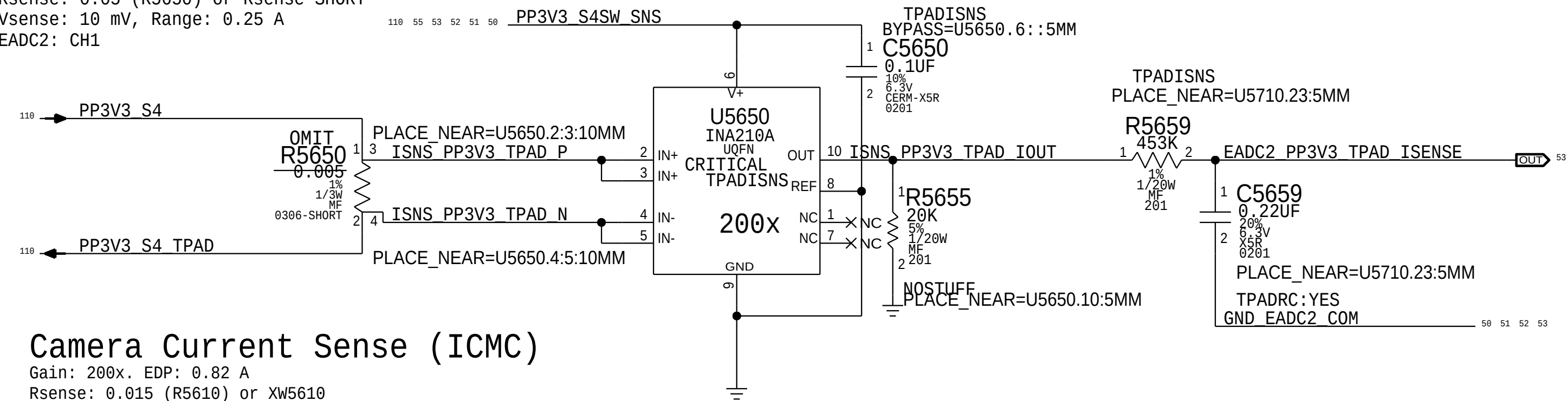
LCD Panel Current Sense (ILDC)

Gain: 200x, EDP: 1 A
RSense: 0.01 (R5620) or Rsense SHORT
Vsense: 5 mV, Range: 1.25 A
EADC2: CH0



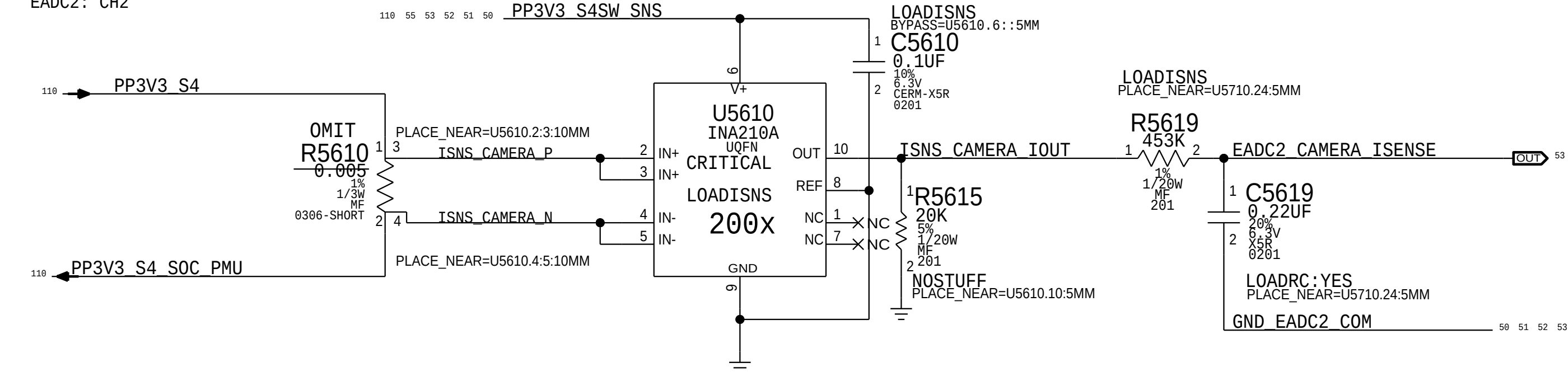
Trackpad 3V Current Sense (IT3C)

Gain: 200x, EDP: 0.2 A
Rsense: 0.05 (R5650) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH1

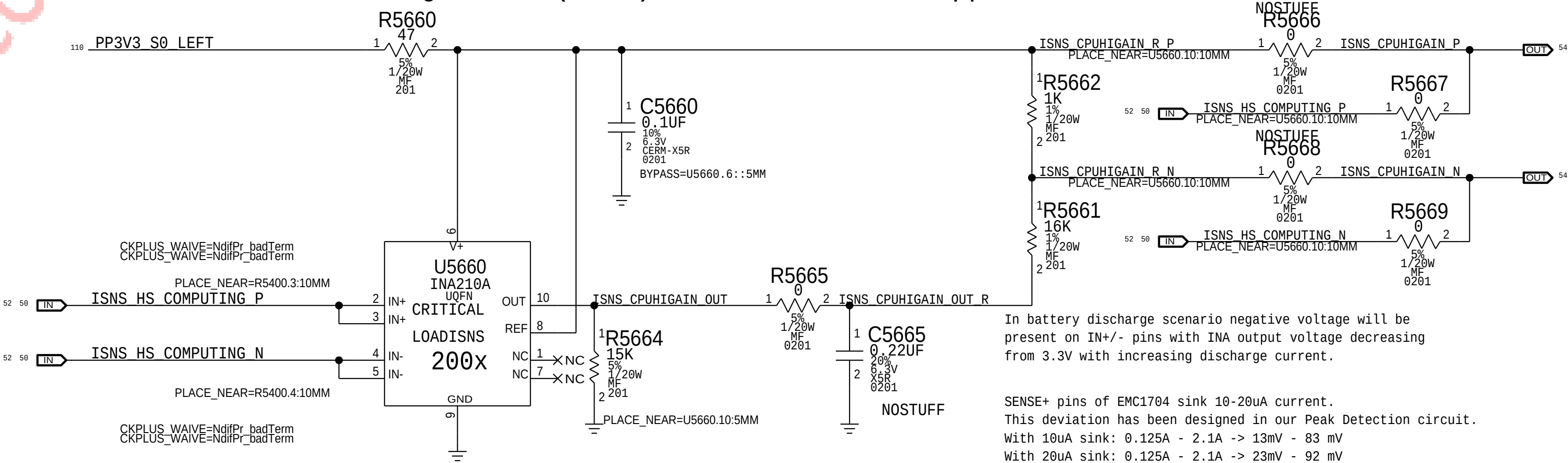


Camera Current Sense (ICMC)

Gain: 200x, EDP: 0.82 A
Rsense: 0.015 (R5610) or XW5610
Vsense: 12.3 mV, Range: 0.83 A
EADC2: CH2



CPU High Side (IC0R) Peak Detection Support

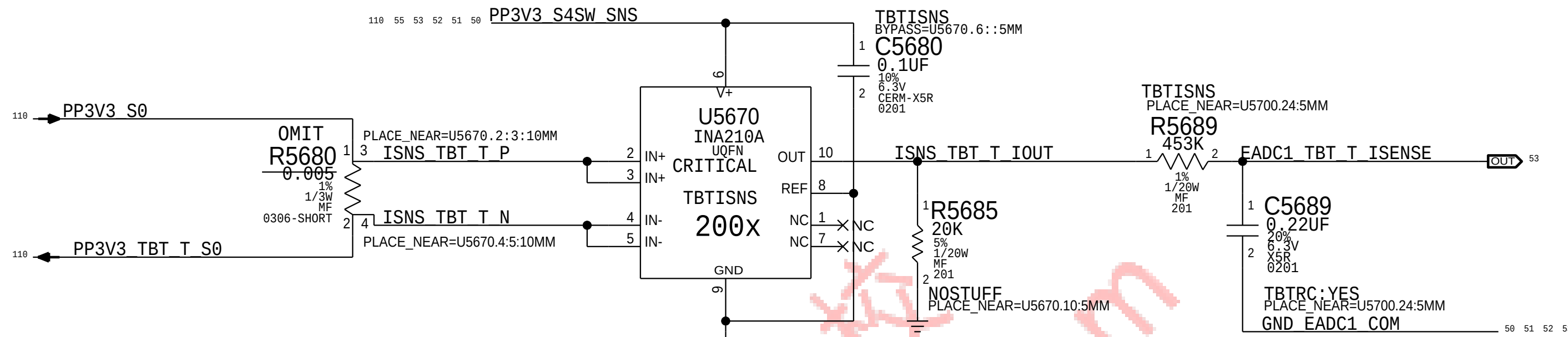


In battery discharge scenario negative voltage will be present on IN+/- pins with INA output voltage decreasing from 3.3V with increasing discharge current.

SENSE+ pins of EMC1704 sink 10-20uA current. This deviation has been designed in our Peak Detection circuit. With 10uA sink: 0.125A - 2.1A -> 13mV - 83 mV With 20uA sink: 0.125A - 2.1A -> 23mV - 92 mV

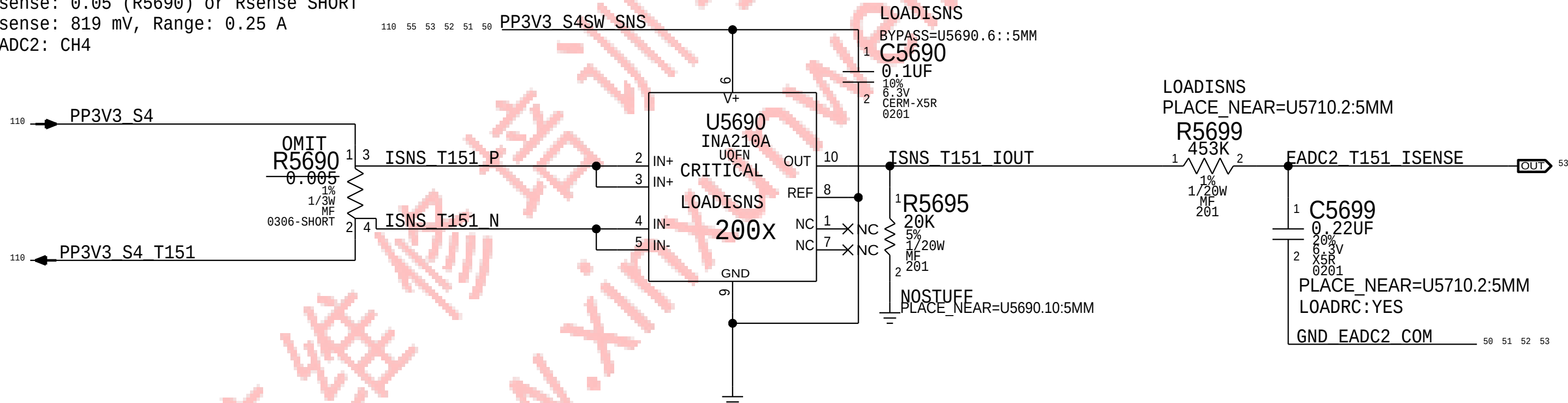
Thunderbolt TBT RIGHT Current Sense (IURC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5680) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH2



T151 Current Sense (IIDC)

Gain: 200x, EDP: 0.1638 A
Rsense: 0.05 (R5690) or Rsense SHORT
Vsense: 819 mV, Range: 0.25 A
EADC2: CH4



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5619,C5629,C5639,C5699		LOADRC:NO
117S0008	2	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5649,C5689		TBTRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5659		TPADRC:NO

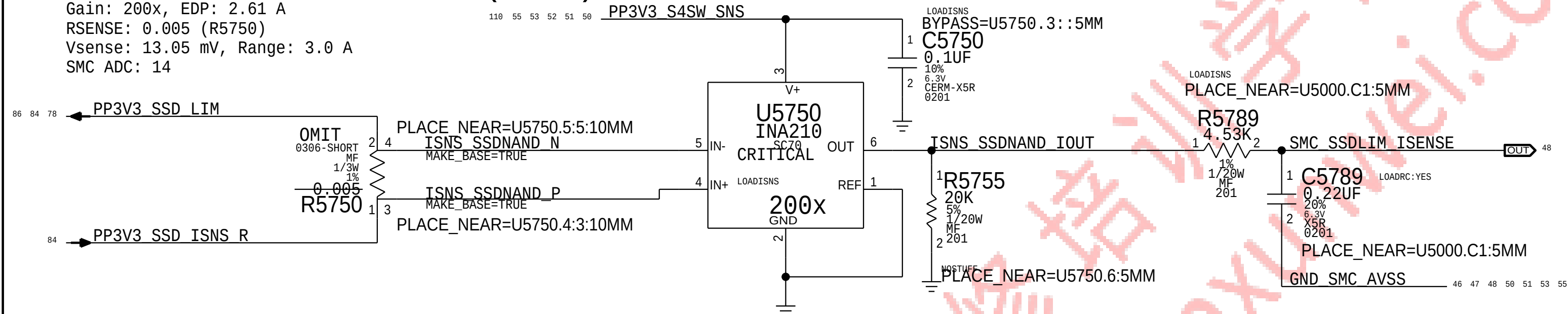
PAGE TITLE			
Power Sensors: Extended			
Apple Inc.		DRAWING NUMBER	051-00647
		REVISION	10.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	dvt-fab10
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	56 OF 145
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	52 OF 121
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

BOM_COST_GROUP=SENSORS



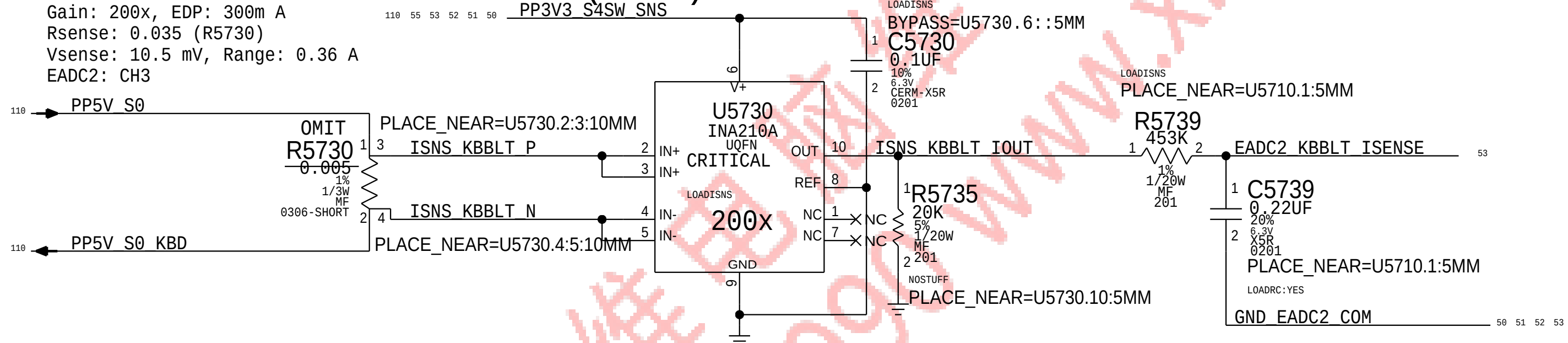
SSDLIM Current Sense (IHDC)

Gain: 200x, EDP: 2.61 A
Rsense: 0.005 (R5750)
Vsense: 13.05 mV, Range: 3.0 A
SMC ADC: 14



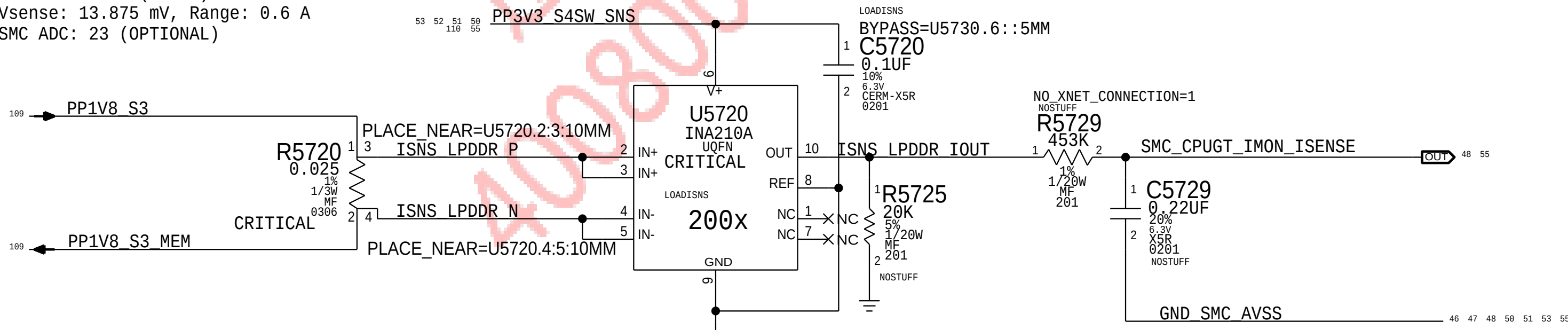
KB backlite Current Sense (IKBC)

Gain: 200x, EDP: 300m A
Rsense: 0.035 (R5730)
Vsense: 10.5 mV, Range: 0.36 A
EADC2: CH3



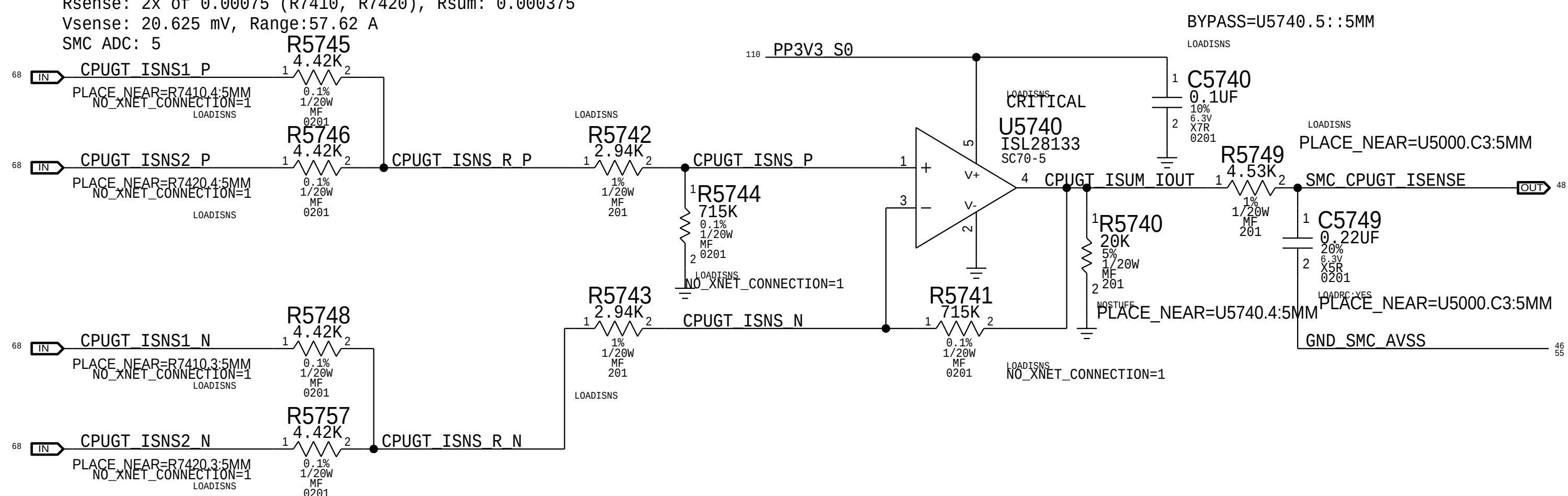
LPDDR 1.8V Current Sense (IM1C)

Gain: 200x, EDP: 0.555 A
Rsense: 0.025 (R5720) or Rsense SHORT
Vsense: 13.875 mV, Range: 0.6 A
SMC ADC: 23 (OPTIONAL)



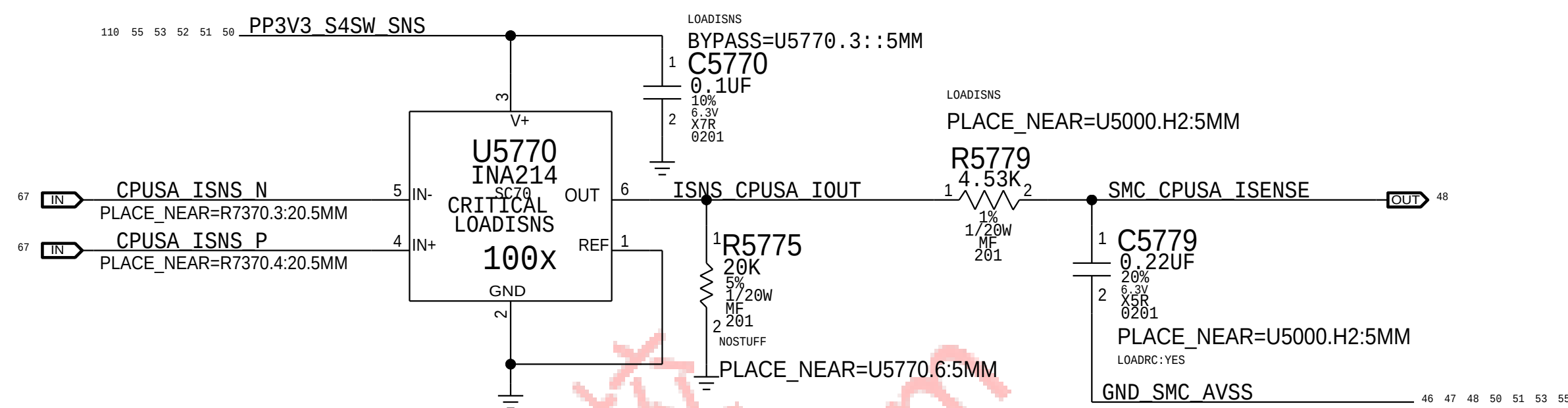
CPU GT Current Sense (ICTC)

Gain: 202.93x, EDP: 55 A
Rsense: 2x of 0.00075 (R7410, R7420), Rsum: 0.000375
Vsense: 20.625 mV, Range: 57.62 A
SMC ADC: 5



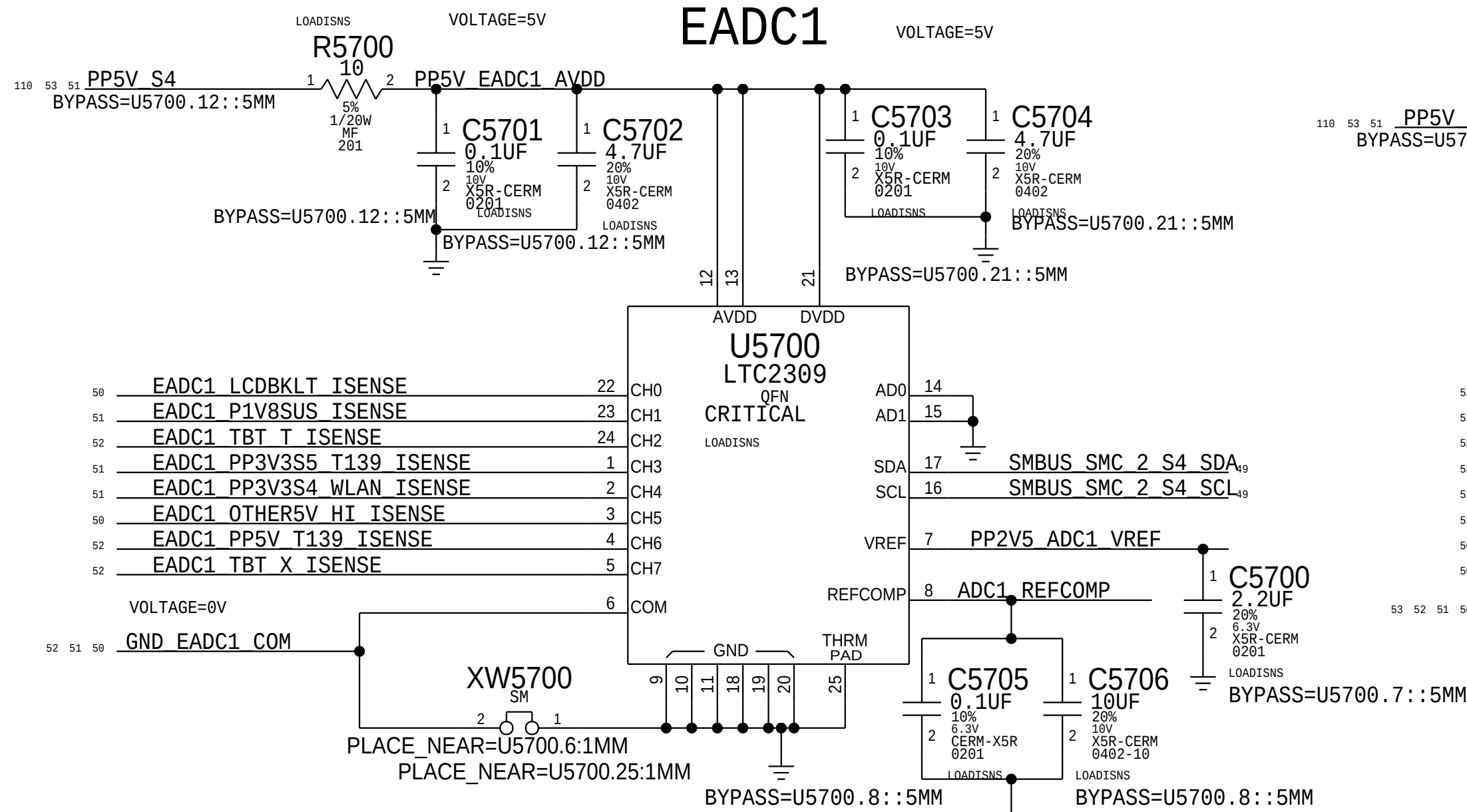
CPU SA Current Sense (ICSC)

Gain: 200x, EDP: 8 A
Rsense: 0.002 (R5750)
Vsense: 16 mV, Range: 7.5 A
SMC ADC: 17

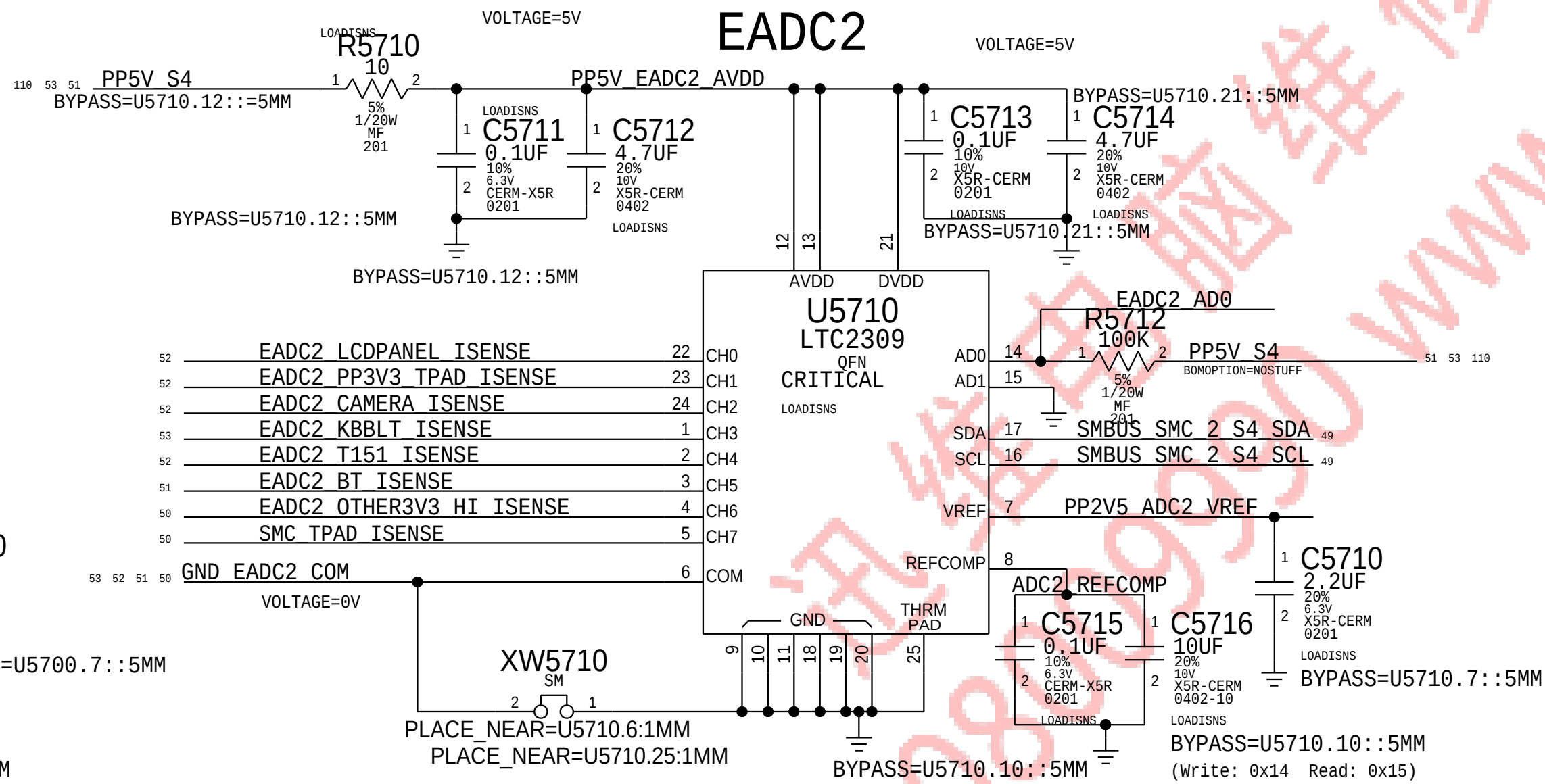


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5739, C5749, C5779, C5789		LOADRC: NO

EADC1

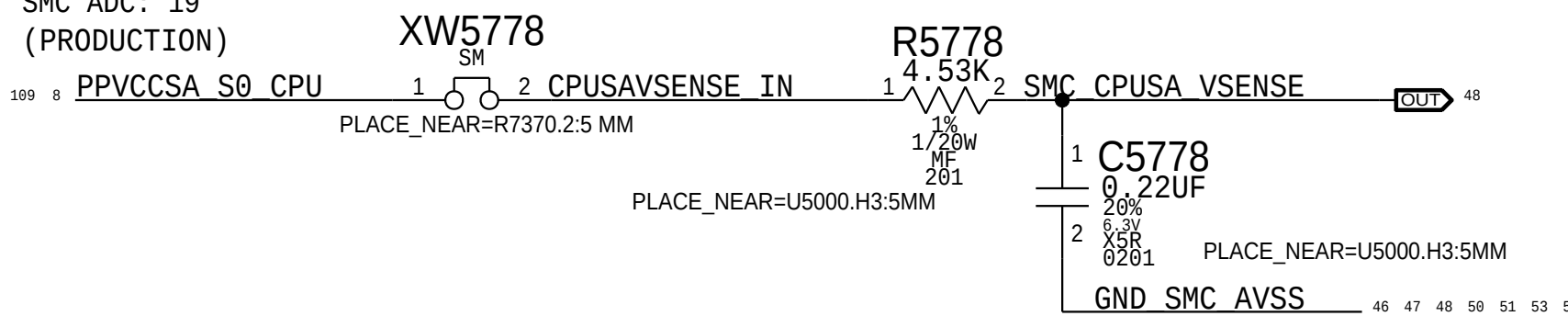


EADC2



CPU SA Voltage Sense (VCSC)

SMC ADC: 19
(PRODUCTION)



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Power Sensors: Extended 2			
		DRAWING NUMBER	051-00647
		REVISION	10.0.0
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		PAGE	57 OF 145
		SHEET	53 OF 121

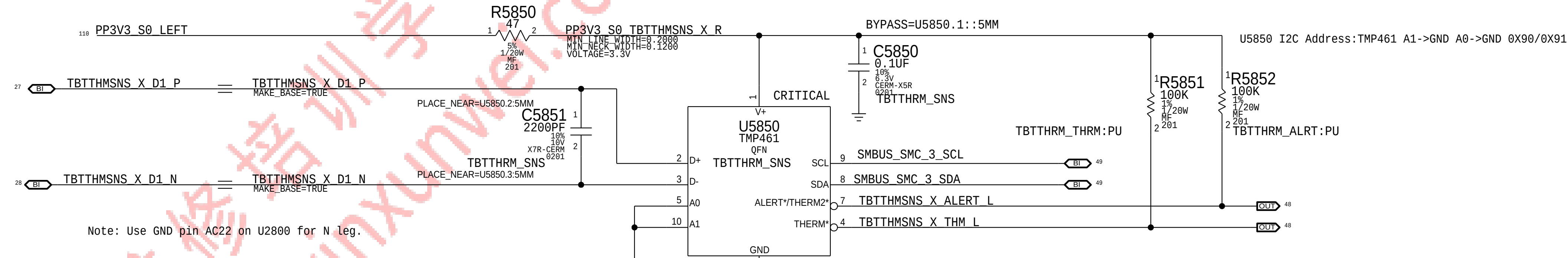
BOM_COST_GROUP=SENSORS

Thermal Sensor A:
Thunderbolt Die, Airflow Left

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (TTLD)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.



Thermal Diode: Airflow Left Proximity (TaLC)

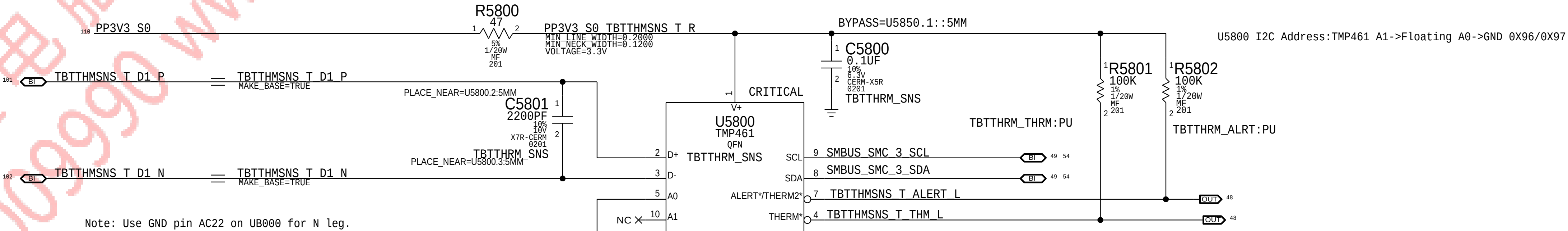
Placement Note:
Place U5850 on the TOP side, on the left portion of the board, 1" to the right of USB connector.

Thermal Sensor C:
Thunderbolt Die, Air Flow Right

I2C Write: 0x98, I2C Read: 0x99

Thermal Diode: TBT Die (TTRD)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AC22.

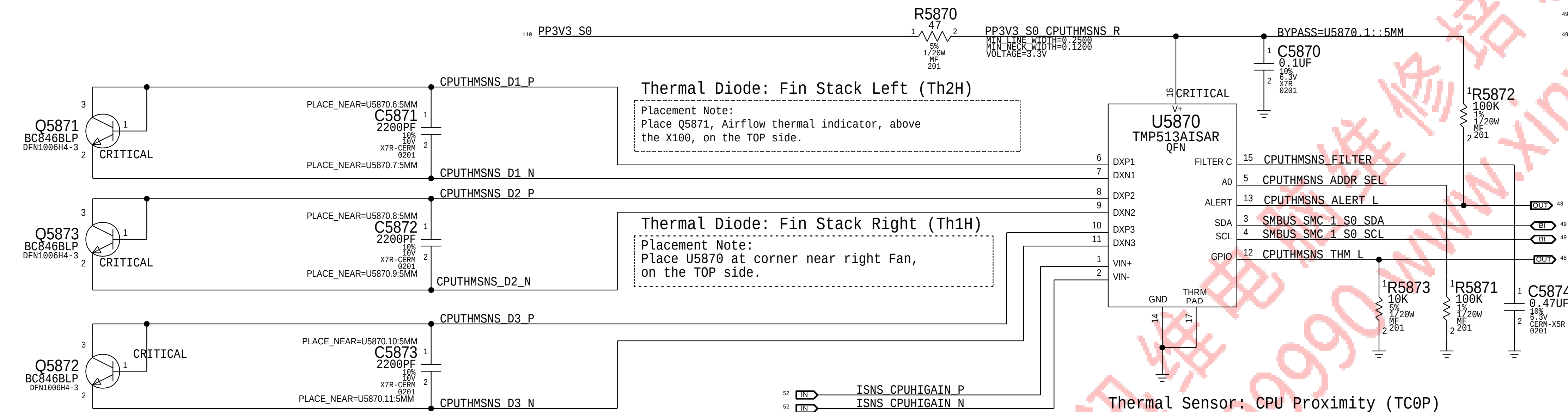


Thermal Diode: Airflow Right Proximity (TaRC)

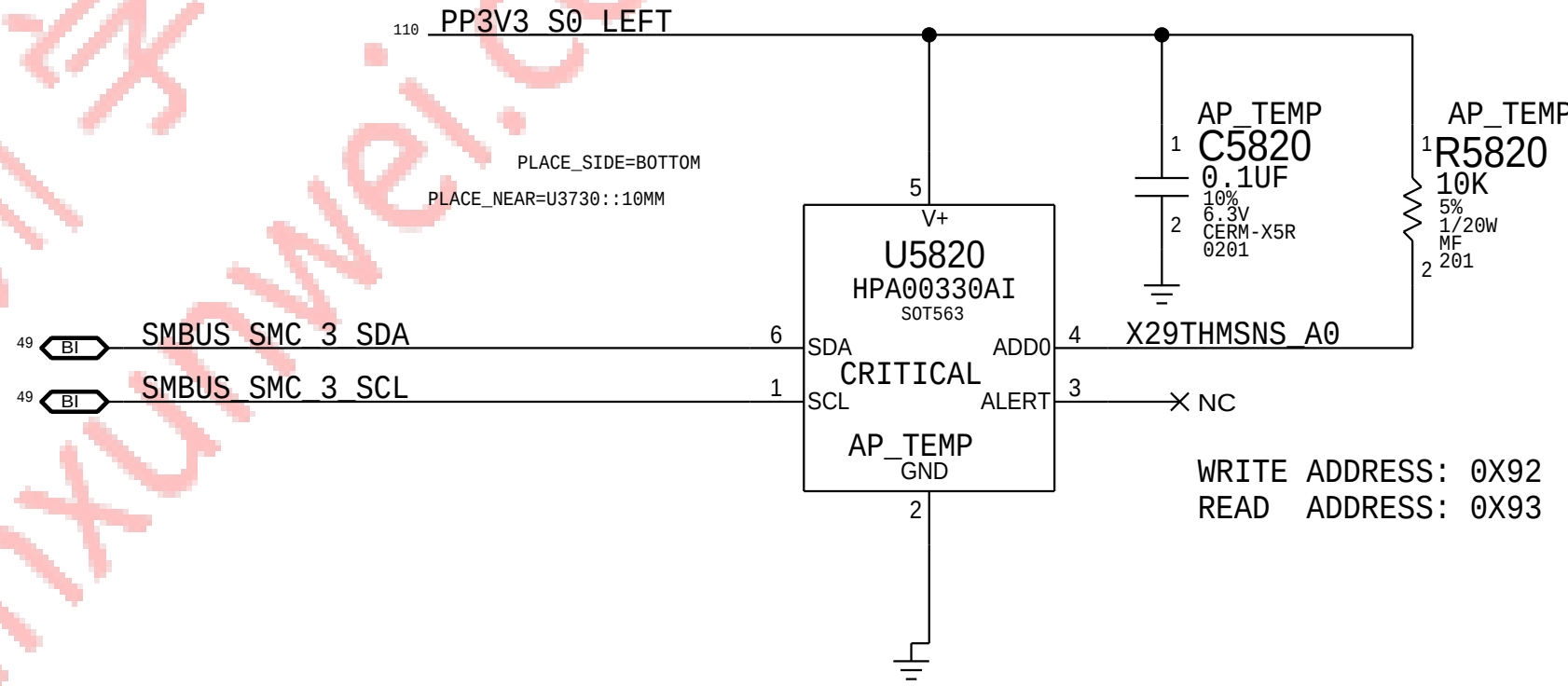
Placement Note:
Place U5800 on the TOP side, on the left portion of the board, 1" to the right of USB connector.

Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Fin Stack Left, Fin Stack Right

I2C Write: 0x98, I2C Read: 0x99



X100 PROXIMITY (TW0P)



Placement note:
PLACE U5820 ON BOTTOM NEAR X100

Thermal Diode: Memory Proximity (TM0P)

Placement Note:
Place Q5872 between two rows of Memory devices, between channel A and B, on the BOTTOM side.

Thermal Sensor: CPU Proximity (TC0P)

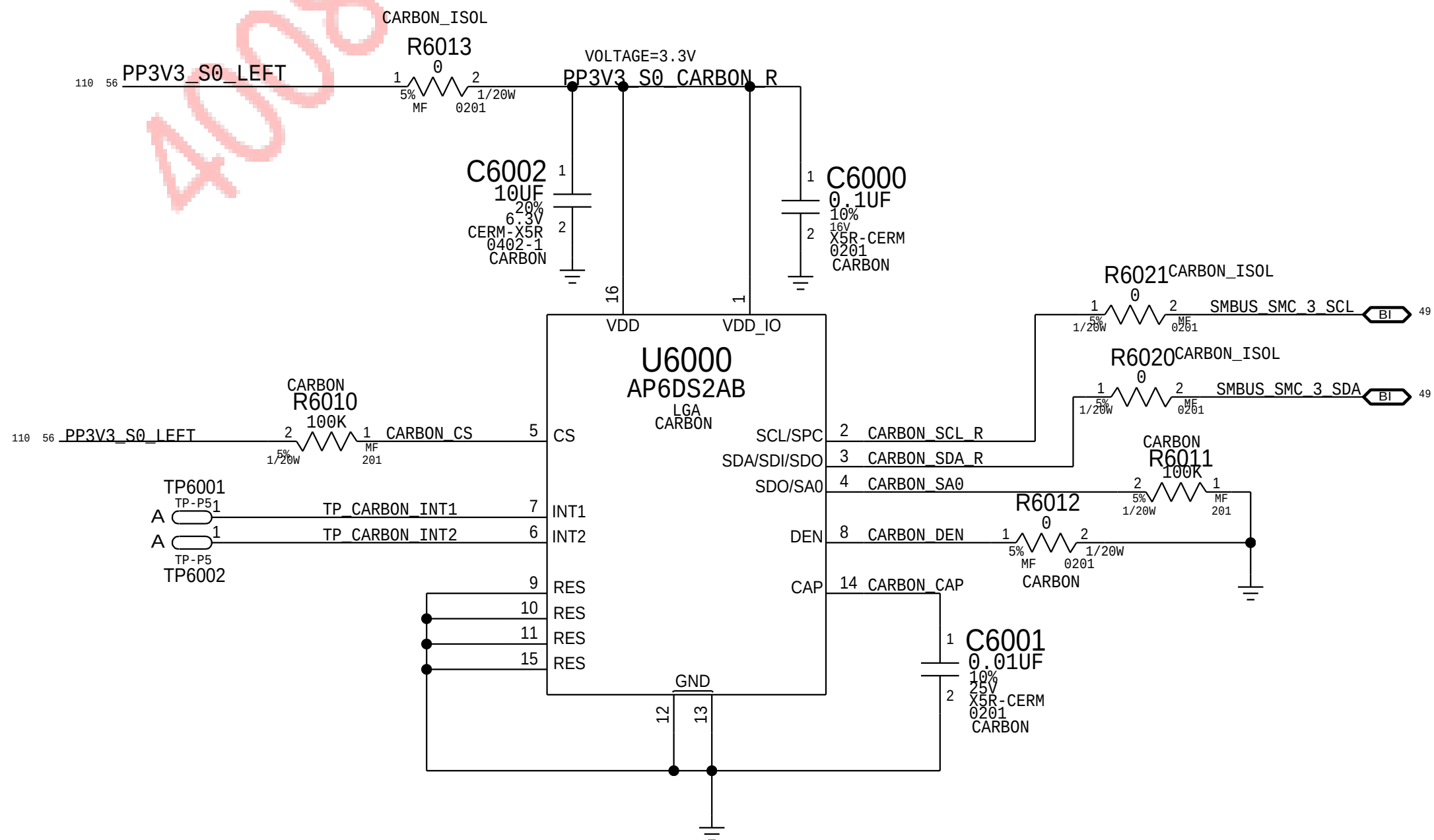
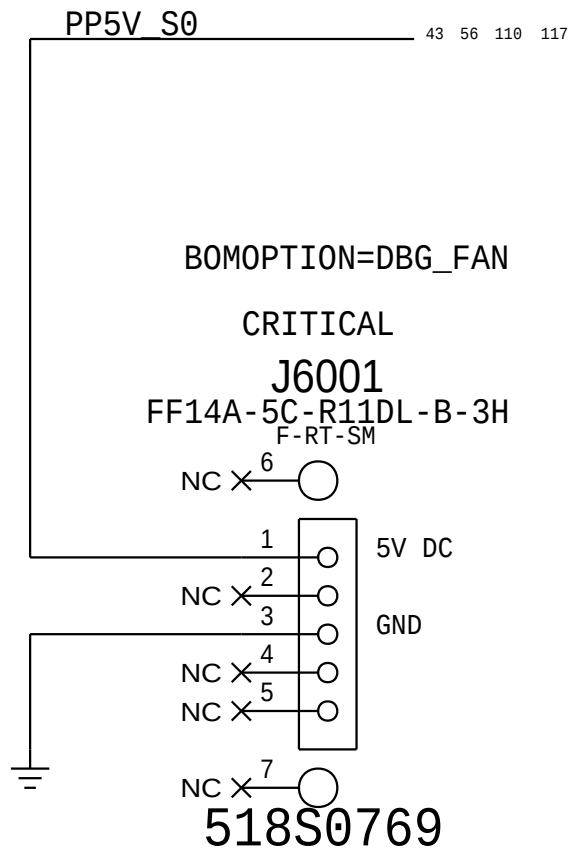
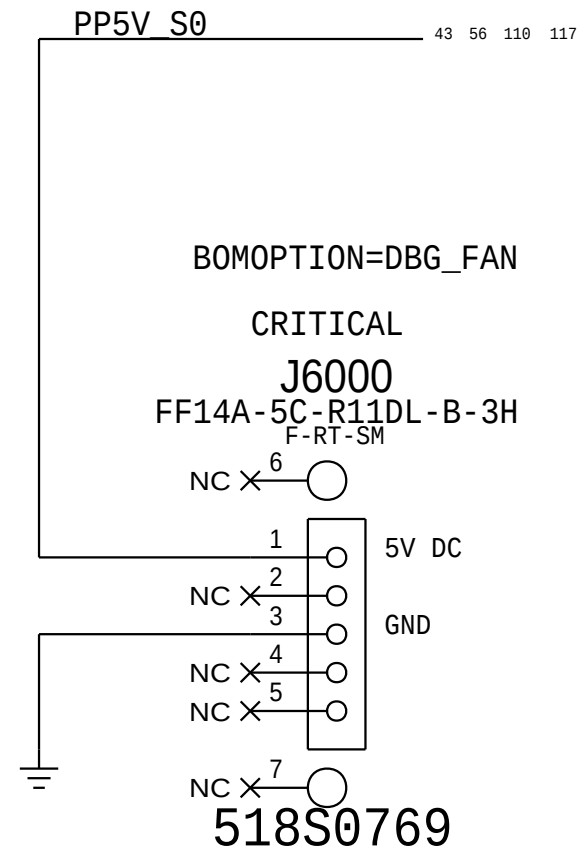
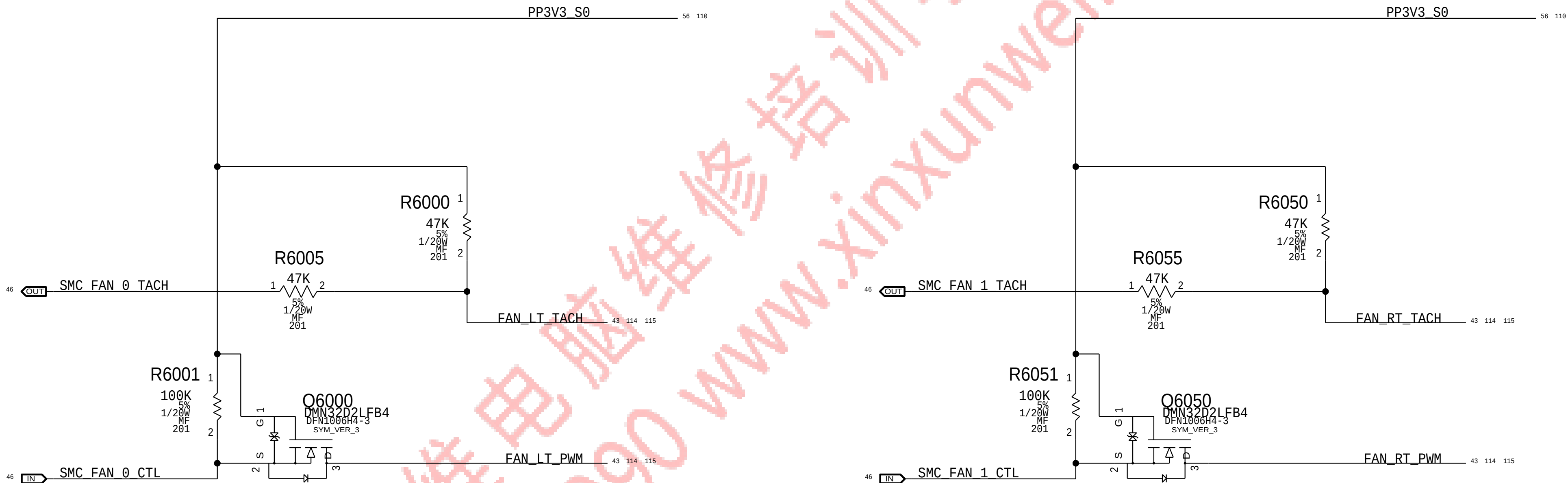
Placement Note:
Place Q5873 under the CPU, on the BOTTOM side.

I2C ADDRESS (U5870): 0XB8/0XB9

PAGE TITLE			
Thermal Sensors			
	DRAWING NUMBER	051-00647	SIZE
	REVISION	10.0.0	D
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FAN CONNECTOR

KEEP THE 5 PIN CONNECTOR FROM D1



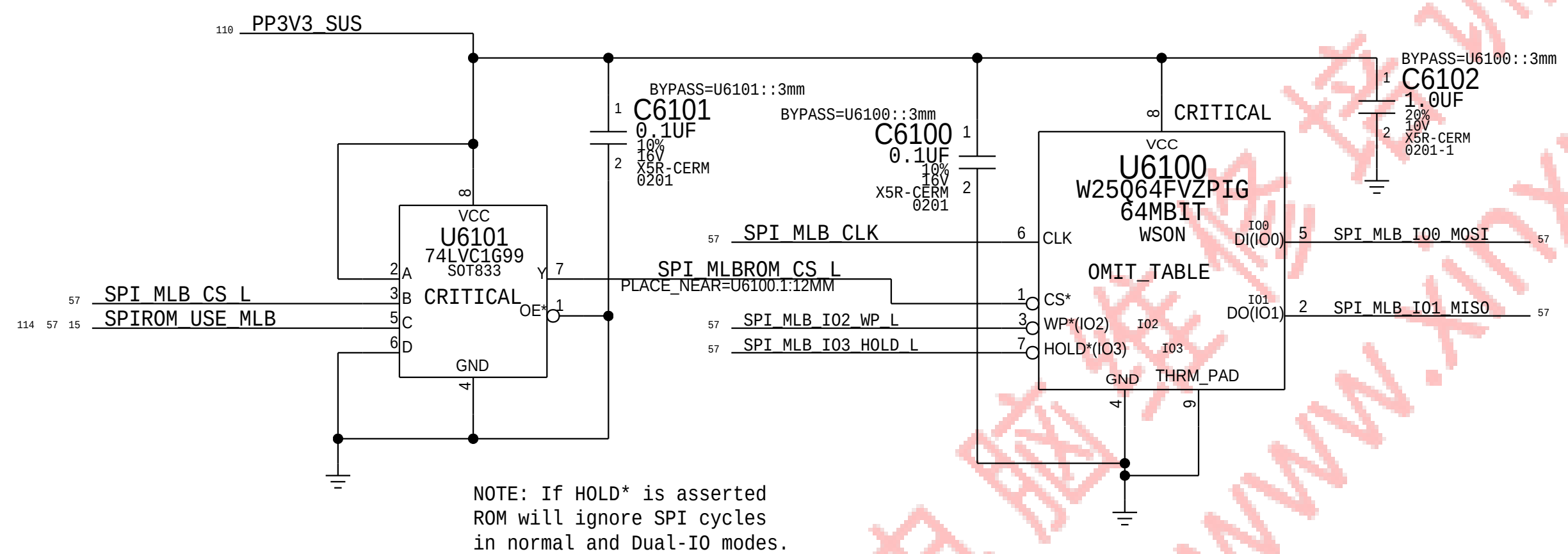
PAGE TITLE		Fans	
 Apple Inc.	DRAWING NUMBER	051-00647	SIZE
	REVISION	10.0.0	D
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BOM_COST_GROUP=FAN



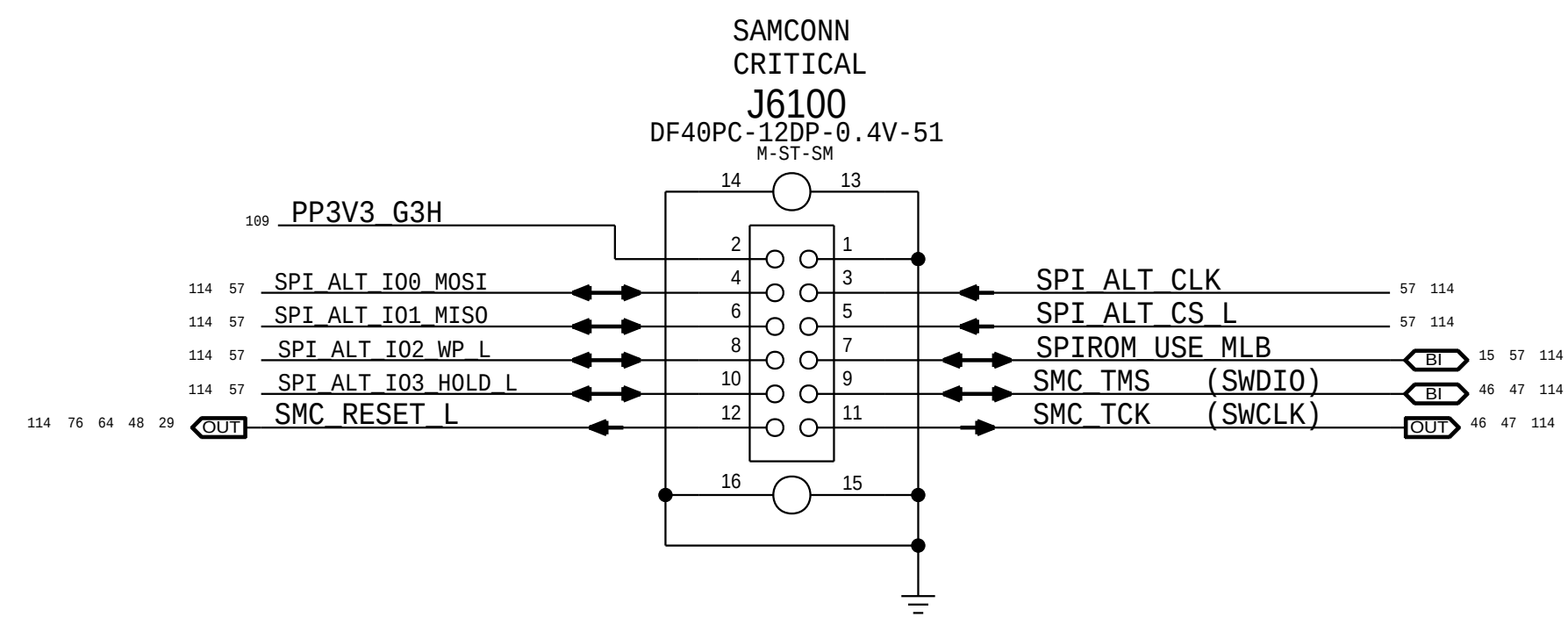
SPI ROM

Quad-I/O Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.



Quad SPI and QPI instructions require the non-volatile Quad Enable bit (QE) in Status Register-2 to be set. When QE=1, the /WP pin becomes IO2 and /HOLD pin becomes IO3.

SPI+SWD SAM Connector

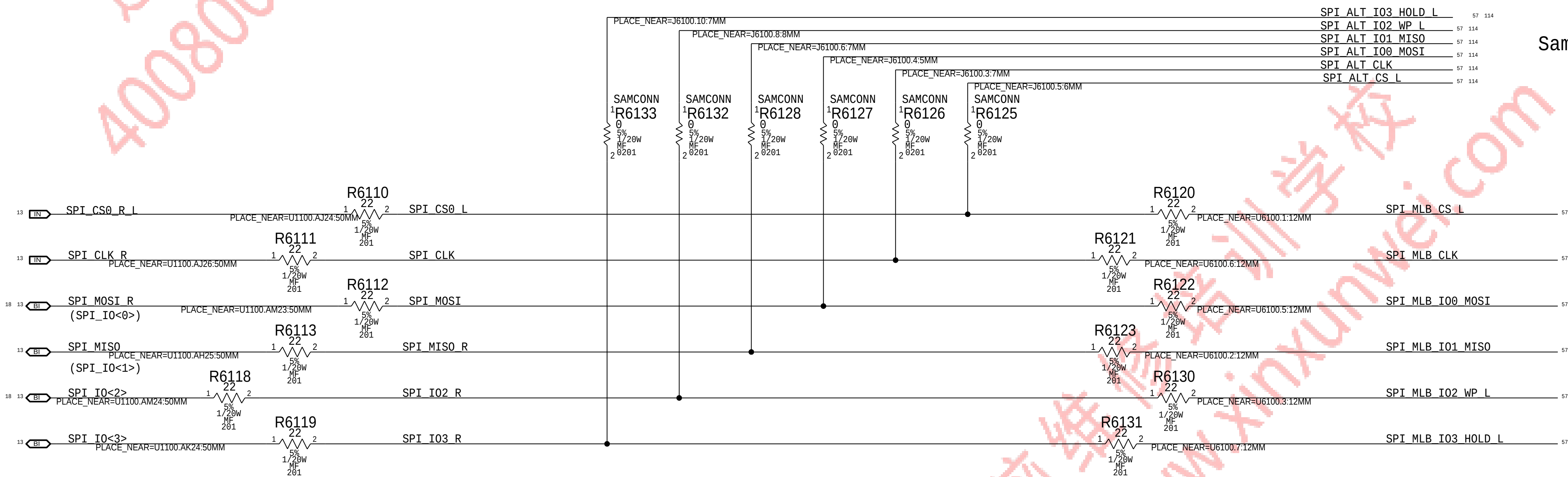


SPI Bus Series Termination (Modified per PDG)

CPU Master

Sam Card ROM Slave

SPI ROM Slave



SYNC_MASTER=380_MLB		SYNC_DATE=11/06/2015	
PAGE TITLE			
SPI Debug Connector			
 Apple Inc.		DRAWING NUMBER	051-00647
		SIZE	D
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		BRANCH	dvt-fab10
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BOM_COST_GROUP=CPU & CHIPSET

D

D

C

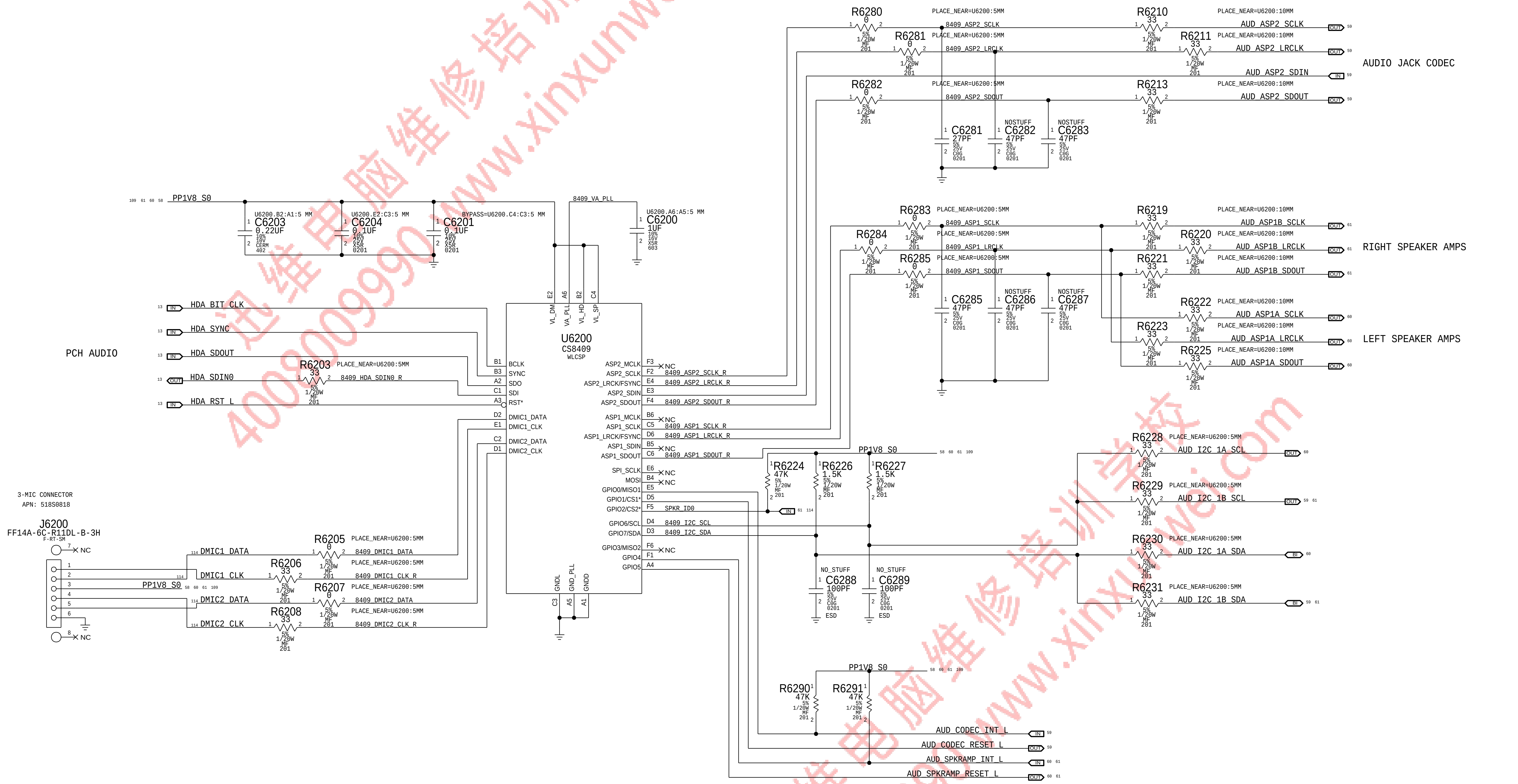
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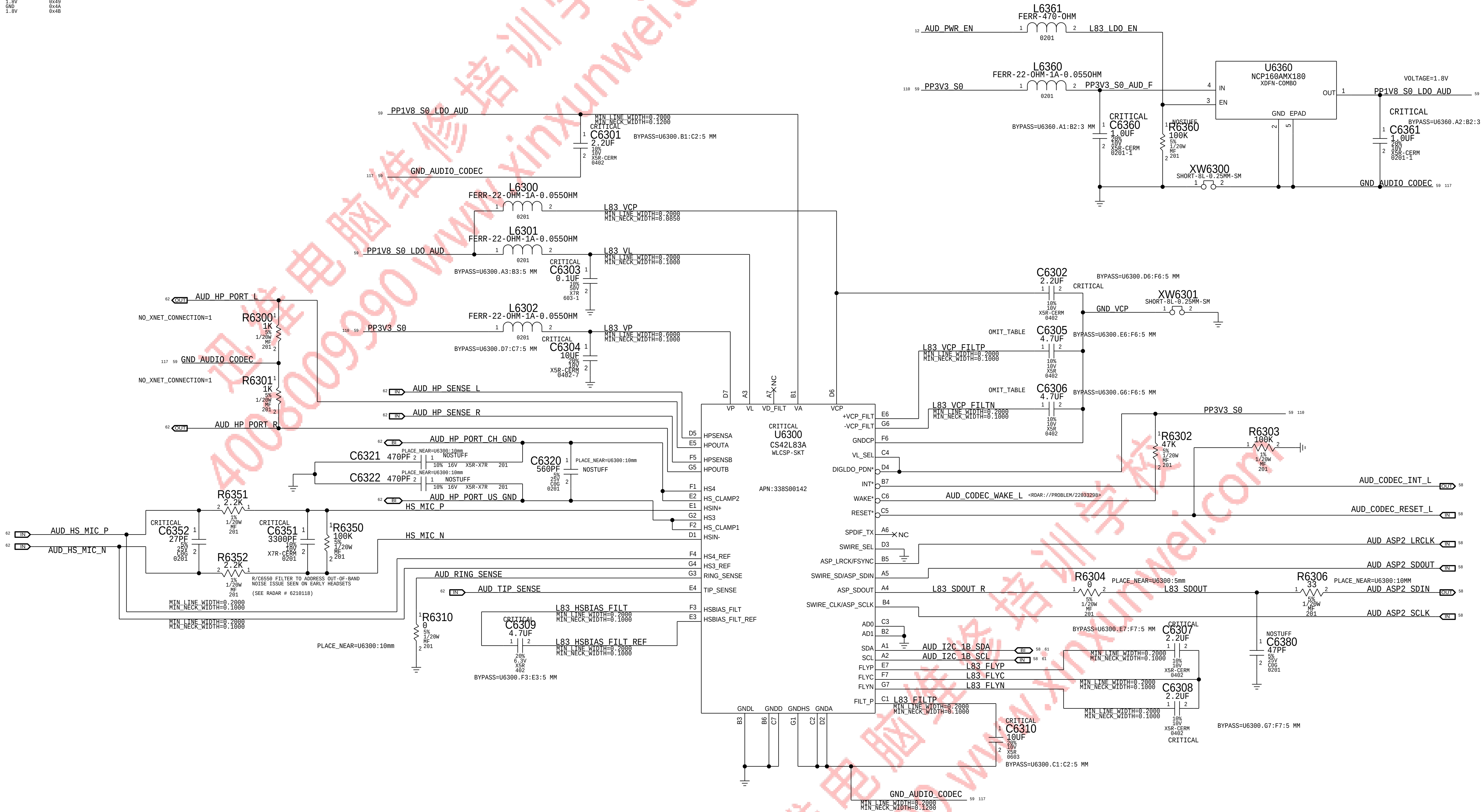
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BOM_COST_GROUP=AUDIO

AUDIO JACK CODEC I2C ADDRESS		
AD1	AD0	ADDRESS
GND	GND	0x48 <--
GND	1.8V	0x49
1.8V	GND	0x4A
1.8V	1.8V	0x4B



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0719	2	CAP, CER, 4.7UF, 20%, 18V, XSR, 0402, MURATA	C6305, C6306	CRITICAL	

BOM_COST_GROUP=AUDIO

PAGE TITLE		
AUDIO JACK CODEC		
 Apple Inc.	DRAWING NUMBER	051-00647
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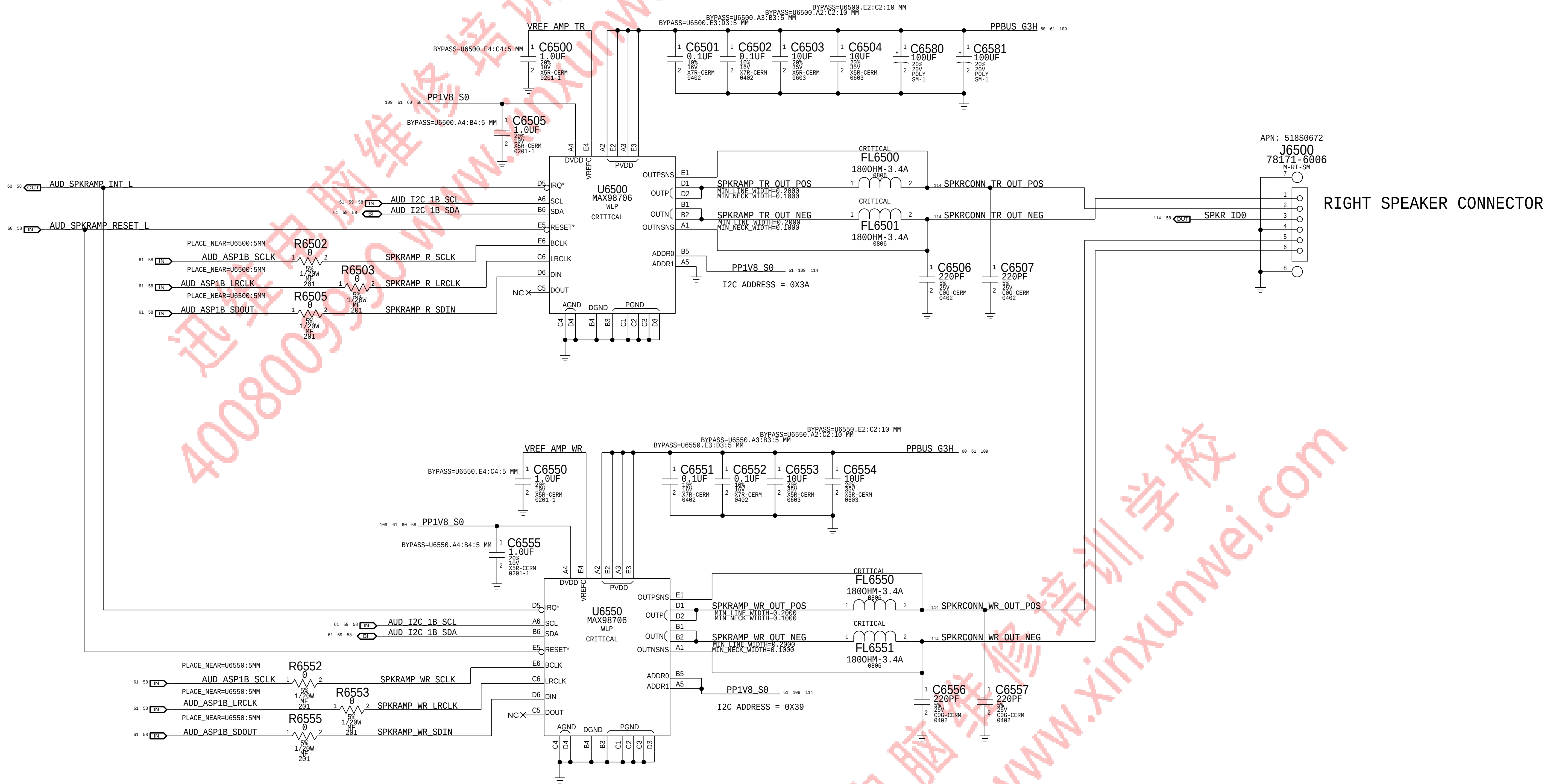
LEFT SPEAKER CONNECTOR

APN: 518S00521
J6410
78171-0004
M-RT-SM

BOM_COST_GROUP=AUDIO

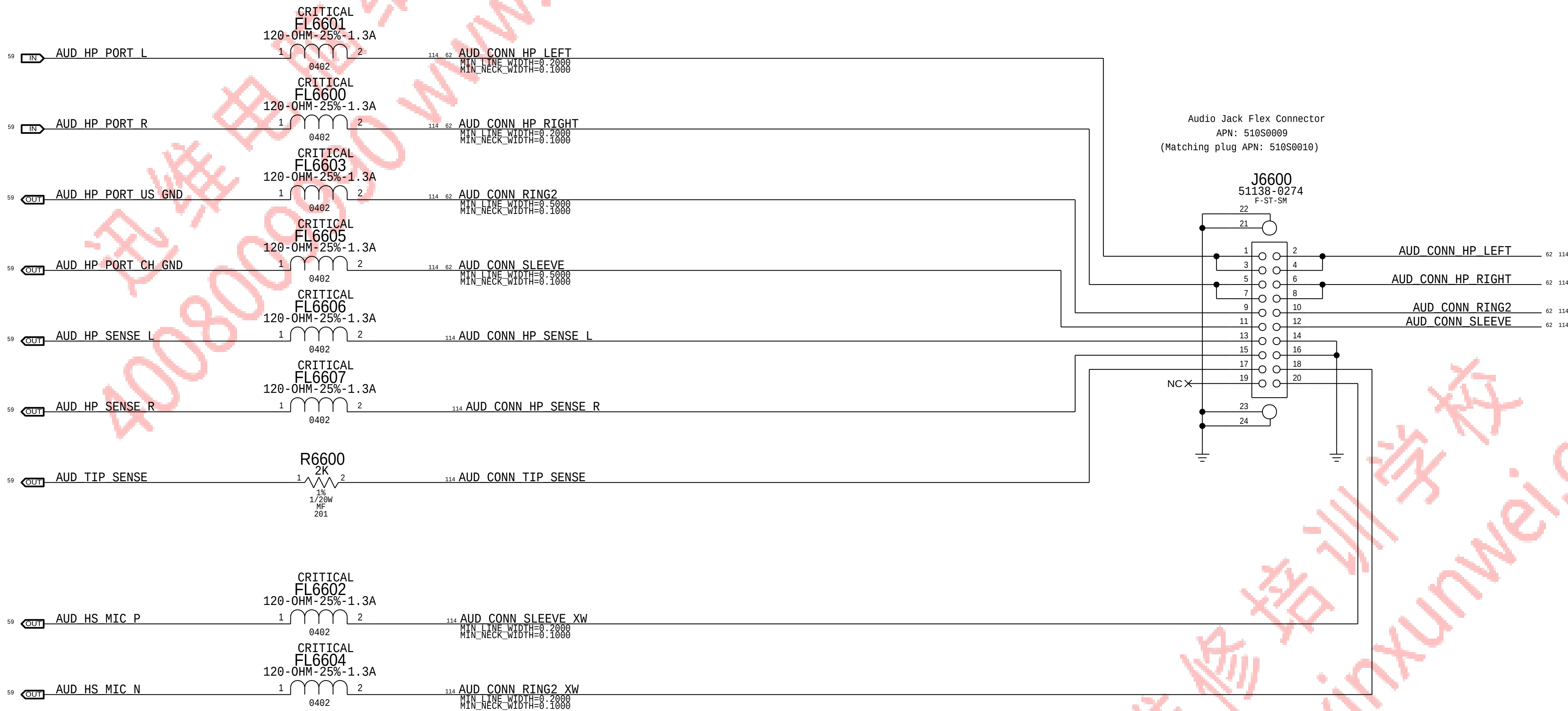
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AUDIO Speaker Amps & Conn			
 Apple Inc.	DRAWING NUMBER	051-00647	SIZE D
	REVISION	10.0.0	
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2X MONO SPEAKER AMPLIFIER
APN: 353S00604
0dBFS = 9VPK



PAGE TITLE			
AUDIO Speaker Amps & Conn			
 Apple Inc.	DRAWING NUMBER	051-00647	SIZE D
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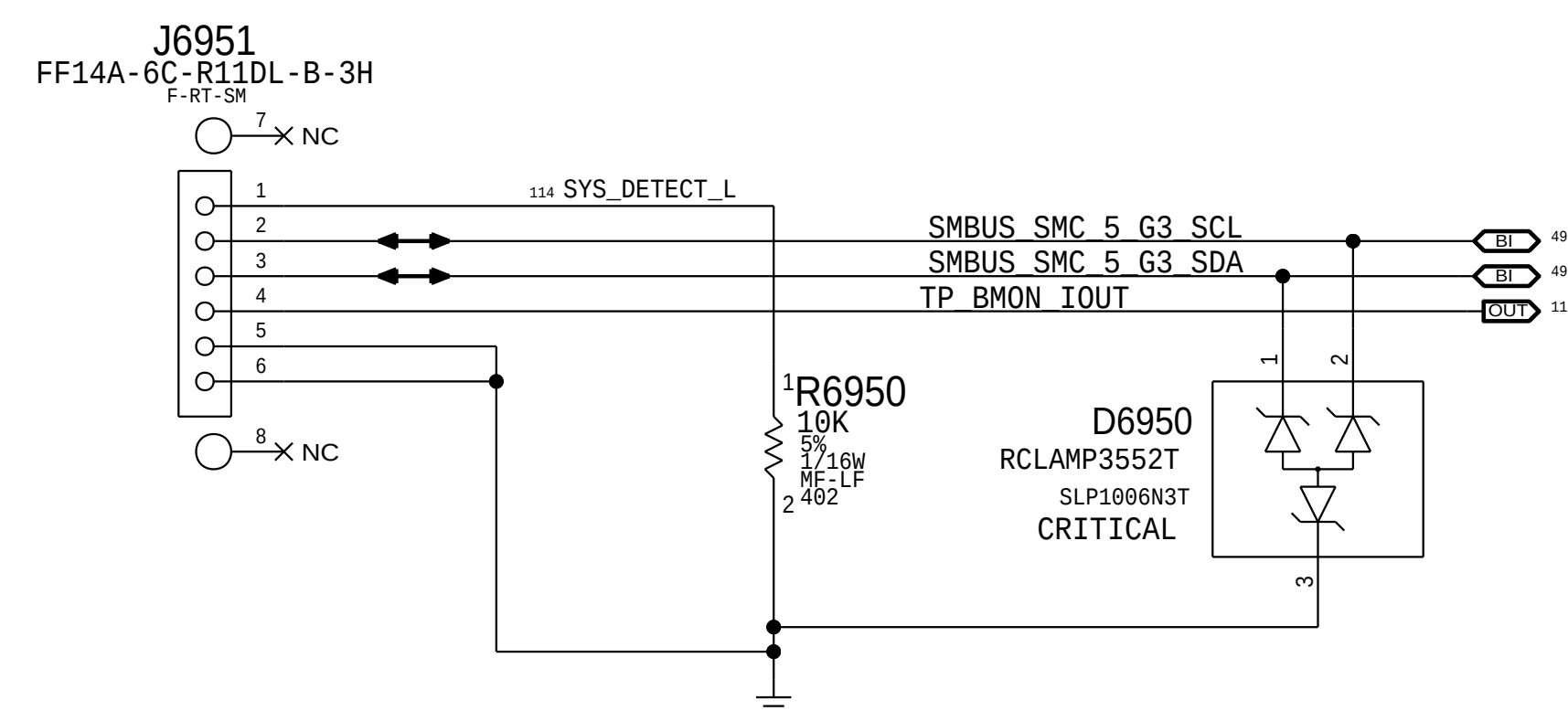
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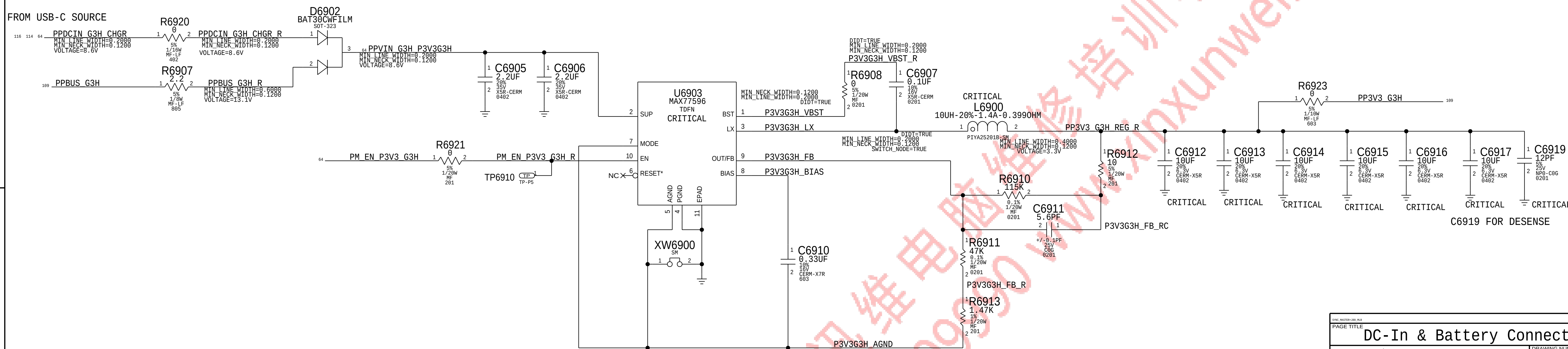
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AUDIO JACK CONNECTOR		
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BOM_COST_GROUP=AUDIO

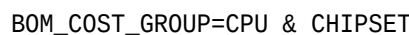
APN: 518S0818



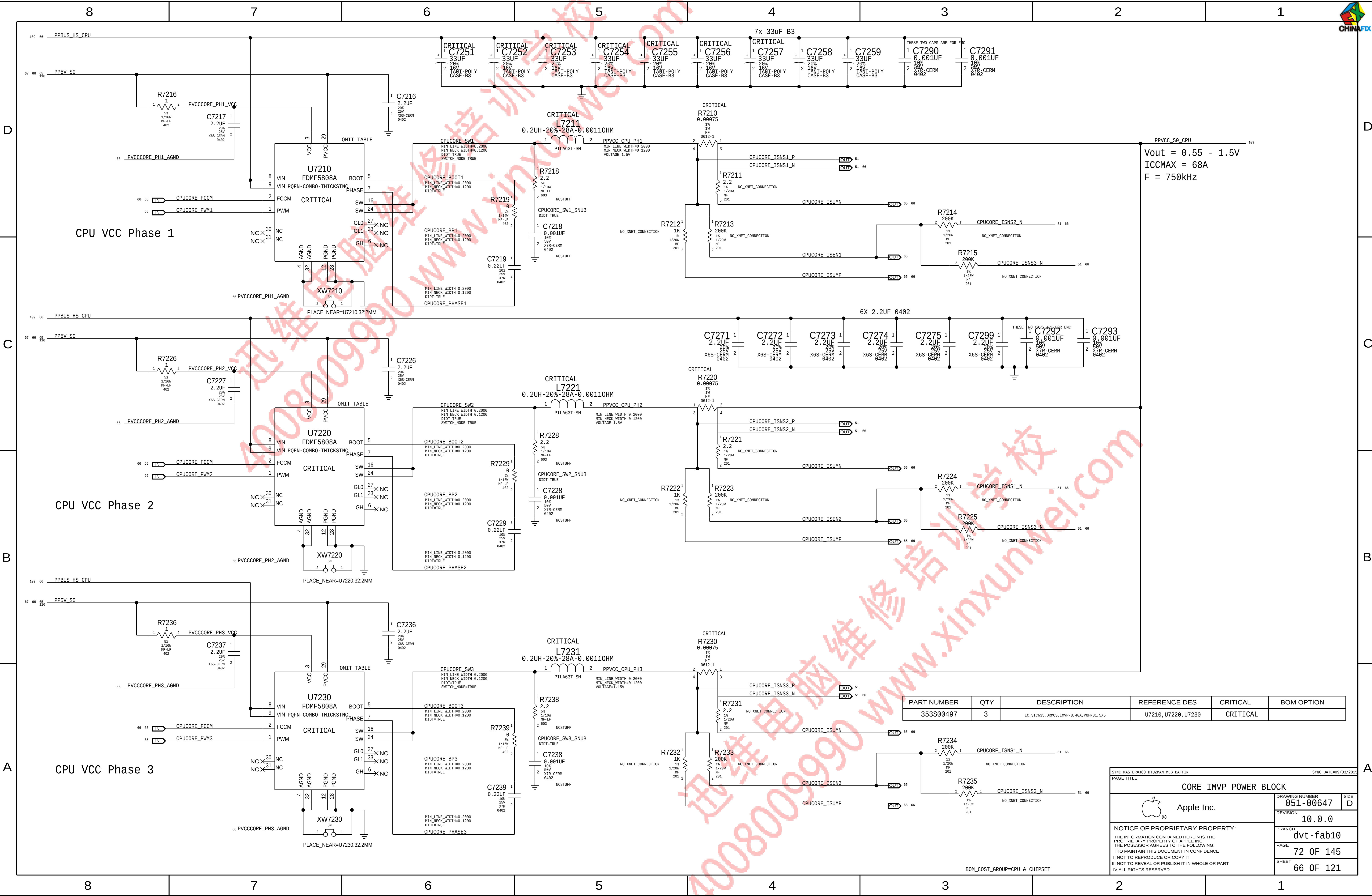
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
632-00862	1	PCBA, FLEX, BMU PWR, X363	J6950	CRITICAL	



1596_00202-001-0010 PAGE TITLE		1596_00202-001-0010	
DC-In & Battery Connectors			
 Apple Inc.	DRAWING NUMBER 051-006647		SIZE D
	REVISION 10.0.0		
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SYNC_MASTER:J98_OT2UWVN_HLB_BAFFIN		SYNC_DATE:11/18/2013	
PAGE TITLE			
CORE & SA IMVP IC			
 Apple Inc.		DRAWING NUMBER 051-000647	SIZE D
		REVISION 10.0.0	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	3	IC, SIC635, DRMOS, IMVP-B, 40A, PQFN31, 5X5	U7210, U7220, U7230	CRITICAL	

SYNC_MASTER=380_DTU2MAN_MLB_BAFFIN		SYNC_DATE=09/03/2015	
PAGE TITLE			
CORE IMVP POWER BLOCK			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00647	D
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		BRANCH	dvt-fab10
		PAGE	72 OF 145
		SHEET	66 OF 121

BOM_COST_GROUP=CPU & CHIPSET

D

C

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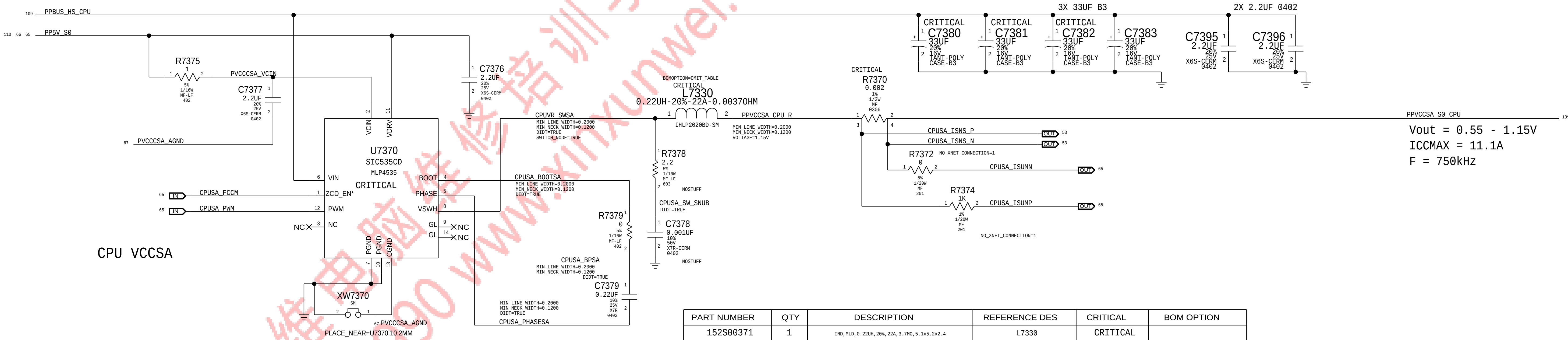
A

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Vout = 0.55 - 1.15V
ICCMAX = 11.1A
F = 750kHz

D

C

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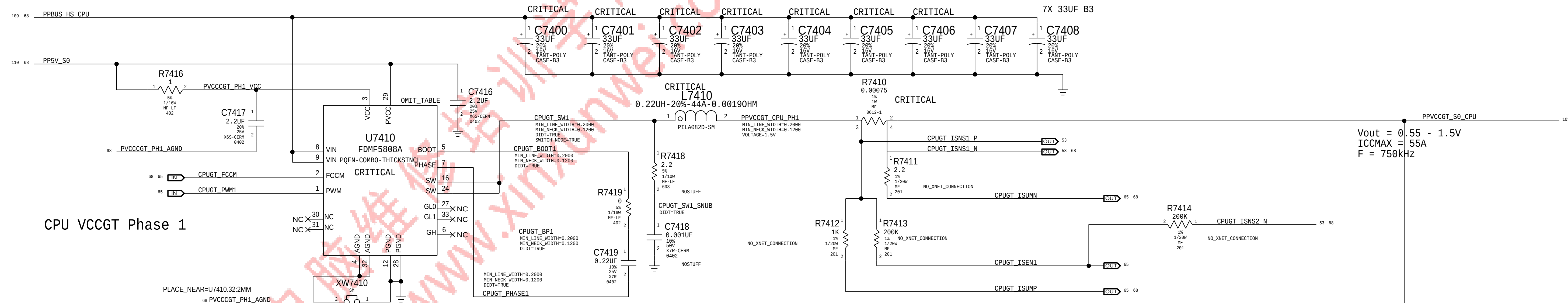
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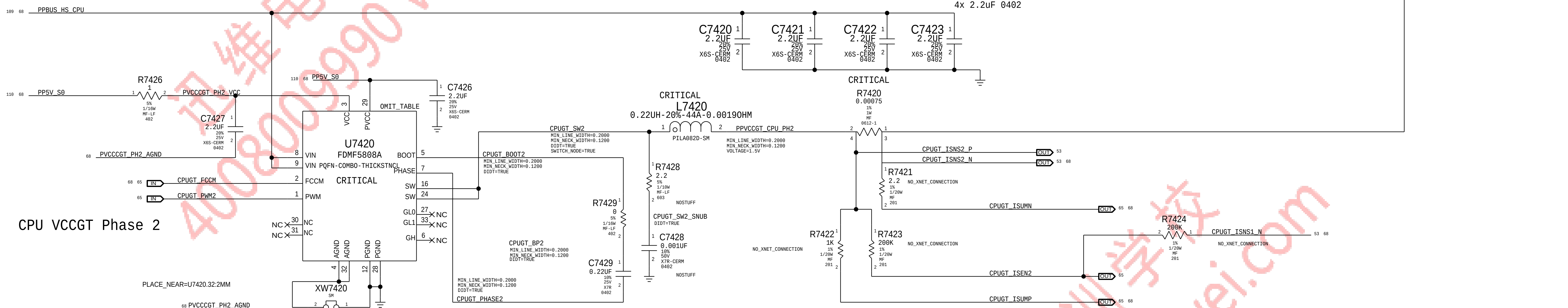
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CPU VCCGT Phase 1



CPU VCCGT Phase 2

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	2	IC, SIC035, DRM05, IMVP-8, 46A, PQFN31, 5X5	U7410, U7420	CRITICAL	

BOM_COST_GROUP=CPU & CHIPSET

DRAWING NUMBER 051-00647		SIZE D
REVISION 10.0.0		BRANCH dvt-fab10
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GT & GTX IMVP POWER BLOCK

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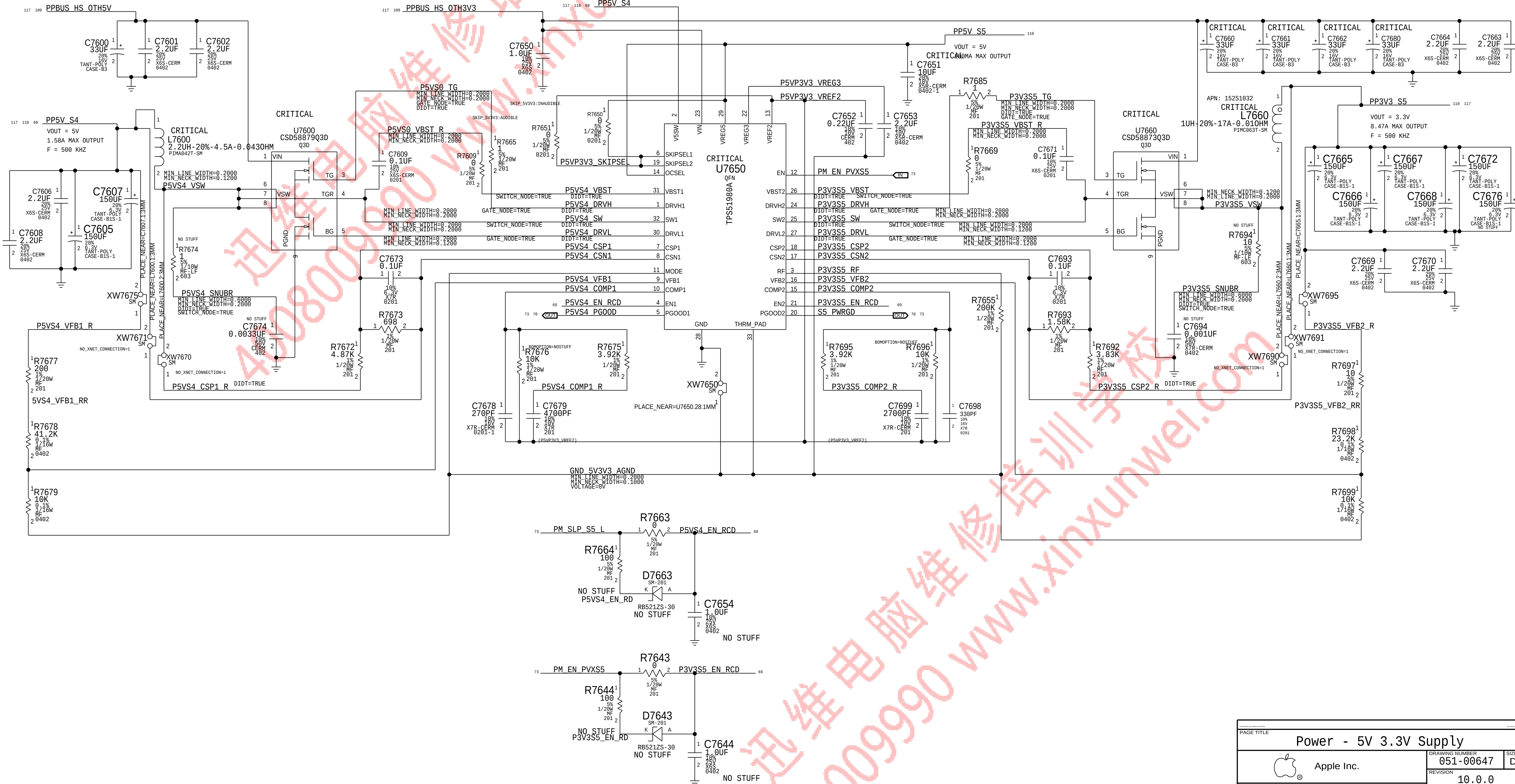
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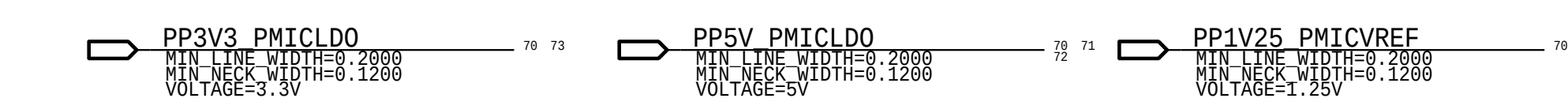
5V S0 - V5

3.3V DSW - V6



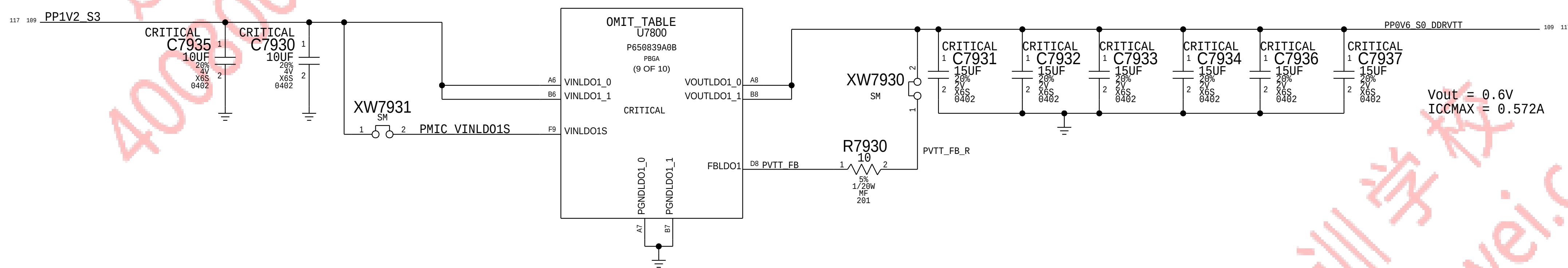
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PAGE TITLE		
Power - 5V 3.3V Supply		
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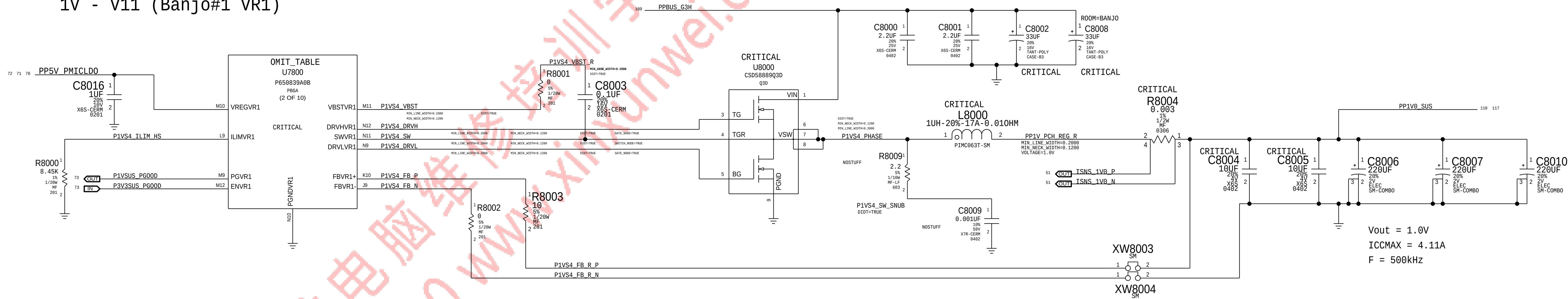
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0.6V VTT - V13 (Banjo#1 LD01)

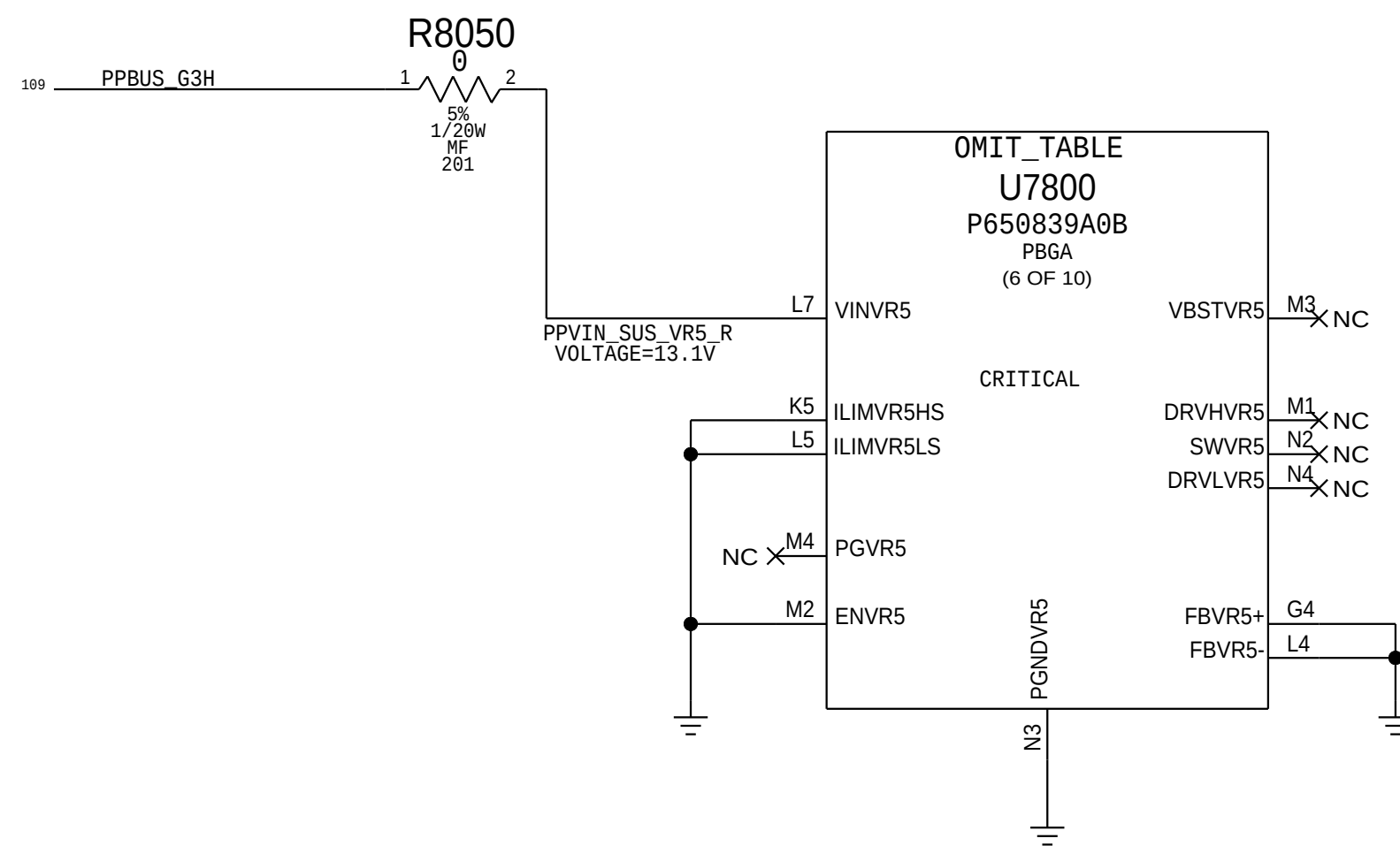
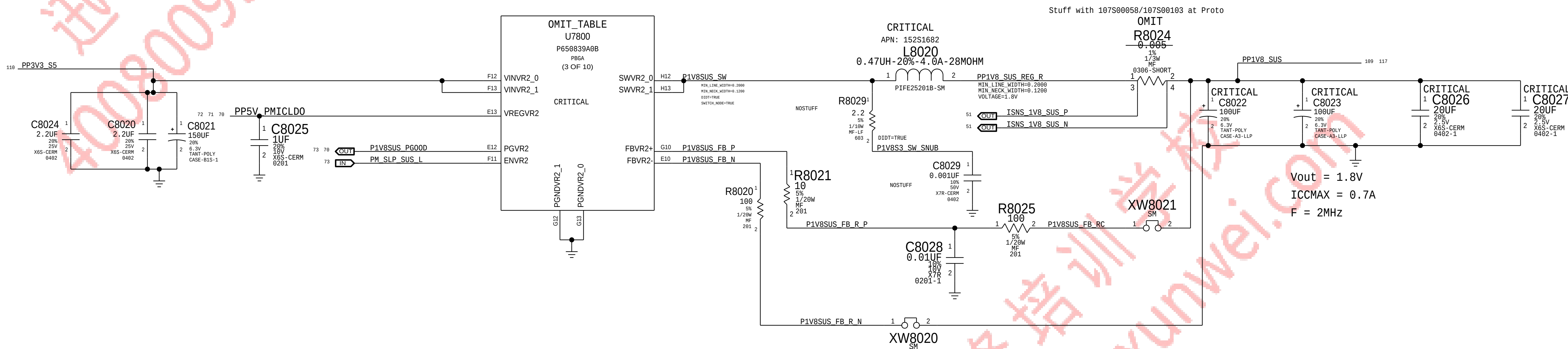
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BOM_COST_GROUP=PLATFORM POWER

1V - V11 (Banjo#1 VR1)



1.8V - V8 (Banjo#1 VR2)



PAGE TITLE			
PMIC-1 1V 1.8V VCCPCH			
	DRAWING NUMBER	051-00647	SIZE
	REVISION	10.0.0	D
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		PAGE	80 OF 145
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D

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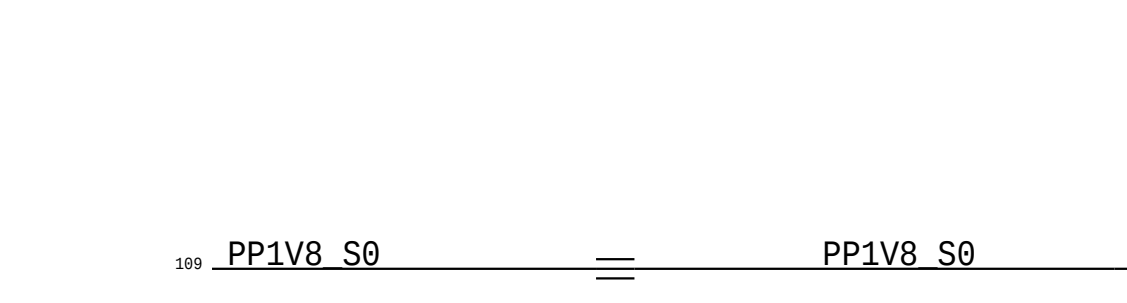
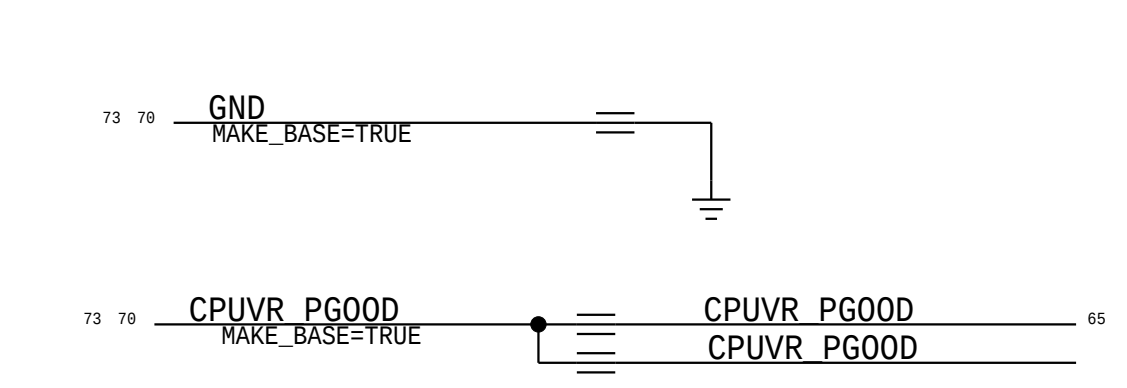
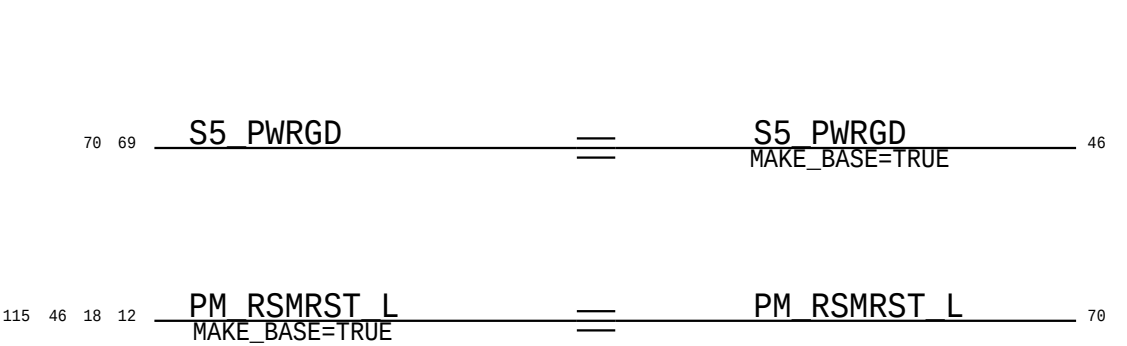
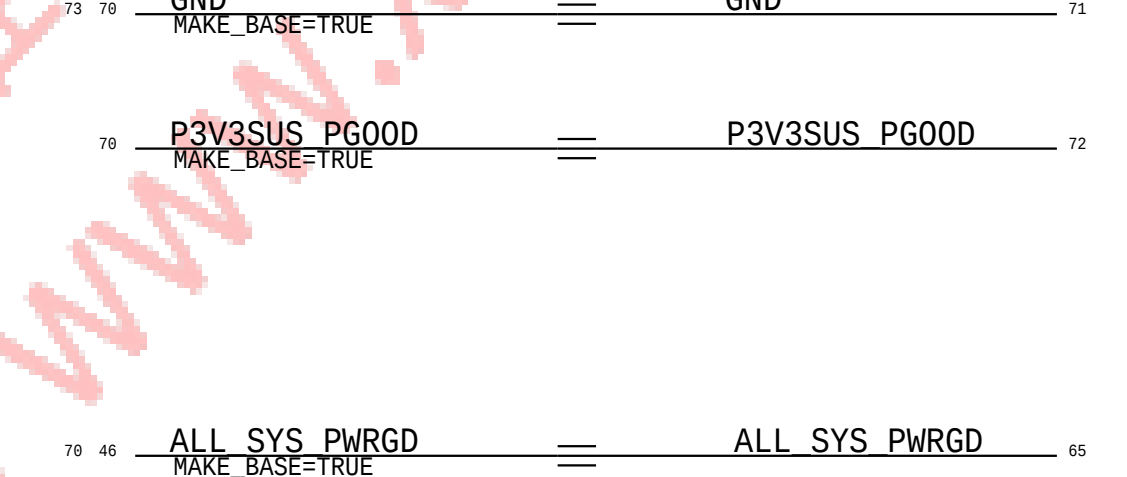
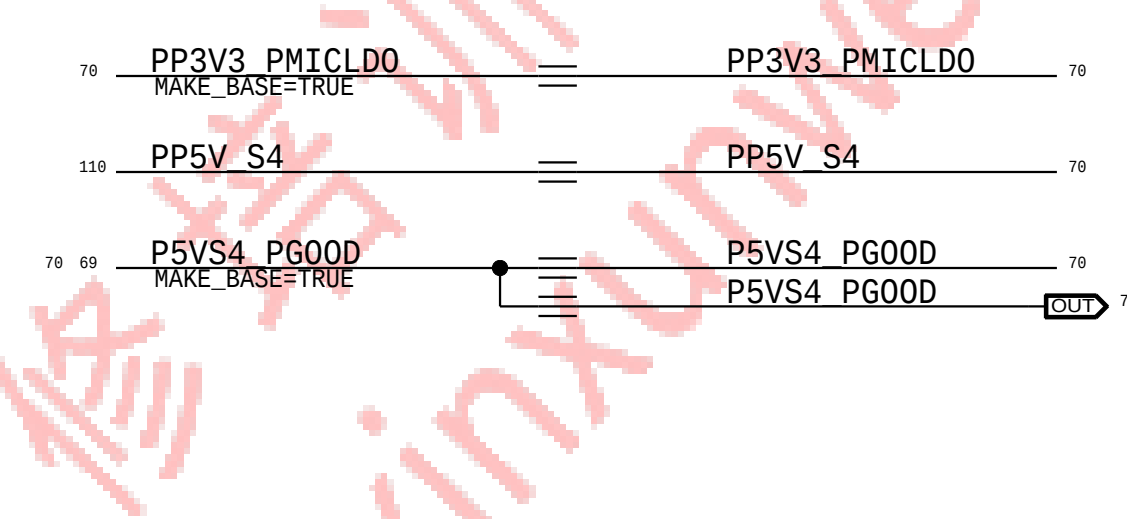
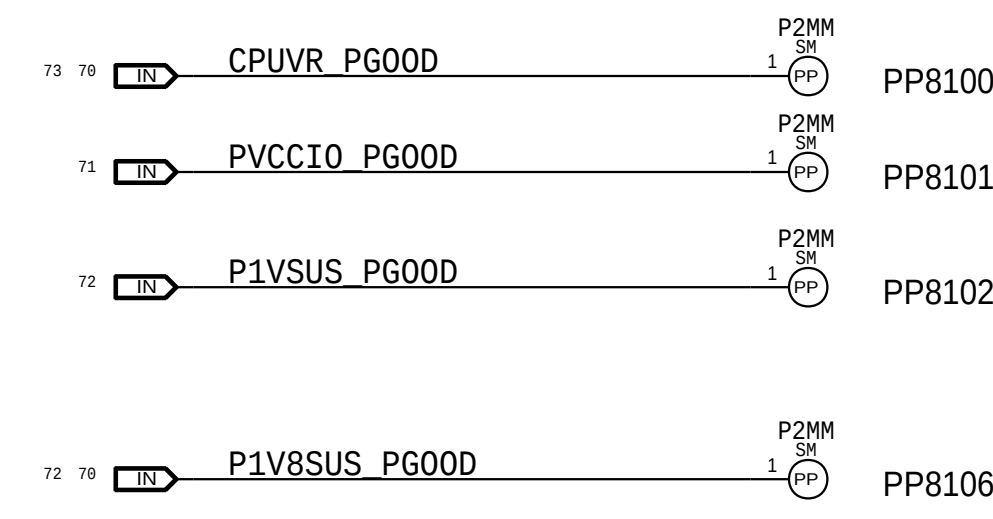
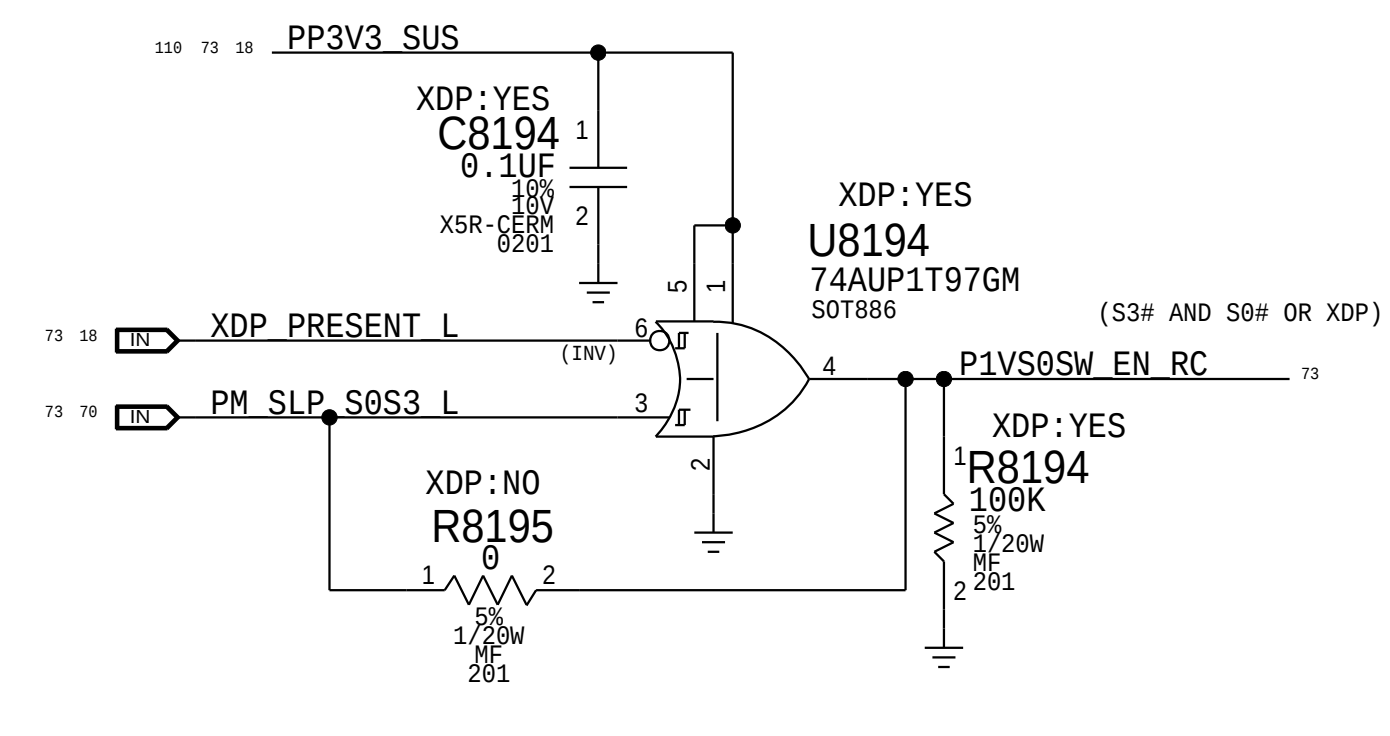
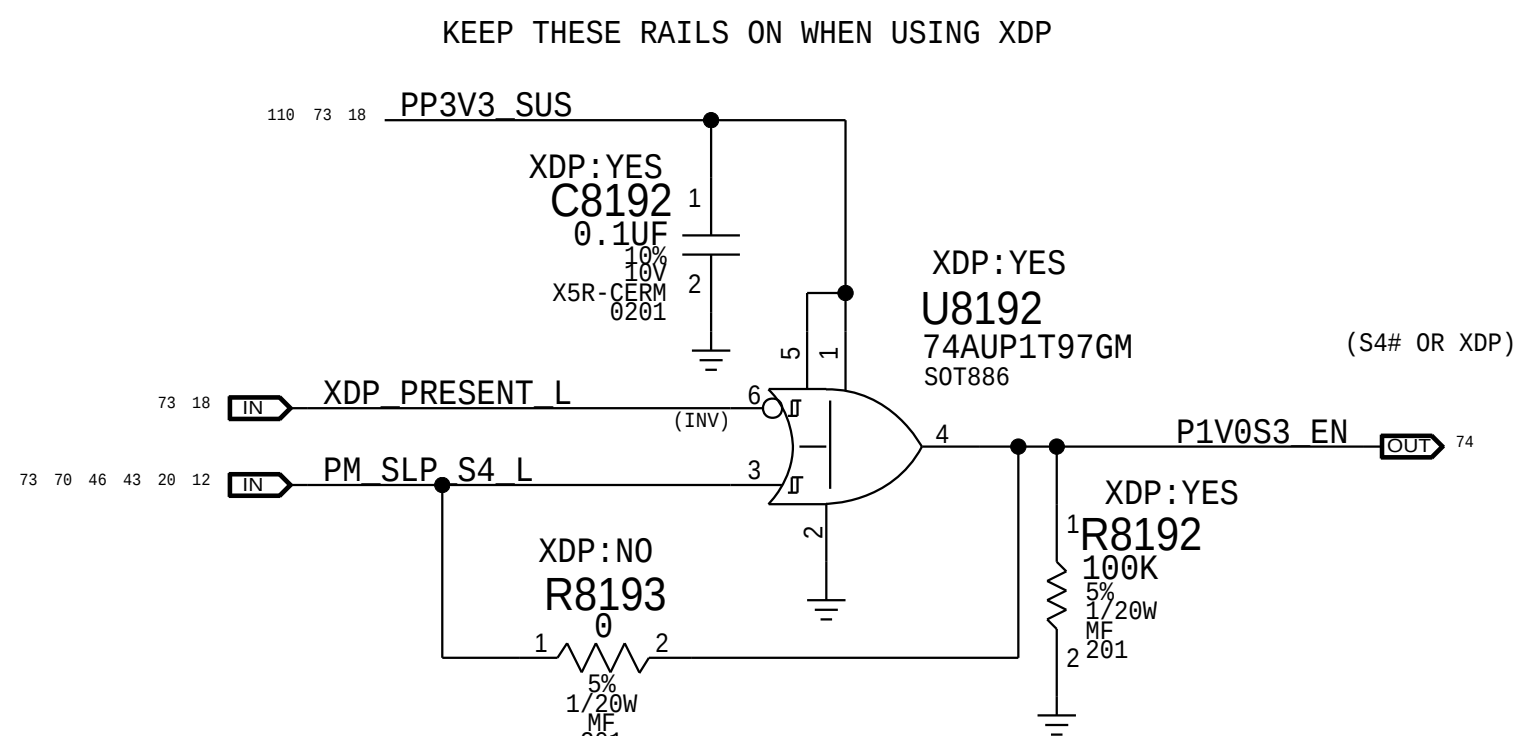
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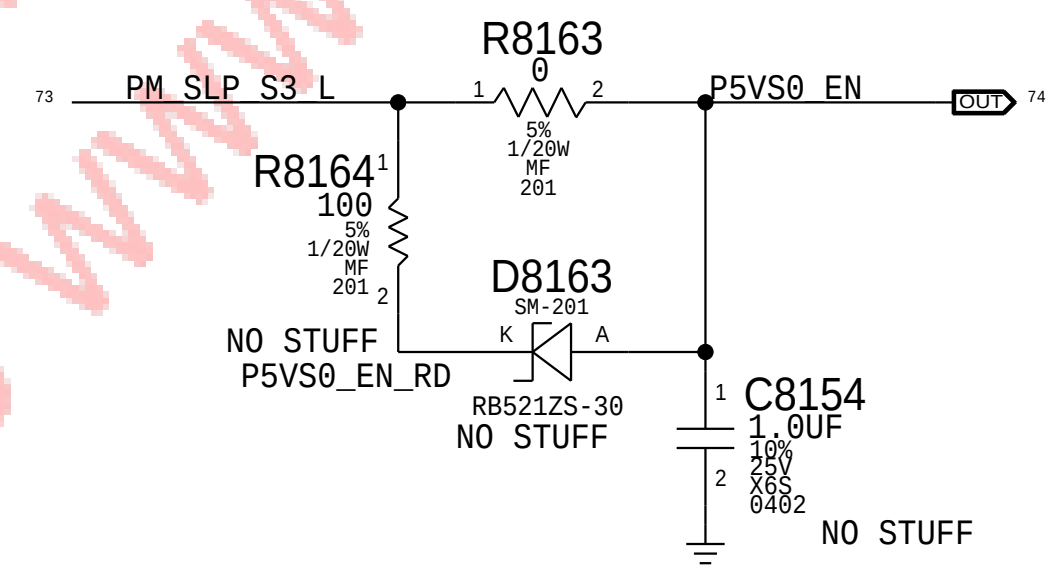
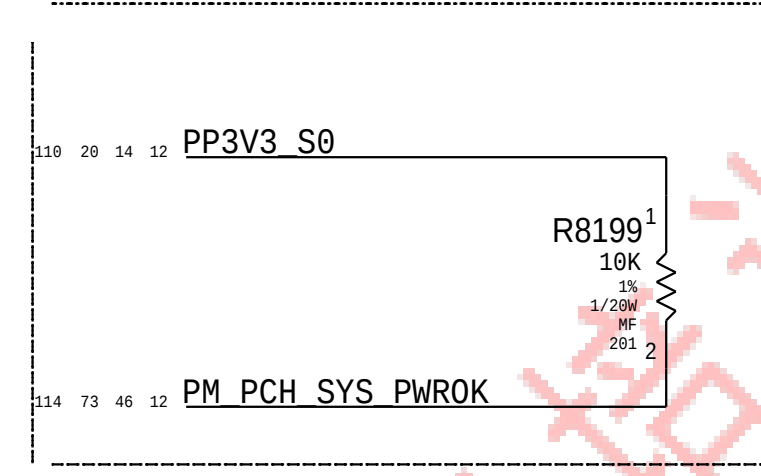
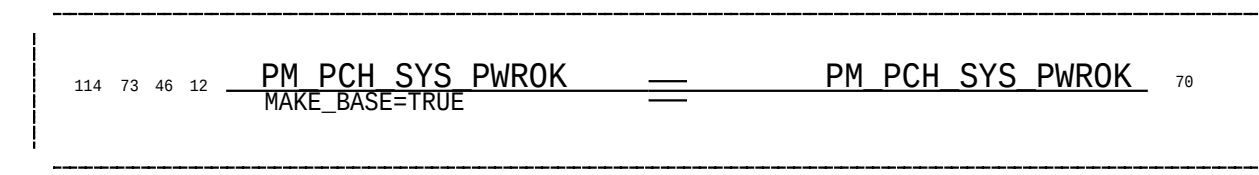
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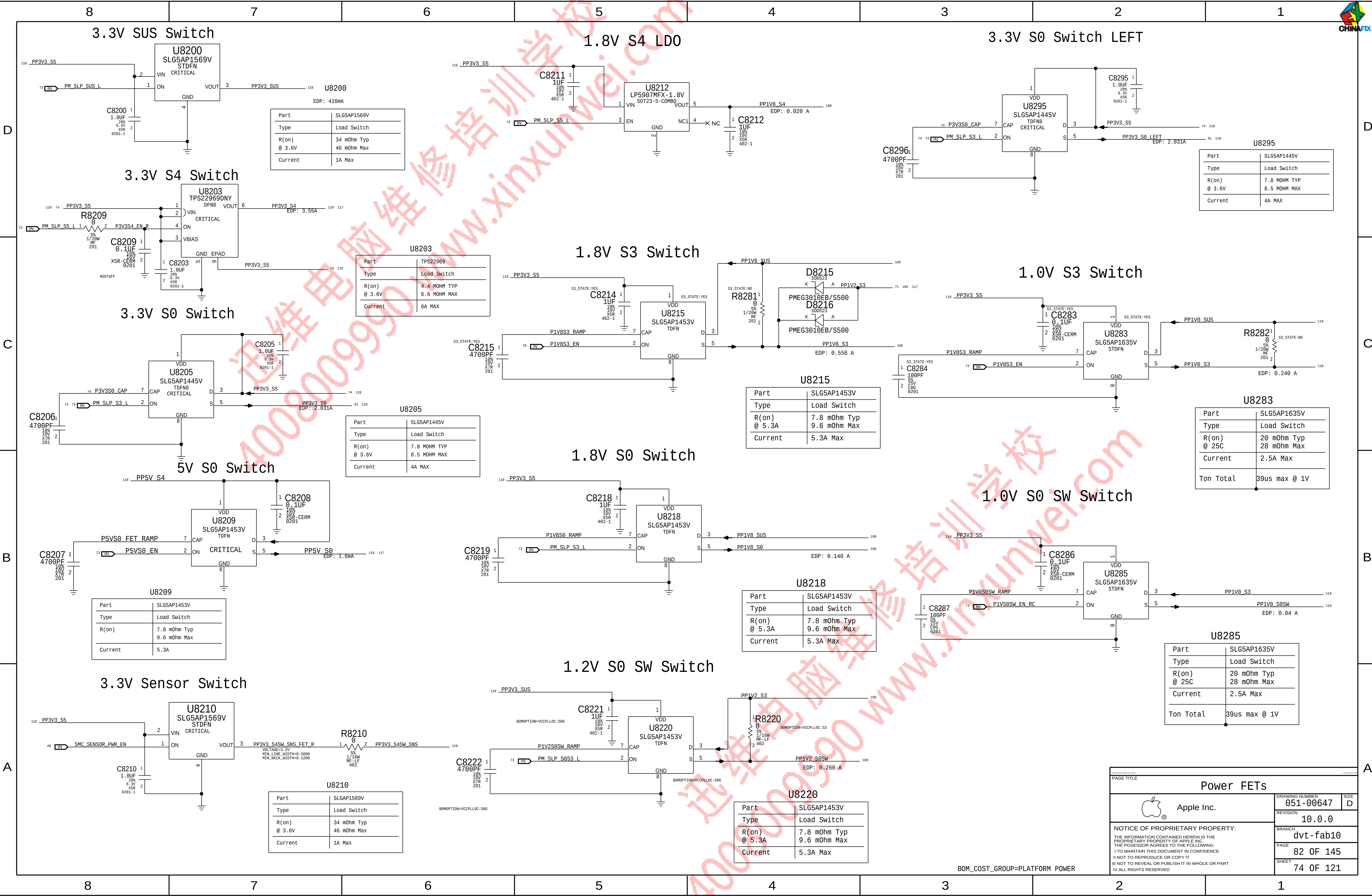


J80G specific

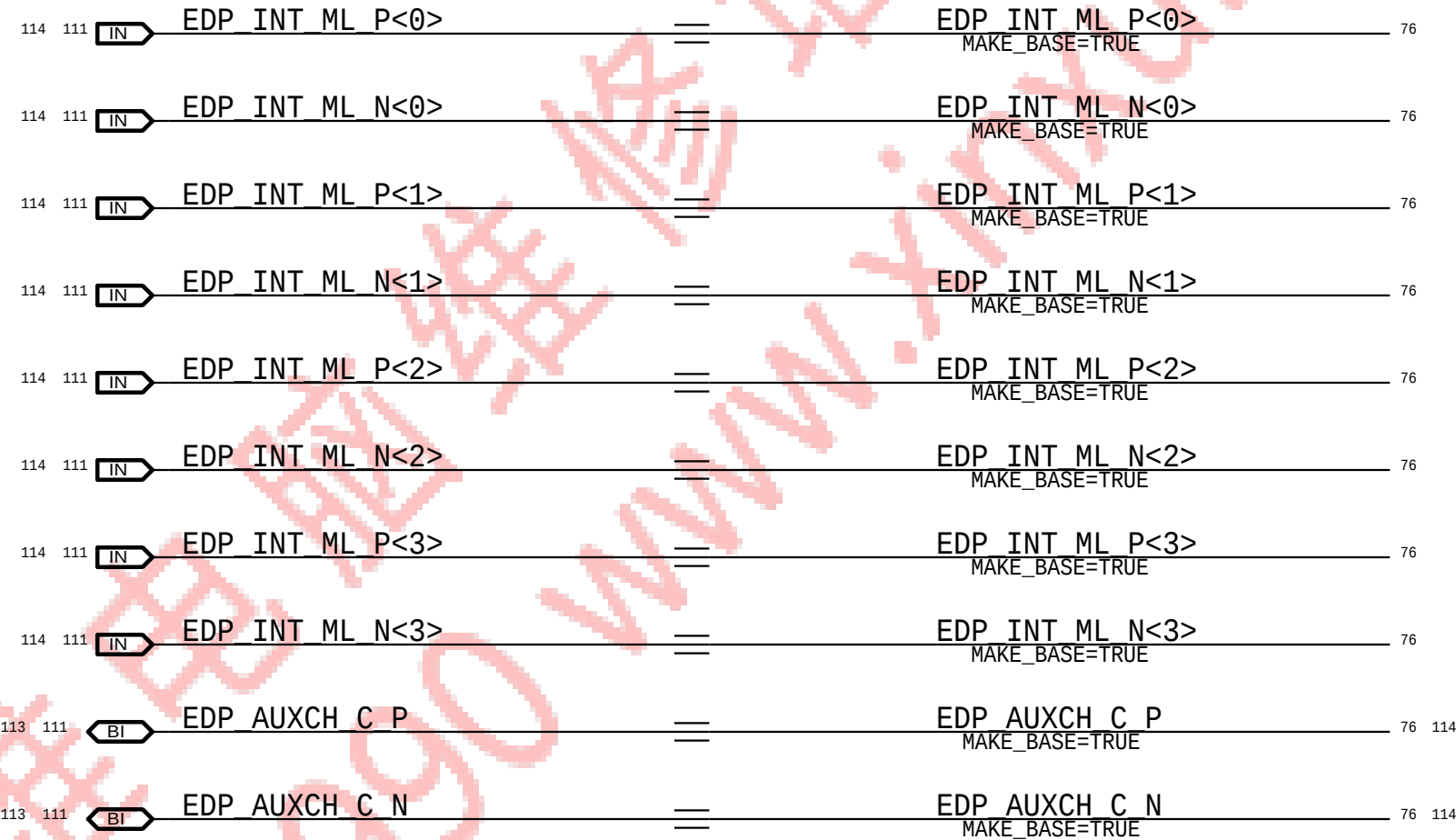
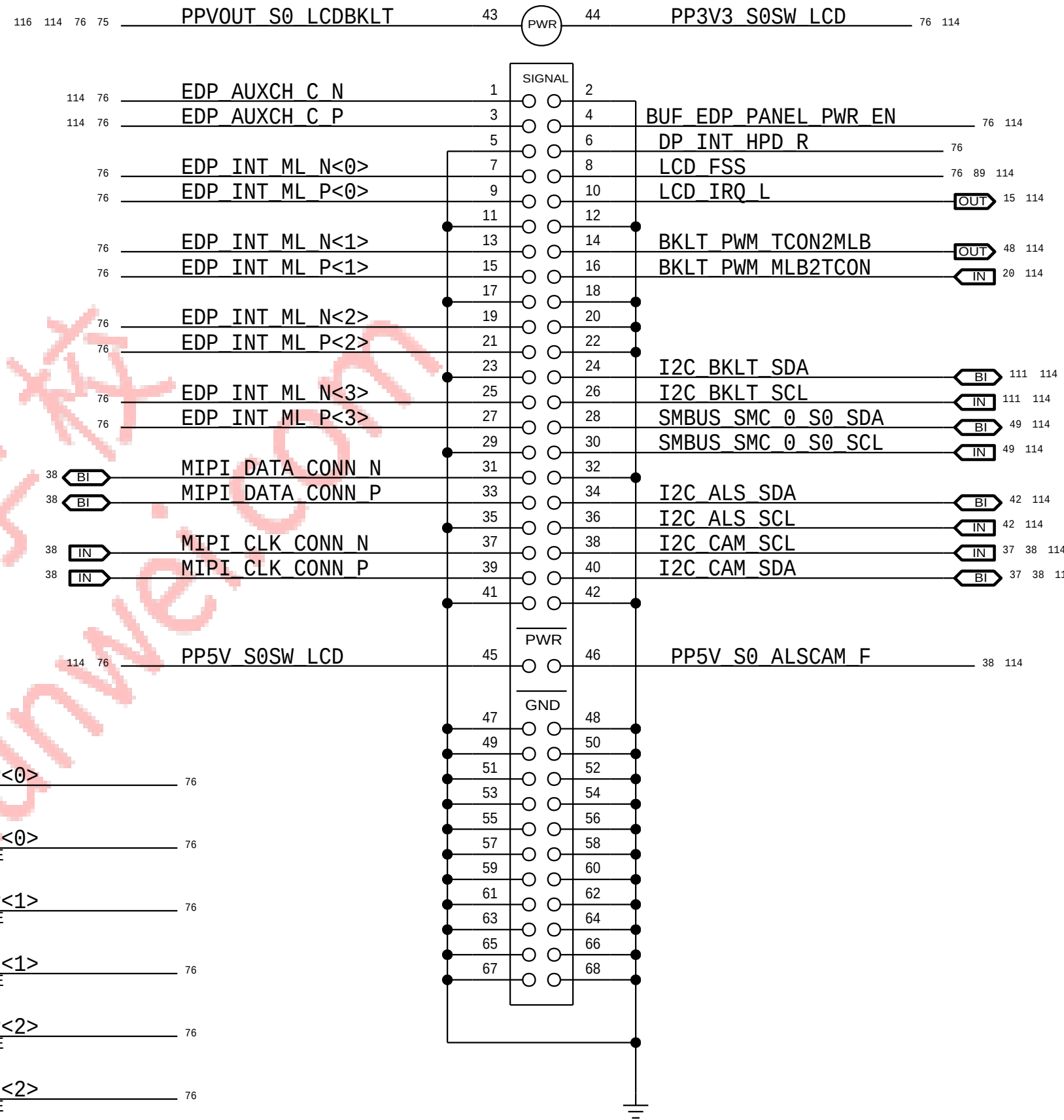
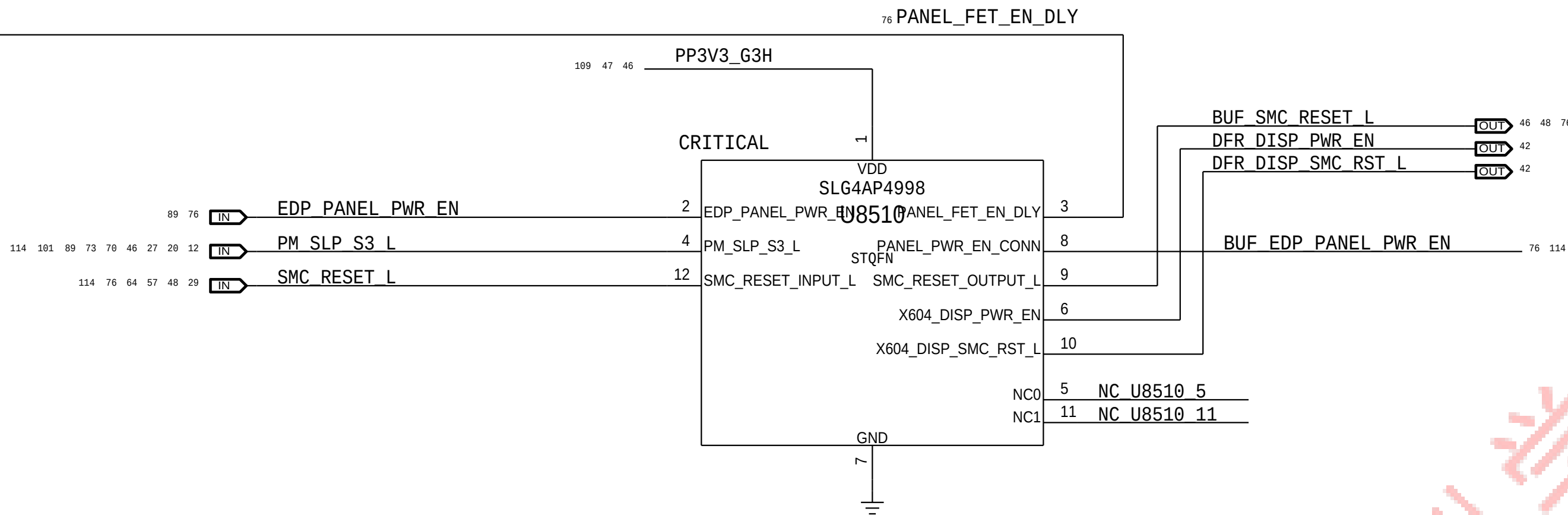
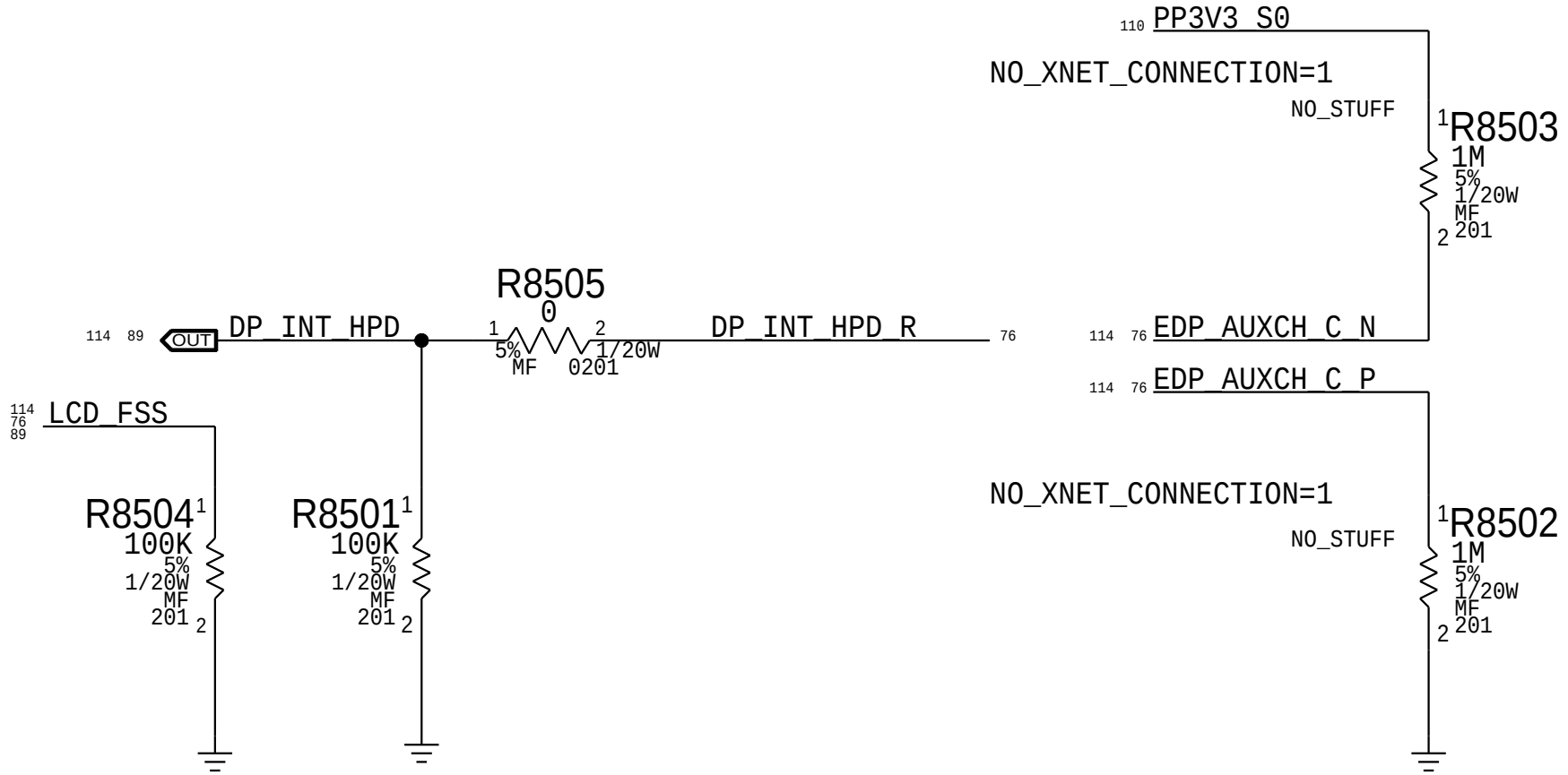
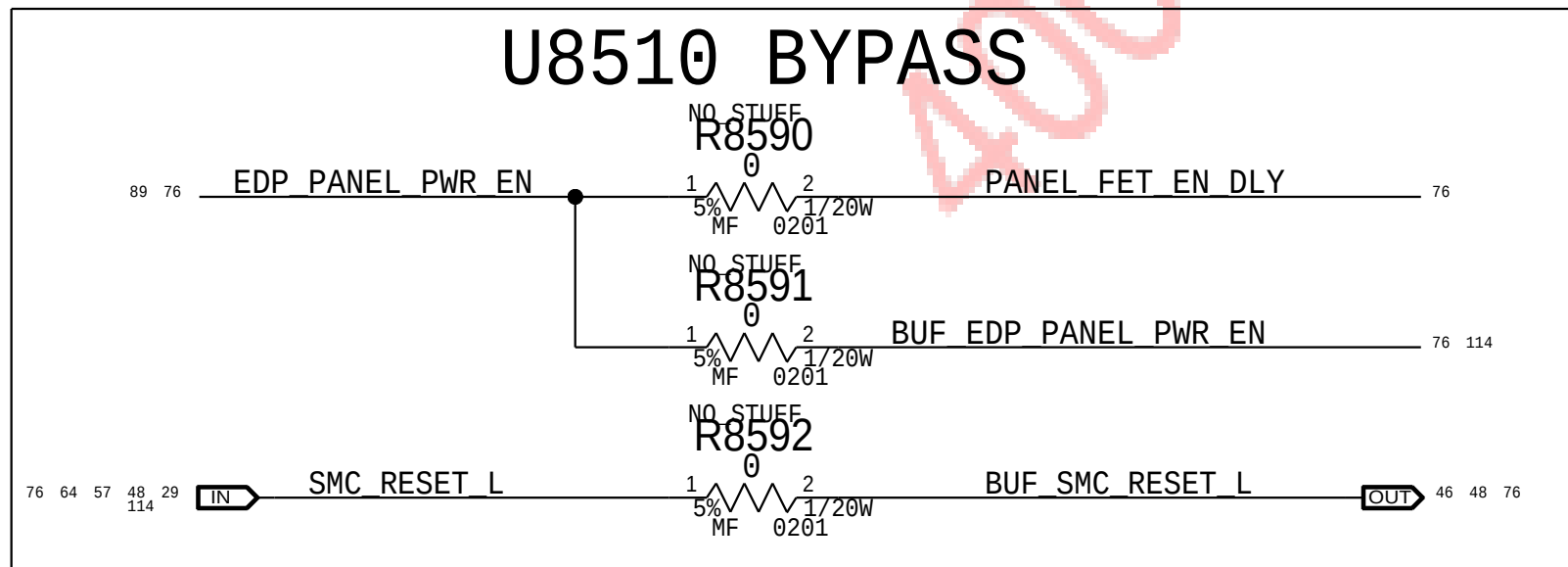
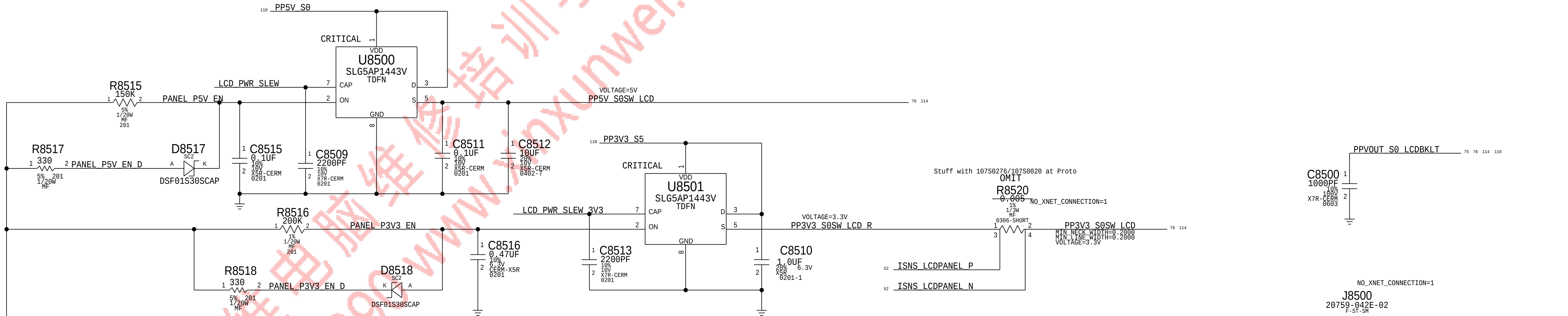


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BOM_COST_GROUP=PLATFORM POWER



LCD PANEL INTERFACE (eDP) + Camera (MIPI)



LCD Panel HPD, FSS & AUX strapping

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eDP Display Connector			
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		SHEET	76 OF 121

BOM_COST_GROUP=SSD

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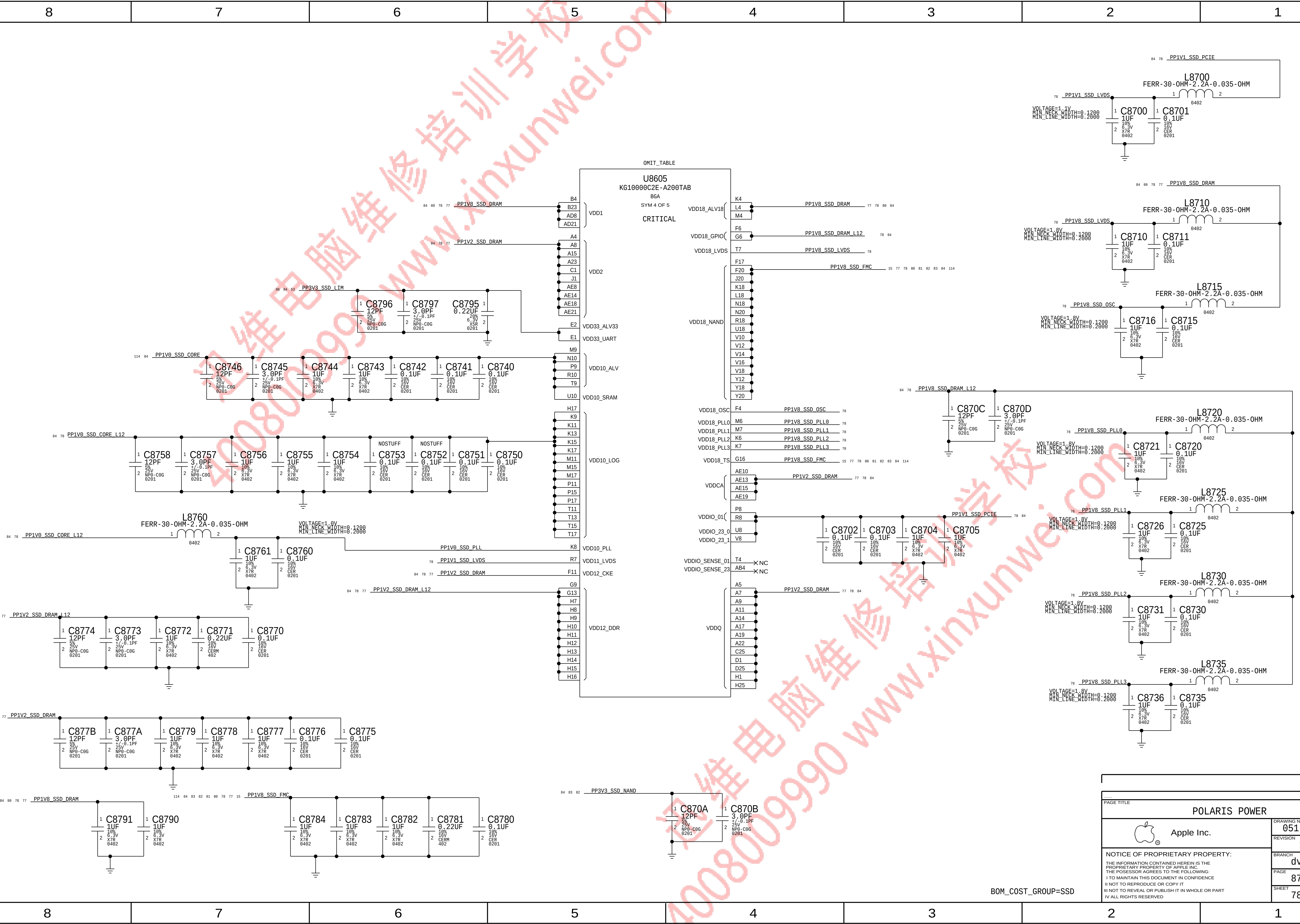
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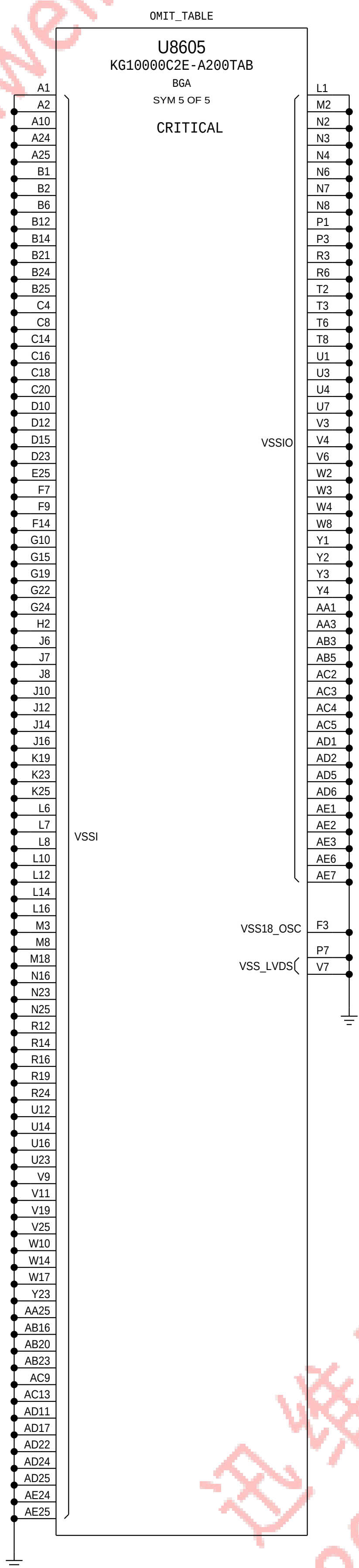
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		dvt-fab10	
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BOM_COST_GROUP=SSD

SSD RELATED BOM Groups

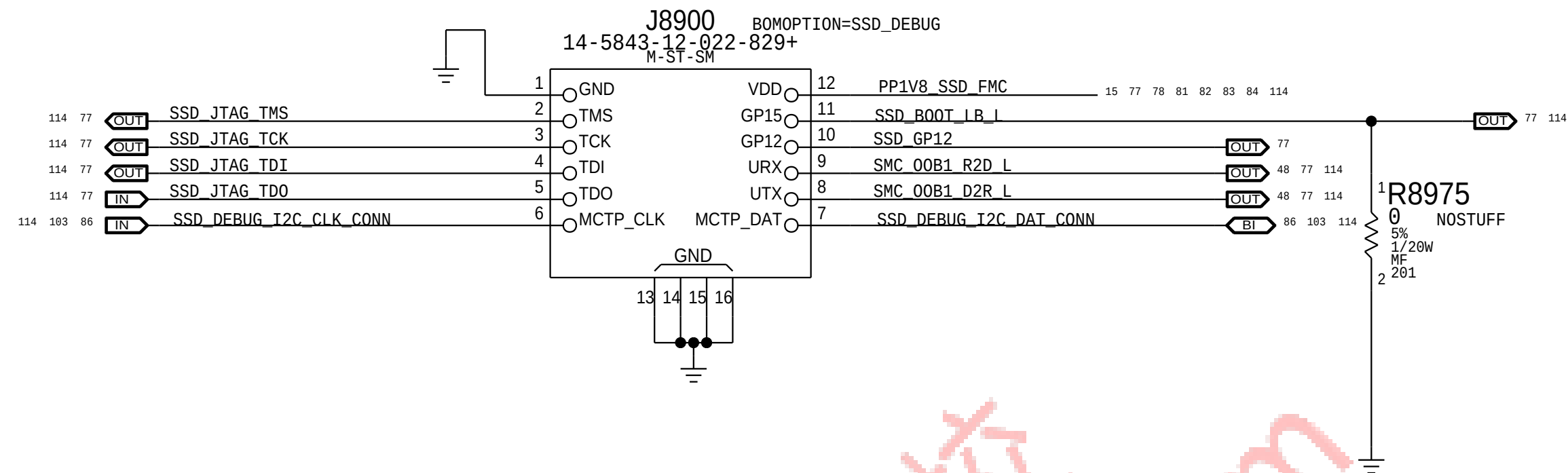
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SSD_CONFIG:256GB	SSD_GPI09, NAND_TYPE: 256GB, SSD_CTRL_TYPE: 4GBIT
SSD_CONFIG:512GB	SSD_GPI08, NAND_TYPE: 512GB, SSD_CTRL_TYPE: 4GBIT
SSD_CONFIG:1TB	NAND_TYPE: 1TB, SSD_CTRL_TYPE: 8GBIT
SSD_CONFIG:2TB	SSD_GPI010, NAND_TYPE: 2TB, SSD_CTRL_TYPE: 8GBIT

NAND Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S00149	4	NAND, 1Z, 64GBM, T06G DDR2, 64G, SS, BGA 168	U9100, U9110, U9200, U9210	CRITICAL	NAND_TYPE: 256GB
335S00204	4	NAND, V3, 128GBM, T06G DDR2, 256G, SSBGA 168	U9100, U9110, U9200, U9210	CRITICAL	NAND_TYPE: 512GB
335S00205	4	NAND, V3, 256GBM, T06G DDR2, 256G, SS, BGA 168	U9100, U9110, U9200, U9210	CRITICAL	NAND_TYPE: 1TB
335S00219	4	NAND, V3, 512GBM, T06G DDR2, 256G, SS, BGA 168	U9100, U9110, U9200, U9210	CRITICAL	NAND_TYPE: 2TB

SSD Controller Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
339S00154	1	POP, POLARIS+4GBIT, SSD_CTRL_, A2, BGA516	U8605	CRITICAL	SSD_CTRL_TYPE: 4GBIT
339S00155	1	POP, POLARIS+8GBIT, SSD_CTRL_, A2, BGA516	U8605	CRITICAL	SSD_CTRL_TYPE: 8GBIT

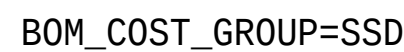


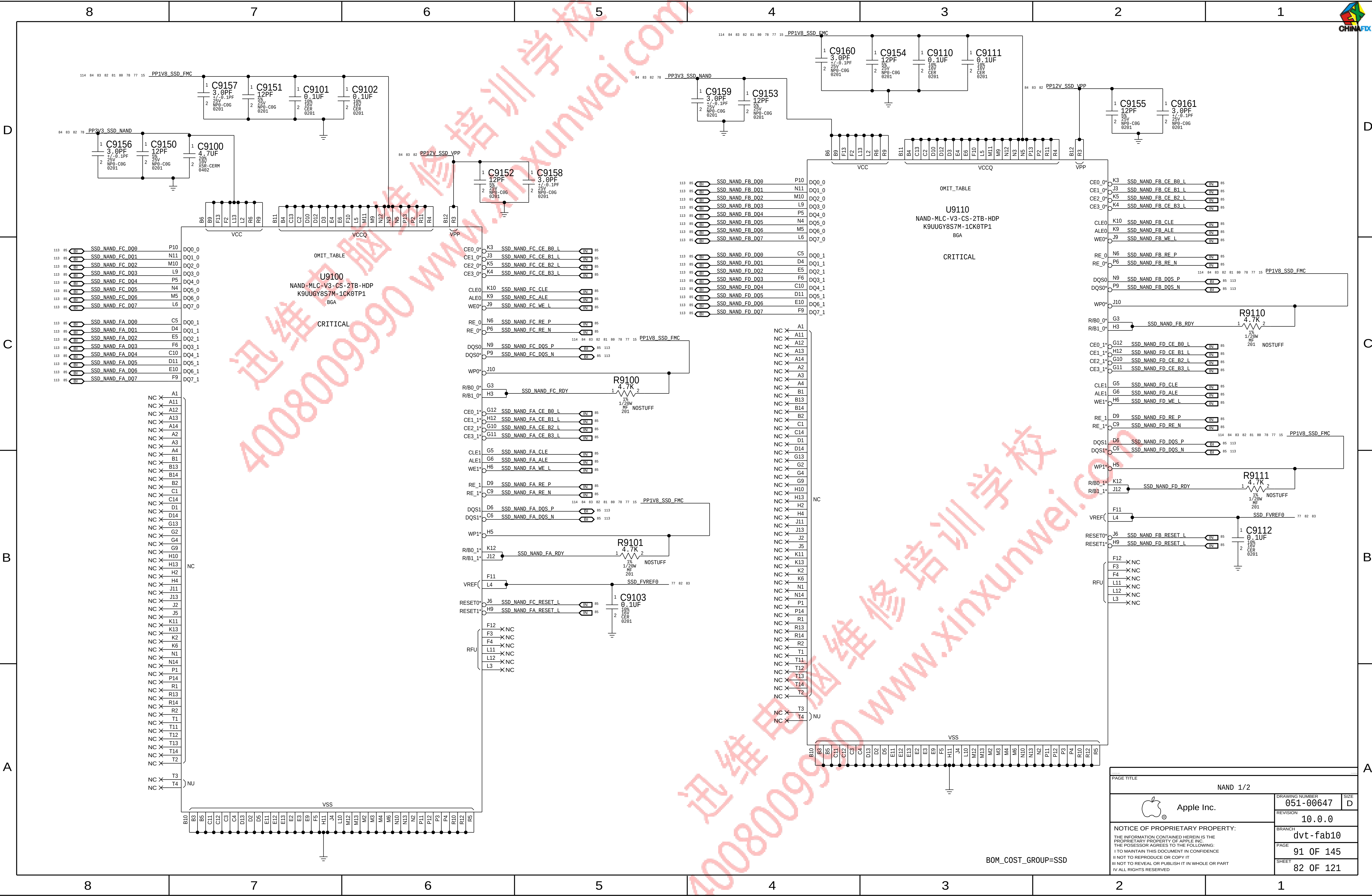
SSD CONFIGURATIONS

CAPACITY	NAND APN	CONTROLLER APN	R8980/GPI08	R8981/GPI09	R8928/GPI010
128 GB	-	-	STUFF-0	STUFF-0	STUFF-0
256 GB	335S00149	339S00154 (A2 4GB DRAM)	NOSTUFF-1	STUFF-0	NOSTUFF-1
512 GB	335S00204	339S00154 (A2 4GB DRAM)	STUFF-0	NOSTUFF-1	NOSTUFF-1
1 TB	335S00205	339S00155 (A2 8GB DRAM)	NOSTUFF-1	NOSTUFF-1	NOSTUFF-1
2 TB	335S00219	339S00155 (A2 8GB DRAM)	NOSTUFF-1	NOSTUFF-1	STUFF-0

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Connector		
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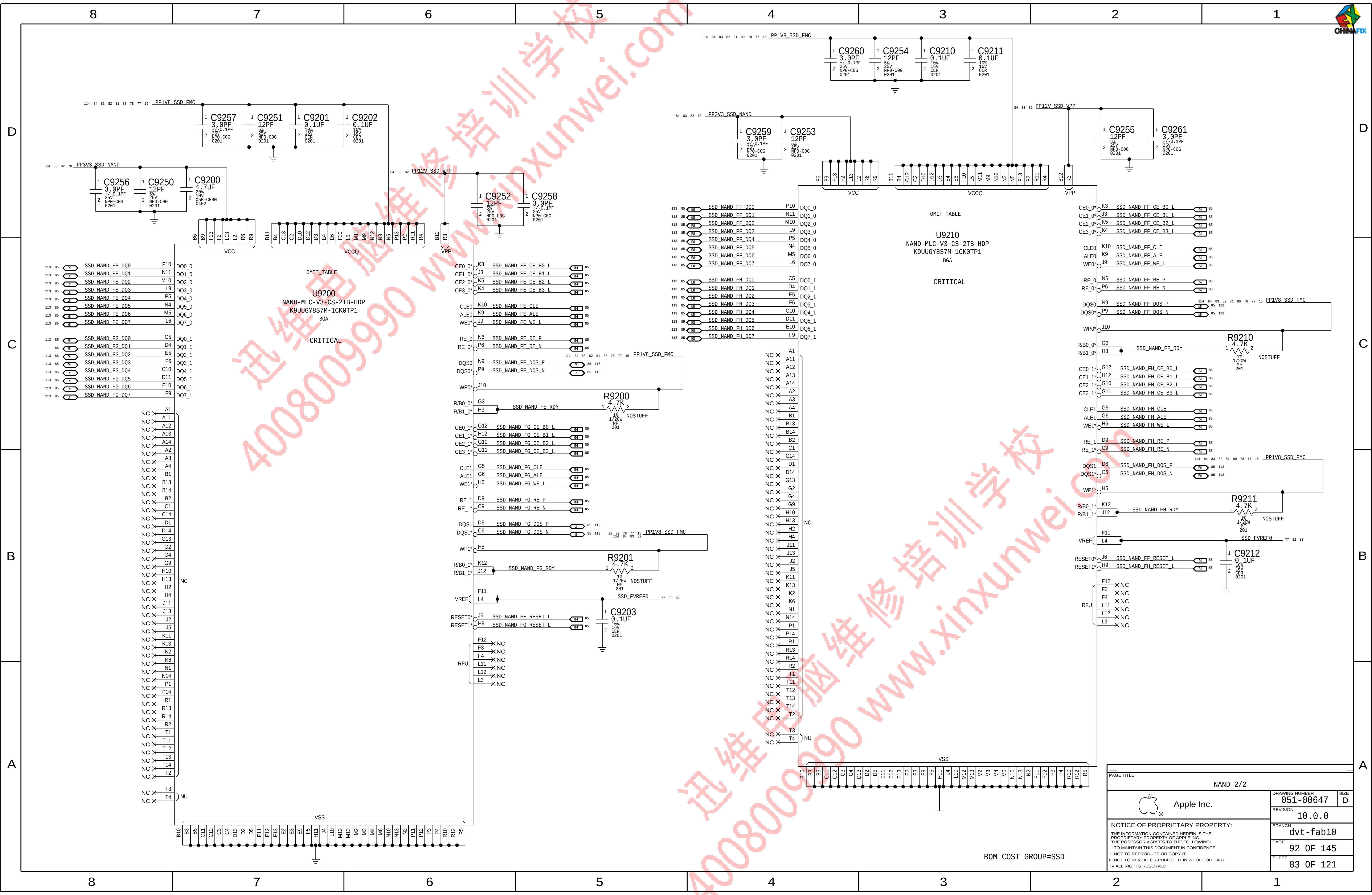
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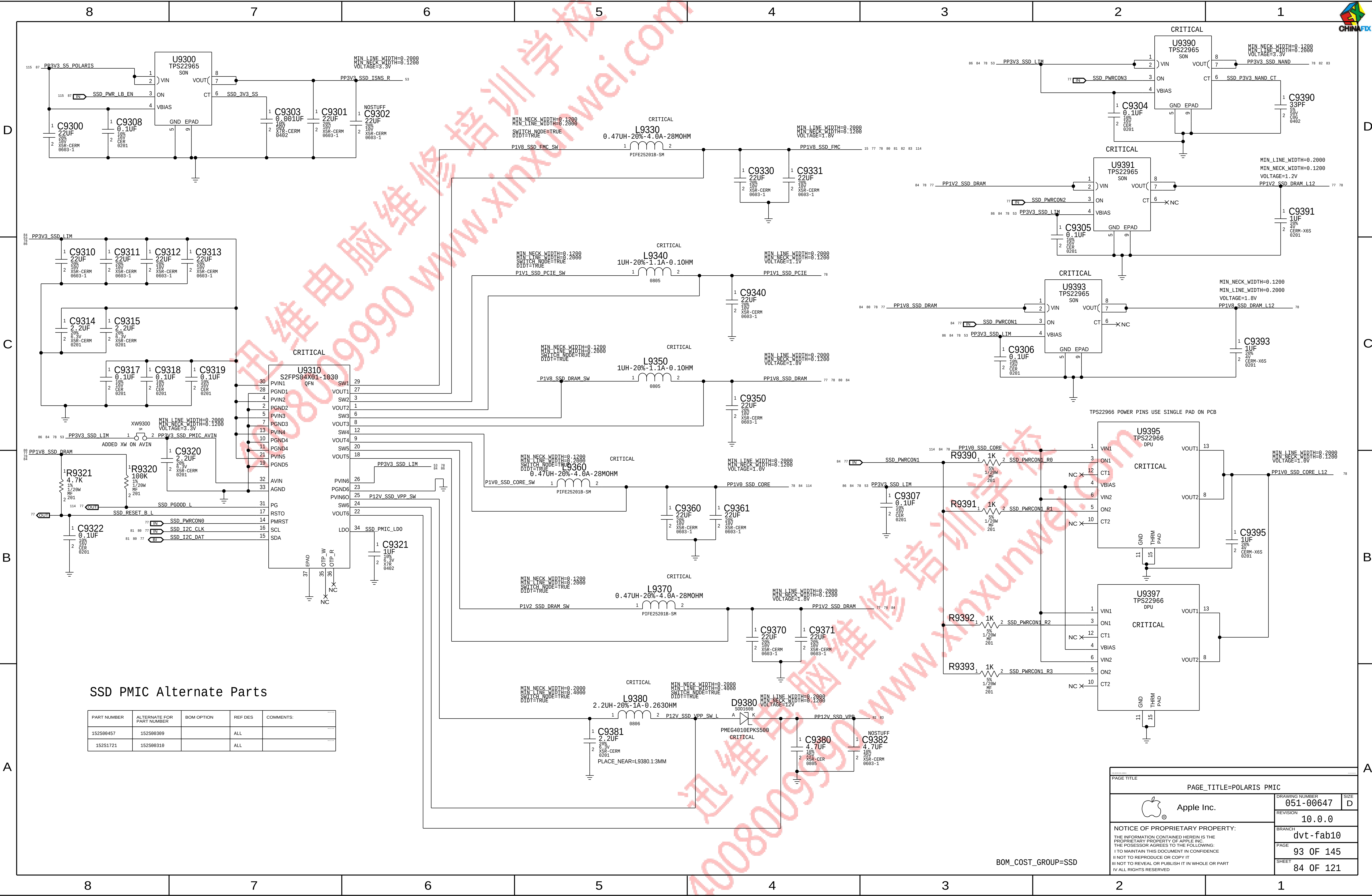




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BOM_COST_GROUP=SSD





SSD PMIC Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S08457	152S08309		ALL	
152S1721	152S08310		ALL	

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		SHEET	84 OF 121

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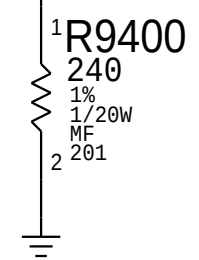
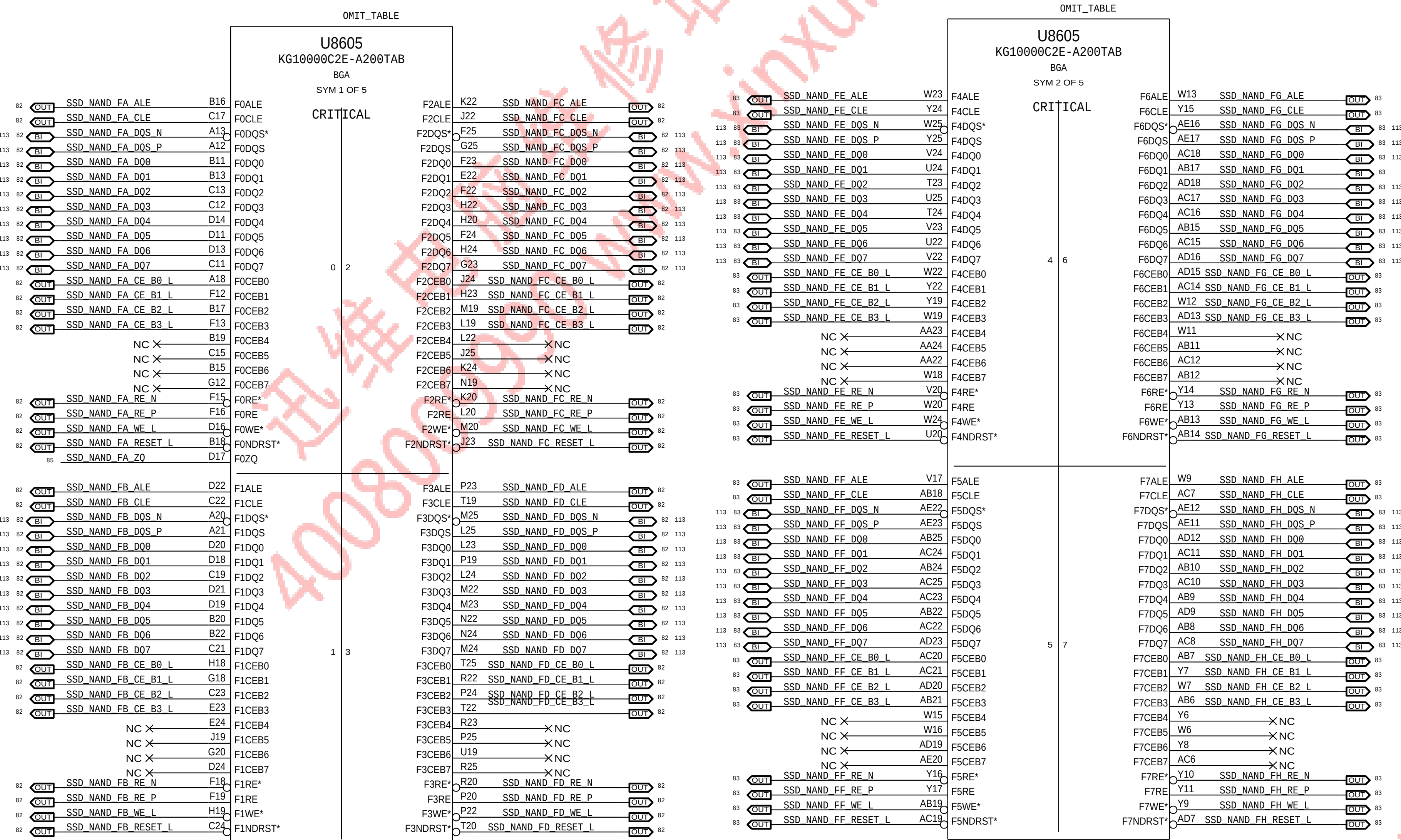
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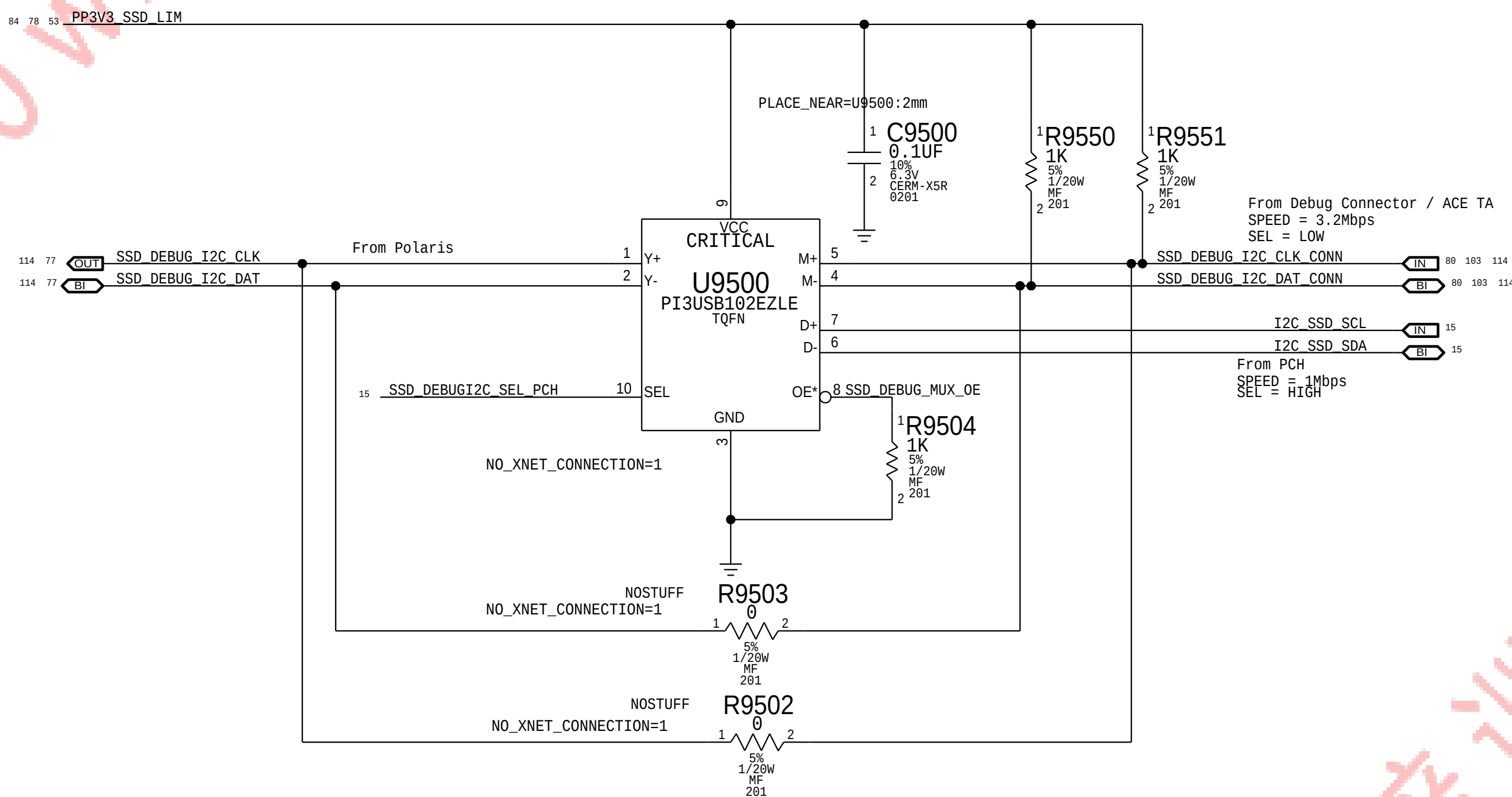
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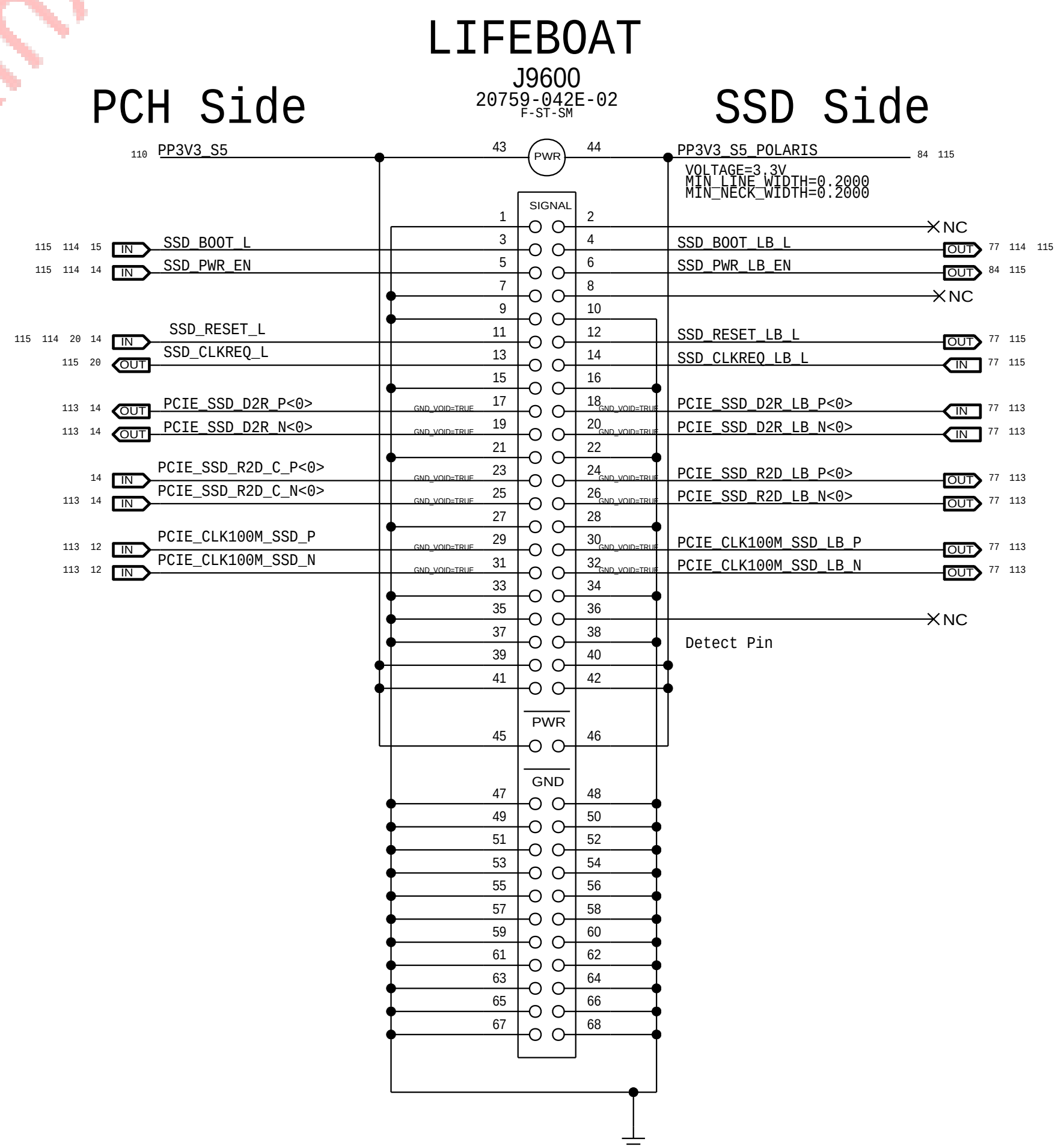
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BOM_COST_GROUP=SSD

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Lifeboat											
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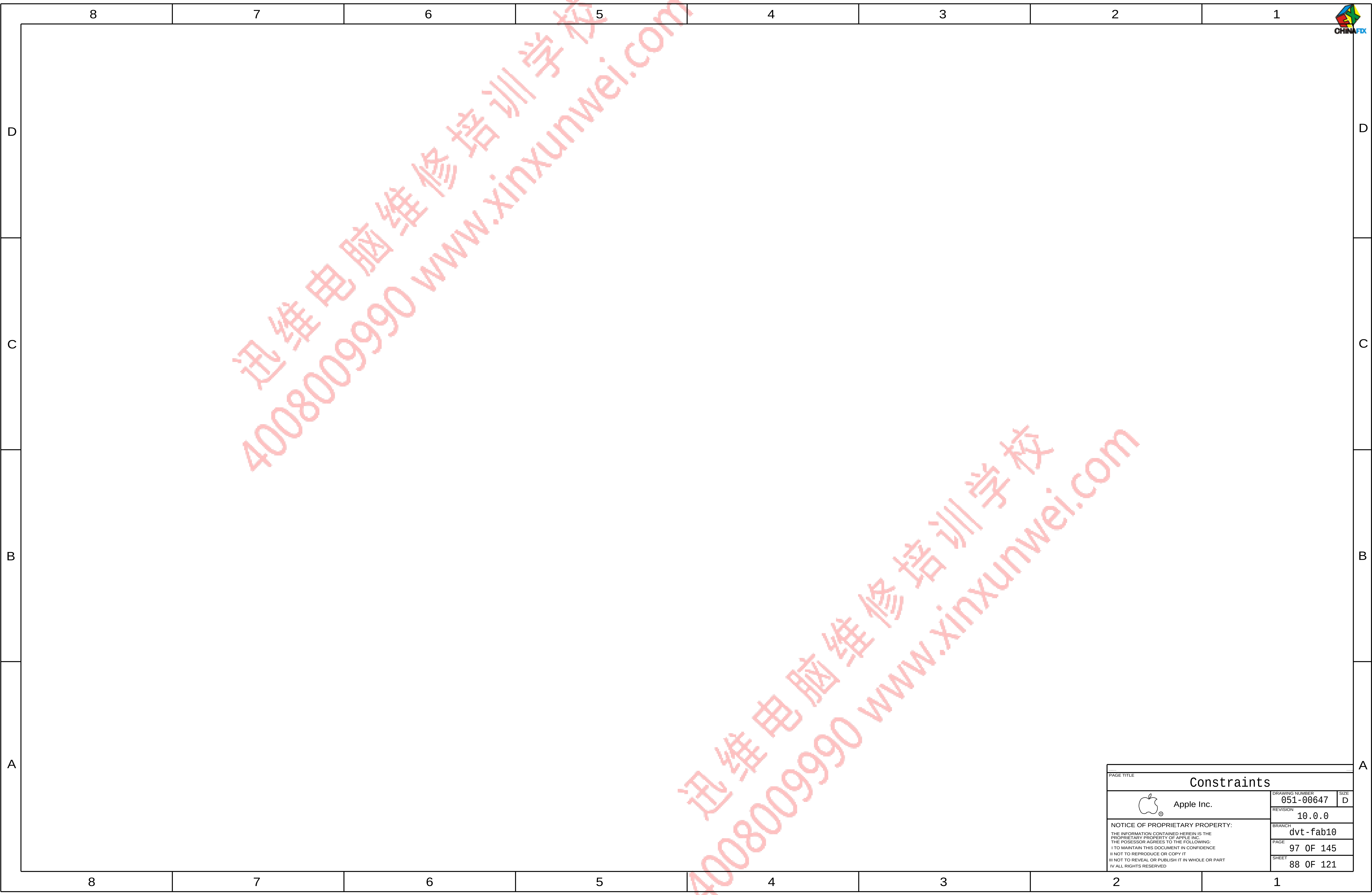
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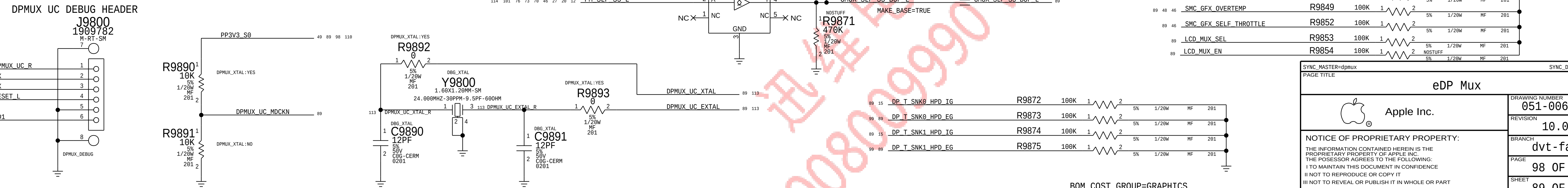
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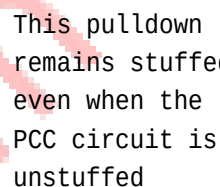
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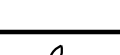


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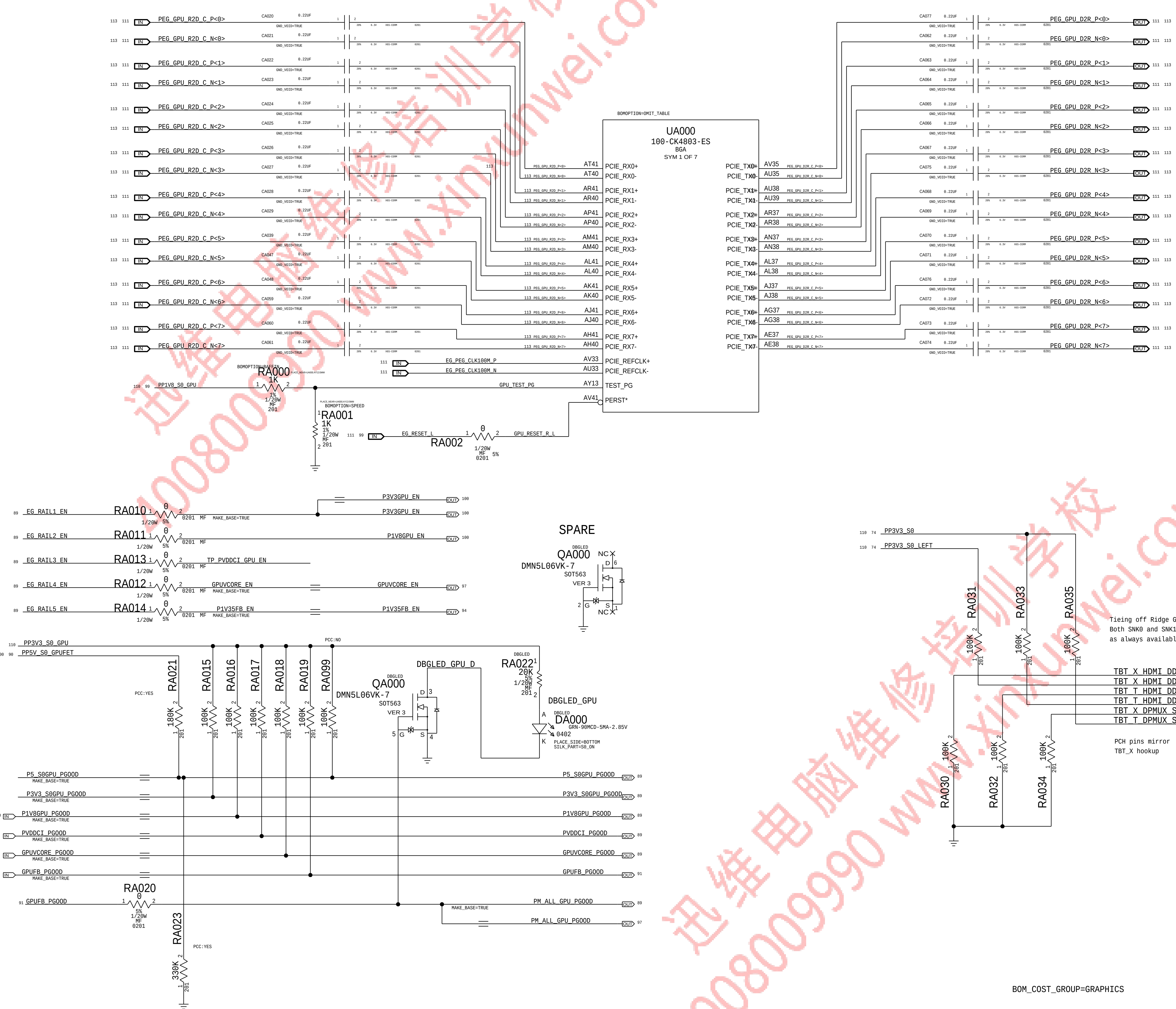
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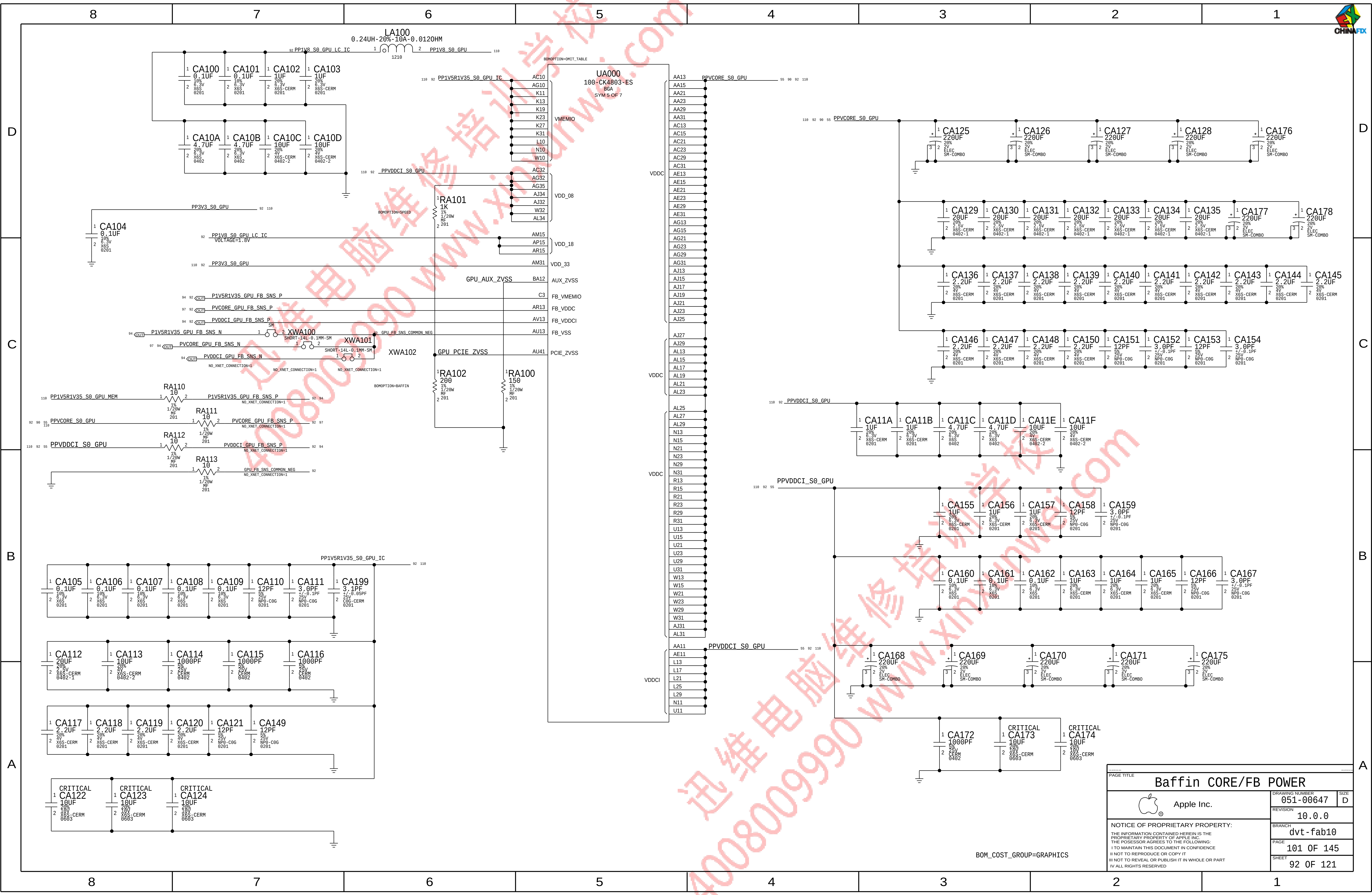
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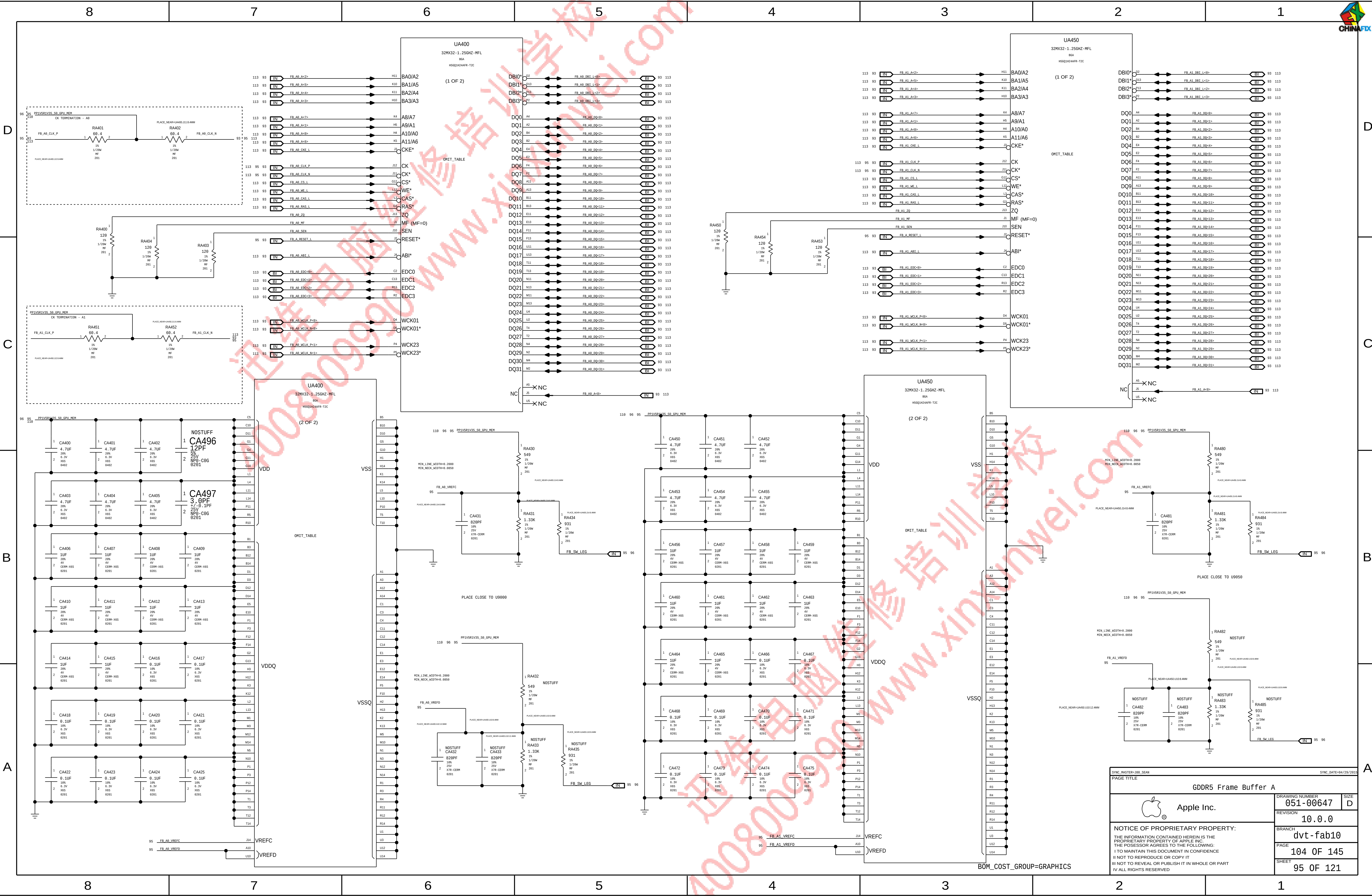
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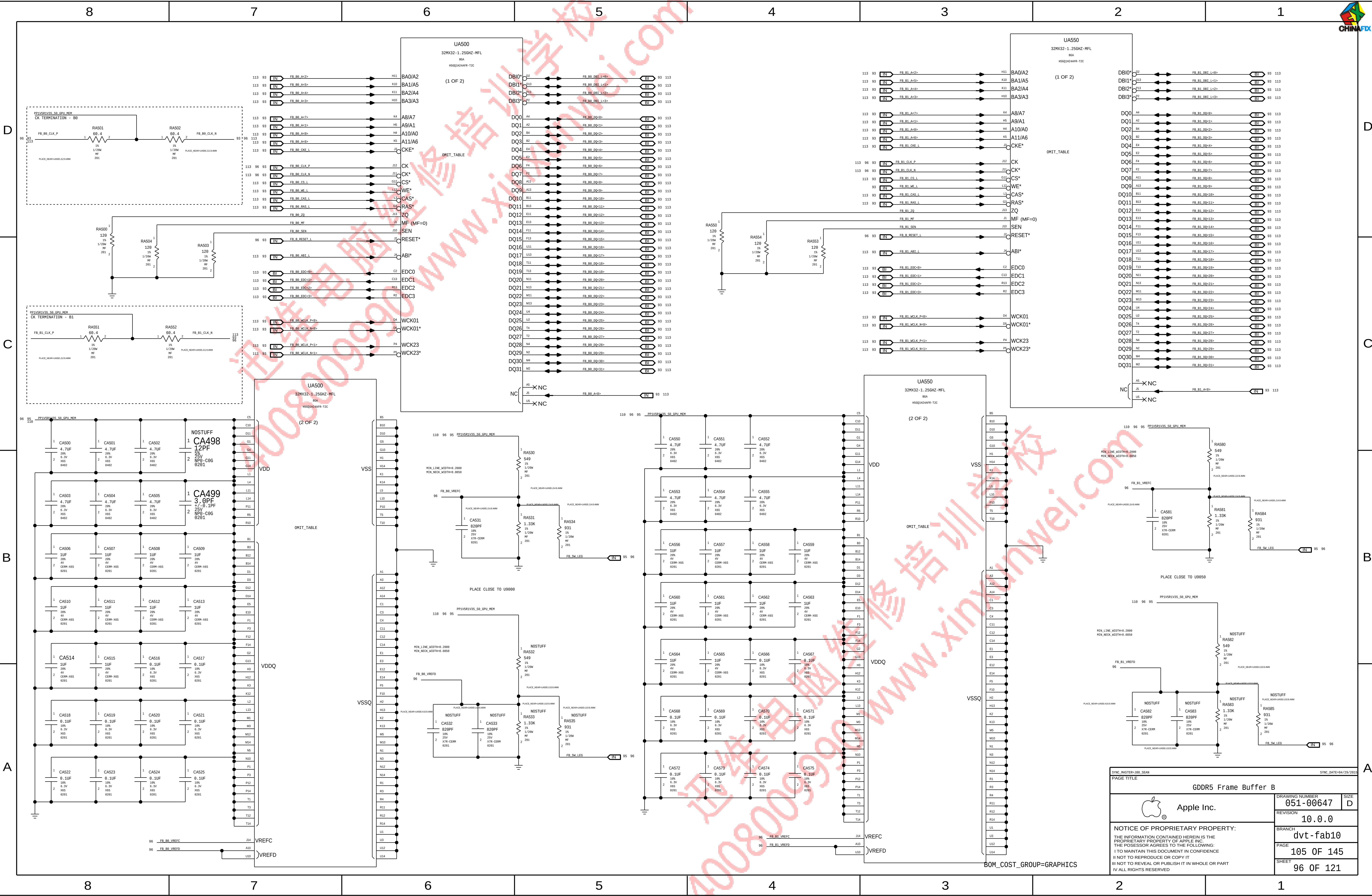
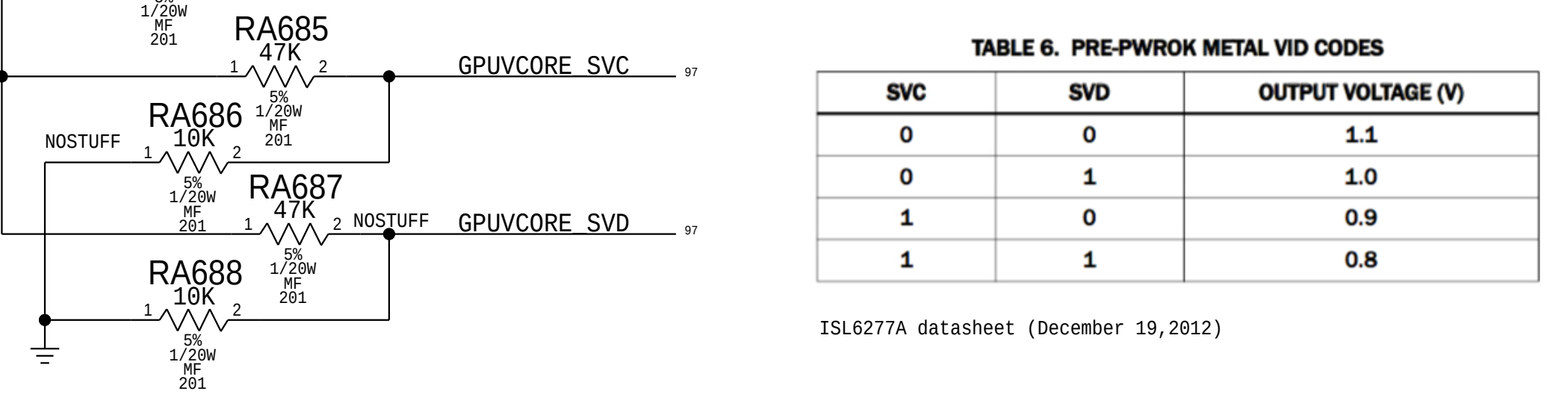


TABLE 6. PRE-PWROK METAL VID CODES

SVC	SVD	OUTPUT VOLTAGE (V)
0	0	1.1
0	1	1.0
1	0	0.9
1	1	0.8

ISL6277A datasheet (December 19, 2012)



SVC	SVD	OUTPUT VOLTAGE (V)
0	0	1.1
0	1	1.0
1	0	0.9
1	1	0.8

ISL6277A datasheet (December 19, 2012)

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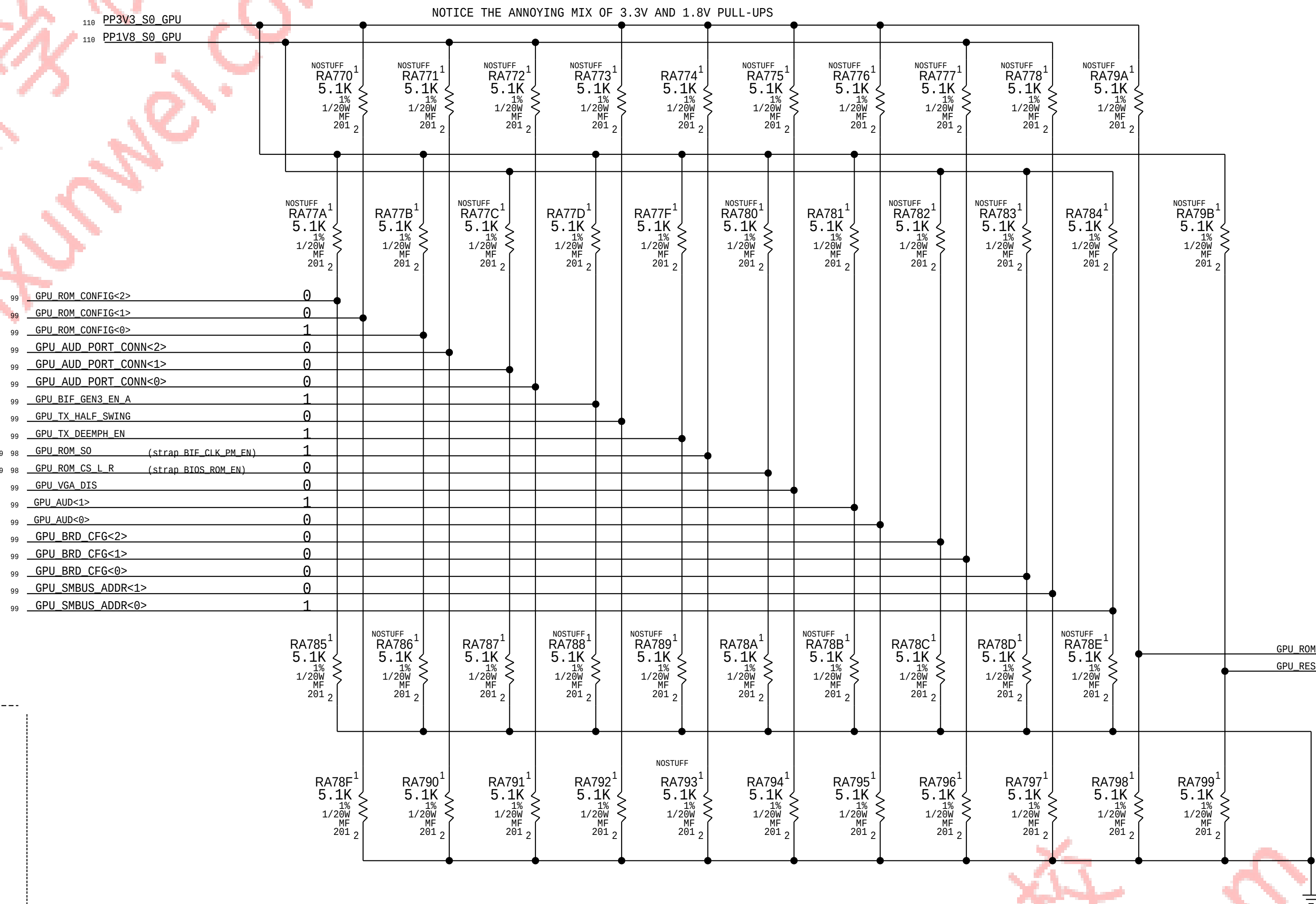
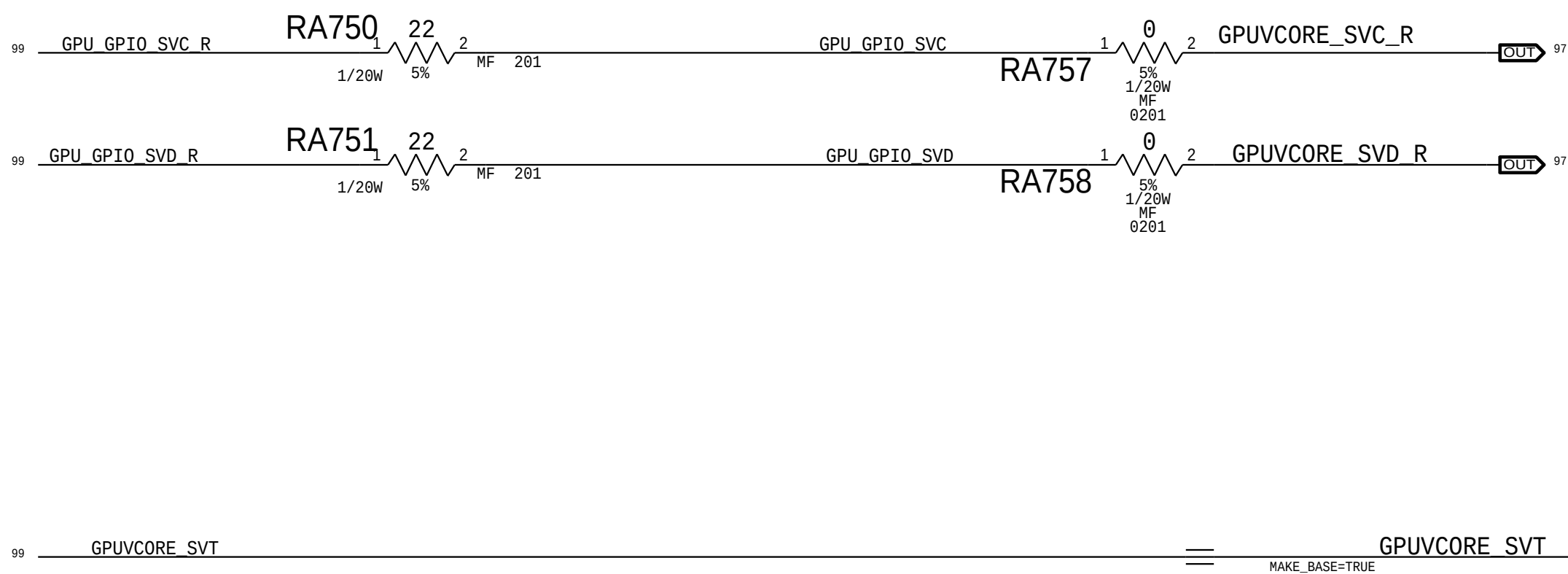
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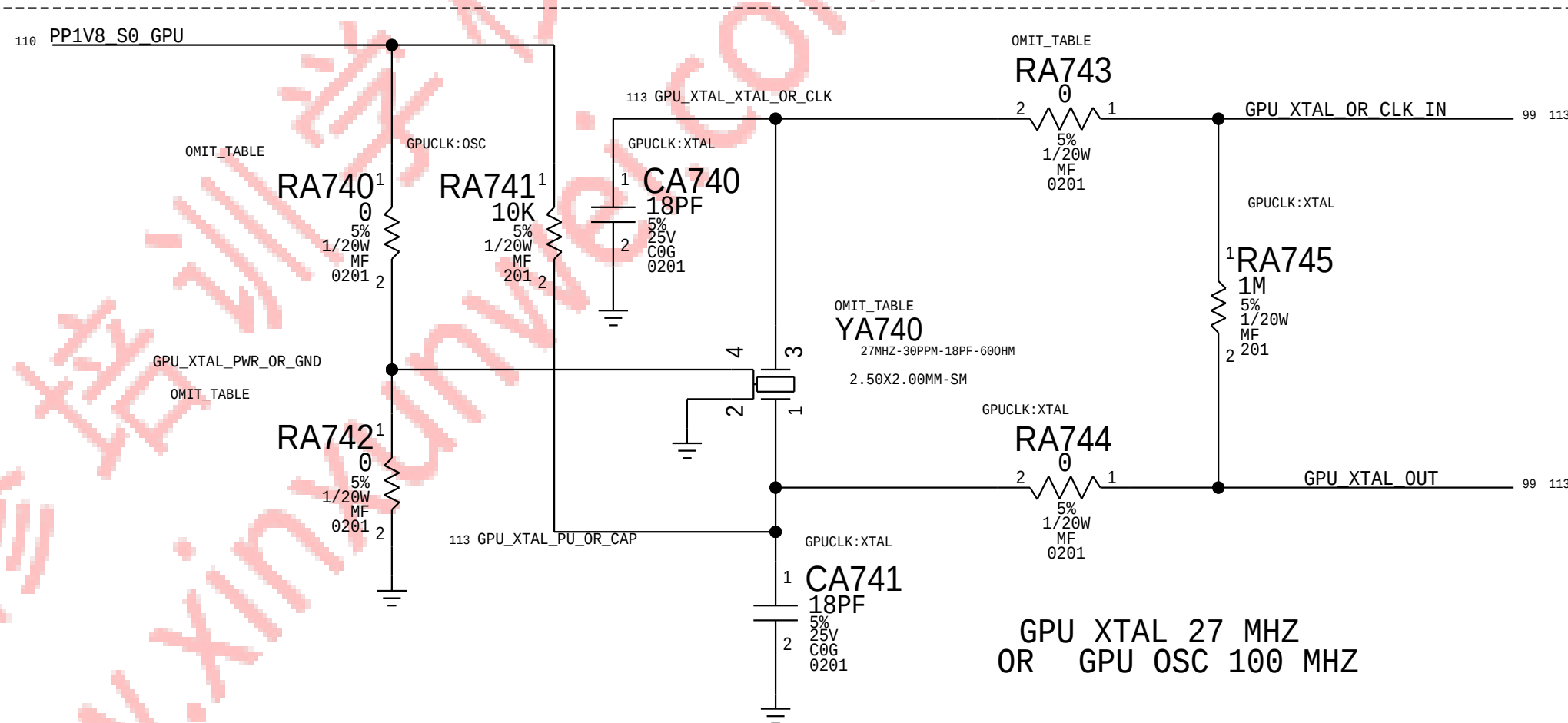
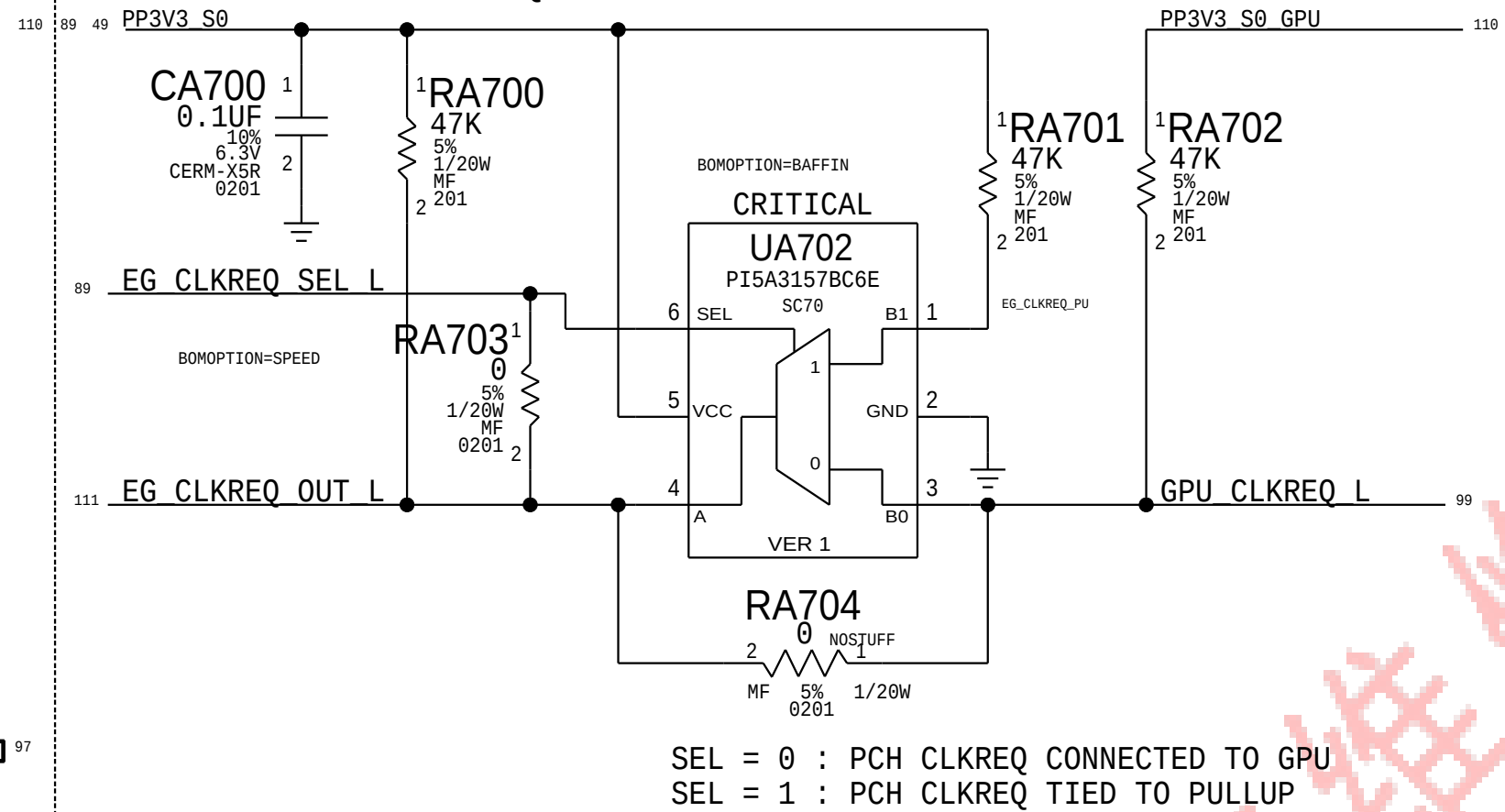
former site of the SPEED MLPS STRAPS
PCIe Gen3 enabled, Half-Swing, TX De-emp enabled
VBIOS disabled, Boot from EFI
VGA enabled
All power audio-capable display output
256MB FB Aperture Size
PS_0: 01001 82nF 8.45k 2k
PS_1: 10001 10nF 8.45k 2k
PS_2: 10000 10nF NC 4.75k
PS_3: 00000 680nF NC 4.75k

BAFFIN STRAPS
PCIe Gen3 enabled, FULL-Swing, TX De-emp enabled
VBIOS disabled, Boot from EFI
VGA enabled, CLKREQ_L enabled
All power audio-capable display output
Audio for DP or HDMI dongle
256MB FB Aperture Size
SMBUS address 41

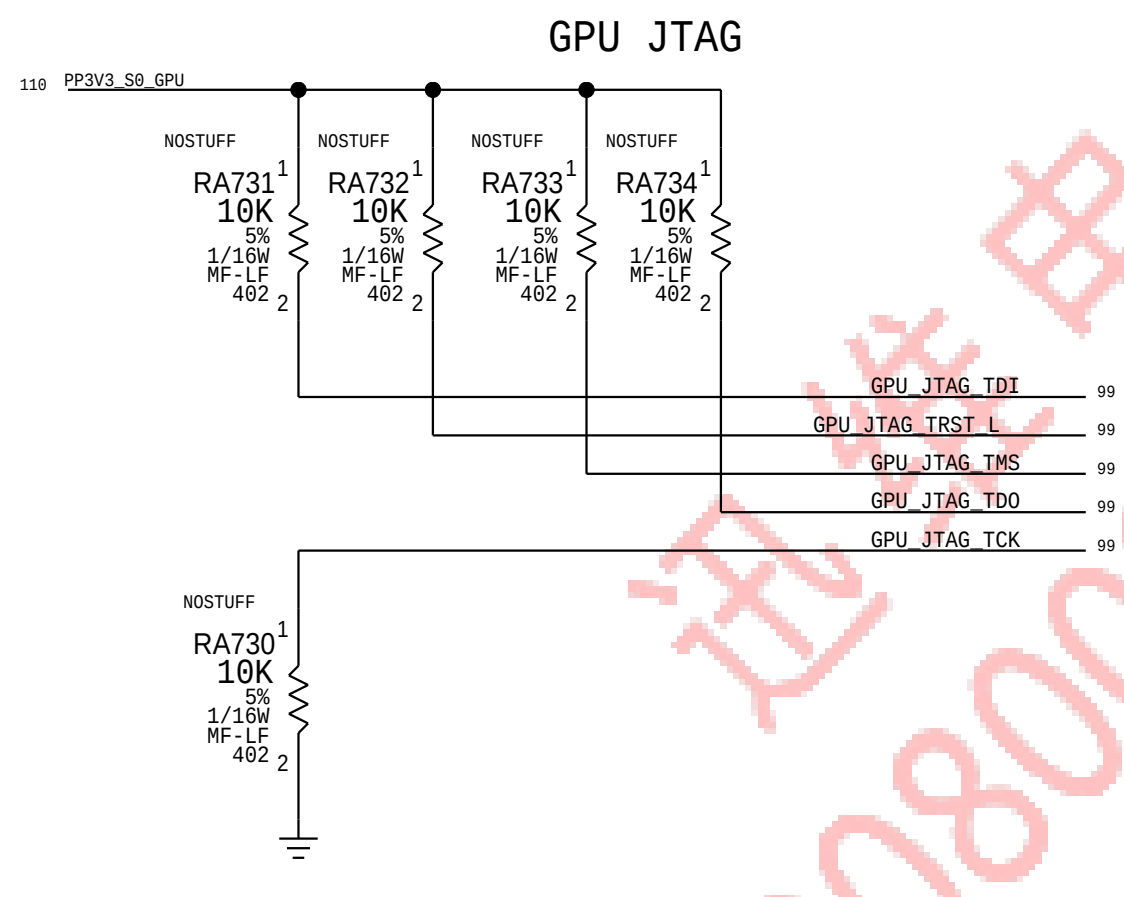
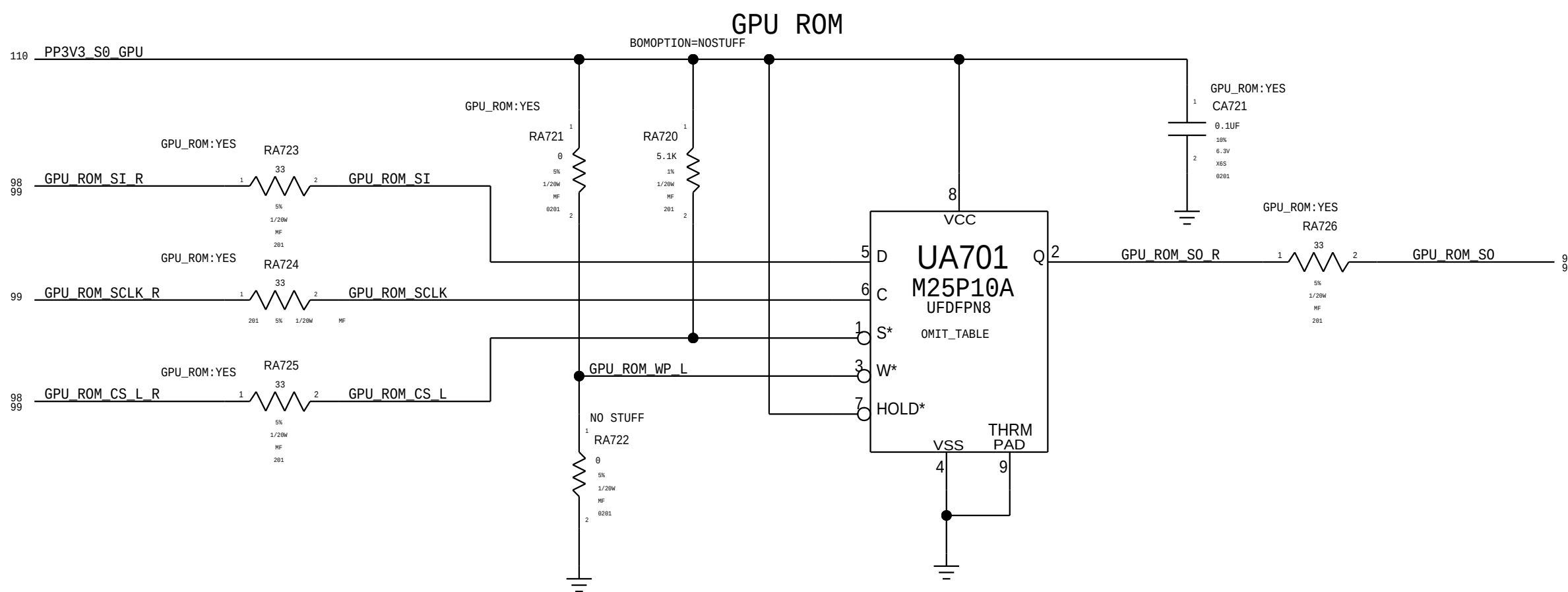
Former site of the GPU SVI2 VOLTAGE TRANSLATION
SPEED ONLY



GPU CLKREQ GATING



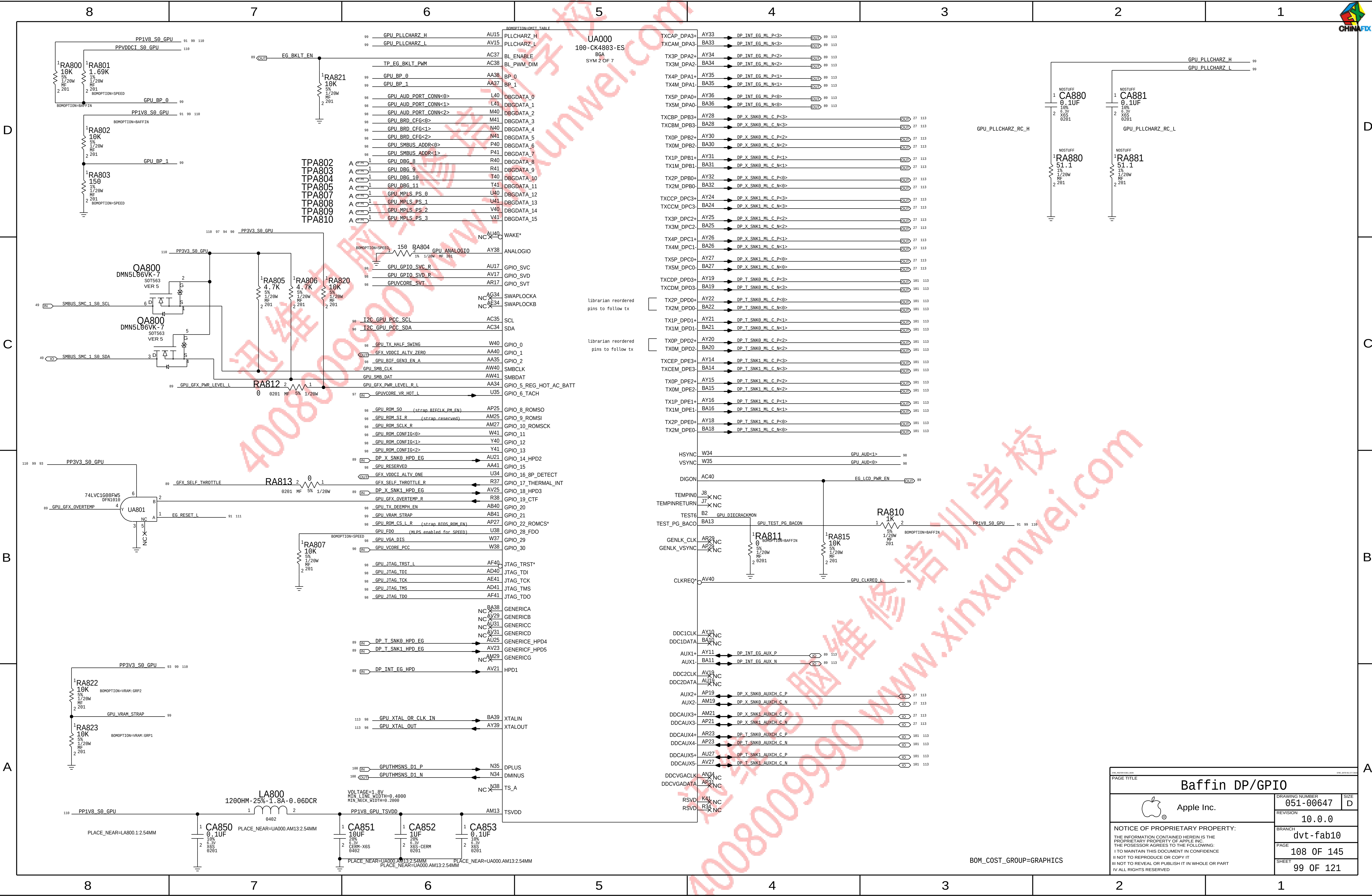
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197S00056	1	OSC, MEMS, 100MHZ, +/-20PPM, 1.8V, 2520	YA740	CRITICAL	GPUCLK:OSC
117S0201	1	RES, 0 OHM, 5%, 0201	RA743	CRITICAL	GPUCLK:XTAL
117S0080	1	RES, 33 OHM, 5%, 0201	RA743	CRITICAL	GPUCLK:OSC
155S0387	1	FERRITE BEAD, 4700HM, 0.1A, 1.5MOHM DCR, 060	RA740	CRITICAL	GPUCLK:OSC
117S0201	1	RES, 0 OHM, 5%, 0201	RA742	CRITICAL	GPUCLK:XTAL
132S0444	1	CAP, CER, XSR, 0.1UF, 18%, 6.3V, 0201	RA742	CRITICAL	GPUCLK:OSC



Baffin GPIOs, CLK & Straps

	DRAWING NUMBER	051-00647	SIZE	D
	REVISION	10.0.0		
	BRANCH	dvt-fab10		
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BOM_COST_GROUP=GRAPHICS



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051-00647				D	
REVISION				10.0.0	
BRANCH				dvt-fab10	
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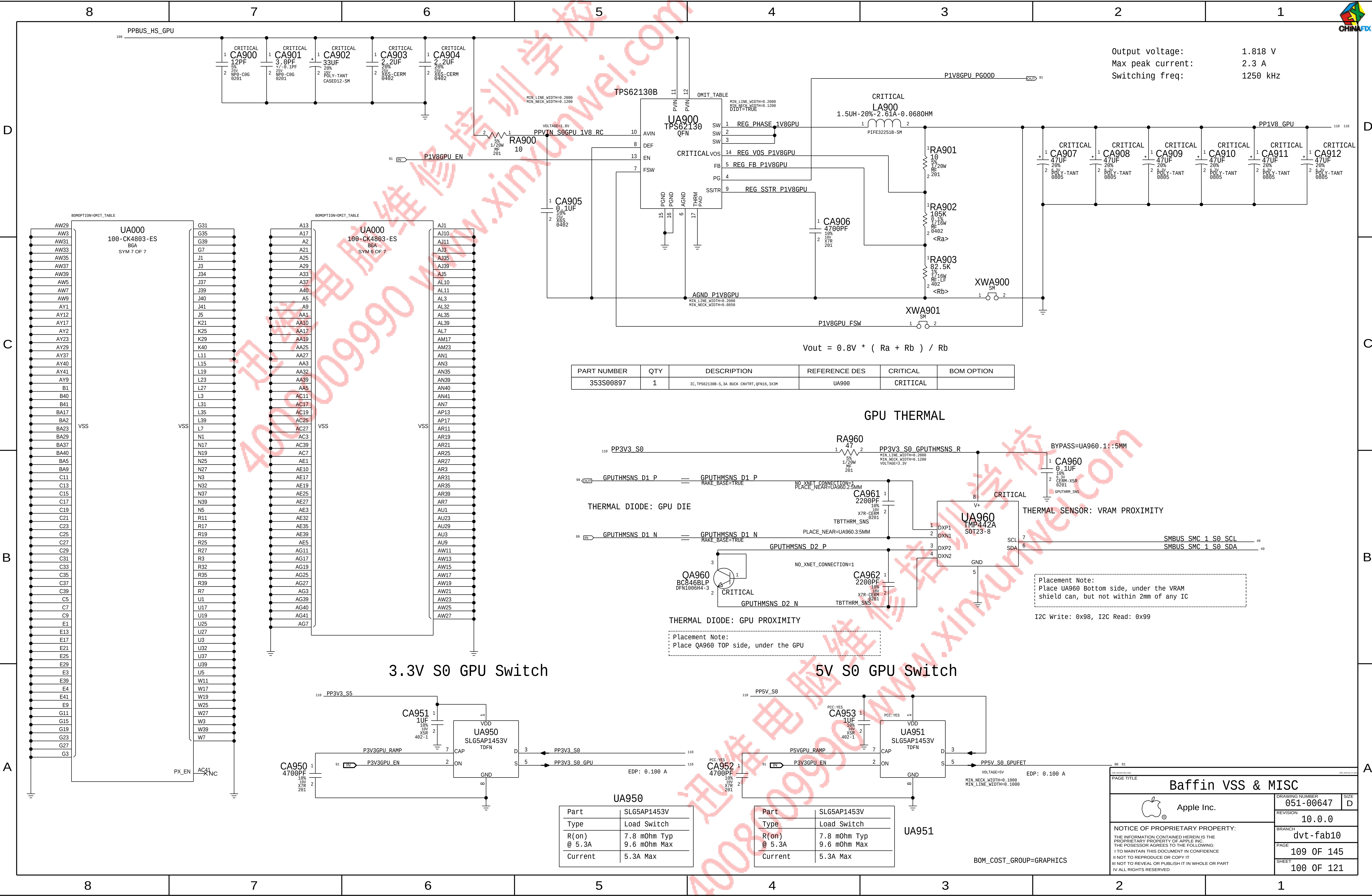
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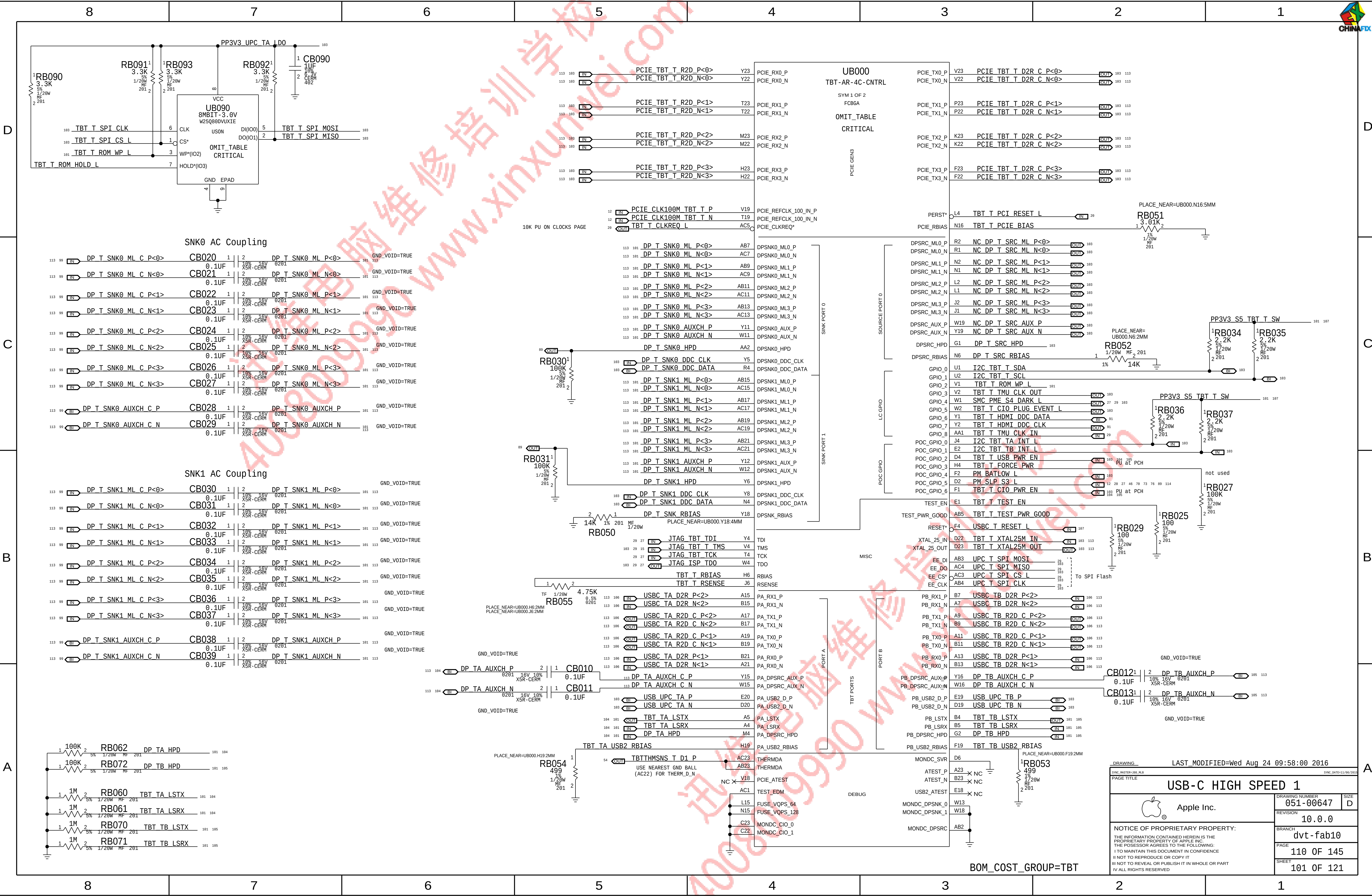
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Baffin DP/GPIO

BOM_COST_GROUP=GRAPHICS







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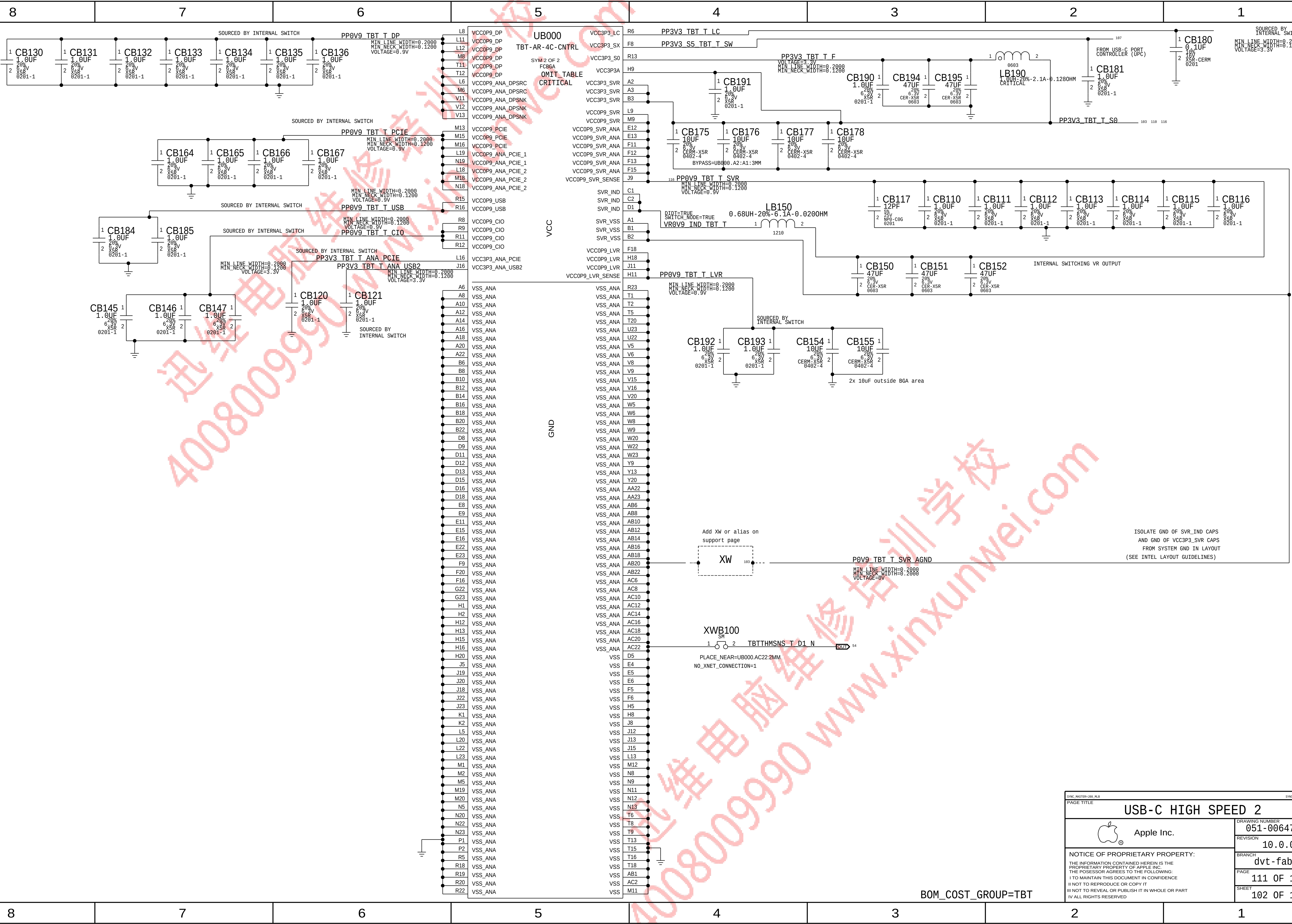
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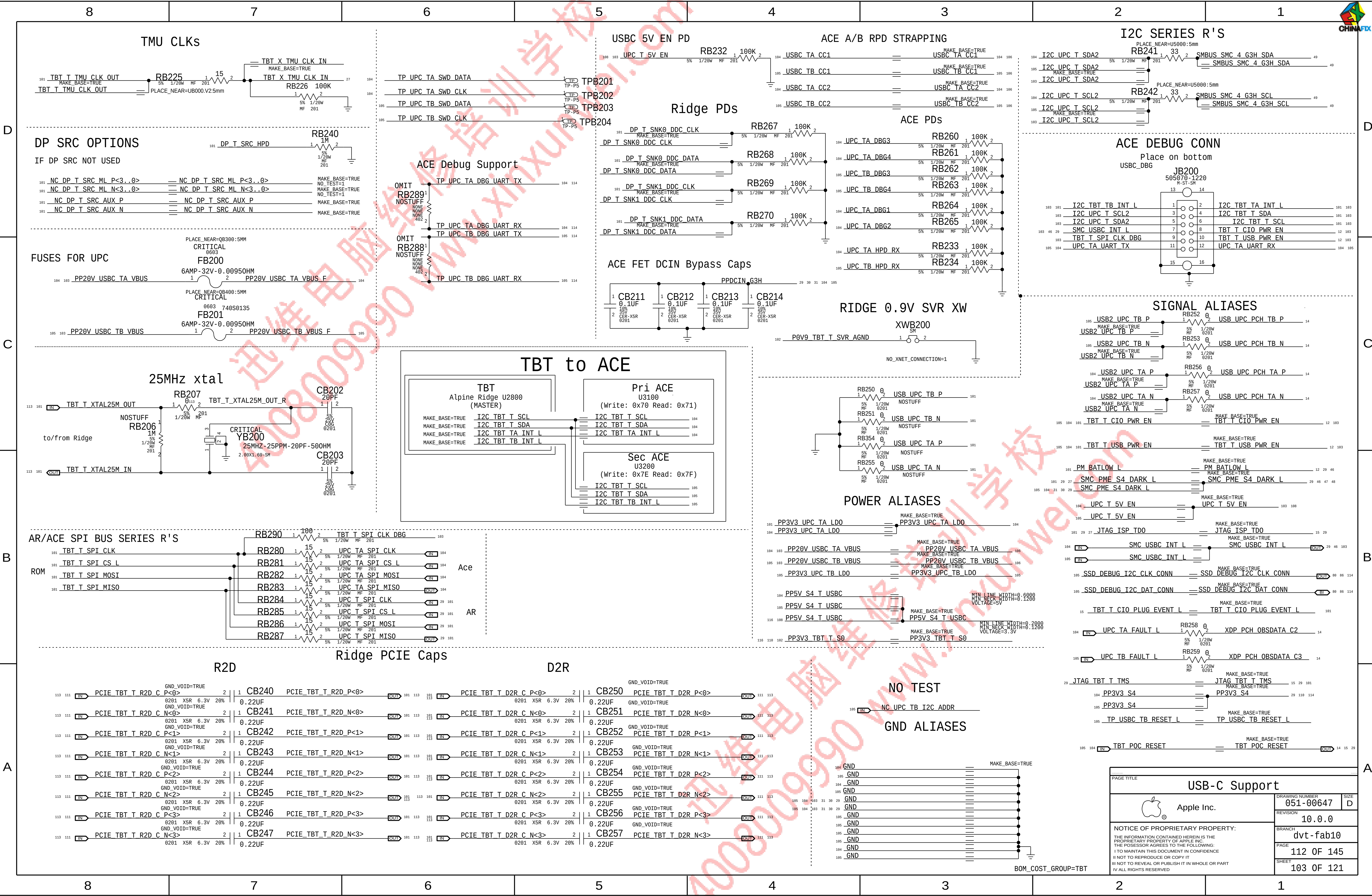
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REVISION		10.0.0		
BRANCH		dvt-fab10		
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BOM_COST_GROUP=TBT



D

D

C

C

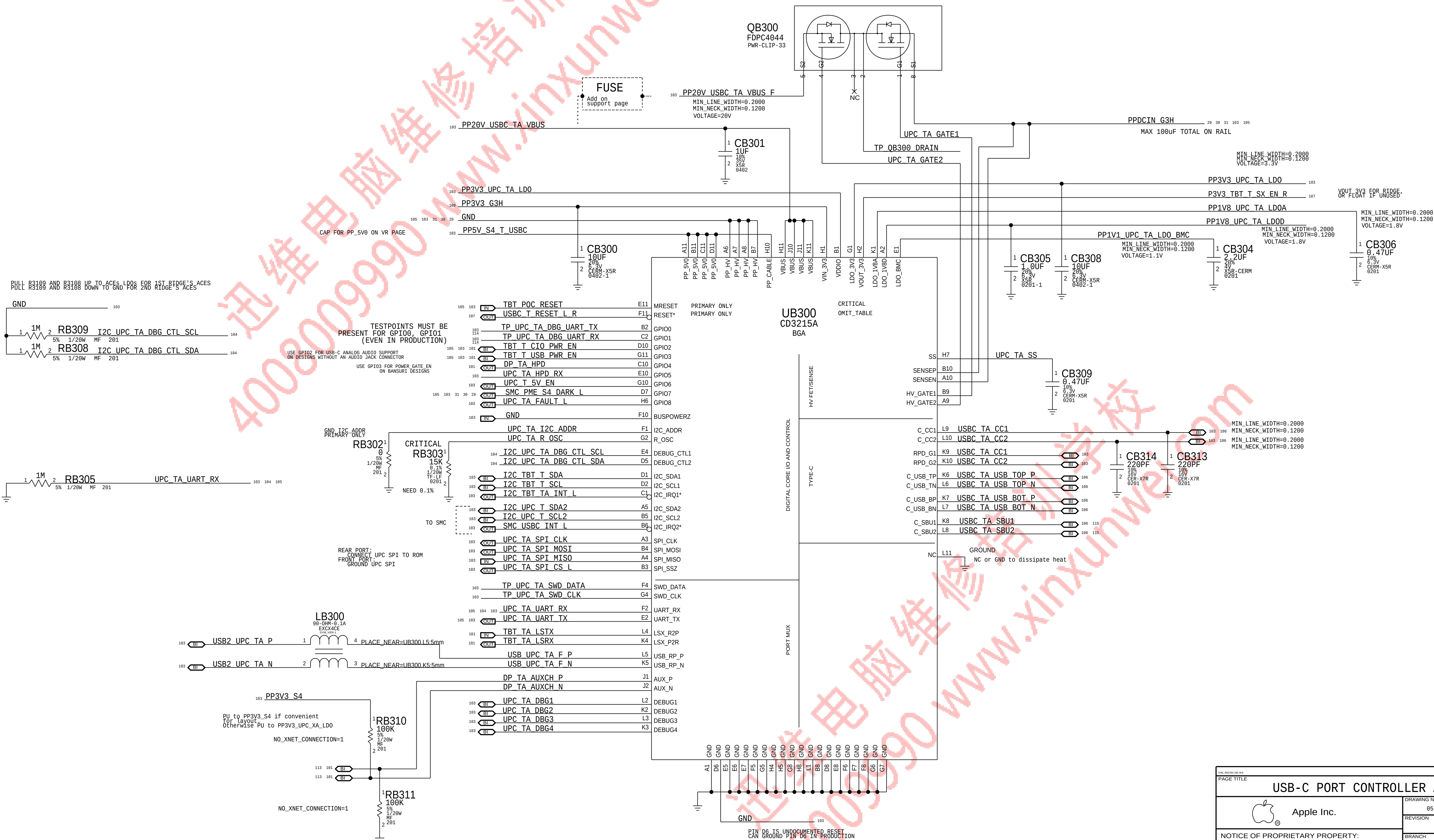
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A

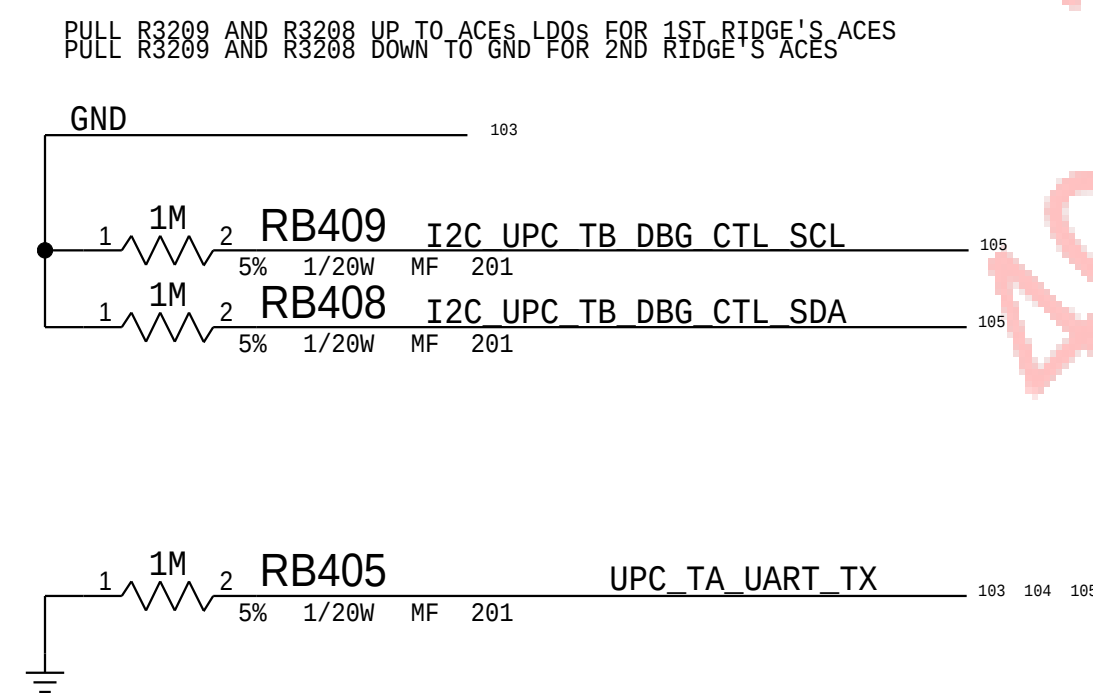
PRIMARY ACE USB-C PORT CONTROLLER (UPC)



BOM_COST_GROUP=USB-C

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Apple Inc.			REVISION		10.0.0
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			PAGE		113 OF 145
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BOM_COST_GROUP=USB-C



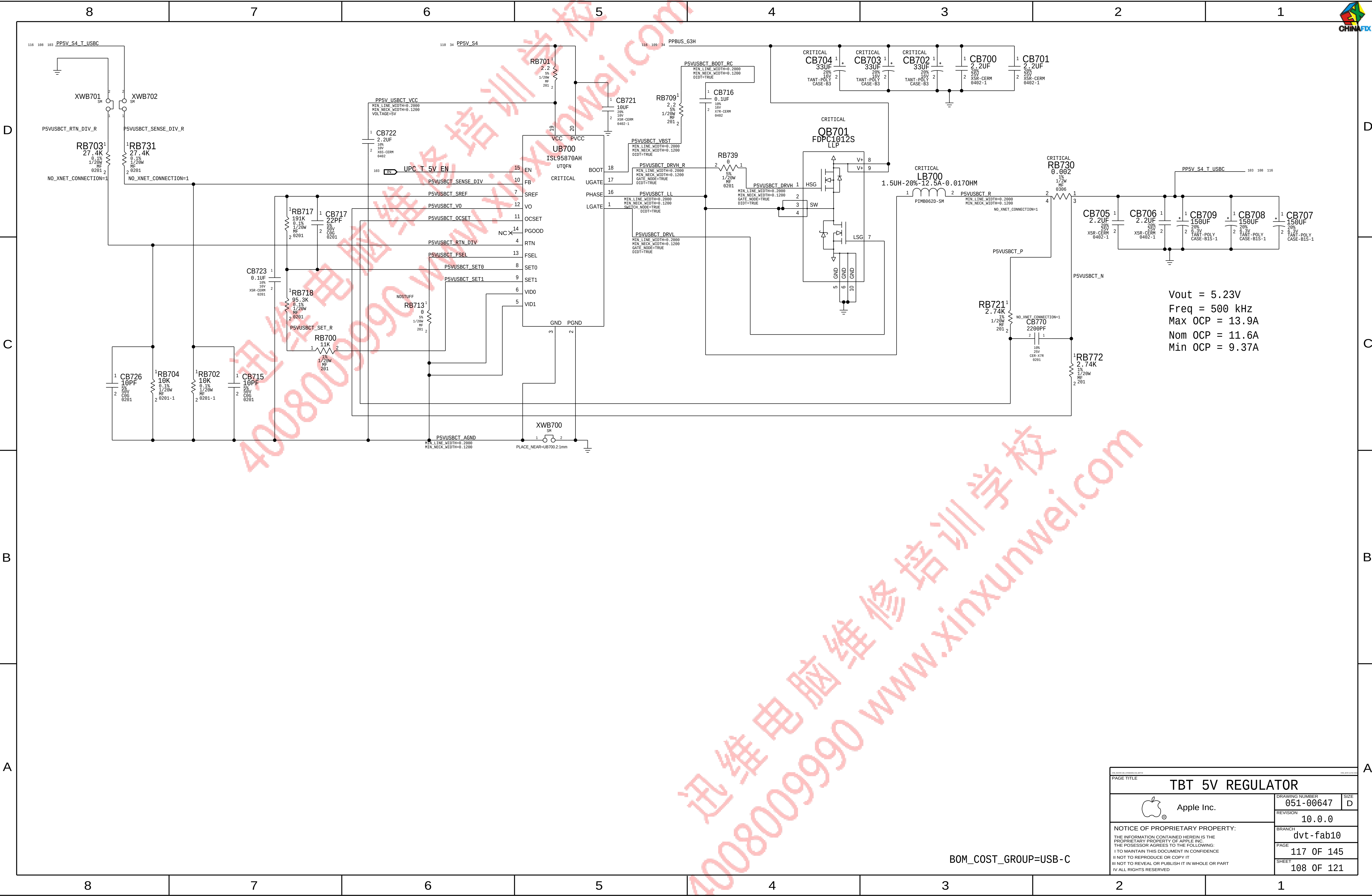
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LAST CHANGE: Wed Apr 1 22:57:37 2015			
PAGE TITLE			
USB-C CONNECTOR A			
 Apple Inc.	DRAWING NUMBER		SIZE
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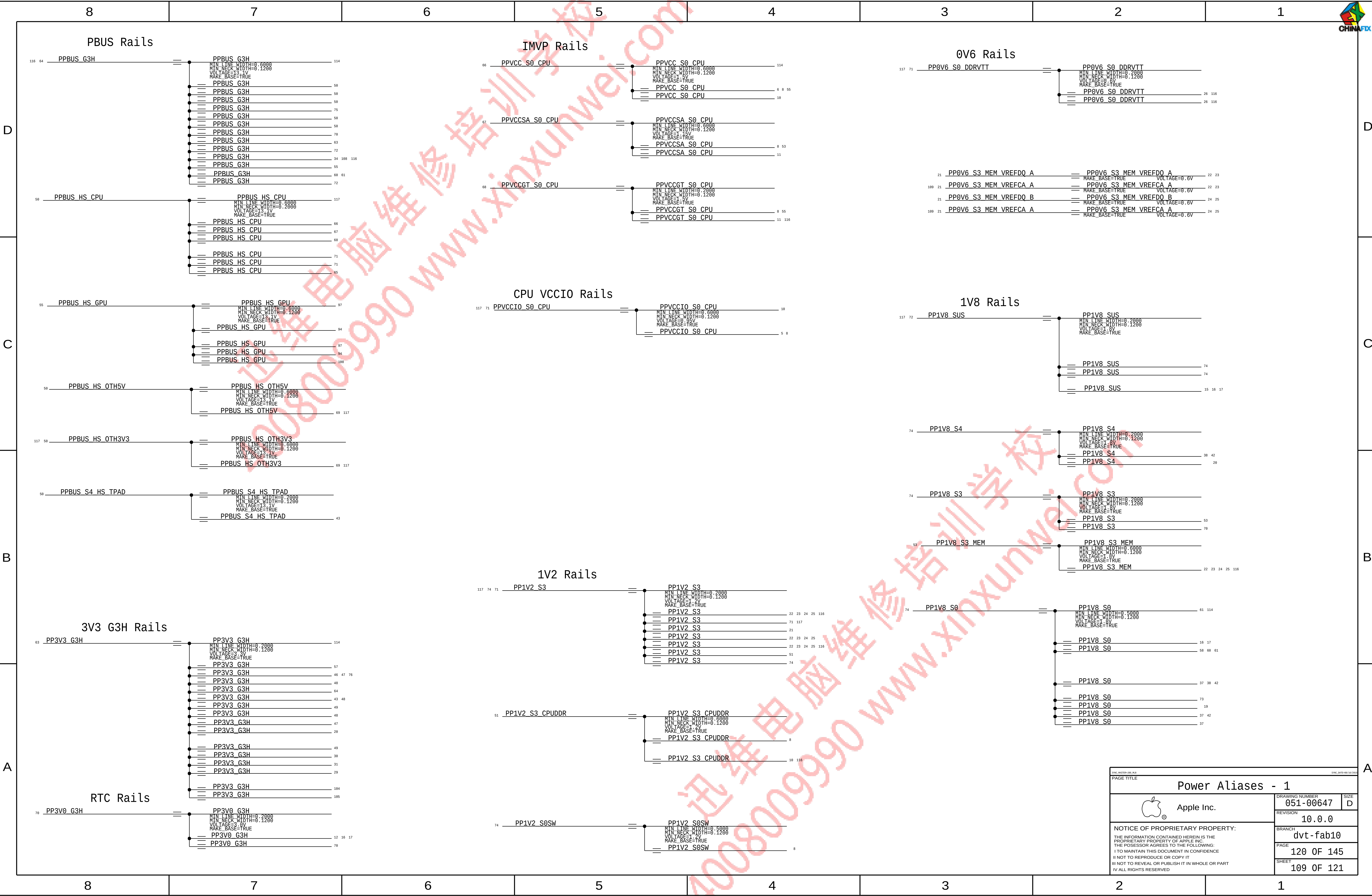


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 Apple Inc.		DRAWING NUMBER 051-00647	SIZE D
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TBT 5V REGULATOR		
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BOM_COST_GROUP=USB-C



SYMC: MASTER-000_RL0

SYMC: DATE: 01/10/2010

PAGE TITLE

Power Aliases - 1



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[illegible]

8		7		6		5		4		3		2		1		CHINAFOX	
Memory Bit/Byte Swizzle																	
D	LPDDR3 COMMAND/ADDRESS																
C																	
B																	
A																	
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BDW-H LPDDR3 NET				BIT SWIZZLE			
MAKE BASE=TRUE							
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MAKE_BASE			
113	7	TRUE	MEM B DQS P<0>
113	7	TRUE	MEM B DQS N<0>
113	7	TRUE	MEM B DQS P<1>
113	7	TRUE	MEM B DQS N<1>
113	7	TRUE	MEM B DQS P<2>
113	7	TRUE	MEM B DQS N<2>
113	7	TRUE	MEM B DQS P<3>
113	7	TRUE	MEM B DQS N<3>
113	7	TRUE	MEM B DQS P<4>
113	7	TRUE	MEM B DQS N<4>
113	7	TRUE	MEM B DQS P<5>
113	7	TRUE	MEM B DQS N<5>
113	7	TRUE	MEM B DQS P<6>
113	7	TRUE	MEM B DQS N<6>
113	7	TRUE	MEM B DQS P<7>
113	7	TRUE	MEM B DQS N<7>

SYNC_MASTER=380_MLB

SYNC_DATE=11/06/2015

PAGE TITLE

Memory Bit/Byte Swizzle

 Apple Inc.

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DRAWING NUMBER

051-00647

SIZE

D

REVISION

10.0.0

BRANCH

dvt-fab10

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High Speed NO_TEST

D

C

B

A

D

C

B

A

8	7	6	5	4	3	2	1
				High Speed NO_TEST			
DMI				CPU/PCH CLK			
DMI_S2N_P<3..0>				CPU_CLK24M_NSSC_CLK_N			
DMI_S2N_N<3..0>				CPU_CLK24M_NSSC_CLK_P			
DMI_N2S_P<3..0>				CPU_CLK100M_PCBCLK_N			
DMI_N2S_N<3..0>				CPU_CLK100M_PCBCLK_P			
DP - CPU/ACE				CPU_CLK100M_BCLK_N			
DP_X_SNK0_ML_C_N<3..0>				CPU_CLK100M_BCLK_P			
DP_X_SNK0_ML_C_P<3..0>				PCIE_CLK100M_SSD_N			
DP_X_SNK0_ML_N<3..0>				PCIE_CLK100M_SSD_P			
DP_X_SNK0_ML_P<3..0>				PCIE_CLK100M_SSD_LB_N			
DP_X_SNK1_ML_C_N<3..0>				PCIE_CLK100M_SSD_LB_P			
DP_X_SNK1_ML_C_P<3..0>				NC_ITPXPDP_CLK100MN			
DP_X_SNK1_ML_N<3..0>				NC_ITPXPDP_CLK100MP			
DP_X_SNK1_ML_P<3..0>				PCIE_CLK100M_AP_P			
DP_T_SNK0_ML_C_N<3..0>				PCIE_CLK100M_AP_N			
DP_T_SNK0_ML_C_P<3..0>							
DP_T_SNK0_ML_N<3..0>							
DP_T_SNK0_ML_P<3..0>							
DP_T_SNK1_ML_C_N<3..0>							
DP_T_SNK1_ML_C_P<3..0>							
DP_T_SNK1_ML_N<3..0>							
DP_T_SNK1_ML_P<3..0>							
DP_X_SNK0_AUXCH_C_P							
DP_X_SNK0_AUXCH_C_N							
DP_T_SNK0_AUXCH_C_P							
DP_T_SNK0_AUXCH_C_N							
DP_X_SNK1_AUXCH_P							
DP_T_SNK1_AUXCH_P							
DP_T_SNK1_AUXCH_N							
CPU/EDP							
EDP_AUXCH_C_N							
EDP_AUXCH_C_P							
PCH/AR							
PCIE_AP_D2R_P							
PCIE_AP_D2R_N							
PCIE_AP_R2D_P							
PCIE_AP_R2D_N							
PCIE_AP_R2D_C_P							
PCIE_AP_R2D_C_N							
PCH/SSD							
PCIE_SSD_D2R_N<3..0>							
PCIE_SSD_D2R_P<3..0>							
PCIE_SSD_R2D_N<3..0>							
PCIE_SSD_R2D_P<3..0>							
PCIE_SSD_D2R_C_N<3..0>							
PCIE_SSD_R2D_C_N<3..0>							
PCIE_SSD_D2R_N<0>							
PCIE_SSD_D2R_P<0>							
PCIE_SSD_D2R_C_P<3..0>							
PCIE_SSD_R2D_C_N<3..0>							
PCIE_SSD_D2R_LB_P<0>							
PCIE_SSD_R2D_LB_N<0>							
PCIE_SSD_D2R_P<3>							
PCIE_SSD_D2R_LB_N<0>							
PCIE_SSD_D2R_N<3>							
PCIE_SSD_R2D_LB_P<0>							
USB-C_X							
USBC_XB_D2R_N<2..1>							
USBC_XB_D2R_P<2..1>							
USBC_XA_D2R_P<2..1>							
USBC_XB_R2D_C_P<2..1>							
DP_XA_AUXCH_P							
USBC_XA_D2R_N<2..1>							
USBC_XB_R2D_C_N<2..1>							
USBC_XA_R2D_C_P<2..1>							
DP_XB_AUXCH_P							
USBC_XA_R2D_C_N<2..1>							
DP_XB_AUXCH_N							
DP_XA_AUXCH_N							
DP_XA_AUXCH_C_N							
DP_XA_AUXCH_C_P							
DP_XB_AUXCH_C_P							
DP_XB_AUXCH_C_N							
PEG							
PEG_GPU_D2R_N<7..0>							
PEG_GPU_D2R_P<7..0>							
PEG_GPU_R2D_C_N<7..0>							
PEG_GPU_R2D_C_P<7..0>							
PCIE_TBT_X_D2R_P<3..0>							
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PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
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PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
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PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							
PCIE_TBT_T_R2D_C_N<3..0>							
PCIE_TBT_T_R2D_C_P<3..0>							

Wifi & SSD fixture Test Points
(row,col)=(4,2) Diameter=0.6mm Pitch=1mm

Right Side - Top Layer

1	TPAC528
1	TPAC519
1	TPAC520
1	TPAC521
1	TPAC522
1	TPAC523
1	TPAC524
1	TPAC535

(row,col)=(2,6) Diameter=0.6mm Pitch=1mm

Left Side - Top Layer

1	TPAC500
1	TPAC501
1	TPAC502
1	TPAC503
1	TPAC504
1	TPAC505
1	TPAC506
1	TPAC507
1	TPAC529
1	TPAC512
1	TPAC536
1	TPAC518

(row,col)=(5,5) Diameter=0.6mm Pitch=1mm

Middle - Top Layer

1	TPAC517
1	TPAC538
1	TPAC539
1	TPAC527
1	TPAC510
1	TPAC511
1	TPAC595
1	TPAC596
1	TPAC598
1	TPAC597
1	TPAC599
1	TPAC525
1	TPAC526
1	TPAC537
1	TPAC530
1	TPAC531
1	TPAC532
1	TPAC533
1	TPAC534
1	TPAC508
1	TPAC509
1	TPAC513
1	TPAC514
1	TPAC515
1	TPAC516

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MESA CONNECTOR		
46	PP3V0_MESA_CONN	FUNC_TEST=TRUE
46	MESA_SPI_MISO_CONN	FUNC_TEST=TRUE
46	MESA_SNSR_INT_CONN	FUNC_TEST=TRUE
46	MESA_BOOST_EN_CONN	FUNC_TEST=TRUE
46	MESA_I2C_SDA	FUNC_TEST=TRUE
46	MESA_I2C_SCL	FUNC_TEST=TRUE
46	PP1V8_MESA_CONN	FUNC_TEST=TRUE
46	MESA_SPI_MOSI_CONN	FUNC_TEST=TRUE
46	MENU_KEY_L	FUNC_TEST=TRUE
46	MESA_SPI_CLK_CONN	FUNC_TEST=TRUE
46	PP16V0_MESA_CONN	FUNC_TEST=TRUE
46	GND	

SPI ROM		
114	PP3V3_G3H	FUNC_TEST=TRUE
57	SPI_ALT_I00_MOSI	FUNC_TEST=TRUE
57	SPI_ALT_I01_MISO	FUNC_TEST=TRUE
57	SPI_ALT_I02_WP_L	FUNC_TEST=TRUE
57	SPI_ALT_I03_HOLD_L	FUNC_TEST=TRUE
57	SMC_RESET_L	FUNC_TEST=TRUE
57	SPI_ALT_CLK	FUNC_TEST=TRUE
57	SPI_ALT_CS_L	FUNC_TEST=TRUE
57	SPIROM_USE_MLB	FUNC_TEST=TRUE
57	SMC_TMS	FUNC_TEST=TRUE
57	SMC_TCK	FUNC_TEST=TRUE
57	GND	

AUDIO AMP		
61	SPKRCONN_TR_OUT_POS	FUNC_TEST=TRUE
61	SPKRCONN_TR_OUT_NEG	FUNC_TEST=TRUE
61	SPKRCONN_WR_OUT_POS	FUNC_TEST=TRUE
61	SPKRCONN_WR_OUT_NEG	FUNC_TEST=TRUE
61	SPKRCONN_TL_OUT_POS	FUNC_TEST=TRUE
61	SPKRCONN_TL_OUT_NEG	FUNC_TEST=TRUE
61	SPKRCONN_WL_OUT_POS	FUNC_TEST=TRUE
61	SPKRCONN_WL_OUT_NEG	FUNC_TEST=TRUE
61	SPKR_I00	FUNC_TEST=TRUE
61	GND	

DFR Touch Conn		
42	DFR_TOUCH_LID	FUNC_TEST=TRUE
42	DFR_TOUCH_GP102	FUNC_TEST=TRUE
42	DFR_TOUCH_SPI_CS_L	FUNC_TEST=TRUE
42	DFR_TOUCH_SPI_MOSI_R	FUNC_TEST=TRUE
42	DFR_TOUCH_ROM_I2C_SCL	FUNC_TEST=TRUE
42	DFR_TOUCH_ROM_I2C_SDA	FUNC_TEST=TRUE
42	DFR_TOUCH_RESET_L	FUNC_TEST=TRUE
114	PP1V8_S0SW_DFR	X2
114	PP5V_S0_T139	
42	DFR_TOUCH_PANEL_DETECT	FUNC_TEST=TRUE
42	DFR_DISP_VSYNC	FUNC_TEST=TRUE
42	DFR_TOUCH_SPI_MISO_R	FUNC_TEST=TRUE
42	DFR_TOUCH_SPI_CLK_R	FUNC_TEST=TRUE
42	DFR_TOUCH_INT_L	FUNC_TEST=TRUE
42	DFR_CLKIN_RESET_L	FUNC_TEST=TRUE
42	DFR_TOUCH_ROM_WC	FUNC_TEST=TRUE
42	GND	X4

DFR Disp Conn		
114	DFR_DISP_VSYNC	FUNC_TEST=TRUE
114	DFR_DISP_TE	FUNC_TEST=TRUE
114	DFR_DISP_INT	FUNC_TEST=TRUE
114	DFR_DISP_RESET_L	FUNC_TEST=TRUE
114	PP3V3_S0SW_DFR	X3
114	MIPID_CLK_P	FUNC_TEST=TRUE
114	MIPID_CLK_N	FUNC_TEST=TRUE
114	MIPID_DATA_P	FUNC_TEST=TRUE
114	MIPID_DATA_N	FUNC_TEST=TRUE
114	PP1V8_S0SW_DFR	X2
114	DFRDRV_I2C_SCL	FUNC_TEST=TRUE
114	DFRDRV_I2C_SDA	FUNC_TEST=TRUE
114	GND	X6

KBD CONNECTOR		
43	GND_FAN	X2
43	PP5V_S0_FAN_CONN	X2
115	FAN_LT_TACH	FUNC_TEST=TRUE
115	FAN_LT_PWM	FUNC_TEST=TRUE
115	PP5V_S0_KBD	X2
115	PP3V3_G3H	FUNC_TEST=TRUE
114	KBD_BLC_GSSOUT	FUNC_TEST=TRUE
114	KBD_BLC_GSLAT	FUNC_TEST=TRUE
114	PP3V3_S4	FUNC_TEST=TRUE
114	KBD_BLC_GSSIN	FUNC_TEST=TRUE
114	KBD_BLC_XBLANK	FUNC_TEST=TRUE
115	FAN_RT_TACH	FUNC_TEST=TRUE
115	FAN_RT_PWM	FUNC_TEST=TRUE
114	KBD_I2C_SDA	FUNC_TEST=TRUE
114	KBD_INT_L	FUNC_TEST=TRUE
114	KBD_I2C_SCL	FUNC_TEST=TRUE
114	SMC_LSOC_RST_L	FUNC_TEST=TRUE
114	KBD_BLC_GSSCK	FUNC_TEST=TRUE
114	GND	X5

TPAD CONNECTOR		
114	KBD_I2C_SCL	FUNC_TEST=TRUE
114	KBD_I2C_SDA	FUNC_TEST=TRUE
114	KBD_INT_L	FUNC_TEST=TRUE
114	SMBUS_SMC_3_SDA	FUNC_TEST=TRUE
114	SMBUS_SMC_3_SCL	FUNC_TEST=TRUE
114	PP3V3_S4_TPAD	FUNC_TEST=TRUE
114	PP5V_S4_TPAD_CONN	FUNC_TEST=TRUE
114	ACT_GND	X4
114	PPVIN_S4_TPAD_FUSE	X4
114	TPAD_SPI_CLK	FUNC_TEST=TRUE
114	TPAD_SPI_IF_EN_CONN	FUNC_TEST=TRUE
114	TPAD_SPI_MISO	FUNC_TEST=TRUE
114	TPAD_SPI_CS_L_CONN	FUNC_TEST=TRUE
114	TPAD_SPI_MOSI	FUNC_TEST=TRUE
114	TPAD_SPI_INT_L_CONN	FUNC_TEST=TRUE
114	SMC_ACTUATOR_DISABLE_L	FUNC_TEST=TRUE
114	SMC_PMC_S4_WAKE_L	FUNC_TEST=TRUE
114	SMC_LID	FUNC_TEST=TRUE
114	KBD_BLC_GSLAT	FUNC_TEST=TRUE
114	KBD_BLC_GSSCK	FUNC_TEST=TRUE
114	KBD_BLC_GSSOUT	FUNC_TEST=TRUE
114	SMC_VIBE_L	FUNC_TEST=TRUE
114	KBD_BLC_GSSIN	X3
114	GND	

eDP		
116	PPVOUT_S0_LCDBKLT	FUNC_TEST=TRUE
116	PP3V3_S0SW_LCD	X2
116	EDP_AUXCH_C_N	FUNC_TEST=TRUE
116	EDP_AUXCH_C_P	FUNC_TEST=TRUE
116	EDP_INT_ML_N<0>	FUNC_TEST=TRUE
116	EDP_INT_ML_P<0>	FUNC_TEST=TRUE
116	EDP_INT_ML_N<1>	FUNC_TEST=TRUE
116	EDP_INT_ML_P<1>	FUNC_TEST=TRUE
116	EDP_INT_ML_N<2>	FUNC_TEST=TRUE
116	EDP_INT_ML_P<2>	FUNC_TEST=TRUE
116	EDP_INT_ML_N<3>	FUNC_TEST=TRUE
116	EDP_INT_ML_P<3>	FUNC_TEST=TRUE
116	MIPIC_CLK_N	FUNC_TEST=TRUE
116	MIPIC_CLK_P	FUNC_TEST=TRUE
116	PP5V_S0SW_LCD	FUNC_TEST=TRUE
116	PP5V_S0_ALSCAM_F	FUNC_TEST=TRUE
116	I2C_ALS_SDA	FUNC_TEST=TRUE
116	I2C_ALS_SCL	FUNC_TEST=TRUE
116	I2C_CAM_SCL	FUNC_TEST=TRUE
116	I2C_CAM_SDA	FUNC_TEST=TRUE
116	I2C_BKLT_SDA	FUNC_TEST=TRUE
116	I2C_BKLT_SCL	FUNC_TEST=TRUE
116	SMBUS_SMC_0_S0_SDA	FUNC_TEST=TRUE
116	SMBUS_SMC_0_S0_SCL	FUNC_TEST=TRUE
116	BKLT_PWM_TCON2MLB	FUNC_TEST=TRUE
116	BKLT_PWM_MLB2TCON	FUNC_TEST=TRUE
116	LCD_FSS	FUNC_TEST=TRUE
116	LCD_IRQ_L	FUNC_TEST=TRUE
116	BUF_EDP_PANEL_PWR_EN	FUNC_TEST=TRUE
116	DP_INT_HPD	FUNC_TEST=TRUE
116	GND	X3
116	SMBUS_SMC_0_S0_SCL	FUNC_TEST=TRUE
116	SMBUS_SMC_0_S0_SDA	FUNC_TEST=TRUE
116	MIPIC_DATA_P	FUNC_TEST=TRUE
116	MIPIC_DATA_N	FUNC_TEST=TRUE

AUDIO JACK		
62	AUD_CONN_HP_LEFT	FUNC_TEST=TRUE
62	AUD_CONN_HP_RIGHT	FUNC_TEST=TRUE
62	AUD_CONN_RING2	FUNC_TEST=TRUE
62	AUD_CONN_SLEEVE	FUNC_TEST=TRUE
62	AUD_CONN_HP_SENSE_L	FUNC_TEST=TRUE
62	AUD_CONN_HP_SENSE_R	FUNC_TEST=TRUE
62	AUD_CONN_TIP_SENSE	FUNC_TEST=TRUE
62	AUD_CONN_SLEEVE_XW	FUNC_TEST=TRUE
62	AUD_CONN_RING2_XW	FUNC_TEST=TRUE
62	GND	X2

BATTERY TUBA		
64	PPVBAT_G3H_CONN	X7
64	SMBUS_SMC_5_G3_SCL	FUNC_TEST=TRUE
64	SMBUS_SMC_5_G3_SDA	FUNC_TEST=TRUE
64	TP_BMON_IOUT	FUNC_TEST=TRUE
64	CHGR_AMON	FUNC_TEST=TRUE
64	CHGR_BMON	FUNC_TEST=TRUE
64	SYS_DETECT_L	FUNC_TEST=TRUE
64	GND	X5
114	SMC_LSOC_RST_L	FUNC_TEST=TRUE
64	TBA_COMP	FUNC_TEST=TRUE
64	TBA_VDDA	FUNC_TEST=TRUE
64	TBA_LX1	FUNC_TEST=TRUE
64	TBA_LX2	FUNC_TEST=TRUE
64	GND	X2
64	PPVBAT_G3H_CHGR_REG	X5
116	PPDCIN_G3H_CHGR	X5

USB_C		
TP_UPC_XA_SWD_DATA	TP3001	
TP_UPC_XA_SWD_CLK	TP3002	
TP_UPC_XB_SWD_DATA	TP3003	
TP_UPC_XB_SWD_CLK	TP3004	
104	TP_UPC_TA_DBG_UART_TX	FUNC_TEST=TRUE
104	TP_UPC_TA_DBG_UART_RX	FUNC_TEST=TRUE
104	TP_UPC_TB_DBG_UART_TX	FUNC_TEST=TRUE
104	TP_UPC_TB_DBG_UART_RX	FUNC_TEST=TRUE
104	UPC_XA_DBG_UART_TX	FUNC_TEST=TRUE
104	UPC_XA_DBG_UART_RX	FUNC_TEST=TRUE
104	TP_UPC_XB_DBG_UART_TX	FUNC_TEST=TRUE
104	TP_UPC_XB_DBG_UART_RX	FUNC_TEST=TRUE
116	PP20V_USBC_TB_VBUS_CONN	X5
116	PP20V_USBC_TA_VBUS_CONN	X5
116	GND	X5
116	PP20V_USBC_XB_VBUS_CONN	X4
116	PP20V_USBC_XA_VBUS_CONN	X5
116	GND	X5

SSD DEBUG CONNECTOR		
115	SSD_BOOT_LB_L	FUNC_TEST=TRUE
115	SSD_BOOT_LB_R	FUNC_TEST=TRUE
115	SSD_DEBUG_I2C_CLK_CONN	FUNC_TEST=TRUE
115	SSD_DEBUG_I2C_DAT_CONN	FUNC_TEST=TRUE
115	SSD_00B1_R2D_L	FUNC_TEST=TRUE
115	SSD_00B1_D2R_L	FUNC_TEST=TRUE
115	PP1V8_SSD_FMC	FUNC_TEST=TRUE
115	GND	

MIC		
98	DMIC1_CLK	FUNC_TEST=TRUE
98	DMIC1_DATA	FUNC_TEST=TRUE
98	DMIC2_CLK	FUNC_TEST=TRUE
98	DMIC2_DATA	FUNC_TEST=TRUE
98	PP1V8_S0	FUNC_TEST=TRUE
98	GND	

USB-C PROBE BLOCK TESTING		
98	TP_J3300_P2	1
98	TP_J3300_P56	1
98	TP_JB500_P2	1
98	TP_JB500_P56	1

ADDITIONAL TPs		
118	PP3V3_S5	1
118	PP3V3_SUS	1

8

7

6

5

4

3

2

1

8

7

6

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4

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2

1



OTHER TEST POINTS / NC

NC with No Testpoint Property

13	NC CLINK CLK	==	1	TRUE	NC CLINK CLK	
13	NC CLINK DATA	==	1	TRUE	NC CLINK DATA	
13	NC CLINK RESET_L	==	1	TRUE	NC_CLINK_RESET_L	
113 18 12	NC ITPXDP_CLK100MN	==	1	TRUE	NC_ITPXDP_CLK100MN	
113 18 12	NC ITPXDP_CLK100MP	==	1	TRUE	NC_ITPXDP_CLK100MP	
13	NC_HDA_SDIN1	==	1	TRUE	NC_HDA_SDIN1	
114	NC_SPI_SMC_MOSI					48
112	NC_SPI_SMC_MISO					48
118	NC_SPI_SMC_CS_L					48
119	NC_SPI_SMC_CLK					48
117	NC_PCH_CLK32K_RTCX2					28

CPU (Refer to CPU pages)

CPU_DC_B2_C1	----->	TP0501
CPU_DC_B38_C38	----->	TP0502
CPU_DC_BR2_BR1	----->	TP0503
CPU_DC_C1_B2	----->	TP0504
CPU_DC_C38_B38	----->	TP0505
CPU_DC_BR1_BR2	----->	TP0610
CPU_DC_BR38_BT36	----->	TP0900
CPU_DC_BT36_BR38	----->	TP0901

PCH (Refer to PCH pages)

XDP_PCH_OBSDATA_A0	----->	TP1883
XDP_PCH_OBSDATA_A1	----->	TP1884
XDP_PCH_OBSDATA_A2	----->	TP1870
XDP_PCH_OBSDATA_A3	----->	TP1871
XDP_PCH_OBSDATA_B0	----->	TP1872
XDP_PCH_OBSDATA_B1	----->	TP1885
XDP_PCH_OBSDATA_B2	----->	TP1886
XDP_PCH_OBSDATA_B3	----->	TP1887
XDP_PCH_OBSDATA_D0	----->	TP1877
XDP_PCH_OBSDATA_D1	----->	TP1878
XDP_PCH_OBSDATA_D2	----->	TP1879
XDP_PCH_OBSDATA_D3	----->	TP1880
XDP_PCH_OBSFN_C0	----->	TP1882
XDP_BPM_L<0>	----->	TP1800
XDP_BPM_L<1>	----->	TP1801
XDP_BPM_L<2>	----->	TP1802
XDP_BPM_L<3>	----->	TP1803
NC_USB_EXT_A_OC_L	----->	TP1873
NC_USB_EXT_B_OC_L	----->	TP1874
NC_USB_EXT_C_OC_L	----->	TP1875
NC_USB_EXT_D_OC_L	----->	TP1876

FAN Test Points

125	TRUE	FAN_LT_PWM	43 56 114
126	TRUE	FAN_LT_TACH	43 56 114
127	TRUE	PP5V_S0	38 110 115
128	TRUE	FAN_RT_PWM	43 56 114
129	TRUE	FAN_RT_TACH	43 56 114
130	TRUE	PP5V_S0	38 110 115

XDP Test-Points

152	TRUE	XDP_CPU_TCK	6 18
153	TRUE	XDP_PCH_TCK	13 18
154	TRUE	XDP_CPU_TDI	6 18
155	TRUE	XDP_CPU_TDO	6 18
156	TRUE	XDP_CPU_TRST_L	6 13 18
157	TRUE	XDP_CPU_TMS	6 18
158	TRUE	XDP_PCH_TMS	13 18
159	TRUE	XDP_PCH_TDI	13 18
160	TRUE	XDP_PCH_TDO	13 18
161	TRUE	XDP_CPU_PREQ_L	6 13 18
162	TRUE	XDP_CPU_PROQ_L	6 13 18
163	TRUE	PM_RSMRST_L	12 18 46 73
164	TRUE	PM_PCH_PWR0K	12 70
165	TRUE	PM_SYSRST_L	12 18 46
166	TRUE	CPU_CFG<3>	6 18

TPs on BOTTOM to check USB-C Installation

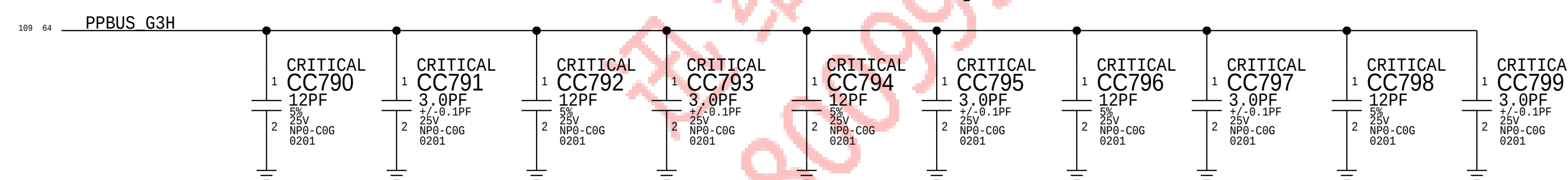
175	USBC_XA_SBU1	FUNC_TEST=TRUE	30 32
176	USBC_XB_SBU1	FUNC_TEST=TRUE	31 32
177	USBC_XA_SBU2	FUNC_TEST=TRUE	30 32
178	USBC_XB_SBU2	FUNC_TEST=TRUE	31 32
179	USBC_TA_SBU1	FUNC_TEST=TRUE	104 106
180	USBC_TB_SBU1	FUNC_TEST=TRUE	105 106
181	USBC_TA_SBU2	FUNC_TEST=TRUE	104 106
182	USBC_TB_SBU2	FUNC_TEST=TRUE	105 106
183	PP20V_USBC_XA_VBUS_CONN	FUNC_TEST=TRUE	32 114 116
184	PP20V_USBC_XB_VBUS_CONN	FUNC_TEST=TRUE	32 114 116
185	PP20V_USBC_TA_VBUS_CONN	FUNC_TEST=TRUE	106 114 116
186	PP20V_USBC_TB_VBUS_CONN	FUNC_TEST=TRUE	106 114 116

TPs to check LifeBoat Installation

188	PP3V3_S5_POLARIS	FUNC_TEST=TRUE	84 87
189	PP3V3_S5	FUNC_TEST=TRUE	110
190	SSD_PWR_EN	FUNC_TEST=TRUE	14 87 114
191	SSD_PWR_LB_EN	FUNC_TEST=TRUE	84 87
192	SSD_BOOT_L	FUNC_TEST=TRUE	15 87 114
193	SSD_BOOT_LB_L	FUNC_TEST=TRUE	77 87 114
194	SSD_RESET_L	FUNC_TEST=TRUE	14 20 87 114
195	SSD_RESET_LB_L	FUNC_TEST=TRUE	77 87
196	SSD_CLKREQ_L	FUNC_TEST=TRUE	20 87
197	SSD_CLKREQ_LB_L	FUNC_TEST=TRUE	77 87

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		PAGE	126 OF 145
		SHEET	115 OF 121

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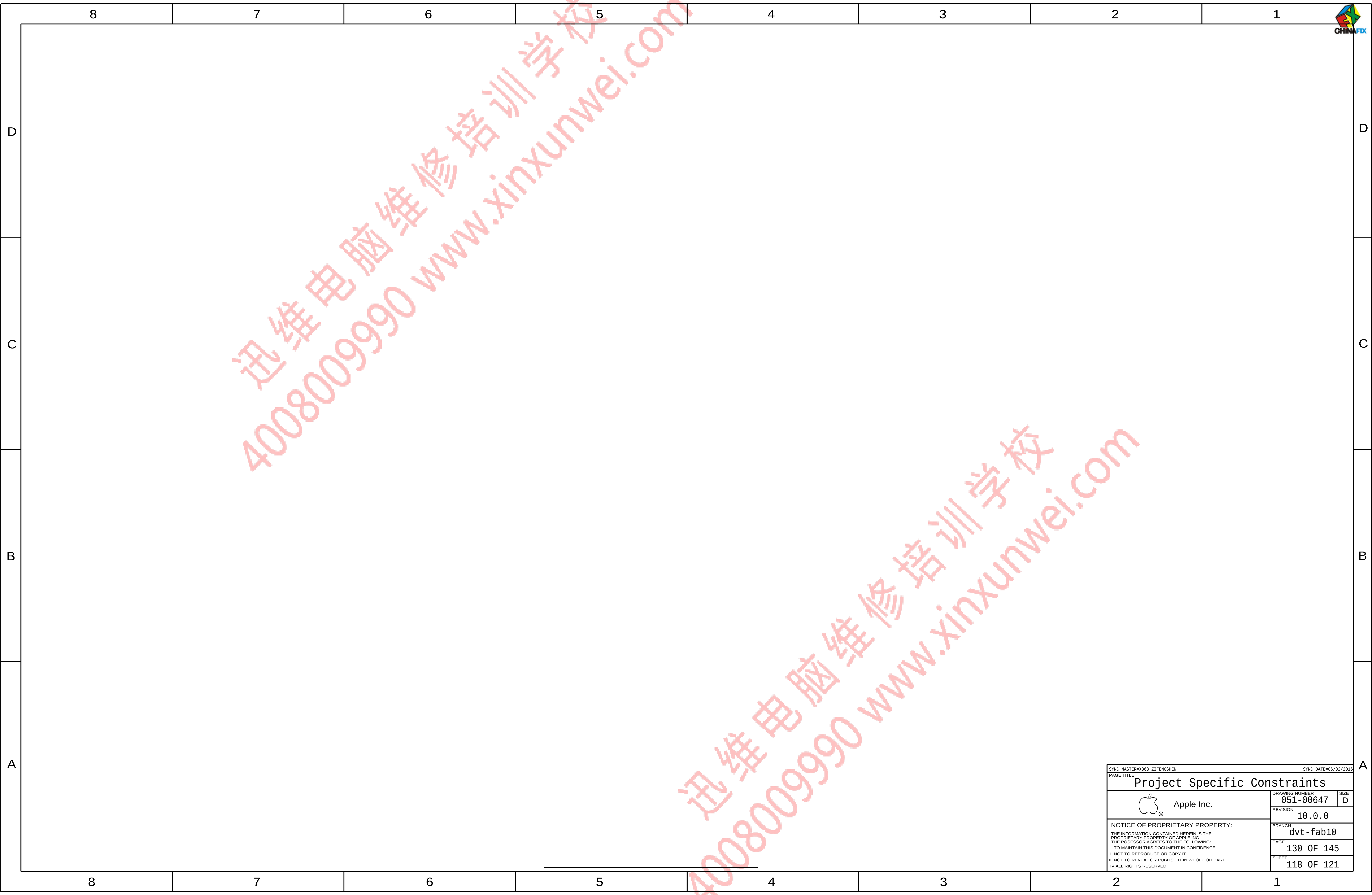
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D	BOM Variants 44 ULA Configs			36 PROA Configs					D																																																																																																																																								
	<table><tr><th>BOM NUMBER</th><th>BOM NAME</th><th>BOM OPTIONS</th></tr><tr><td>639-02212</td><td>MLB, 2.6G, MC-16, ULA-MC, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02120</td><td>MLB, 2.6G, MC-16, ULA-MC, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02591</td><td>MLB, 2.6G, MC-16, ULA-MC, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-02214</td><td>MLB, 2.6G, MC-16, ULA-MC, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-02587</td><td>MLB, 2.6G, SM-16, ULA-MC, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02116</td><td>MLB, 2.6G, SM-16, ULA-MC, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02589</td><td>MLB, 2.6G, SM-16, ULA-MC, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-02586</td><td>MLB, 2.6G, SM-16, ULA-MC, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01615</td><td>MLB, 2.7G, MC-16, ULA-MC, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02592</td><td>MLB, 2.7G, MC-16, ULA-MC, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01614</td><td>MLB, 2.7G, MC-16, ULA-MC, SSD-512, X36G3</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01603</td><td>MLB, 2.7G, SM-16, ULA-MC, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02594</td><td>MLB, 2.7G, SM-16, ULA-MC, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01602</td><td>MLB, 2.7G, SM-16, ULA-MC, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01639</td><td>MLB, 2.9G, MC-16, ULA-MC, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-01637</td><td>MLB, 2.9G, MC-16, ULA-MC, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02596</td><td>MLB, 2.9G, MC-16, ULA-MC, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01638</td><td>MLB, 2.9G, MC-16, ULA-MC, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01627</td><td>MLB, 2.9G, SM-16, ULA-MC, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-01625</td><td>MLB, 2.9G, SM-16, ULA-MC, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02598</td><td>MLB, 2.9G, SM-16, ULA-MC, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01626</td><td>MLB, 2.9G, SM-16, ULA-MC, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-02154</td><td>MLB, 2.6G, MC-16, ULA-SM, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02119</td><td>MLB, 2.6G, MC-16, ULA-SM, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02590</td><td>MLB, 2.6G, MC-16, ULA-SM, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-02213</td><td>MLB, 2.6G, MC-16, ULA-SM, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-02211</td><td>MLB, 2.6G, SM-16, ULA-SM, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02115</td><td>MLB, 2.6G, SM-16, ULA-SM, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02588</td><td>MLB, 2.6G, SM-16, ULA-SM, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-02585</td><td>MLB, 2.6G, SM-16, ULA-SM, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01612</td><td>MLB, 2.7G, MC-16, ULA-SM, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02593</td><td>MLB, 2.7G, MC-16, ULA-SM, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01611</td><td>MLB, 2.7G, MC-16, ULA-SM, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01600</td><td>MLB, 2.7G, SM-16, ULA-SM, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-02595</td><td>MLB, 2.7G, SM-16, ULA-SM, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01599</td><td>MLB, 2.7G, SM-16, ULA-SM, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01636</td><td>MLB, 2.9G, MC-16, ULA-SM, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-01634</td><td>MLB, 2.9G, MC-16, ULA-SM, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02597</td><td>MLB, 2.9G, MC-16, ULA-SM, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01635</td><td>MLB, 2.9G, MC-16, ULA-SM, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr><tr><td>639-01624</td><td>MLB, 2.9G, SM-16, ULA-SM, SSD-1TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB</td></tr><tr><td>639-01622</td><td>MLB, 2.9G, SM-16, ULA-SM, SSD-256, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB</td></tr><tr><td>639-02599</td><td>MLB, 2.9G, SM-16, ULA-SM, SSD-2TB, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB</td></tr><tr><td>639-01623</td><td>MLB, 2.9G, SM-16, ULA-SM, SSD-512, X363G</td><td>BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB</td></tr></table>			BOM NUMBER	BOM NAME	BOM OPTIONS	639-02212	MLB, 2.6G, MC-16, ULA-MC, SSD-1TB, X363G		BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02120	MLB, 2.6G, MC-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02591	MLB, 2.6G, MC-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-02214	MLB, 2.6G, MC-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-02587	MLB, 2.6G, SM-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02116	MLB, 2.6G, SM-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02589	MLB, 2.6G, SM-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-02586	MLB, 2.6G, SM-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01615	MLB, 2.7G, MC-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02592	MLB, 2.7G, MC-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01614	MLB, 2.7G, MC-16, ULA-MC, SSD-512, X36G3	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01603	MLB, 2.7G, SM-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02594	MLB, 2.7G, SM-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01602	MLB, 2.7G, SM-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01639	MLB, 2.9G, MC-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-01637	MLB, 2.9G, MC-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02596	MLB, 2.9G, MC-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01638	MLB, 2.9G, MC-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01627	MLB, 2.9G, SM-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-01625	MLB, 2.9G, SM-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02598	MLB, 2.9G, SM-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01626	MLB, 2.9G, SM-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-02154	MLB, 2.6G, MC-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02119	MLB, 2.6G, MC-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02590	MLB, 2.6G, MC-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-02213	MLB, 2.6G, MC-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-02211	MLB, 2.6G, SM-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02115	MLB, 2.6G, SM-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02588	MLB, 2.6G, SM-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-02585	MLB, 2.6G, SM-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01612	MLB, 2.7G, MC-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02593	MLB, 2.7G, MC-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01611	MLB, 2.7G, MC-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01600	MLB, 2.7G, SM-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-02595	MLB, 2.7G, SM-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01599	MLB, 2.7G, SM-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01636	MLB, 2.9G, MC-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-01634	MLB, 2.9G, MC-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02597	MLB, 2.9G, MC-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01635	MLB, 2.9G, MC-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB	639-01624	MLB, 2.9G, SM-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB	639-01622	MLB, 2.9G, SM-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB	639-02599	MLB, 2.9G, SM-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB	639-01623	MLB, 2.9G, SM-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB						
	BOM NUMBER	BOM NAME	BOM OPTIONS																																																																																																																																														
	639-02212	MLB, 2.6G, MC-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02120	MLB, 2.6G, MC-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																														
	639-02591	MLB, 2.6G, MC-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-02214	MLB, 2.6G, MC-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-02587	MLB, 2.6G, SM-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02116	MLB, 2.6G, SM-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																														
	639-02589	MLB, 2.6G, SM-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-02586	MLB, 2.6G, SM-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01615	MLB, 2.7G, MC-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02592	MLB, 2.7G, MC-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-01614	MLB, 2.7G, MC-16, ULA-MC, SSD-512, X36G3	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01603	MLB, 2.7G, SM-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02594	MLB, 2.7G, SM-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-01602	MLB, 2.7G, SM-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01639	MLB, 2.9G, MC-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-01637	MLB, 2.9G, MC-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																														
	639-02596	MLB, 2.9G, MC-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-01638	MLB, 2.9G, MC-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01627	MLB, 2.9G, SM-16, ULA-MC, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-01625	MLB, 2.9G, SM-16, ULA-MC, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																														
	639-02598	MLB, 2.9G, SM-16, ULA-MC, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-01626	MLB, 2.9G, SM-16, ULA-MC, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_MC,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-02154	MLB, 2.6G, MC-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02119	MLB, 2.6G, MC-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																														
	639-02590	MLB, 2.6G, MC-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-02213	MLB, 2.6G, MC-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-02211	MLB, 2.6G, SM-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02115	MLB, 2.6G, SM-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																														
	639-02588	MLB, 2.6G, SM-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-02585	MLB, 2.6G, SM-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01612	MLB, 2.7G, MC-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02593	MLB, 2.7G, MC-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-01611	MLB, 2.7G, MC-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01600	MLB, 2.7G, SM-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
	639-02595	MLB, 2.7G, SM-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																														
	639-01599	MLB, 2.7G, SM-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.7, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																														
	639-01636	MLB, 2.9G, MC-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																														
639-01634	MLB, 2.9G, MC-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																															
639-02597	MLB, 2.9G, MC-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																															
639-01635	MLB, 2.9G, MC-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, MICRON_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																															
639-01624	MLB, 2.9G, SM-16, ULA-SM, SSD-1TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:1TB																																																																																																																																															
639-01622	MLB, 2.9G, SM-16, ULA-SM, SSD-256, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:256GB																																																																																																																																															
639-02599	MLB, 2.9G, SM-16, ULA-SM, SSD-2TB, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:2TB																																																																																																																																															
639-01623	MLB, 2.9G, SM-16, ULA-SM, SSD-512, X363G	BASE_BOM,DEVEL_BOM,CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 4GB_SM,BAFFIN,BAFFIN_ULA,SSD_CONF1G:512GB																																																																																																																																															
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		REVISION	10.0.0	
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8		7		6		5		4		3		2		1				
BOM Variants																		
48 LEA Configs																		
D	BOM NUMBER		BOM NAME				BOM OPTIONS											
	639-02621		MLB, 2.6G, MC-16, LEA-HY, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02618		MLB, 2.6G, MC-16, LEA-HY, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02622		MLB, 2.6G, MC-16, LEA-HY, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02620		MLB, 2.6G, MC-16, LEA-HY, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02608		MLB, 2.6G, SM-16, LEA-HY, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02602		MLB, 2.6G, SM-16, LEA-HY, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02611		MLB, 2.6G, SM-16, LEA-HY, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02605		MLB, 2.6G, SM-16, LEA-HY, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02645		MLB, 2.9G, MC-16, LEA-HY, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02643		MLB, 2.9G, MC-16, LEA-HY, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02646		MLB, 2.9G, MC-16, LEA-HY, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
C	639-02644		MLB, 2.9G, MC-16, LEA-HY, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02633		MLB, 2.9G, SM-16, LEA-HY, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02631		MLB, 2.9G, SM-16, LEA-HY, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02634		MLB, 2.9G, SM-16, LEA-HY, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02632		MLB, 2.9G, SM-16, LEA-HY, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_HY_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02647		MLB, 2.6G, MC-16, LEA-MC, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02616		MLB, 2.6G, MC-16, LEA-MC, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02619		MLB, 2.6G, MC-16, LEA-MC, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02617		MLB, 2.6G, MC-16, LEA-MC, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02607		MLB, 2.6G, SM-16, LEA-MC, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02601		MLB, 2.6G, SM-16, LEA-MC, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02610		MLB, 2.6G, SM-16, LEA-MC, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
B	639-02604		MLB, 2.6G, SM-16, LEA-MC, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02641		MLB, 2.9G, MC-16, LEA-MC, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02639		MLB, 2.9G, MC-16, LEA-MC, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02642		MLB, 2.9G, MC-16, LEA-MC, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02640		MLB, 2.9G, MC-16, LEA-MC, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02629		MLB, 2.9G, SM-16, LEA-MC, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02627		MLB, 2.9G, SM-16, LEA-MC, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02630		MLB, 2.9G, SM-16, LEA-MC, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02628		MLB, 2.9G, SM-16, LEA-MC, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_MC_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02614		MLB, 2.6G, MC-16, LEA-SM, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02612		MLB, 2.6G, MC-16, LEA-SM, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02615		MLB, 2.6G, MC-16, LEA-SM, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
A	639-02613		MLB, 2.6G, MC-16, LEA-SM, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02606		MLB, 2.6G, SM-16, LEA-SM, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02600		MLB, 2.6G, SM-16, LEA-SM, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02609		MLB, 2.6G, SM-16, LEA-SM, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02603		MLB, 2.6G, SM-16, LEA-SM, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.6, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02637		MLB, 2.9G, MC-16, LEA-SM, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02635		MLB, 2.9G, MC-16, LEA-SM, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02638		MLB, 2.9G, MC-16, LEA-SM, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
	639-02636		MLB, 2.9G, MC-16, LEA-SM, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, MICRON_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB											
	639-02625		MLB, 2.9G, SM-16, LEA-SM, SSD-1TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:1TB											
	639-02623		MLB, 2.9G, SM-16, LEA-SM, SSD-256, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:256GB											
	639-02626		MLB, 2.9G, SM-16, LEA-SM, SSD-2TB, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:2TB											
639-02624		MLB, 2.9G, SM-16, LEA-SM, SSD-512, X363G				BASE_BOM, DEVEL_BOM, CPU_SKL:2.9, RAM_16G, SAMSUNG_2133, 2GB_SM_BAFFIN, BAFFIN_LEA, SSD_CONFIG:512GB												
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SYNC_MASTER=J80_MLB		SYNC_DATE=07/23/2015	
PAGE TITLE			
639 BOM Configuration 2			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00647	D
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		BRANCH	dvt-fab10
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Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
740S0144	740S0118		ALL	
740S00003	740S0135		ALL	
376S1009	376S1128		ALL	
376S1000	376S0020		ALL	
376S00006	376S0761		ALL	
376S00074	376S0055		ALL	
376S00014	376S0761		ALL	
372S0106	372S0185		ALL	
371S0713	371S0558		ALL	
311S00060	311S0273		ALL	
311S00004	311S0370		ALL	
155S0014	155S0097		ALL	
155S0094	155S0387		ALL	
155S0060	155S0513		ALL	
155S00154	155S0398		ALL	
155S00007	155S0667		ALL	
138S0063	138S0053		ALL	
138S0775	138S0060		ALL	
138S0703	138S0048		ALL	
132S00064	132S0409		ALL	
128S0325	128S0397		ALL	
128S00029	128S00007		ALL	
128S00026	128S00011		ALL	
128S00070	128S00007		ALL	
128S00009	128S00007		ALL	
107S0249	107S0251		ALL	
107S00071	107S00053		ALL	
107S00070	107S0005		ALL	
107S00033	107S00034		ALL	
107S00015	107S00011		ALL	
376S1106	376S0678		ALL	Fairchild alt to Vishay
138S0738	138S1101		ALL	Samsung alt to Murata
138S0846	138S0011		ALL	Samsung alt to Murata
376S1053	376S0604		ALL	Diodes alt to Fairchild
152S000359	152S00253		ALL	Chillisin alt to Cyttec
740S00027	740S0159		ALL	Bourns alt to Little Fuse
371S0704	371S00077		ALL	NXP alt to Diodes
138S00032	138S0031		ALL	
138S00049	138S0031		ALL	

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<rdar://problem/23564405> for EG
<rdar://problem/23110001> for IG
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BLC

BLC

990-04070	990-04071		OFF	Hynix alt to SS
128S00010	128S00011		ALL	
128S00031	128S00011		ALL	
138S00004	138S00060		ALL	
155S00155	155S0441		OFF	
155S00190	155S0097		ALL	
353S00107	353S3239		ALL	
353S00525	353S4471		ALL	
376S1193	376S00037		OFF	
740S00028	740S0118		ALL	
152S00309	152S00268		ALL	Cyntec w/ NEC
128S0296	128S0407		ALL	NEC w/ pana
128S00012	128S0407		ALL	NEC w/ Rohm
155S00180	155S0275		ALL	Murata w/ Taiyo
155S00018	155S0664		ALL	Murata w/ Taiyo
152S00388	152S00182		ALL	
107S0240	107S0255		ALL	
128S00062	128S00067		ALL	NEC for Panasonic
138S0660	138S0604		ALL	
155S00204	155S0731		ALL	

] 220uF 2.0V D-case, 128S00044 DQ

- T208

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150uF 6.3V B12

<rdar://problem/24316249>

<rdar://problem/24310037>

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0271	311S00008		ALL	NXP w/ Diodes
197S00046	197S00036		ALL	Epson w/ TXC
197S00047	197S00036		ALL	Kyocera w/ TXC
197S00048	197S00036		ALL	Murata w/ TXC
197S00053	197S00050		ALL	Kyocera w/ TXC
197S00054	197S00050		ALL	NDK w/ TXC
197S00055	197S00050		ALL	Murata w/ TXC
311S0596	311S0593		ALL	NXP w/ Diodes
107S0276	107S00020		ALL	Cyntec w/ TFT
107S00021	107S0284		ALL	TFT w/ Yageo
132S00012	132S0401		ALL	Taiyo w/ Murata&TDK
152S00343	152S1102		ALL	NXP w/ Diodes
107S00007	107S00029		ALL	TFT w/ Yageo
128S00057	128S00018		ALL	NEC w/ Vishay
128S00058	128S00018		ALL	NEC w/ Rohm
128S0364	128S0264		ALL	Kemet w/ Panasonic
138S0041	138S0700		ALL	Murata w/ SS&Taiyo
138S0730	138S0706		ALL	NEC w/ Vishay
138S0945	138S0706		ALL	NEC w/ Rohm
152S00358	152S00208		ALL	Murata w/ Chillisin
152S00309	152S00241		OFF	Cyntec w/ Vishay
152S00390	152S00265		OFF	Cyntec w/ Vishay
152S00400	152S1072		ALL	Murata w/ Cyntec
152S1072	152S00361		ALL	Murata w/ Cyntec
155S00034	155S0706		ALL	Taiyo w/ Murata

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371S00002	371S00046		ALL	On-Semi w/ Diodes
376S00146	376S1001		ALL	NXP w/ Diodes
353S00711	353S2073		ALL	On Semi w/ TI

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740S00019	740S00007		ALL	Bourns w/ Polytronics
155S00109	155S00342		ALL	Murata w/ Taiyo
132S0438	132S0428		ALL	Murata w/ Taiyo&TDK
138S0714	138S0713		ALL	Murata w/ Samsung
138S0715	138S0732		ALL	Murata w/ Samsung
107S00000	107S00056		ALL	TFT w/ Cyntec
138S00075	138S00078		ALL	Taiyo w/ Mur&SS
138S0706	138S0705		ALL	Murata w/ Samsung
155S0302	155S0059		ALL	Murata w/ TDK
152S2052	152S1954		ALL	Taiyo w/ Cyntec
152S2015	152S1950		ALL	Taiyo w/ Cyntec
128S00055	128S00002		ALL	Kemet w/ Panasonic
138S0748	138S0751		ALL	Murata w/ SS
138S00102	138S0773		ALL	Murata w/ Taiyo
138S0709	138S0941		ALL	Murata w/ SS
107S00101	107S00005		ALL	Cyntec w/ Yageo
107S00102	107S00017		ALL	Cyntec w/ Yageo
107S00100	107S00057		ALL	Cyntec w/ TFT
107S00103	107S00058		ALL	Cyntec w/ Yageo
107S00104	107S00061		ALL	Cyntec w/ Yageo
107S00105	107S00062		ALL	Cyntec w/ Yageo
152S00403	152S00322		ALL	Murata w/ Chillisin
353S00052	353S4262		ALL	TI w/ OnSemi

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138S00104	138S0978		ALL	
128S00069	128S00067		ALL	Rohm for Panasonic
138S0759	138S0762		ALL	
377S0077	377S0103		ALL	
152S00363	152S00048		ALL	
138S00111	138S00036		ALL	
138S00097	138S0750		ALL	
155S00203	155S0094		ALL	
116S00006	116S0175		ALL	
311S00104	311S00091		ALL	

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311S00013	311S00008		ALL	Part will be stuffed in production
311S00118	311S0409		ALL	
311S00072	311S0657		ALL	
311S00090	311S00020		ALL	
353S00750	353S00077		ALL	
353S00078	353S00599		ALL	
311S00105	311S0233		ALL	
353S00080	353S3452		ALL	
311S00007	311S0426		ALL	
128S0445	128S0392		ALL	
152S00415	152S00140		ALL	
128S0436	128S0392		ALL	
343S00135	343S00136		ALL	T208
343S00137	343S00136		ALL	T208
343S00138	343S00136		ALL	T208
138S00105	138S00037		ALL	
152S00434	152S1029		ALL	
353S3527	353S3528		ALL	
353S3526	353S3528		ALL	
353S00135	353S00034		ALL	
353S2220	353S00034		ALL	
353S00769	353S4390		ALL	
353S00079	353S00754		ALL	
104S00012	155S0390		ALL	
353S4342	353S00054		ALL	
152S00543	152S00404		ALL	
371S00089	371S00005		ALL	
107S00111	107S00110		ALL	
335S00213	335S00008		ALL	
311S0437	311S00112		ALL	
377S0178	377S00031		ALL	
371S00091	371S00003		ALL	
197S00060	197S00000		ALL	
138S00090	138S00113		ALL	
152S00461	152S00112		ALL	
371S00074	371S00002		ALL	
197S00002	197S00001		ALL	
353S00091	353S00020		ALL	
131S00134	131S00041		ALL	

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